37 SENSOR KIT TUTORIAL

FOR UNO AND MEGA

V1.0.18.10.25

Preface

Our Company

Established in 2011, Elegoo Inc. is a thriving technology company dedicated to research & development, production, and marketing of open-source hardware. Located in Shenzhen, the Silicon Valley of China, we have grown to over 150+ employees with a 10,763+ square ft.factory.

Our product lines include DuPont wires, UNO R3 boards and complete starter kits designed for customers of any level to learn Arduino knowledge. In addition, we also sell Raspberry Pi accessories like TFT touch screens. Additionally, we plan to expand our offerings to include other technologies, including products related to 3D printing. All of our products comply with international quality standards and have been praised by our customers in a variety of different marketplaces throughout the world.

Official website:

http://www.elegoo.com

US Amazon storefront: http://www.amazon.com/shops/A2WWHQ25ENKVJ1

CA Amazon storefront: http://www.amazon.ca/shops/A2WWHQ25ENKVJ1

UK Amazon storefront: http://www.amazon.co.uk/shops/AZF7WYXU5ZANW

DE Amazon storefront: http://www.amazon.de/shops/AZF7WYXU5ZANW

FR Amazon storefront: http://www.amazon.fr/shops/AZF7WYXU5ZANW

ES Amazon storefront: http://www.amazon.es/shops/AZF7WYXU5ZANW

IT Amazon storefront: http://www.amazon.it/shops/AZF7WYXU5ZANW

Our Tutorial

This tutorial is designed for beginners. You will learn all the basic information about how to use the Arduino controller board, sensors and components. If you want to study Arduino in more depth, we recommend that you read the book "Arduino Cookbook" by Michael Margolis.

Some code in this tutorial has been edited by Simon Monk. Simon Monk is th e author of a number of books relating to Open Source Hardware. Some of his works include "Programming Arduino", "30 Arduino Projects for the Evil Genius" and "Programming the Raspberry Pi" and can be found on Amazon

Customer Service

As a continuously and quickly growing technology company, we strive to of fer you excellent products and quality service. You can reach out to us by email at service@elegoo.com or EUservice@elegoo.com. We look forward to hearing from you and any comments or suggestions are of great value to us.

Any problems or questions that you have with our products will be promptly answered by our experienced engineers within 12 hours (24hrs during holiday)

Packing List


HEARTBEAT
2PCS LIGHT CUP
HALL MAGNETIC
7 COLOR FLASH
HYDRARGYRUM -SWITCH
18B20 TEMP
LASER EMISSION
TILT-SWITCH
ANALOG TEMP
BUTTON
PHOTOR-RESISTOR
IR EMISSION
SHOCK
TEMP AND HUMIDITY
IR RECEIVER
ROTARY ENCODERS
ANALOG HALL
TAP MODULE
PHOTO INTERRUPTER

Lesson Listing

Lesson 0 Installing IDE

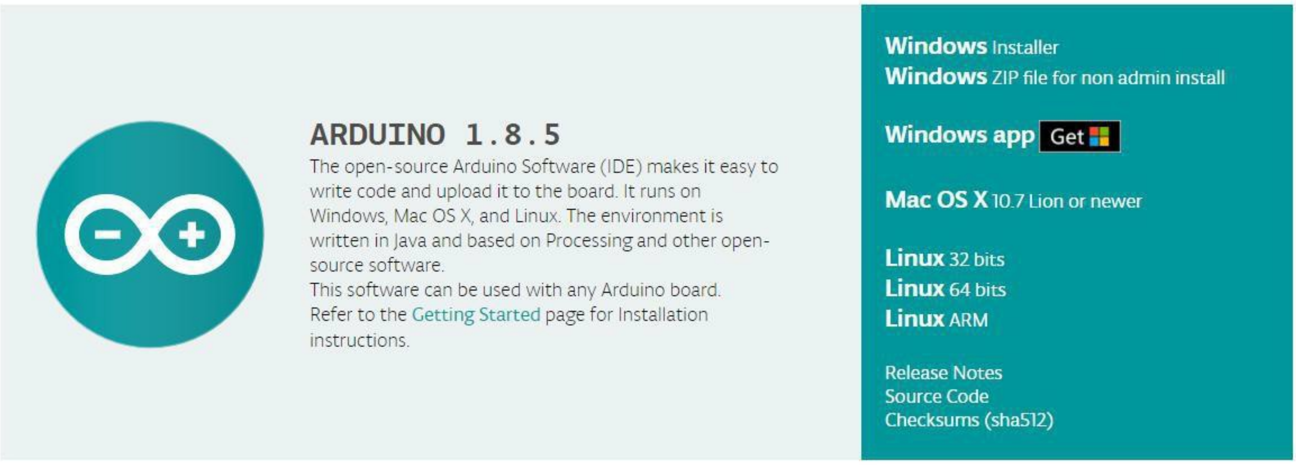
Introduction

The Arduino Integrated Development Environment (IDE) is the software side of the Arduino platform.

In this lesson, you will learn how to setup your computer to use Arduino and how to set about the lessons that follow.

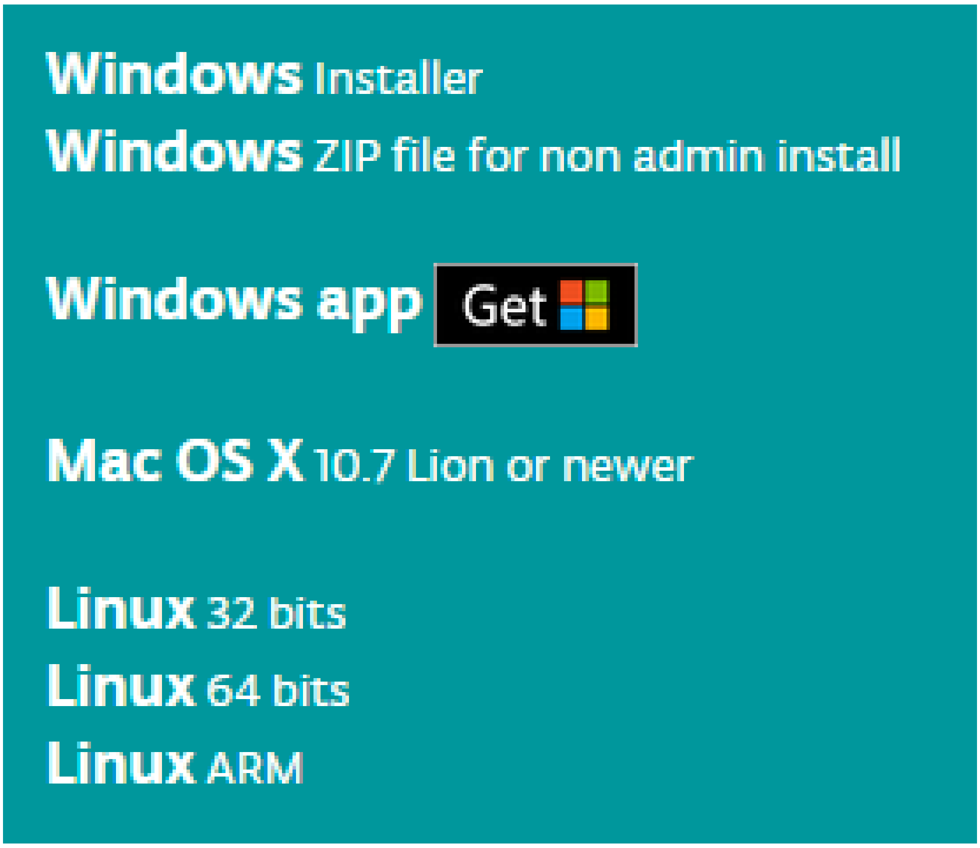
The Arduino software that you will use to program your Arduino is available for Windows, Mac and Linux. The installation process is different for all three platforms and unfortunately there is a certain amount of manual work to install the software.

STEP 1: Go to https://www.arduino.cc/en/Main/Software and find below page.



The version available at this website is usually the latest version, and the actual version may be newer than the version in the picture.

STEP2： Download the development software that is compatible with the operating system of your computer. <u>Take Windows as an example here.</u>

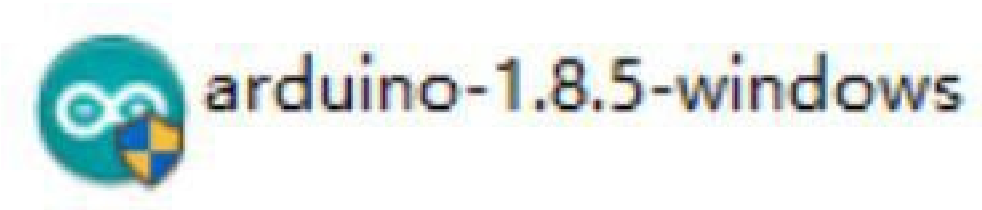


Click *WindowsInstaller*.

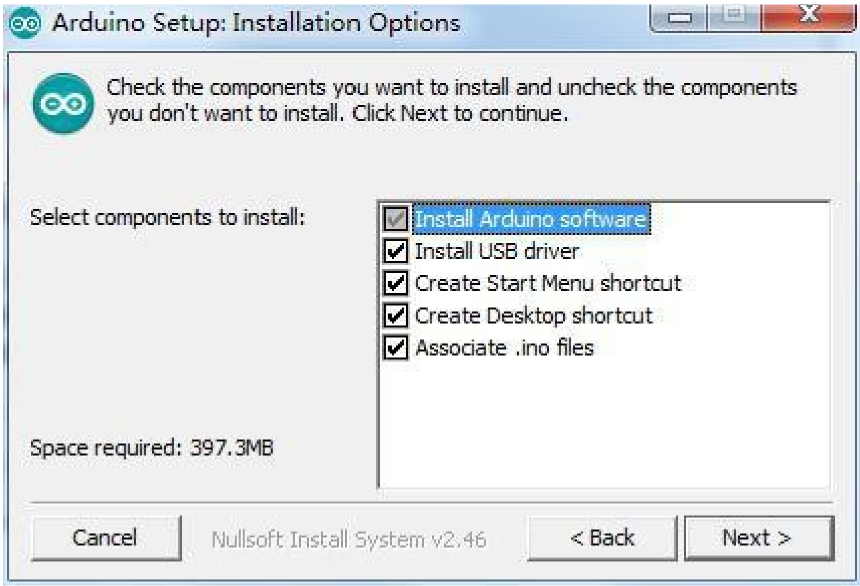
Click *JUST DOWNLOAD*.

Installing Arduino (Windows)

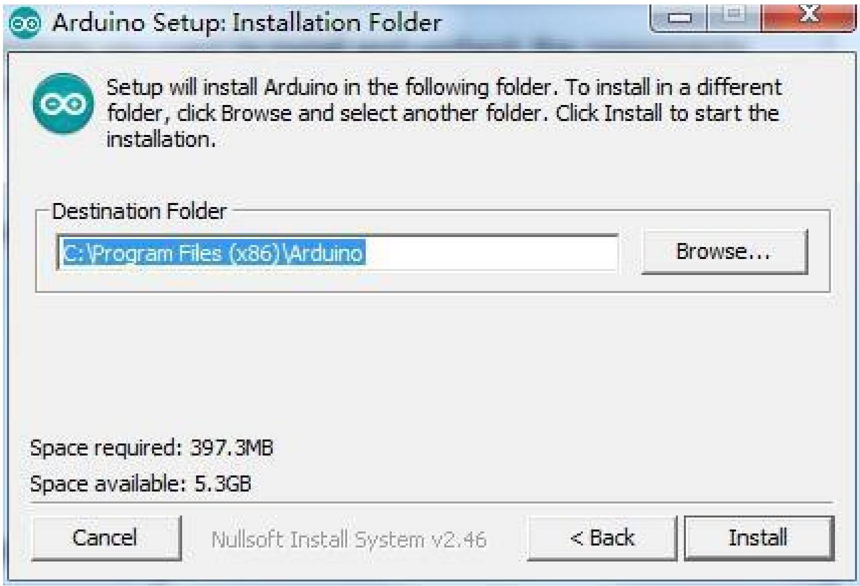
Install Arduino with the exe. Installation package.



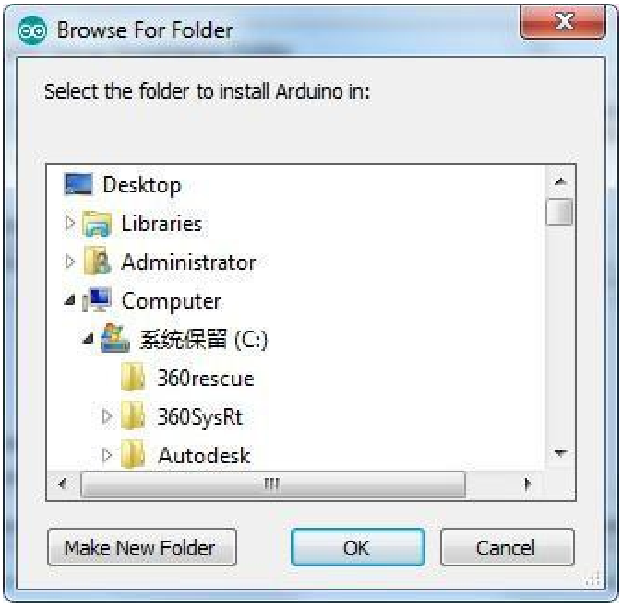
Click *I Agree* to see the following interface



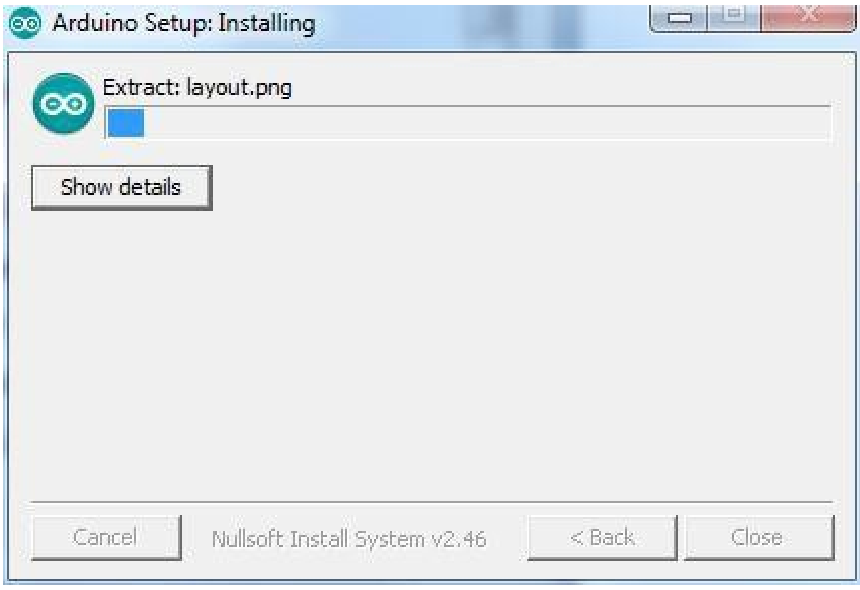
Click *Next*



You can press Browse… to choose an installation path or directly type in the directory you want.



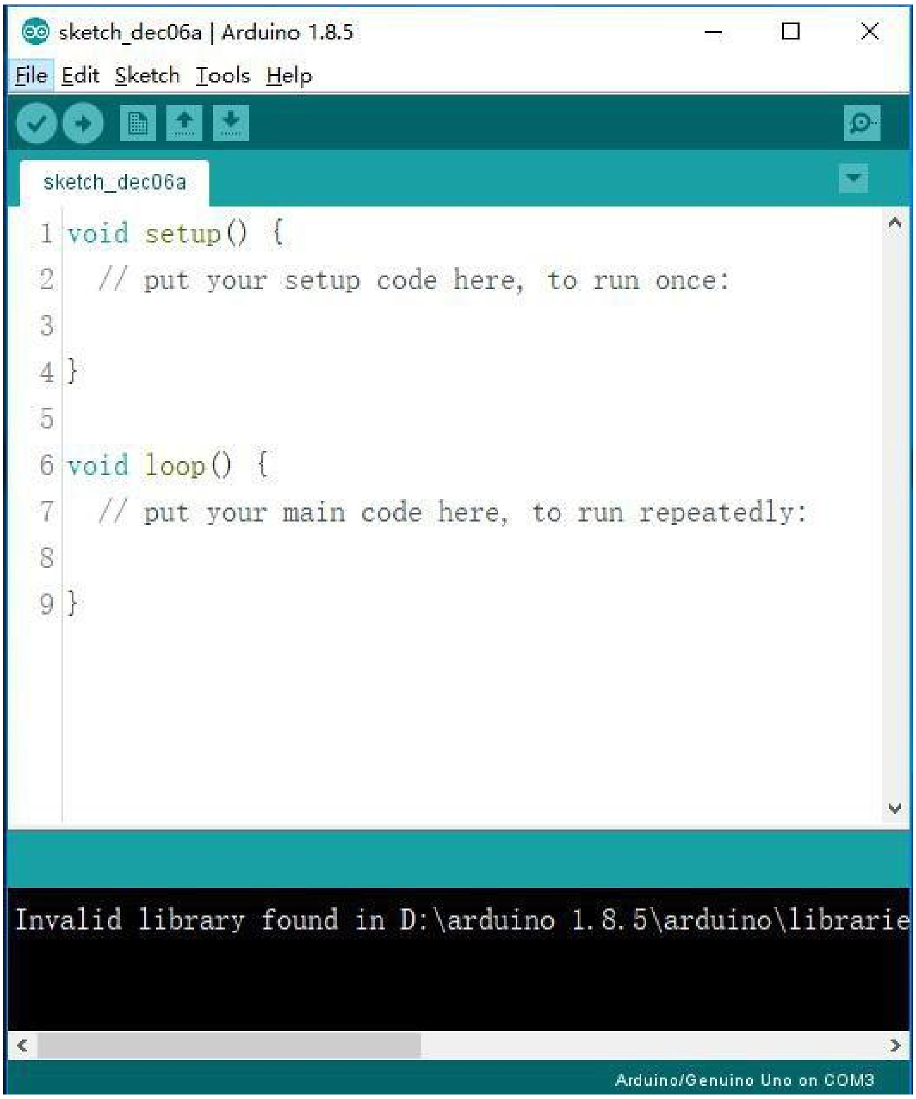
Click *Install* to initiateinstallation



Finally, the following interface appears, click *Install* to finish the installation.

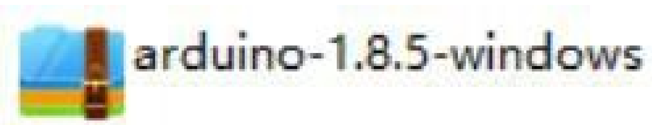
Next, the following icon appears on thedesktop

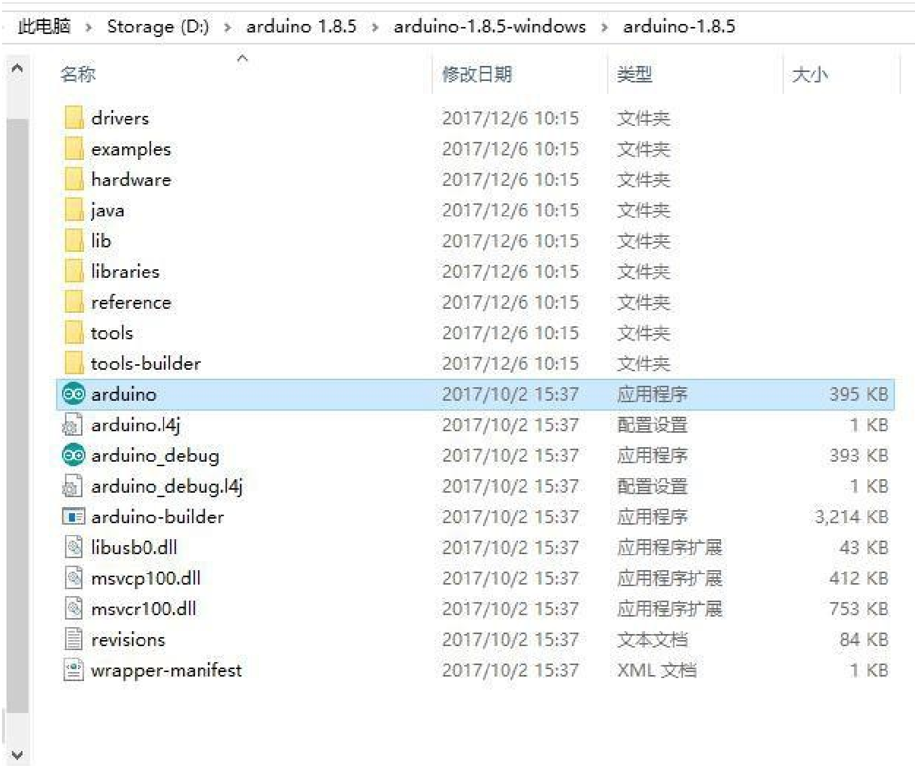
Double-click to enter the desired development environment

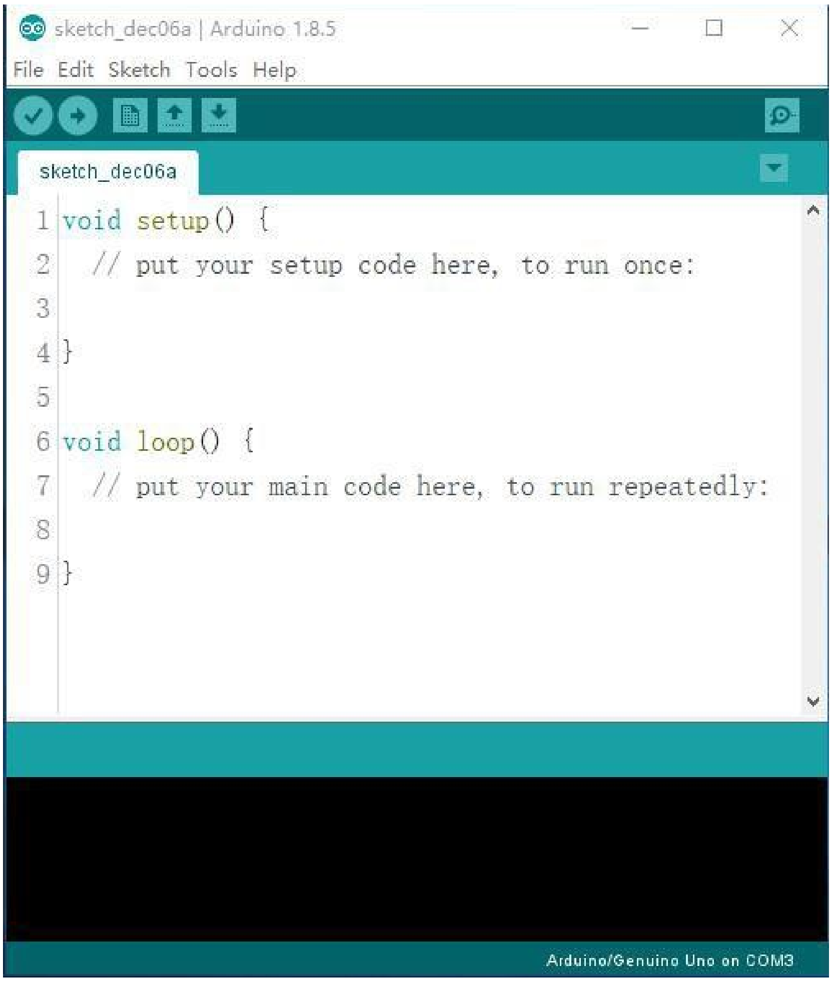


You may directly choose the installation package for installation and skip the contents below and jump to the next section. But if you want to learn some methods other than the installation package, please continue to read the section.

Unzip the zip file downloaded, Double-click to open the program and enter the desired development environment




此电脑 › Storage (D:) › arduino 1.8.5 › arduino-1.8.5-windows › arduino-1.8.5
名称
修改日期
类型
大小
drivers 2017/12/6 10:15 文件夹
examples 2017/12/6 10:15 文件夹
hardware 2017/12/6 10:15 文件夹
java 2017/12/6 10:15 文件夹
lib 2017/12/6 10:15 文件夹
libraries 2017/12/6 10:15 文件夹
reference 2017/12/6 10:15 文件夹
tools 2017/12/6 10:15 文件夹
tools-builder 2017/12/6 10:15 文件夹
arduino 2017/10/2 15:37 应用程序 395 KB
arduino.l4j 2017/10/2 15:37 配置设置 1 KB
arduino_debug 2017/10/2 15:37 应用程序 393 KB
arduino_debug.l4j 2017/10/2 15:37 配置设置 1 KB
arduino-builder 2017/10/2 15:37 应用程序 3,214 KB
libusb0.dll 2017/10/2 15:37 应用程序扩展 43 KB
msvcp100.dll 2017/10/2 15:37 应用程序扩展 412 KB
msvcr100.dll 2017/10/2 15:37 应用程序扩展 753 KB
revisions 2017/10/2 15:37 文本文档 84 KB
wrapper-manifest 2017/10/2 15:37 XML 文档 1 KB


sketch_dec06a | Arduino 1.8.5
File Edit Sketch Tools Help
sketch_dec06a
void setup() {
// put your setup code here, to run once:
}
void loop() {
// put your main code here, to run repeatedly:
}
Arduino/Genuino Uno on COM3

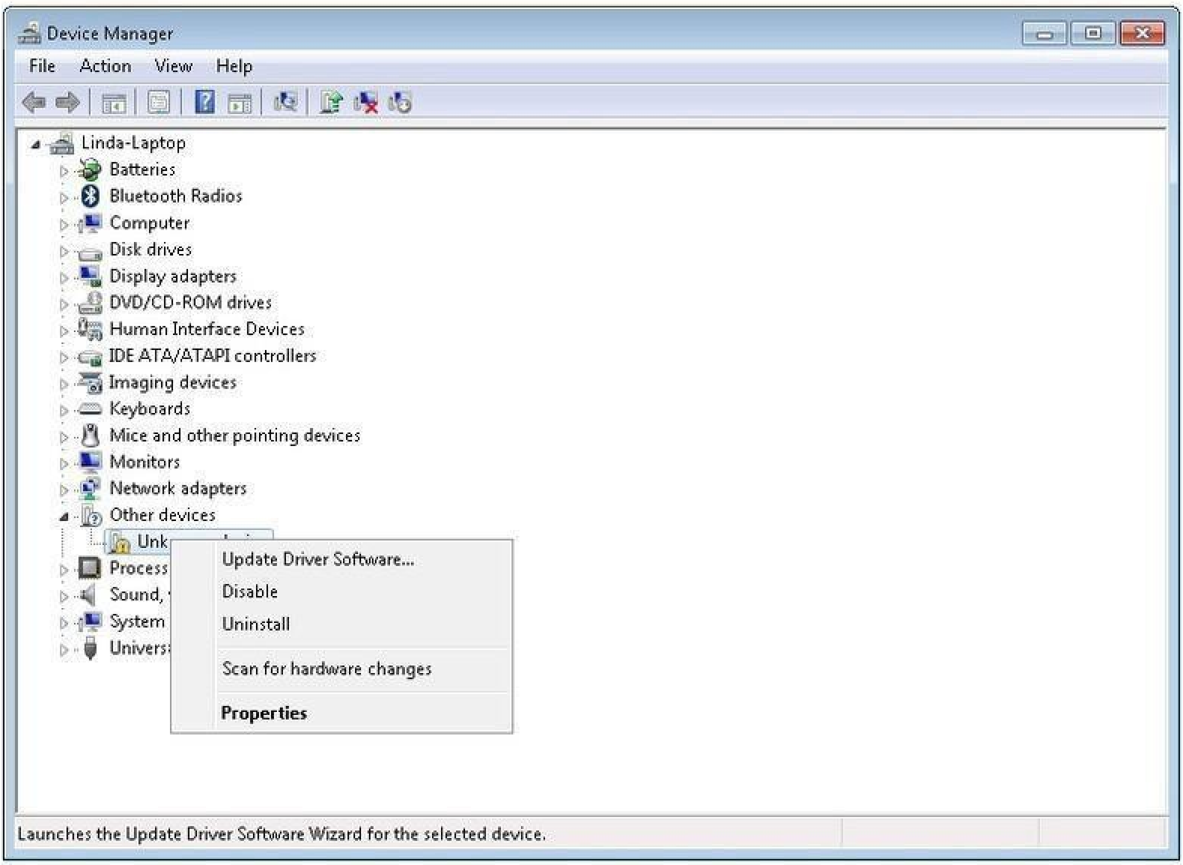
However, this installation method needs separate installation ofdriver.

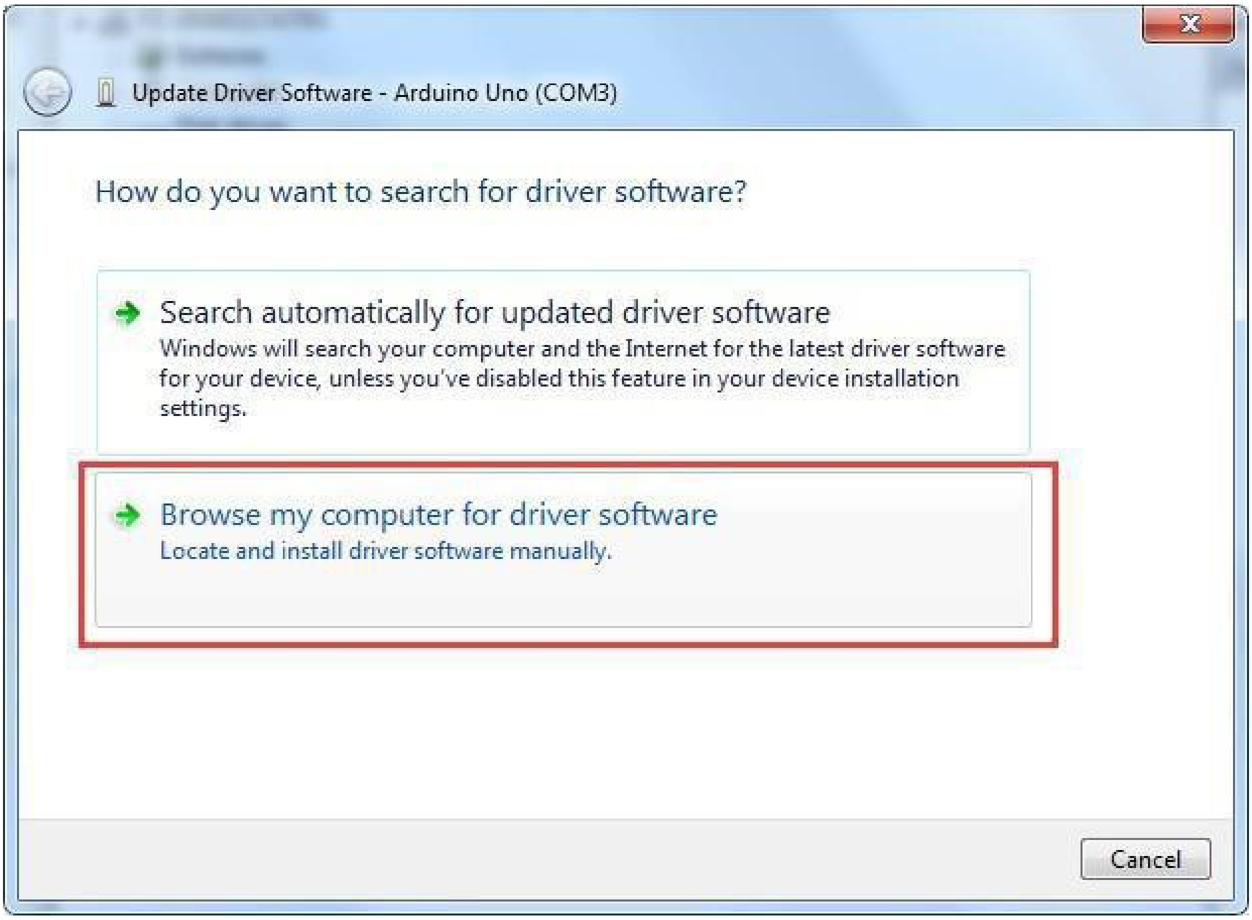
The Arduino folder contains both the Arduino program itself and the drivers that allow the Arduino to be connected to your computer by a USB cable. Before we launch the Arduino software, you are going to install the USB drivers.

Plug one end of your USB cable into the Arduino and the other into a USB socket on your computer. The power light on the LED will light up and you may get a 'Found New Hardware' message from Windows. Ignore this message and cancel any attempts that Windows makes to try and install drivers automatically for you.

The most reliable method of installing the USB drivers is to use the Device Manager. This is accessed in different ways depending on your version of Windows. In Windows 7, you first have to open the Control Panel, then select the option to view Icons, and you should find the Device Manager in the list.

Under 'Other Devices', you should see an icon for 'unknown device' with a little yellow warning triangle next to it. This is your Arduino.





Right-click on the device and select the top menu option (Update Driver Software...). You will then be prompted to either 'Search Automatically for updated driver software' or 'Browse my computer for driver software'. Select the option to browse and navigate to theX\arduino1.8.5\drivers.

Click 'Next' and you may get a security warning, if so, allow the software to be installed. Once the software has been installed, you will get a confirmation message.

Windows users may skip the installation directions for Mac and Linux systems and jump to Lesson 1. Mac and Linux users may continue to read this section.

Install Arduino (Mac OS X)

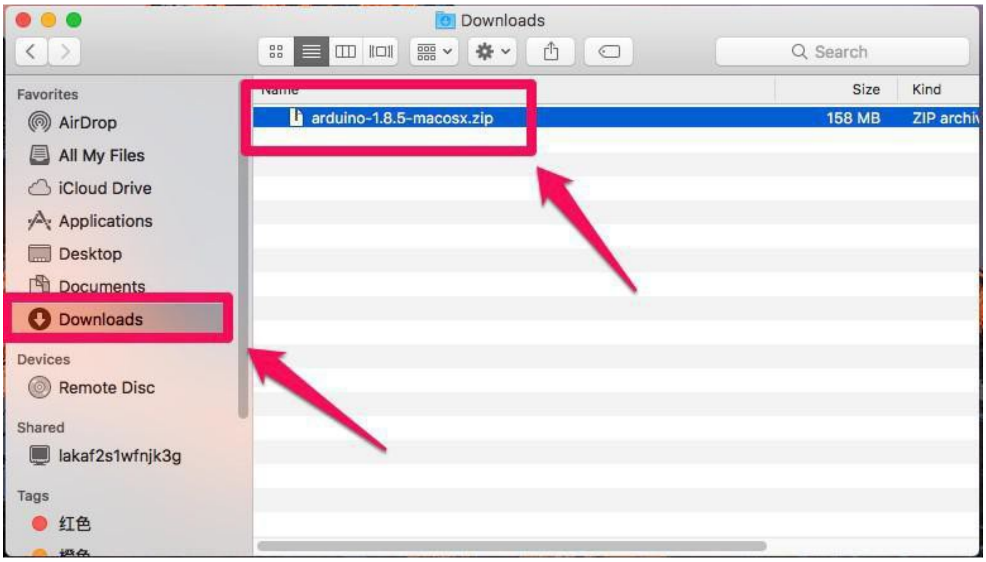
Step One: Download Arduino Software (IDE)

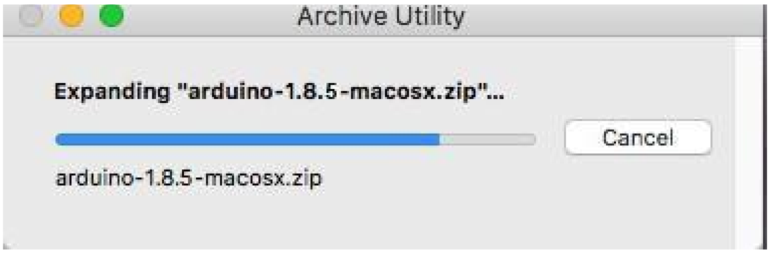
- Open the URL: https://www.arduino.cc/en/Main/Software with browser

- Click Mac OSX 10.7 Lion ornewer

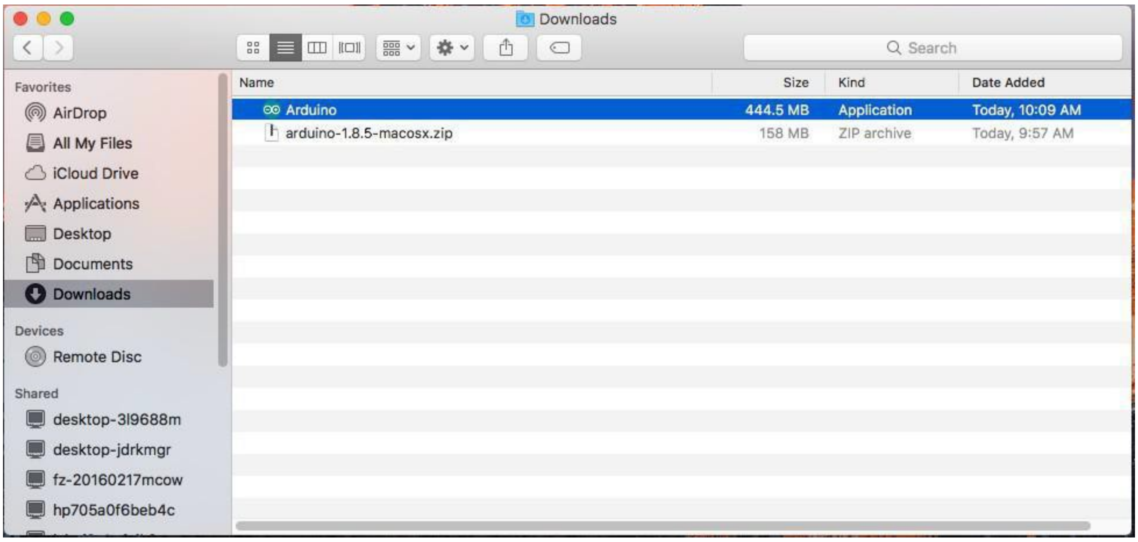
- Click JUST DOWNLOAD

Download and Unzip the zip file, double click the Arduino. app to enter Arduino IDE; the system will ask you to install Java runtime library if you don't have it in your computer. Oncethe installation is complete you can run the Arduino IDE.

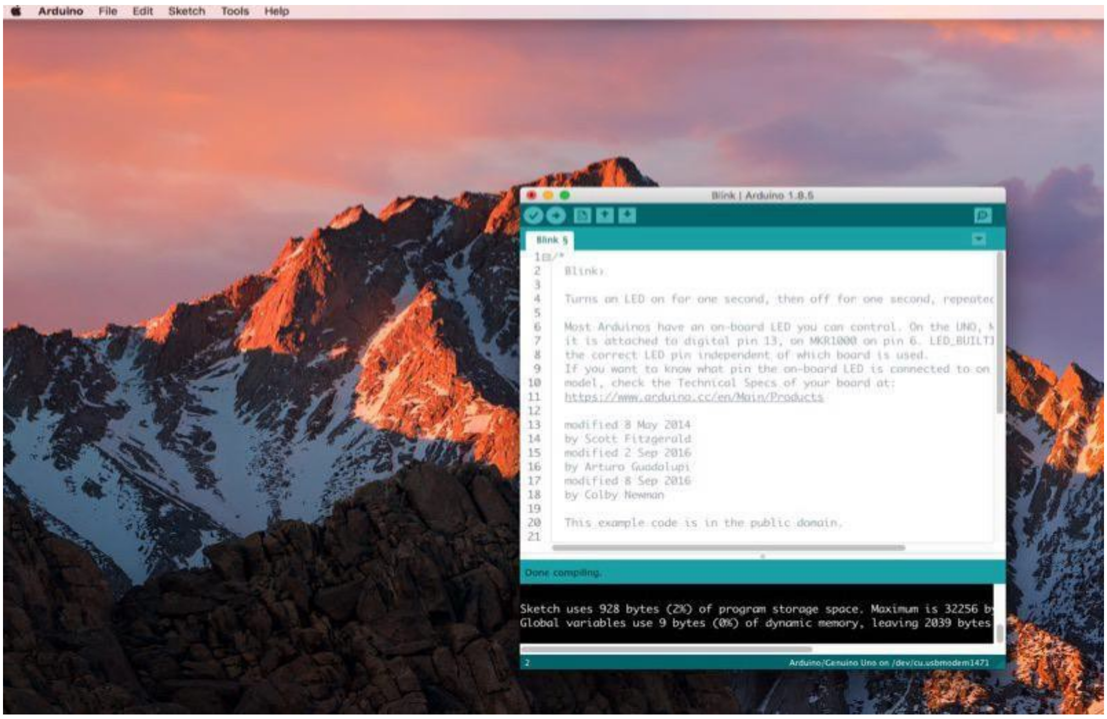




• arduino file is the Arduino application, just double click to open it

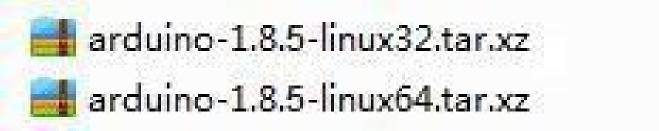


• Arduino IDE



Installing Arduino (Linux)

You will have to use the "make install" command. If you are using the Ubuntu system, it is recommended to install Arduino IDE from the Ubuntu Software Center.



TIPS: If you have problems in installing the drivers, please refer to the UNO R3, MEGA, NANO DRIVER FAQ.

Lesson 1 Add Libraries

Installing Additional ArduinoLibraries

Once you are comfortable with the Arduino software and using the built-in functions, you may want to extend the ability of your Arduino with additional libraries.

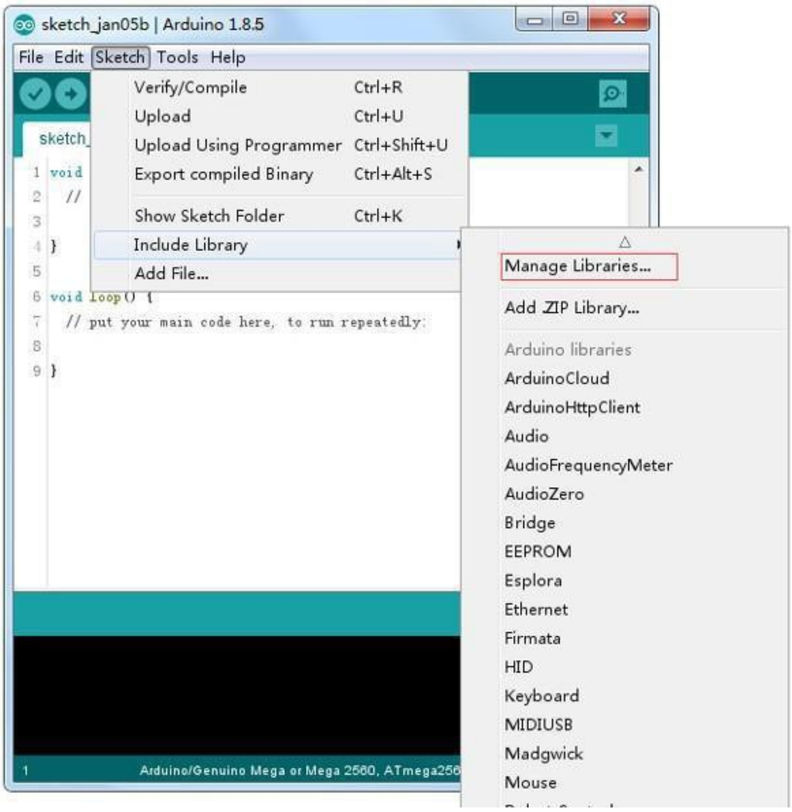
What are Libraries?

Libraries are a collection of code that makes it easy for you to connect to a sensor, display, module, etc. For example, the built-in Liquid Crystal library makes it easy to talk to character LCD displays. There are hundreds of additional libraries available on the Internet for download. The built-in libraries and some of these additional libraries are listed in the reference. To use the additional libraries, you will need to install them.

How to Install a Library

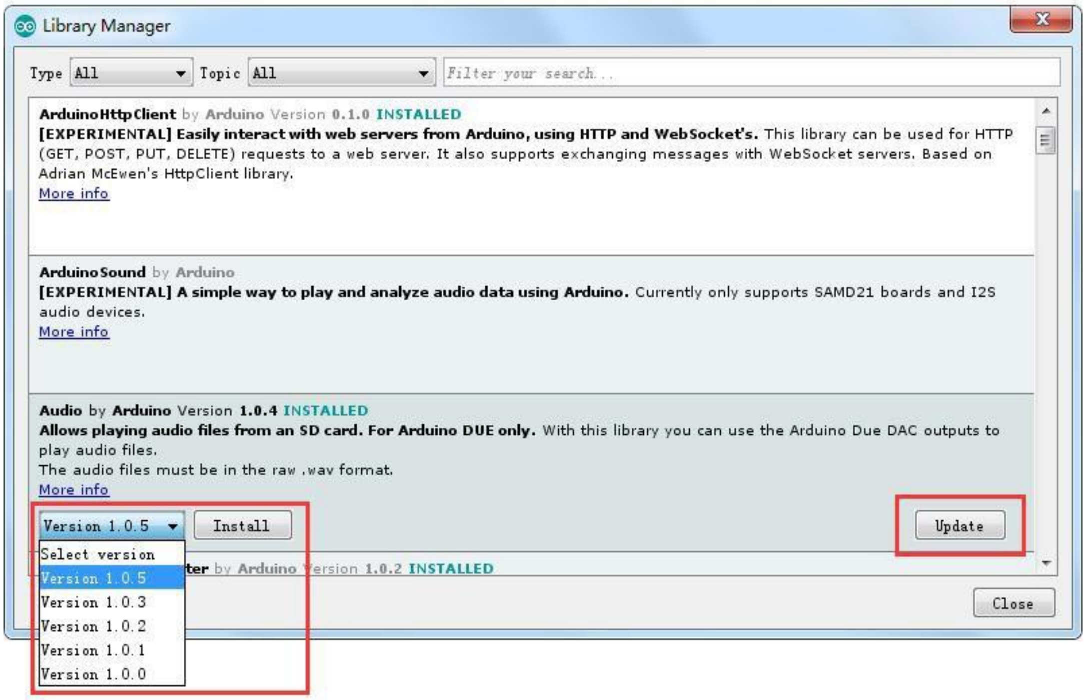
Using the LibraryManager

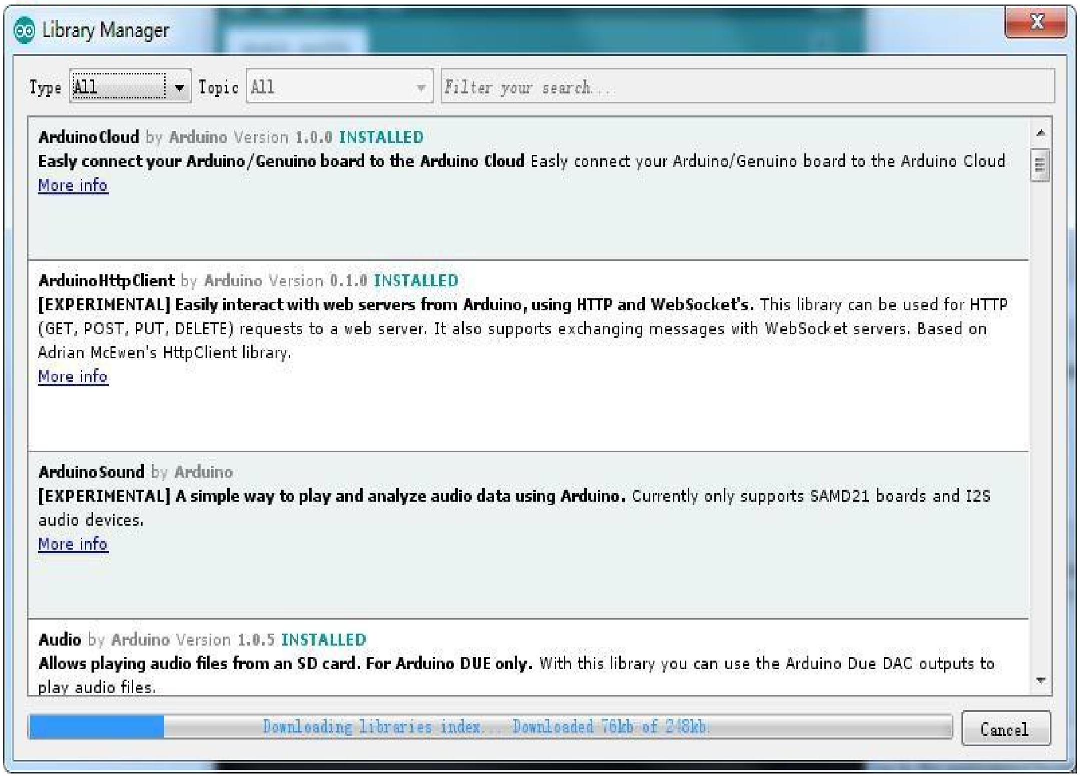
To install a new library into your Arduino IDE you can use the Library Manager (available from IDE version 1.8.5). Open the IDE and click to the "Sketch" menu and then Include Library > Manage Libraries.



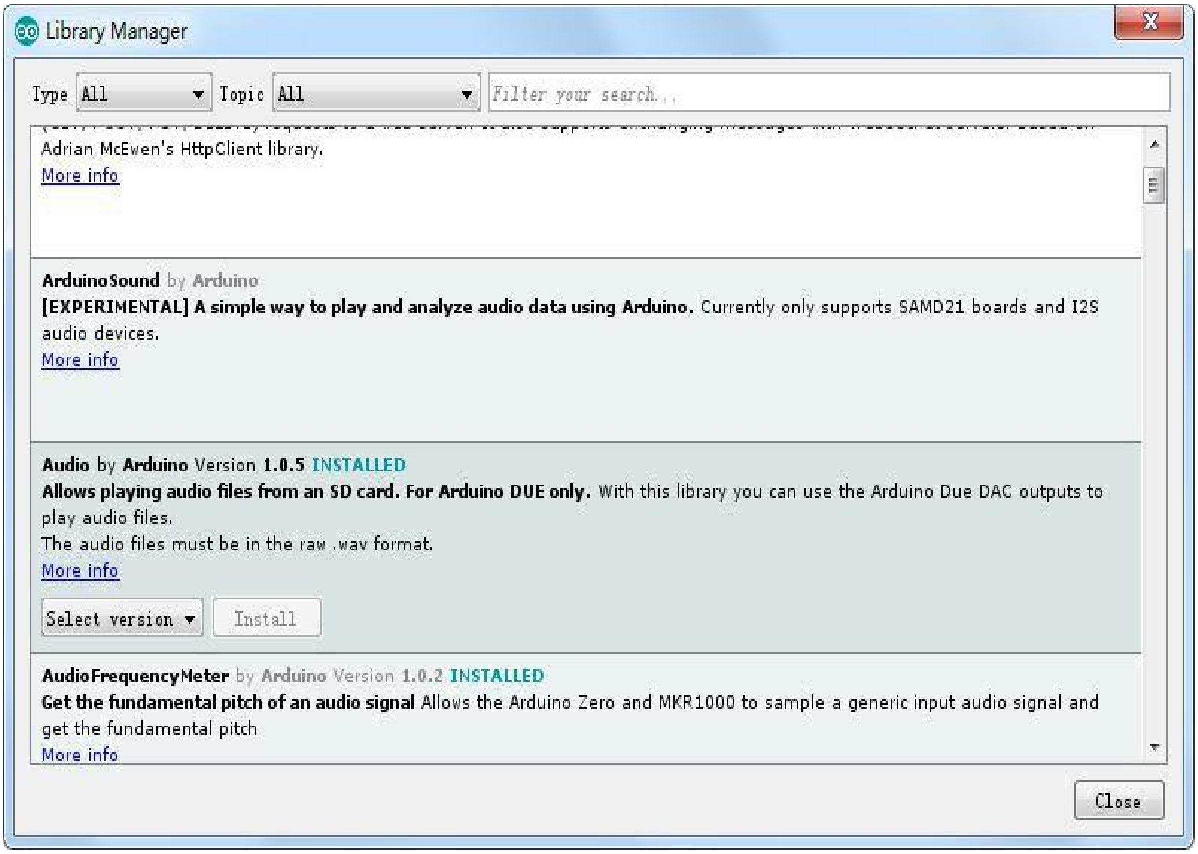
Then the library manager will open and you will find a list of libraries that are already installed or ready for installation. In this example we will install the Bridge library. Scroll the list to find it, then select the version of the library you want to install. Sometimes only one version of the library is available. If the version selection menu does not appear, don't worry: it is normal.

There are times you have to be patient with it, just as shown in the figure. Please refresh it and wait.





Finally click on install and wait for the IDE to install the new library. Downloading may take time depending on your connection speed. Once it has finished, an Installed tag should appear next to the Bridge library. You can close the librarymanager.

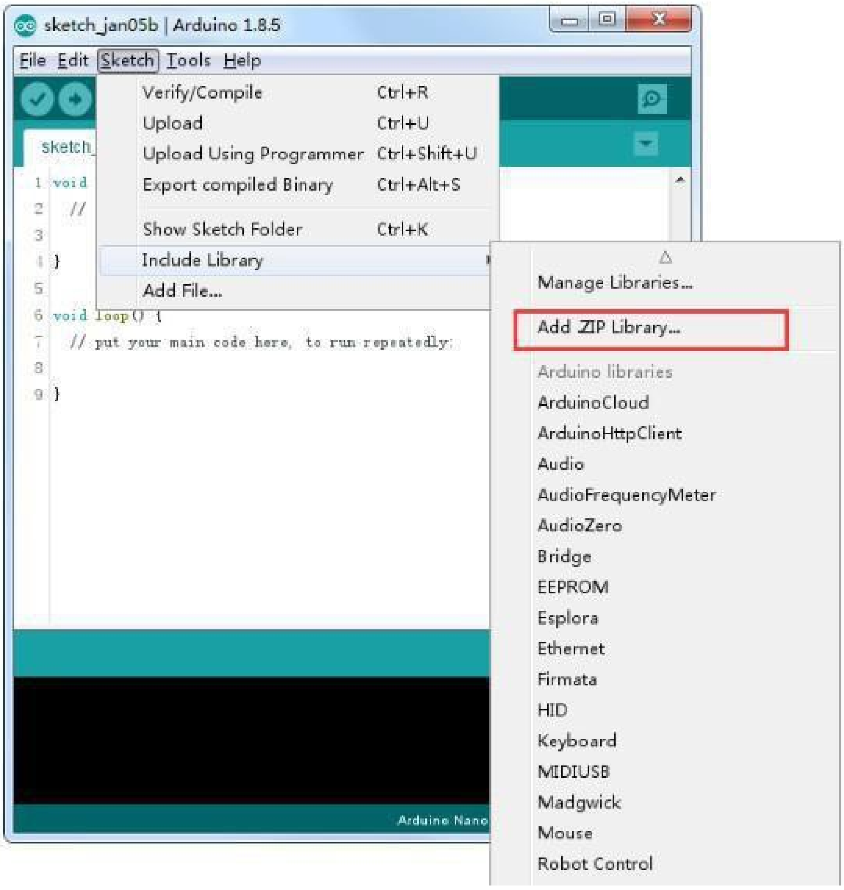


You can now find the new library available in the Include Library menu. If you want to add your own library open a new issue on github.

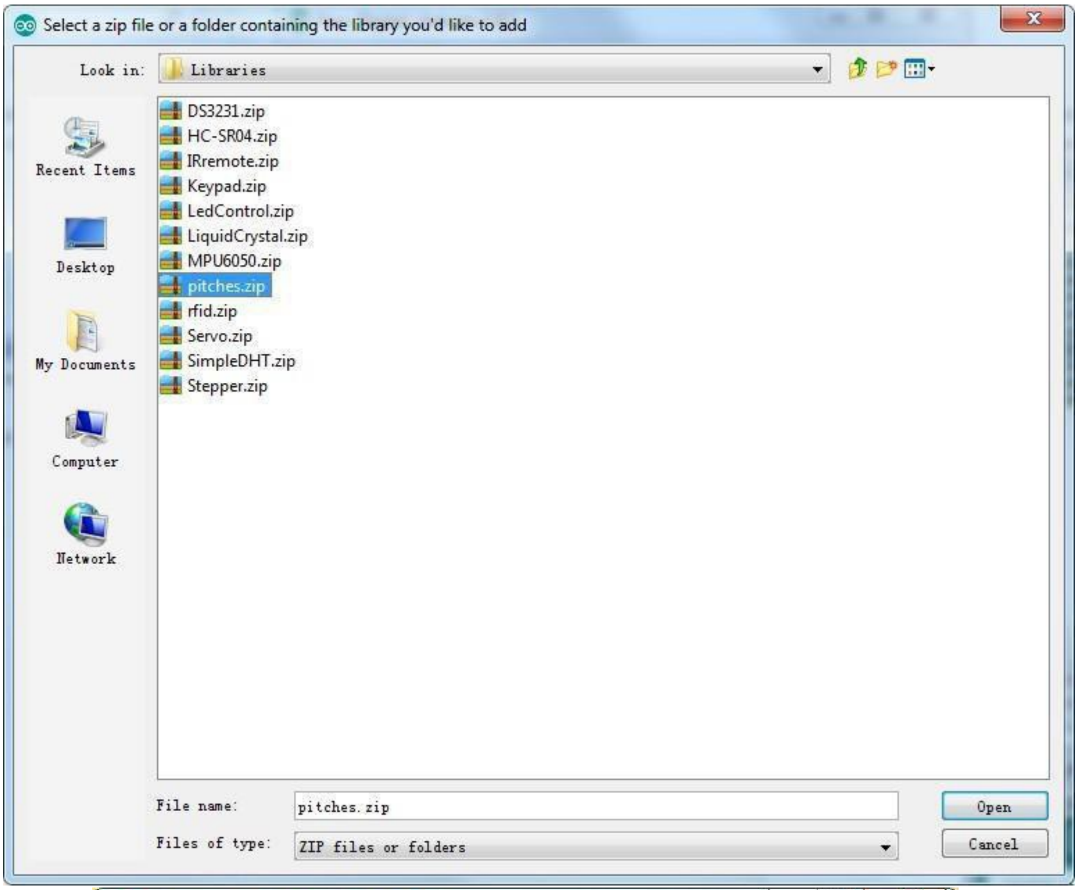
Importing a .zip Library

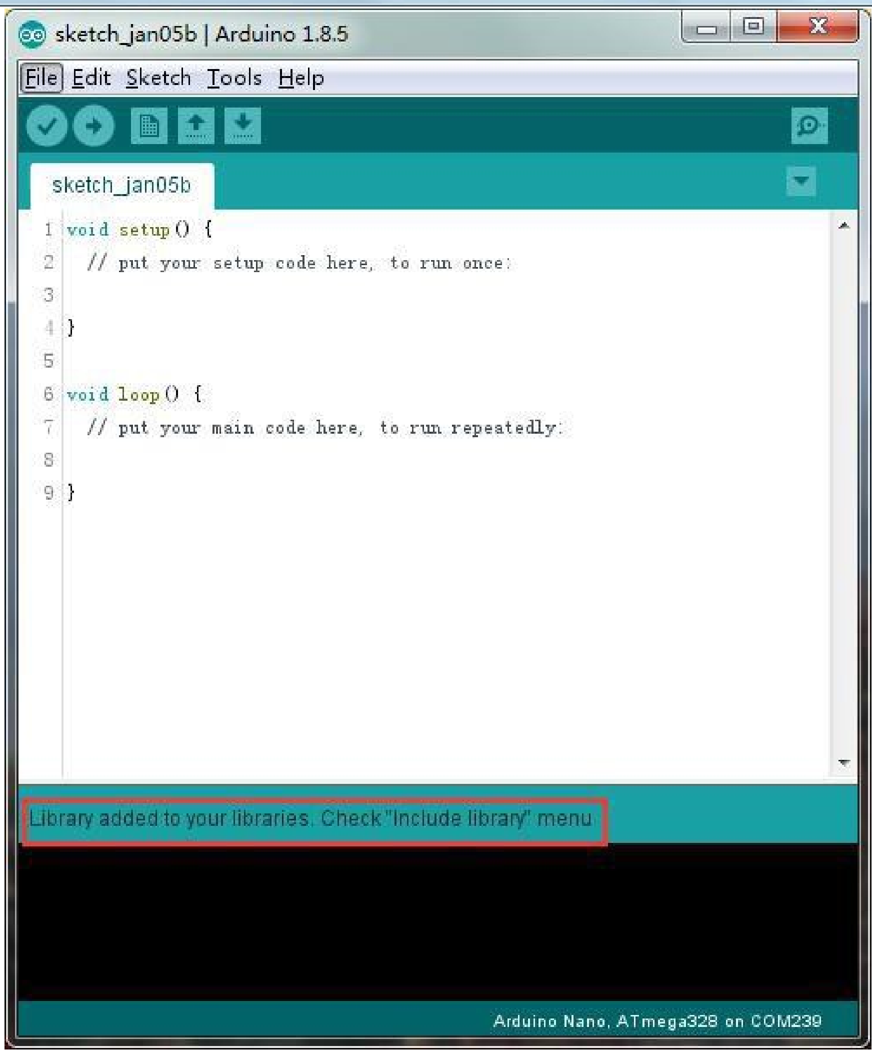
Libraries are often distributed as a ZIP file or folder. The name of the folder is the name of the library. Inside the folder will be a. cpp file, a .h file and often a key words .txt file, examples folder, and other files required by the library. Starting with version 1.0.5, you can install 3rd party libraries in the IDE. Do not unzip the downloaded library, leave it as is.

In the Arduino IDE, navigate to Sketch > Include Library. At the top of the drop down list, select the option to "Add .ZIP Library''.



You will be prompted to select the library you would like to add. Navigate to the .zip file's location and open it.


Select a zip file or a folder containing the library you'd like to add
Look in:
Libraries
Recent Items
Desktop
My Documents
Computer
Network
DS3231.zip
HC-SR04.zip
IRremote.zip
Keypad.zip
LedControl.zip
LiquidCrystal.zip
MPU6050.zip
pitches.zip
rfid.zip
Servo.zip
SimpleDHT.zip
Stepper.zip
File name:
pitches.zip
Files of type:
ZIP files or folders
Open
Cancel


sketch_jan05b | Arduino 1.8.5
File Edit Sketch Tools Help
sketch_jan05b
void setup() {
// put your setup code here, to run once:
}
void loop() {
// put your main code here, to run repeatedly:
}
Library added to your libraries. Check "Include library" menu
Arduino Nano, ATmega328 on COM239

Return to the Sketch > Import Library menu. You should now see the library at the bottom of the drop-down menu. It is ready to be used in your sketch. The zip file will have been expanded in the libraries folder in your Arduino sketches directory.

NB: the Library will be available to use in sketches, but examples for the library will not be exposed in the File > Examples until after the IDE has restarted.

Those two are the most common approaches. MAC and Linux systems can be handled likewise. The manual installation to be introduced below as an alternative may be seldom used and users with no needs may skipit.

Manualinstallation

To install the library, first, quit the Arduino application. The nun zip the ZIP file containing the library. For example, if you're installing a library called "Arduino Party", uncompressed Arduino Party .zip. It should contain a folder called Arduino Party, with files like Arduino Party. cpp and Arduino Party .h inside. (If the.cpp and.h files aren't in a folder, you'll need to create one. In this case, you'd make a folder called "Arduino Party" and move into it all the files that we rein the ZIP file, like Arduino Party .cpp and Arduino Party .h.)

Drag the Arduino Party folder into this folder (your libraries folder). Under Windows, it will likely be called "My Documents\Arduino\libraries". For Mac users, it will likely be called "Documents/Arduino/libraries". On Linux, it will be the "libraries" folder in your sketchbook.

Your Arduino library folder should now look like this (on Windows):

My Documents\Arduino\libraries\ArduinoParty\ArduinoParty.cpp

My Documents\Arduino\libraries\ArduinoParty\ArduinoParty.h

My Documents\Arduino\libraries\ArduinoParty\examples

....

or like this (on Mac and Linux):

Documents/Arduino/libraries/ArduinoParty/ArduinoParty .cpp

Documents/Arduino/libraries/Arduino Party/Arduino Party .h

Documents/Arduino/libraries/Arduino Party/examples

....

There may be more files than just the .cpp and .h files so make sure they're all there. (The library won't work if you put the .cpp and .h files directly into the libraries folder or if they're nested in an extra folder. For example: Documents\Arduino\libraries\ArduinoParty.cpp and Documents\Arduino\libraries\ArduinoParty\ArduinoParty\ArduinoParty.cpp won't work.)

Restart the Arduino application. Make sure the new library appears in the Sketch > Import Library menu. That's it. You've installed a library!

Lesson 2 Temperature and Humidity Module

Overview

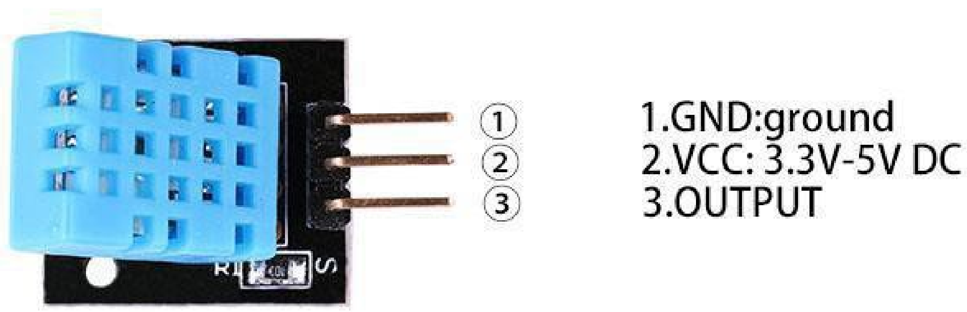
In this tutorial we will learn how to use a DHT11 Temperature and Humidity Sensor.

It's accurate enough for most projects that need to keep track of humidity and temperature readings.

Again we will be using a Library specifically designed for these sensors that will make our code short and easy to write.

Temp and Humidity

A module with a temperature/humidity sensor type DHT11, Temperature range: -20 - 60°C (+/-1°C), Rel. humidity: 5-95% (+/5%), Supply voltage: 3 to 5.5V. With a built-in 10 K ohm pull up resistor. Library:DHT .h



Component Required:

(1)x Elegoo Uno R3

(1)x usb cable

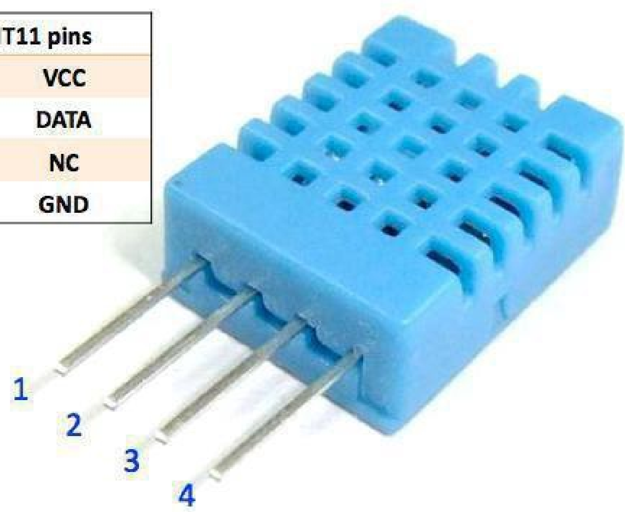
(1)x DHT11 module

(x)x F-M wires

Component Introduction

Temp and humidity sensor:

DHT11 pins	
1	VCC
2	DATA
3	NC
4	GND



DHT11 digital temperature and humidity sensor is a composite sensor contain sac a liberated digital signal output of the temperature and humidity. Application of a dedicated digital modules collection technology and the temperature and humidity sensing technology, to ensure that the product has high reliability and excellent long-term stability. The sensor includes a capacitive sense of wet components and an NTC temperature measurement devices, and connected with a high-performance 8-bit microcontroller

Applications: HVAC, dehumidifier, testing and inspection equipment, consumer goods, automotive, automatic control, data loggers, weather stations, home appliances, humidity regulator, medical and other humidity measurement and control.

Product parameters

Relative humidity: Resolution:

16 Bit

Repeatability: ±1% RH Accuracy: At 25℃
±5% RH

Interchangeability: fully interchangeable Response time:
1 / e (63%) of 25℃ 6s

1 m/sari 6s Hysteresis:

<±0.3% RH

Long-term stability :< ± 0.5% RH/yrin Temperature:

Resolution: 16 Bit

Repeatability: ±0.2℃

Range: At 25℃ ±2℃ Response

Time: 1/e (63%) 10 S Electrical

Characteristics Power supply:

DC3.5 ～5.5V

Supply Current: measurement 0.3 mA standby 60 μA Sampling period: more than 2

seconds

Pin Description: 1, the VDD power supply

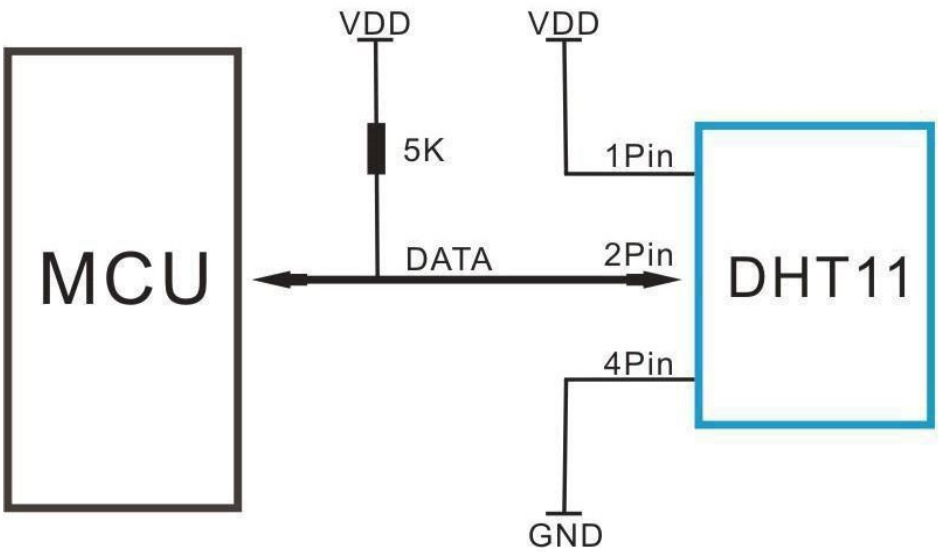
3.5 ～5.5V DC

2 DATA serial data, as ingle bus

3, NC,empty pin4, GND ground, the

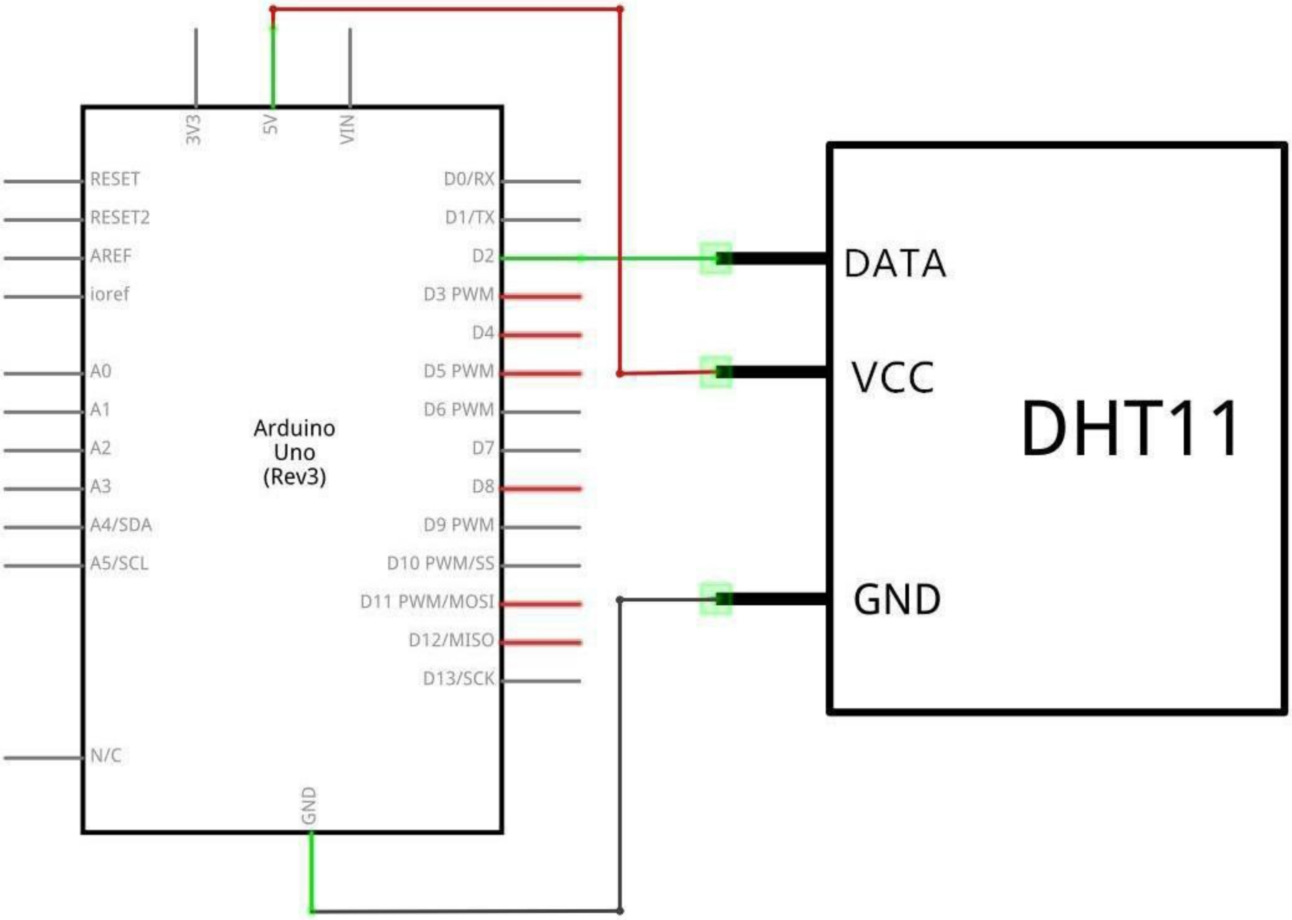
negative power

Typical Application

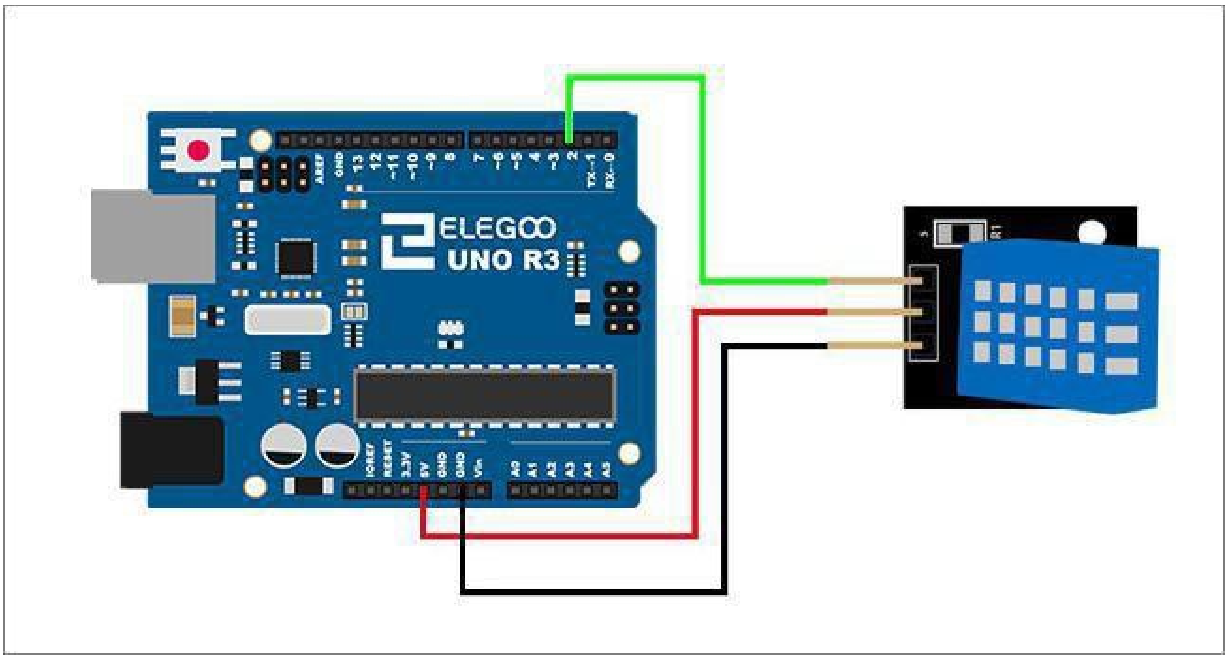


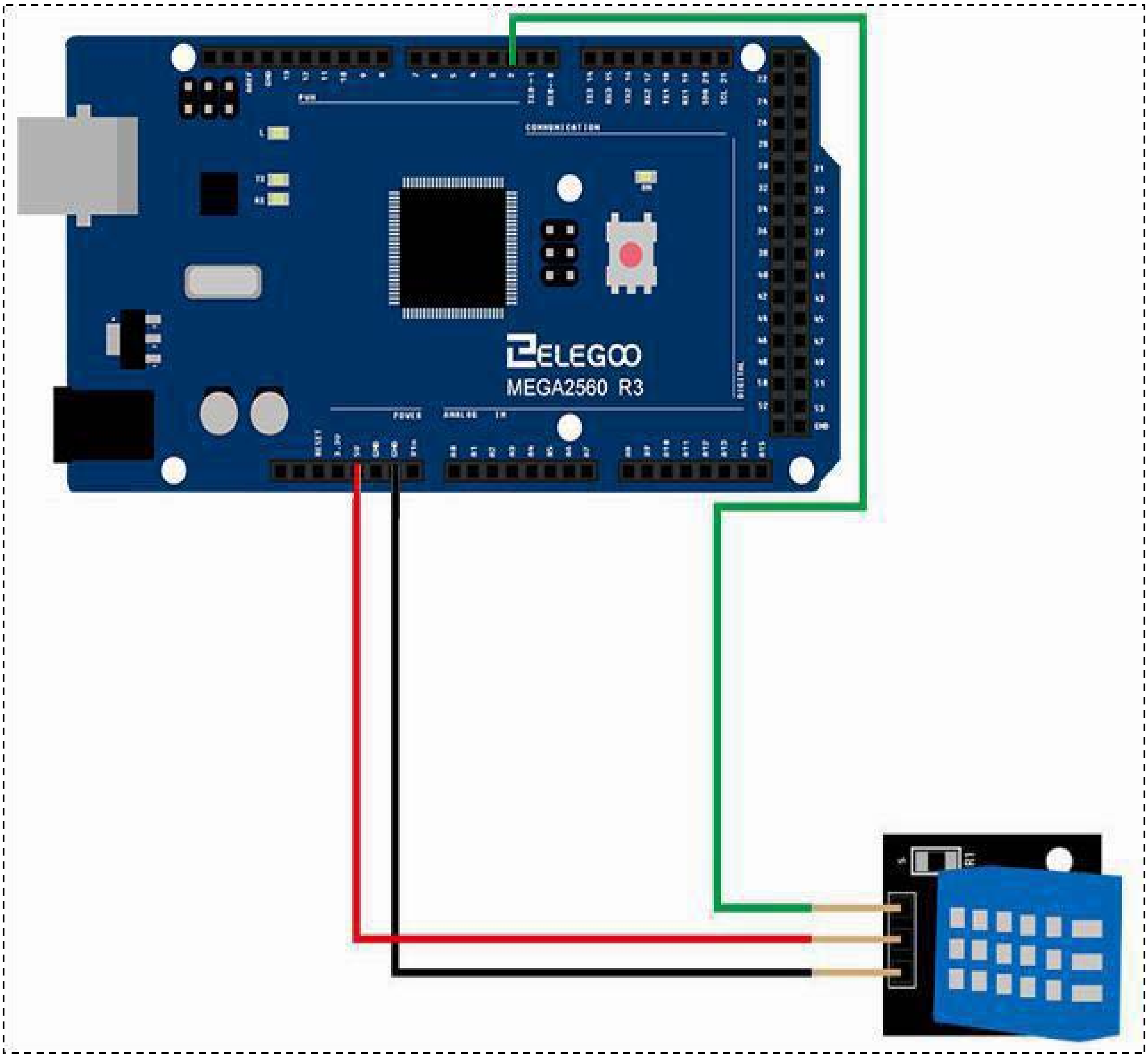
Connection

Schematic

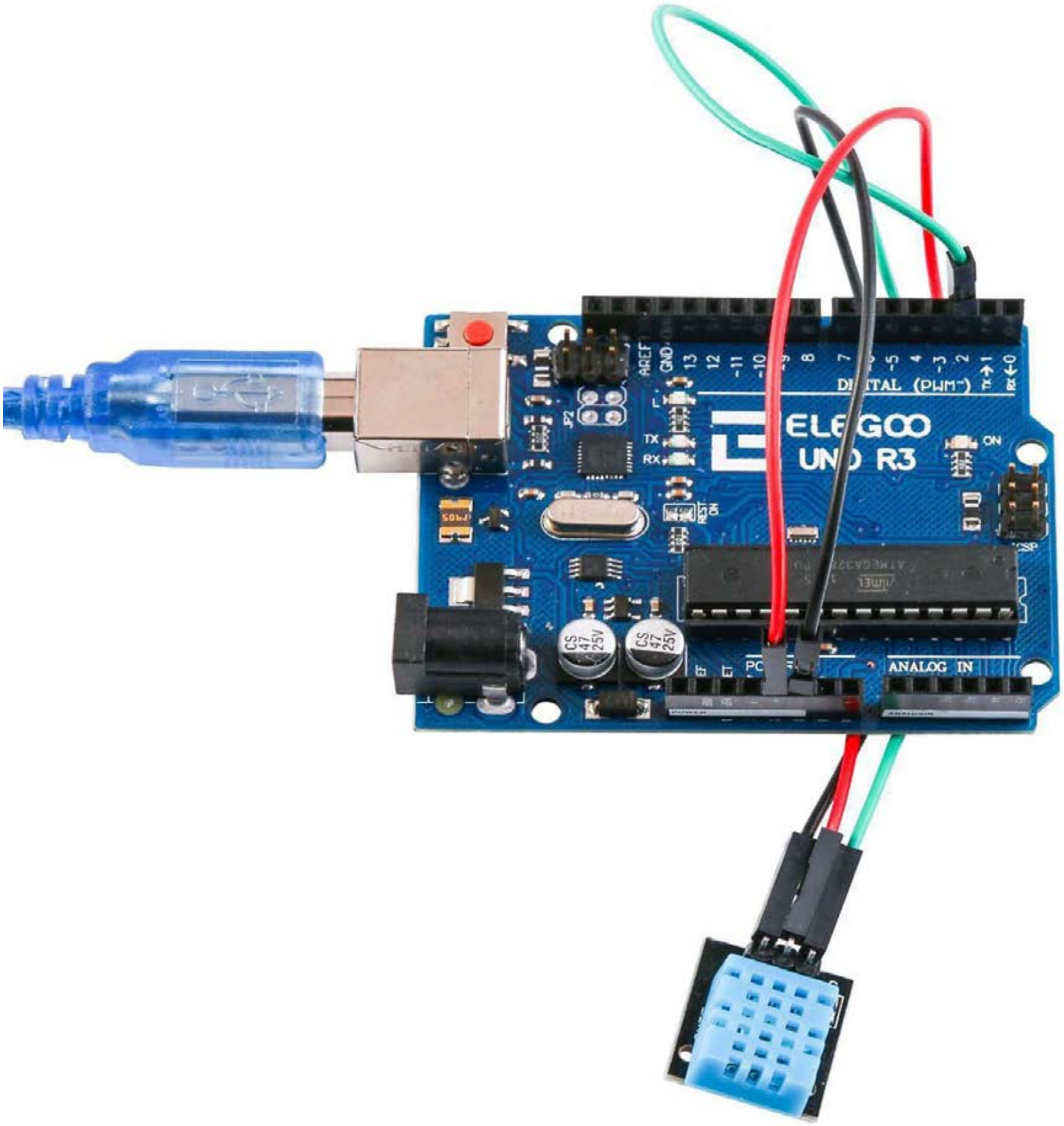


wiring diagram

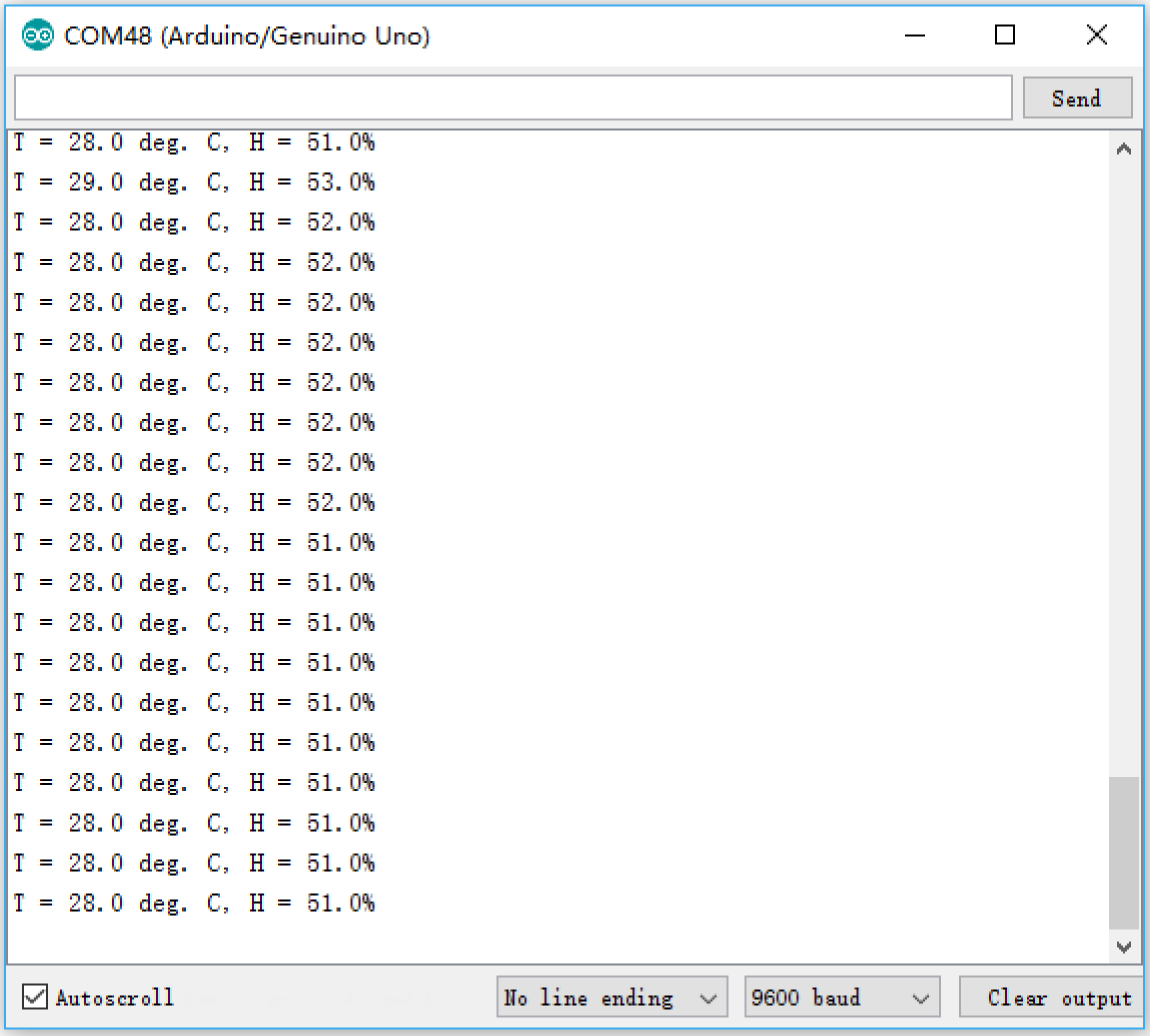



ELEGOO
MEGA2560 R3

Result



Upload the program then open the monitor, we can see the data as below :(It shows the temperature of the environment, we can see it is 28 degree)


COM48 (Arduino/Genuino Uno)
Send
T = 28.0 deg. C, H = 51.0%
T = 29.0 deg. C, H = 53.0%
T = 28.0 deg. C, H = 52.0%
T = 28.0 deg. C, H = 52.0%
T = 28.0 deg. C, H = 52.0%
T = 28.0 deg. C, H = 52.0%
T = 28.0 deg. C, H = 52.0%
T = 28.0 deg. C, H = 52.0%
T = 28.0 deg. C, H = 52.0%
T = 28.0 deg. C, H = 52.0%
T = 28.0 deg. C, H = 51.0%
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T = 28.0 deg. C, H = 51.0%
T = 28.0 deg. C, H = 51.0%
T = 28.0 deg. C, H = 51.0%
T = 28.0 deg. C, H = 51.0%
T = 28.0 deg. C, H = 51.0%
T = 28.0 deg. C, H = 51.0%
T = 28.0 deg. C, H = 51.0%
T = 28.0 deg. C, H = 51.0%
Autoscroll
No line ending
9600 baud
Clear output

Lesson 3 DS18B20 Temperature Sensor Module

Overview

In this experiment, we will learn how to use DS18B20 module test the environmental temperature and make a thermometer.

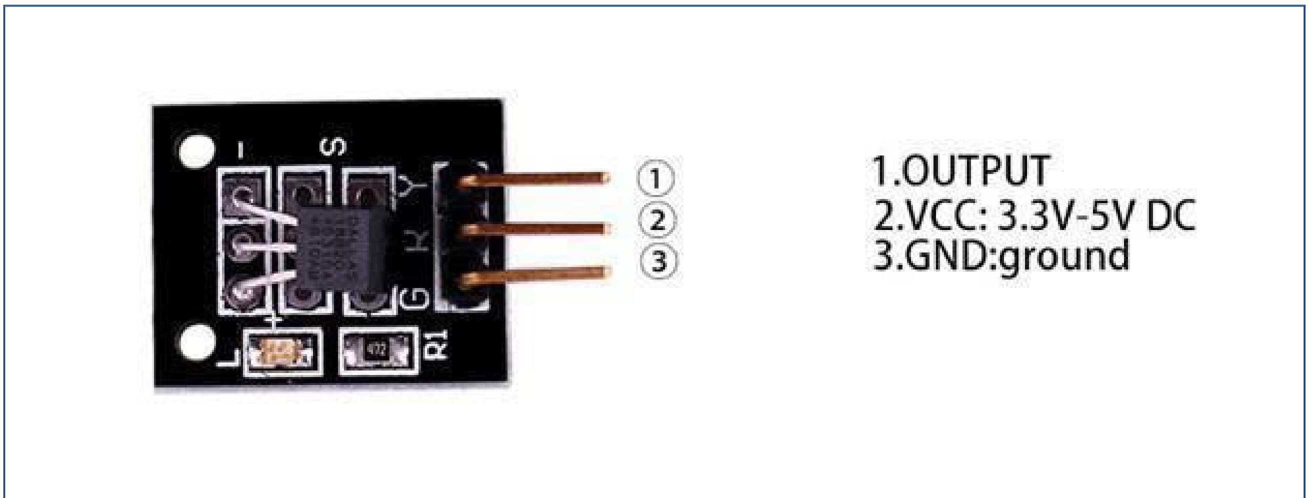
Since the previous temperature sensor output is analog. So we need to add additional A/D and D/A chip into the line transformation. More -over, the Arduino external port is not rich resources and the utilization rate is not high.so we are create the Ds18b20 module.

The new DS18B20 Temperature Sensor Module is very good solve the problem. It have the characteristic of the economy, unique 1-wire bus and it can fully apply the Arduino platform.

Users can easily form a sensor net work through using this module.

Product parameters

- Temperature range: -55 to +125°C
- Typical accuracy: 0.5°C
- Resolution: 9-12 Bit, depending on the program



Component Required:

(1) x Elegoo Uno R3

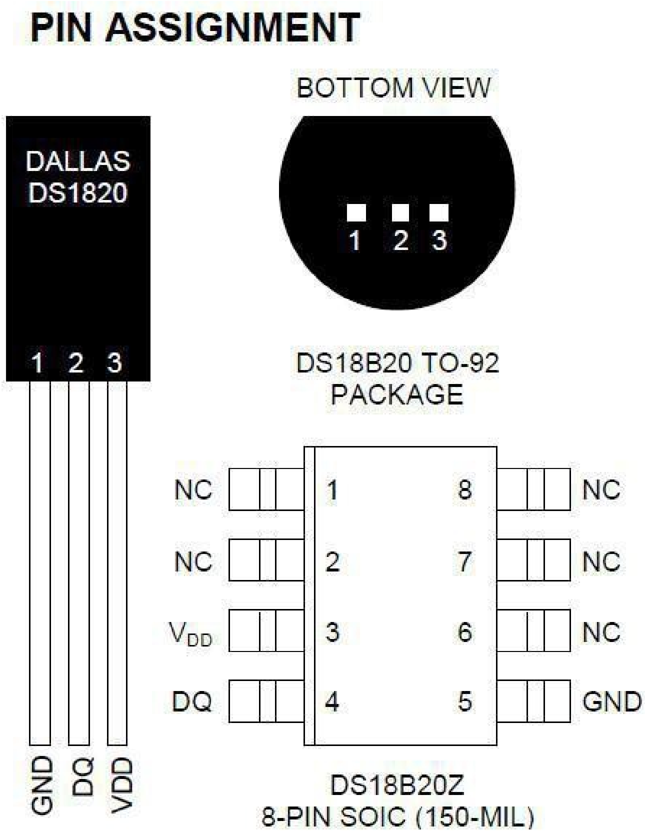
(1)x USB cable

(1)x DS18B20 module

(x) x F-M wires

Component Introduction

DS18B20:



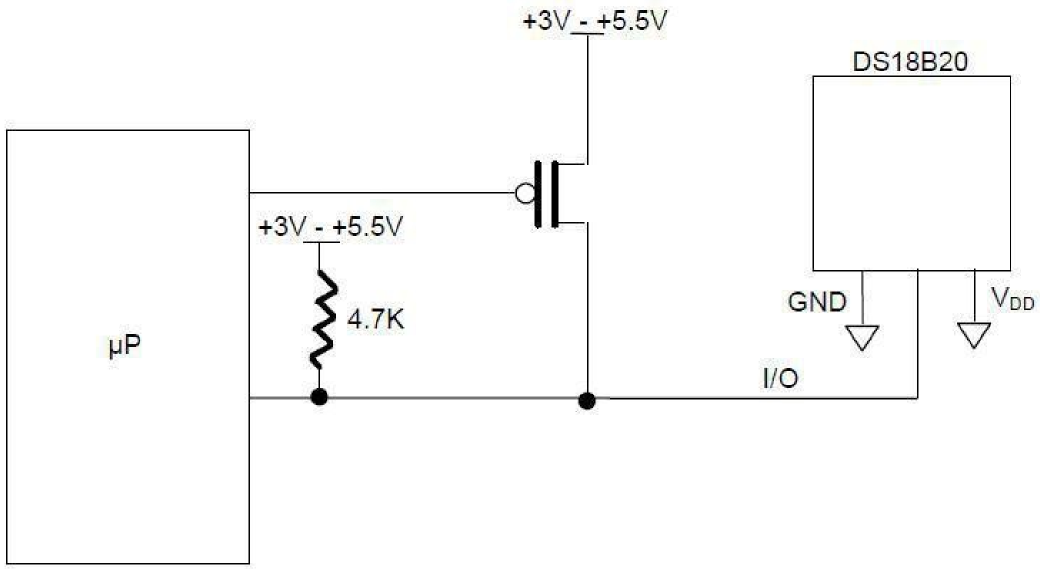
PIN DESCRIPTION

GND - Ground

DQ - Data In/Out

V_{DD} - Power Supply Voltage

NC - No Connect



Principle

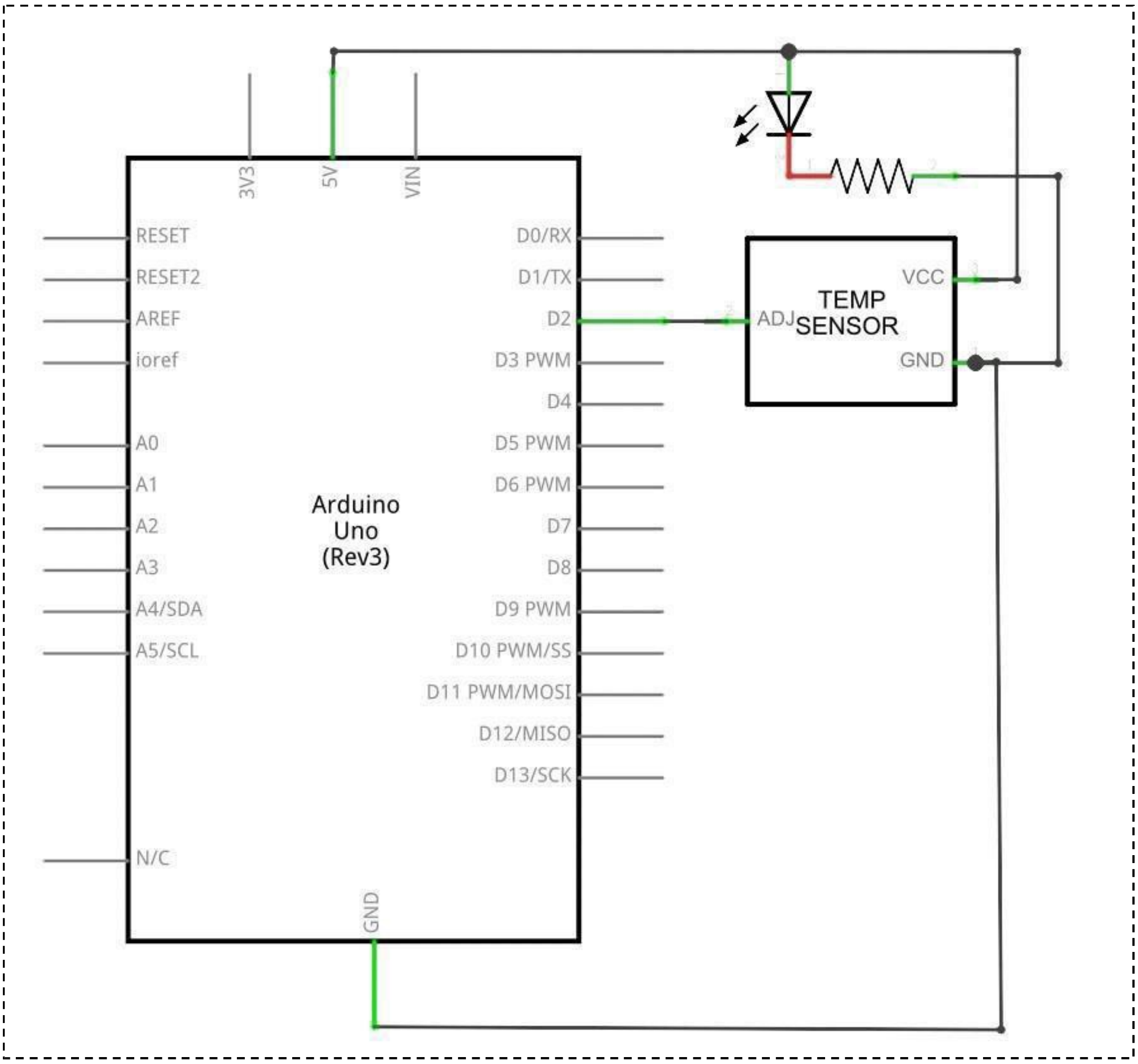
DS18B20 module is using a single bus. The power supply voltage range of 3.0 V to 5.5 V and no standby power supply. It can Measure temperature range for-55 degree to +125 degree with accuracyof+/-0.5 °C.

The programmable DPI of temperature sensor is from 9 to 12. Temperature conversion is 12 digits lattice type. Maximum is 750 milliseconds. Families can be defined non-volatile temperature a l a r m Settings.

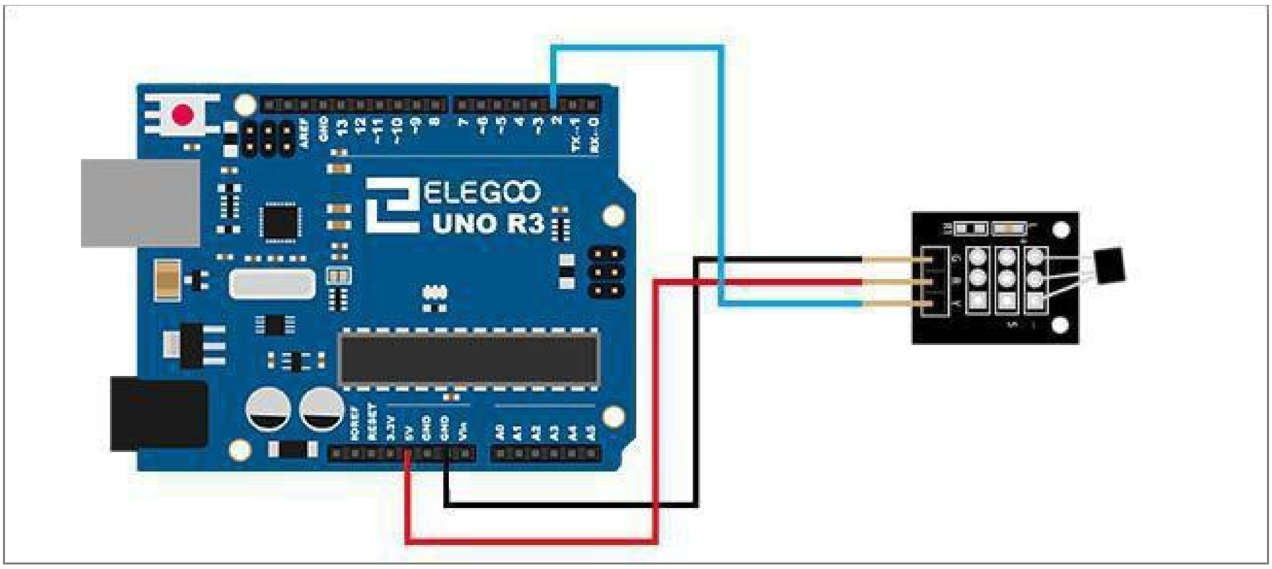
Each DS18B20 contains a unique serial number so that multiple ds18b20s can ex is tin abus.

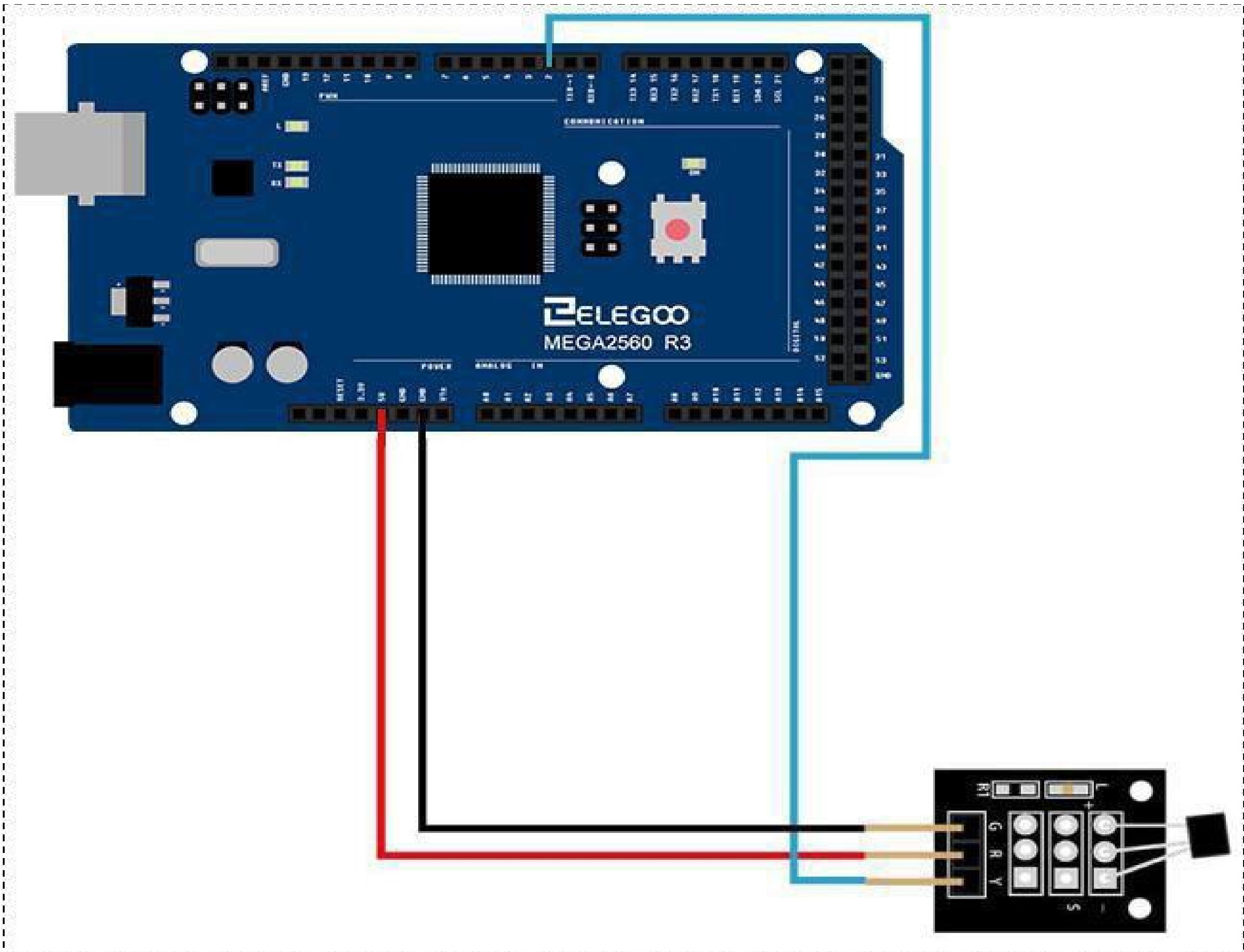
Connection

Schematic

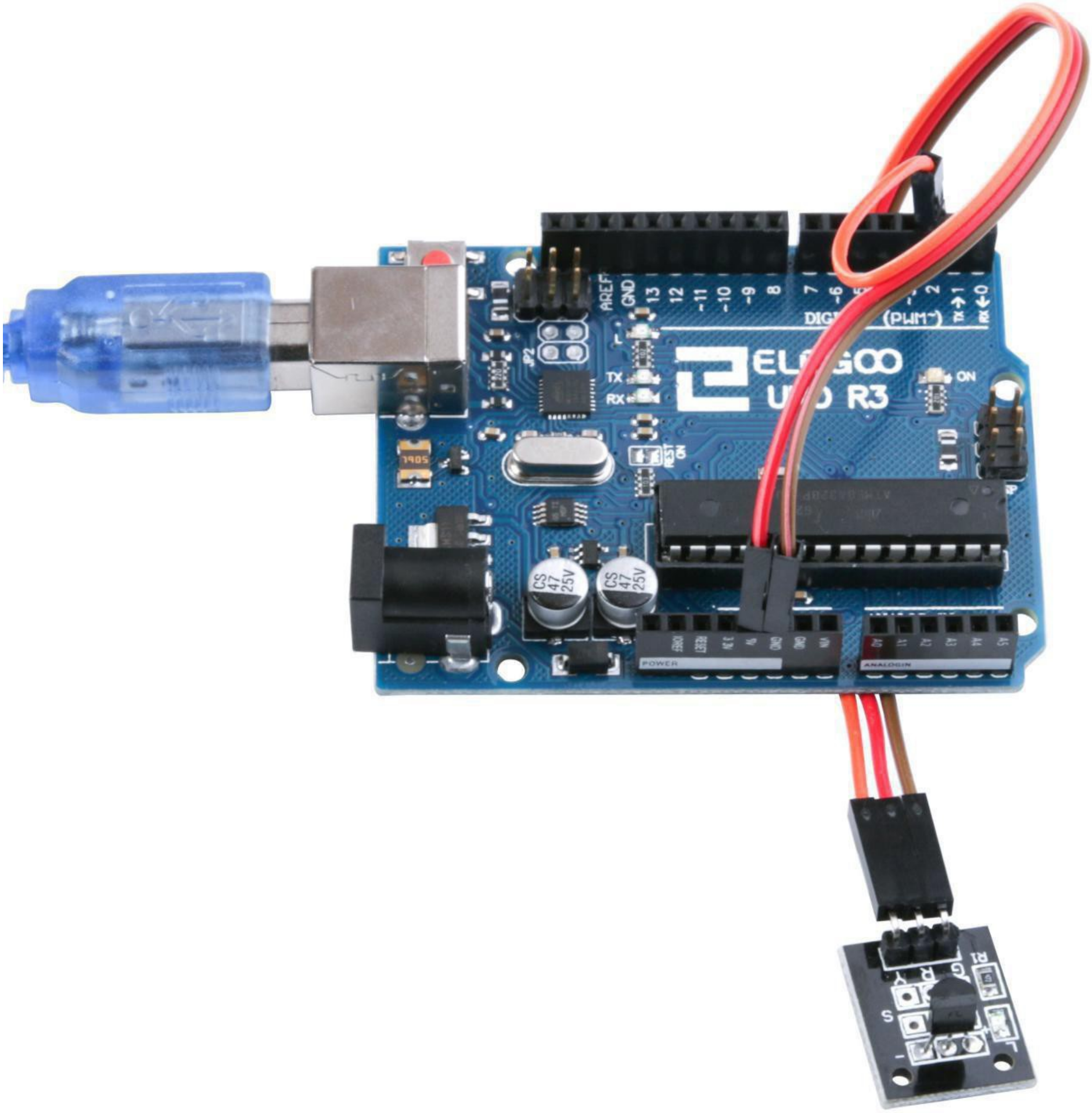


wiring diagram



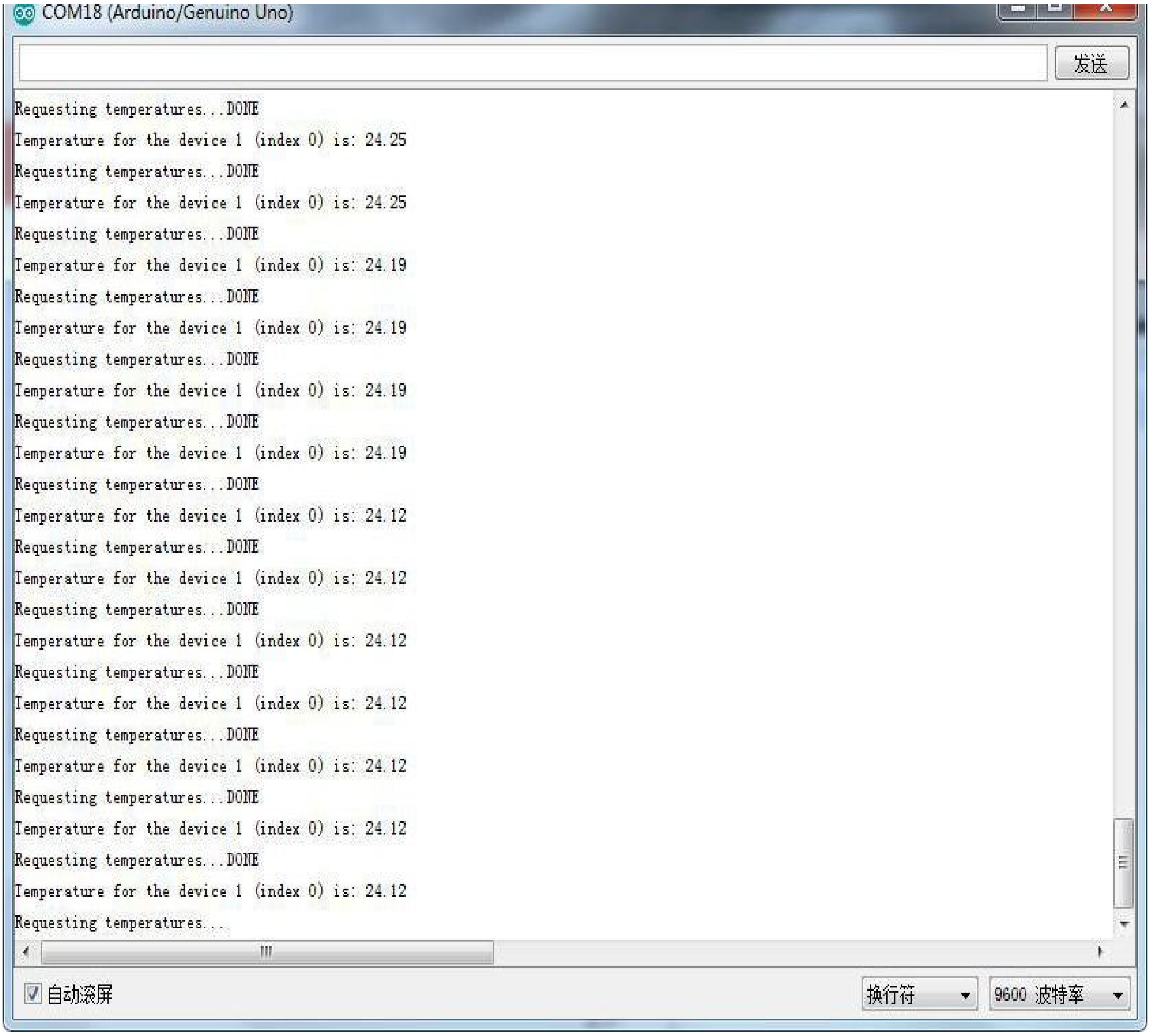

ELEGOO
MEGA2560 R3

Result



Temperature sensor can detect temperature in numbers of different places at the same time.

Upload the program then open the monitor, we can see the data as below:


COM18 (Arduino/Genuino Uno)
发送
Requesting temperatures...DONE
Temperature for the device 1 (index 0) is: 24.25
Requesting temperatures...DONE
Temperature for the device 1 (index 0) is: 24.25
Requesting temperatures...DONE
Temperature for the device 1 (index 0) is: 24.19
Requesting temperatures...DONE
Temperature for the device 1 (index 0) is: 24.19
Requesting temperatures...DONE
Temperature for the device 1 (index 0) is: 24.19
Requesting temperatures...DONE
Temperature for the device 1 (index 0) is: 24.19
Requesting temperatures...DONE
Temperature for the device 1 (index 0) is: 24.12
Requesting temperatures...DONE
Temperature for the device 1 (index 0) is: 24.12
Requesting temperatures...DONE
Temperature for the device 1 (index 0) is: 24.12
Requesting temperatures...DONE
Temperature for the device 1 (index 0) is: 24.12
Requesting temperatures...DONE
Temperature for the device 1 (index 0) is: 24.12
Requesting temperatures...DONE
Temperature for the device 1 (index 0) is: 24.12
Requesting temperatures...DONE
Temperature for the device 1 (index 0) is: 24.12
Requesting temperatures...
自动滚屏
换行符
9600 波特率

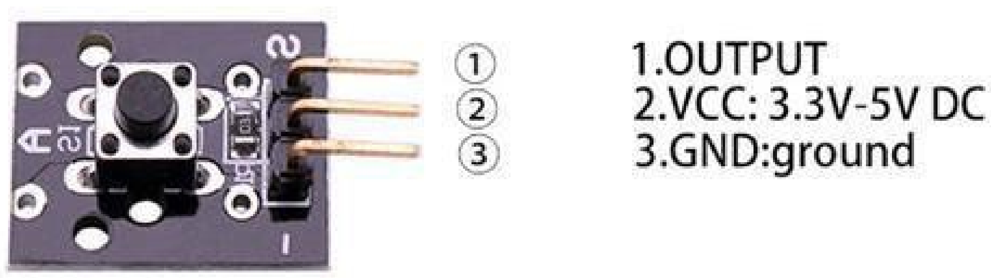
Lesson 4 Button Switch Module

Overview

In this experiment, we will learn how to use button switch.

PCB mounted push Button

A built-in 10 K ohm resistor is connected between the center pin and the 'S' pin and can be used as a pull up or pull down resistor. The push button connects the two outer pins.



Component Required:

(1)x Elegoo Uno R3

(1)x USB cable

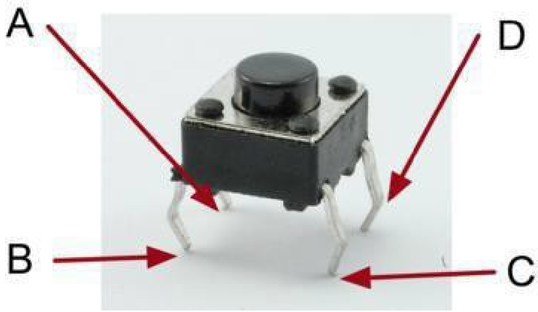
(1)xButton module

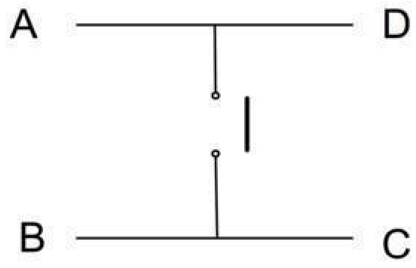
(x) x F-M wires

Component Introduction

PUSH SWITCHES:

Switches are really simple components. When you press a button or flip a lever, they connect two contacts to get her so that electricity can flow through them.





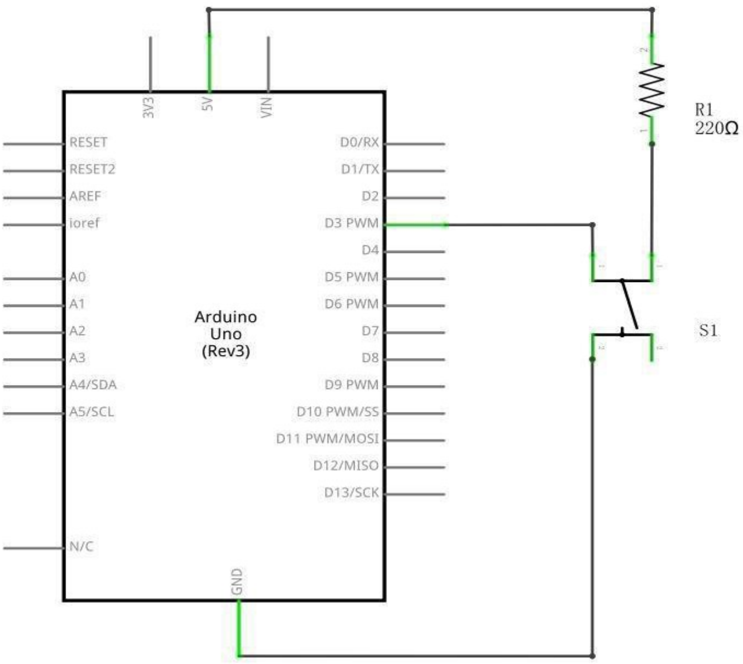
Actually, there are only really two electrical connections, as inside the switch package pins B and C are connected together, as are A and D.

Principle

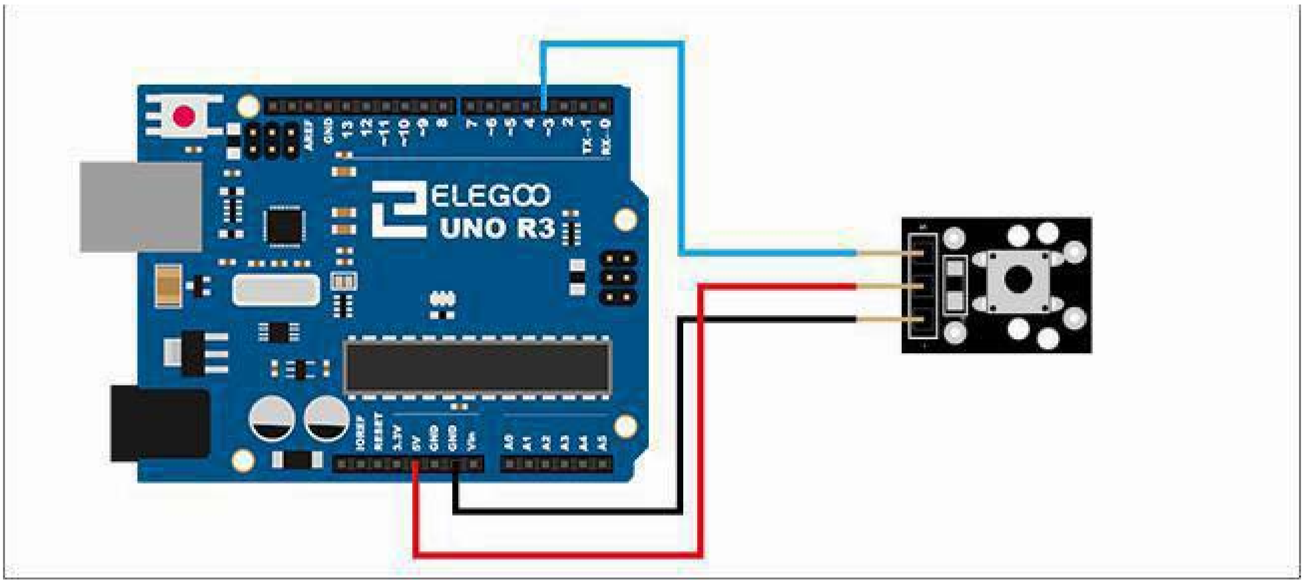
Button switch and number 13 port have the built-in LED simple circuit. To produce a switch flasher, we can use connect the digital port 13 to the built-in LED and connect the button switch Sport to number 3 port of Elegoo Uno board .When the switch sensing, LED twinkle light to the switch signal.

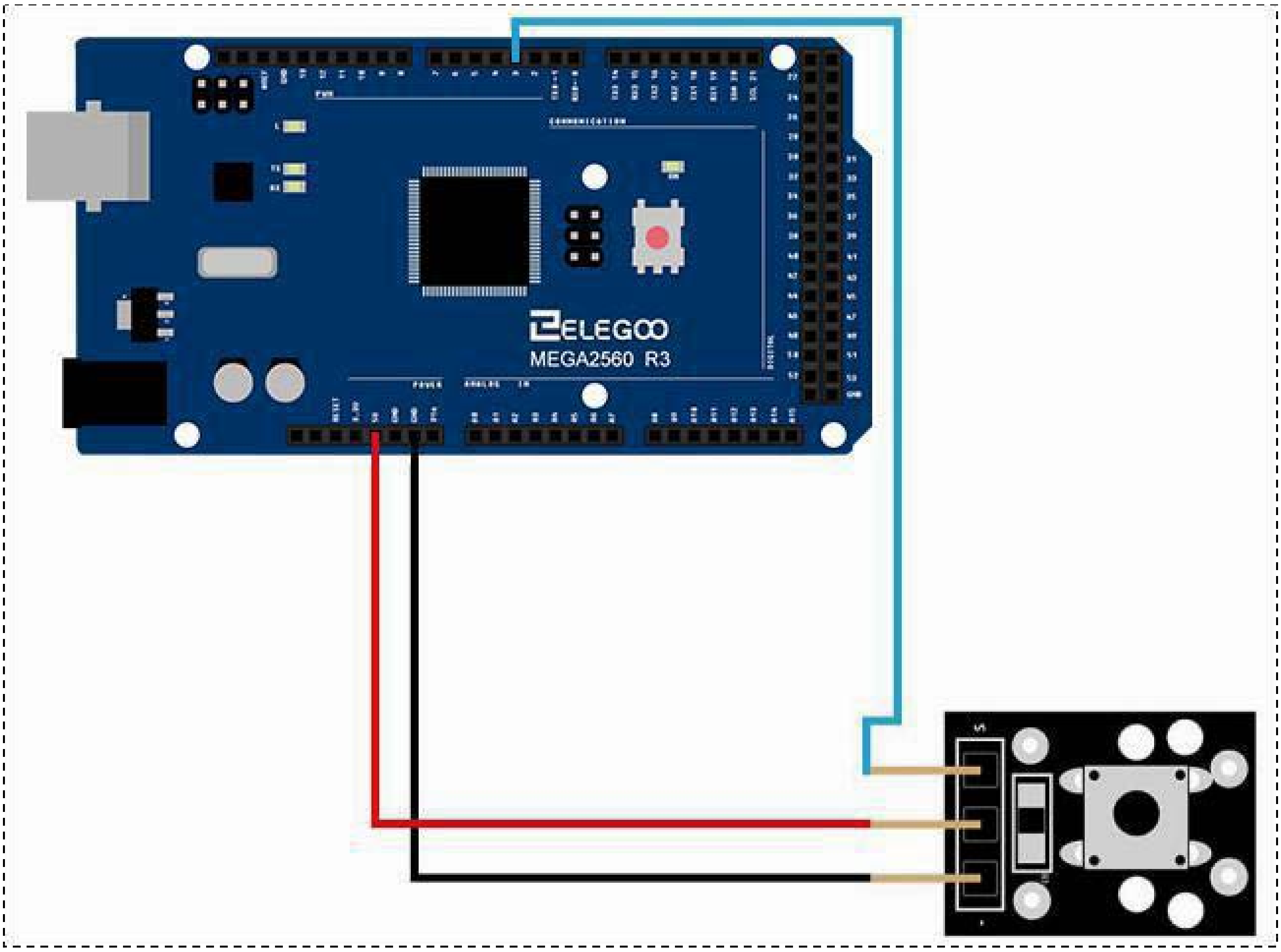
Connection

Schematic

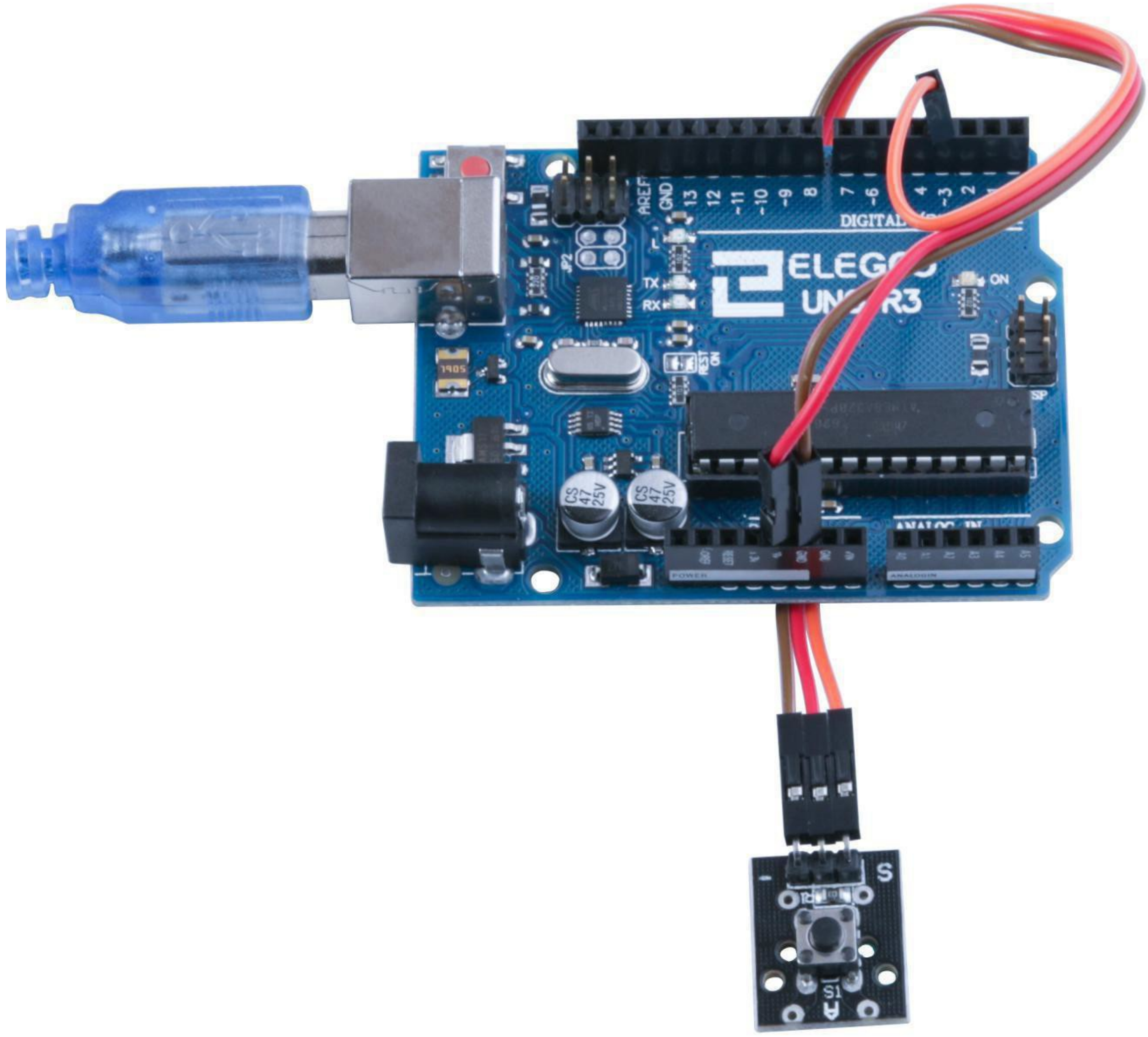


Wiring diagram





Result



Connect the circuit as above and upload theprogram. Then push the button, you can see led on and off.

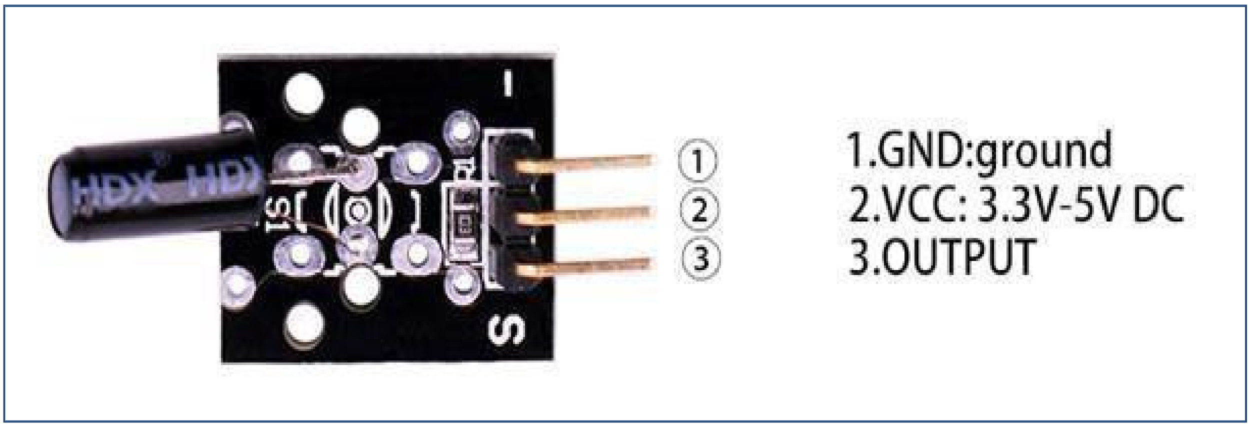
Lesson 5 Two Shock Switch Module

Overview

In this experiment, we will learn how to use switch modules. Including shock switch module and knock switch module.

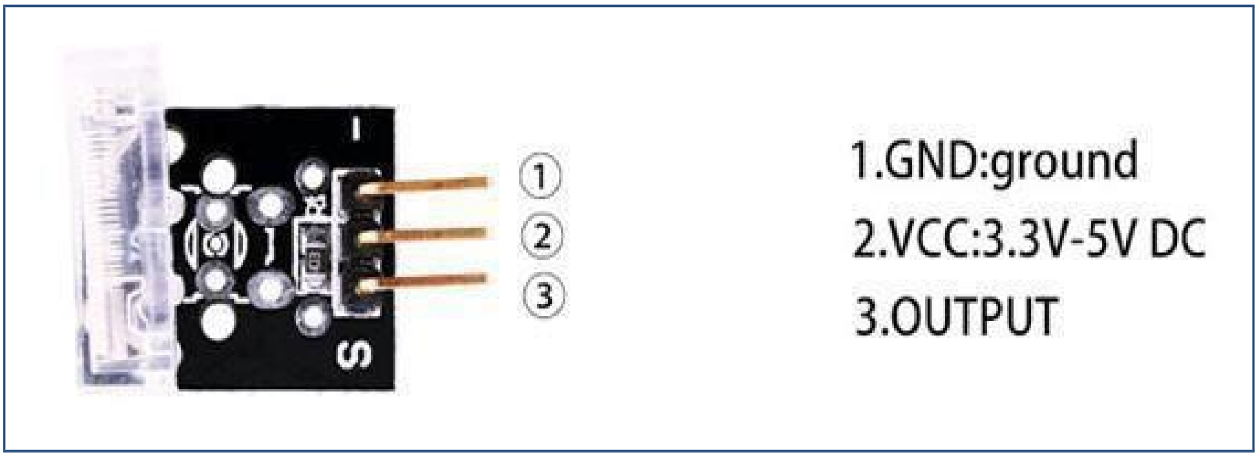
Tilt Switchmodule

A built-in 10 K ohm resistor is connected between the center pin and the 'S'pin and can be used as a pull up or pull down resistor. The switch contacts connect to the two outer pins.



Hit SensorModule

Vibration sensor module. The momentary switch contacts are connected between the two outer pins.



Component Required:

(1) x Elegoo Uno R3

(1) x USB cable

(1) x Shock switch module

(1) x Knock switch module

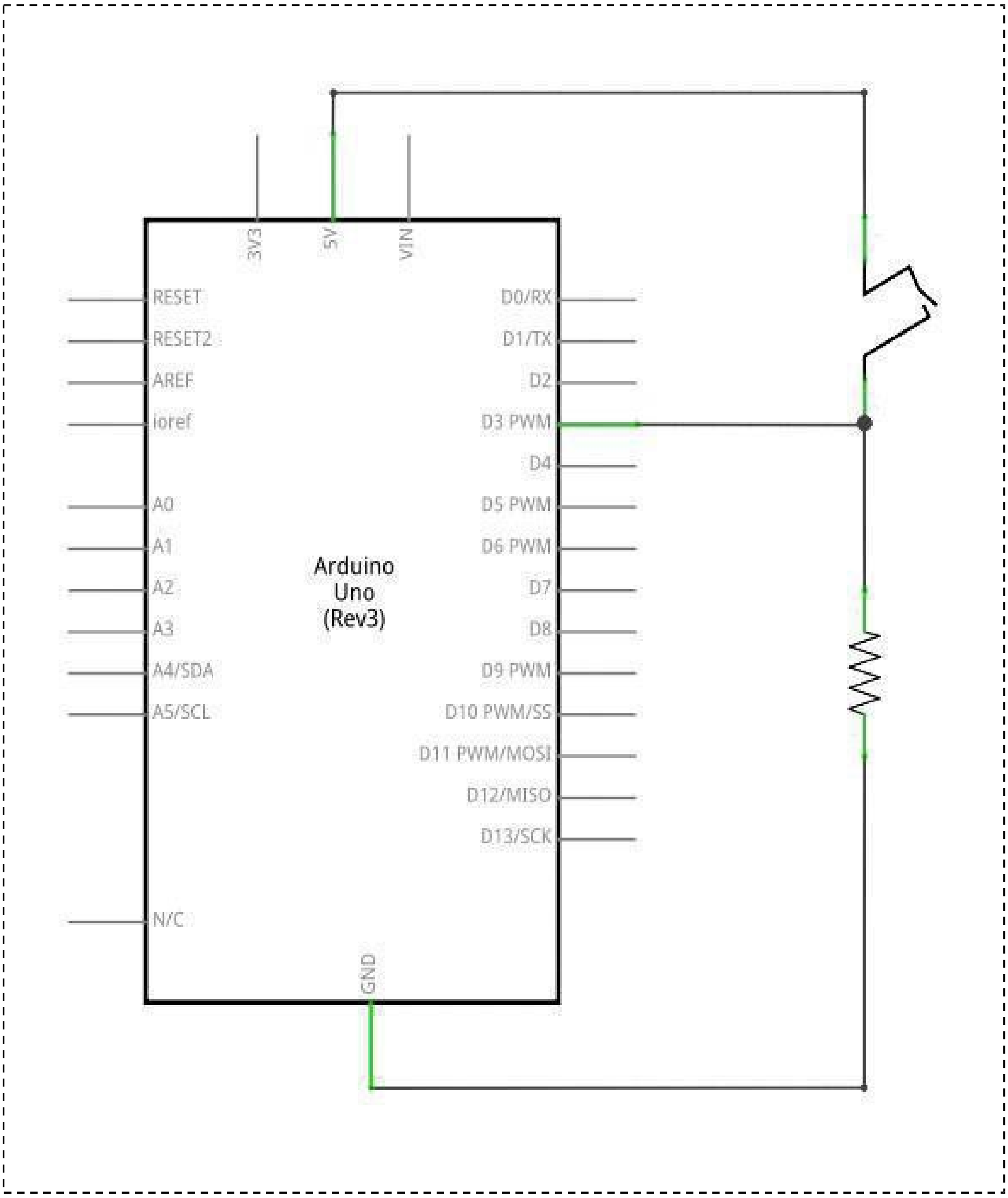
(x) x F-M wires

Principle

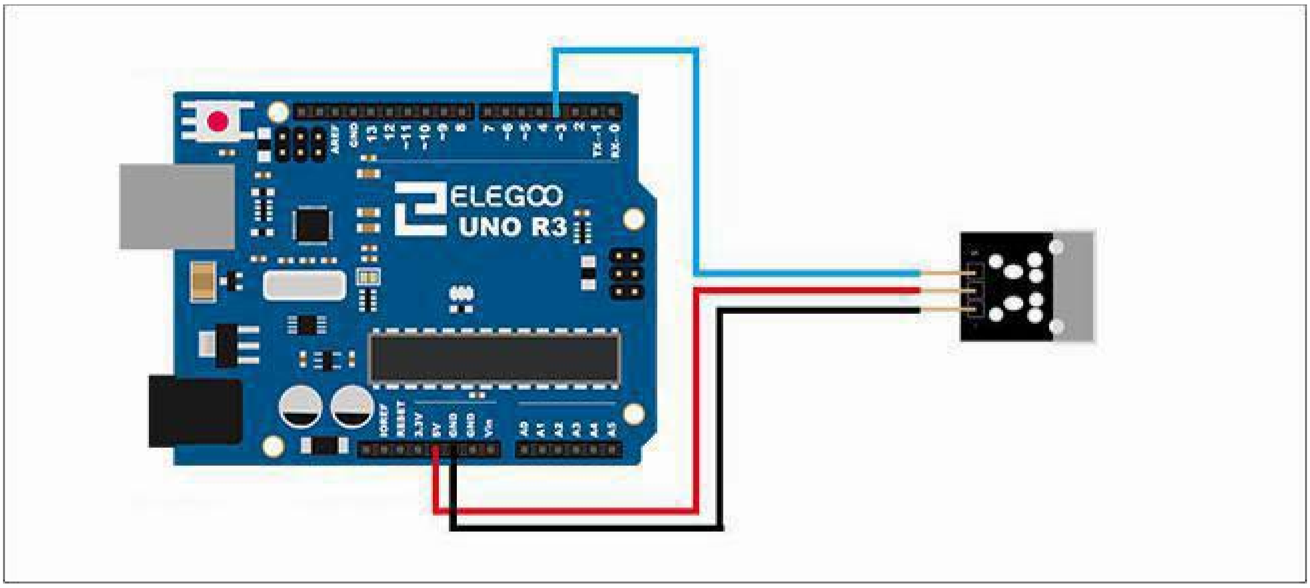
Switch and number 13 port have the built-in LED simple circuit. To produce a switch flasher, we can use connect the digital port 13 to the built-in LED and connect the switch sport to number 3 port of Elegoo Uno board. When the SWITCH sensing, LED twinkle light to the switch signal.

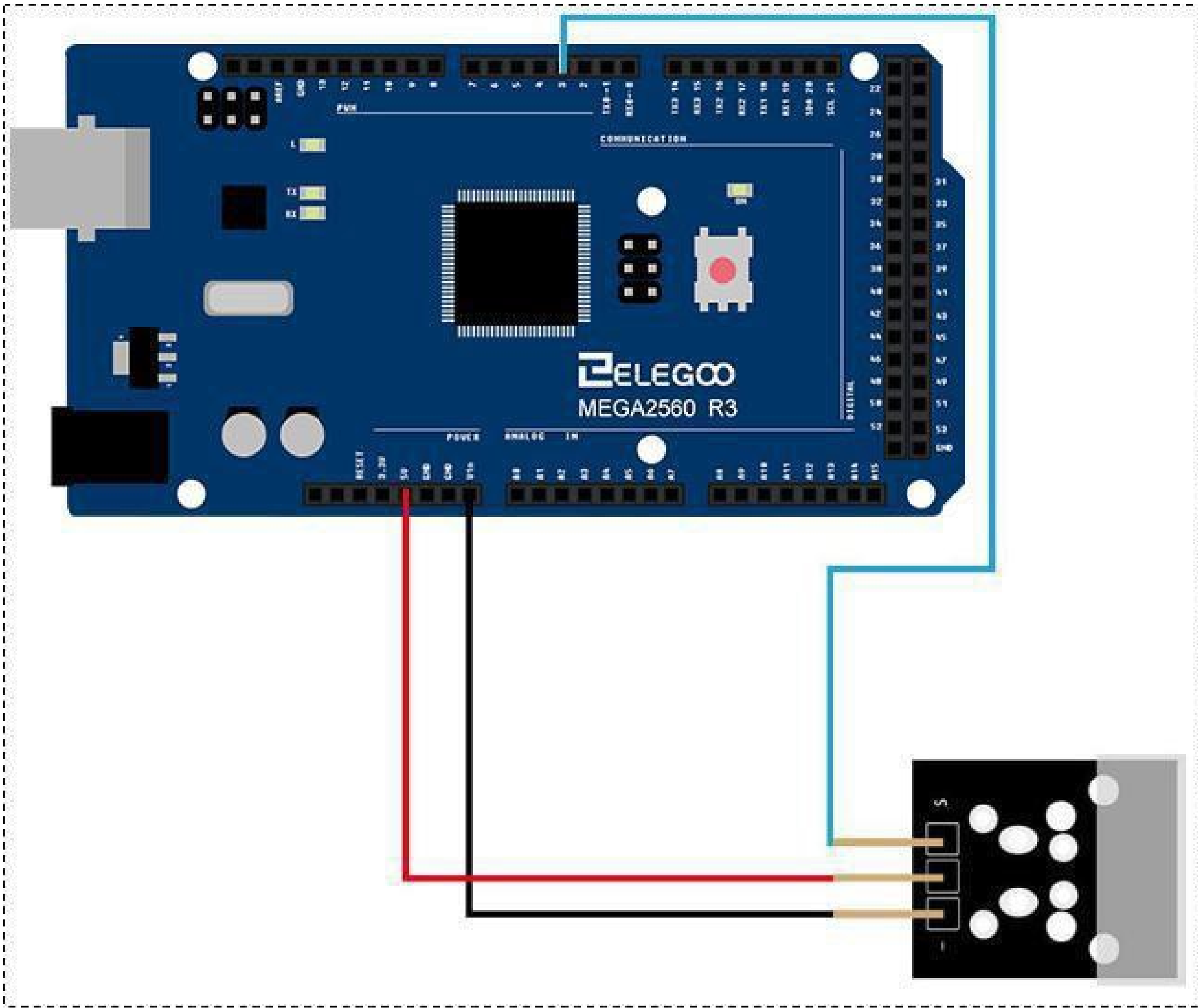
Connection

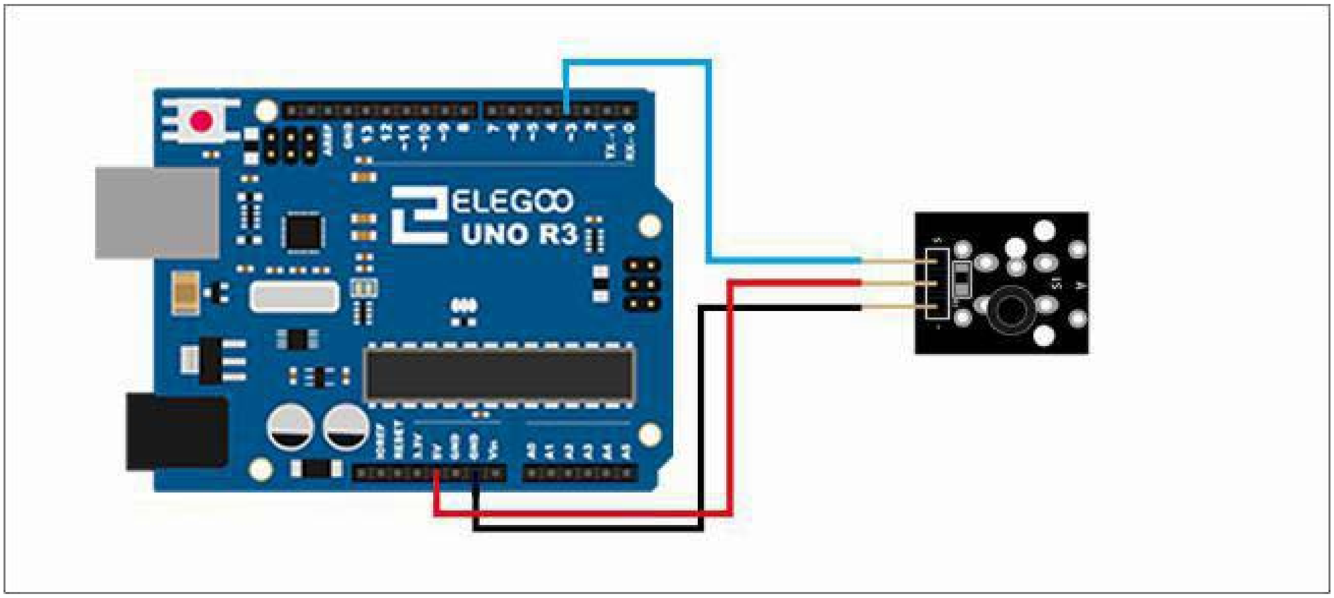
Schematic

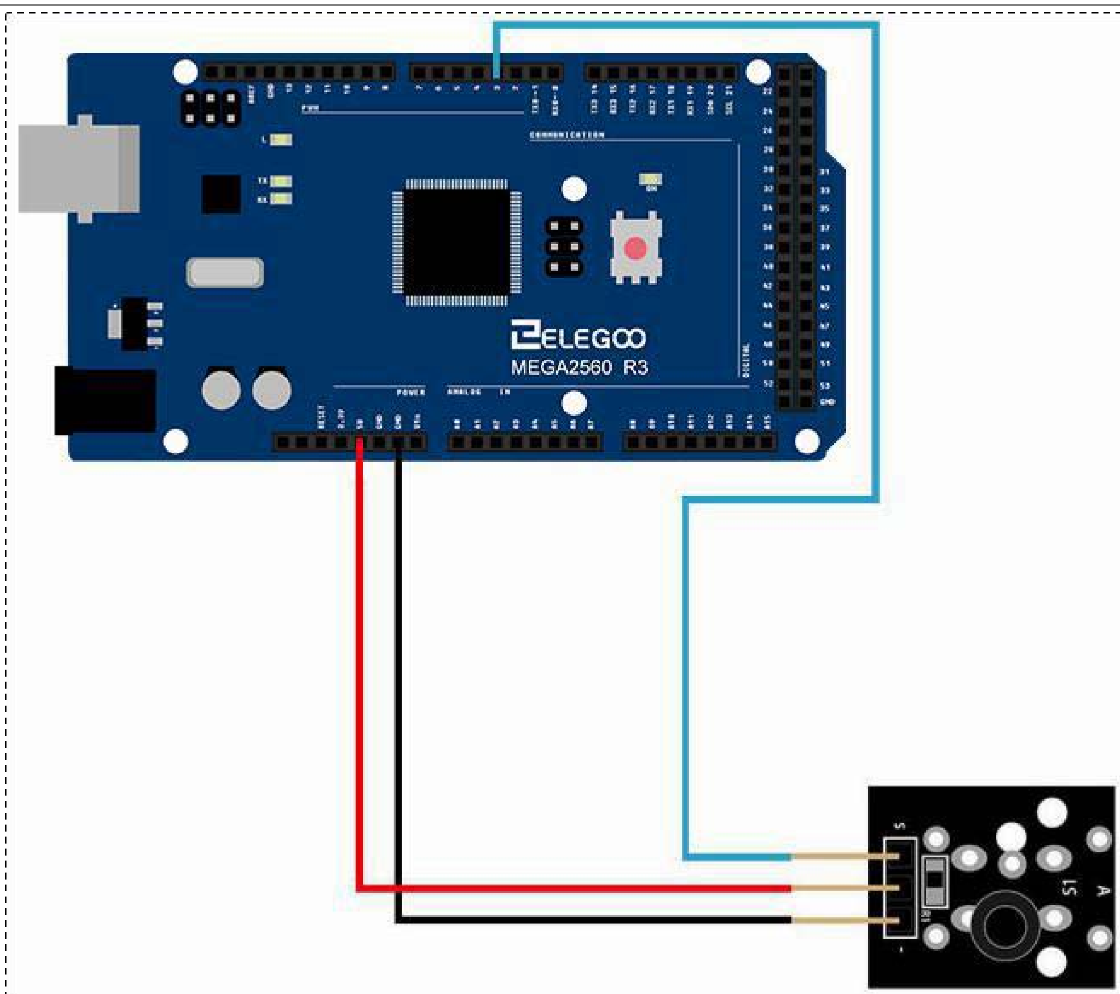


Wiring diagram:

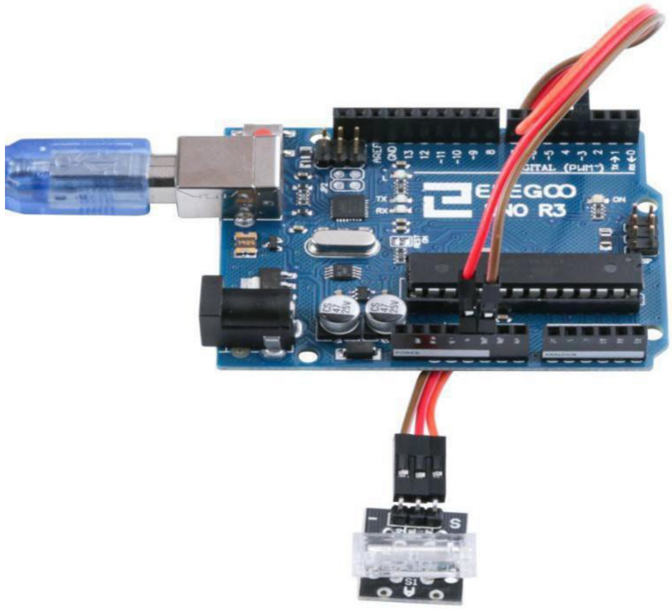


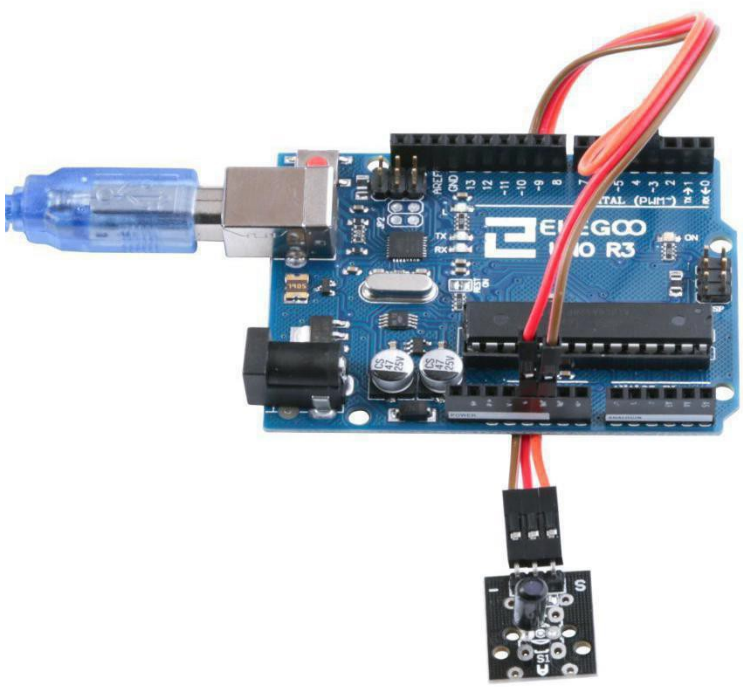



ELEGOO
UNO R3


ELEGOO
MEGA2560 R3

Result





Connect the circuit as above and upload the program, you can see led on and off.

Lesson 6 Two Tilt Switch Module

Overview

In this experiment, we will learn how to use switch modules.

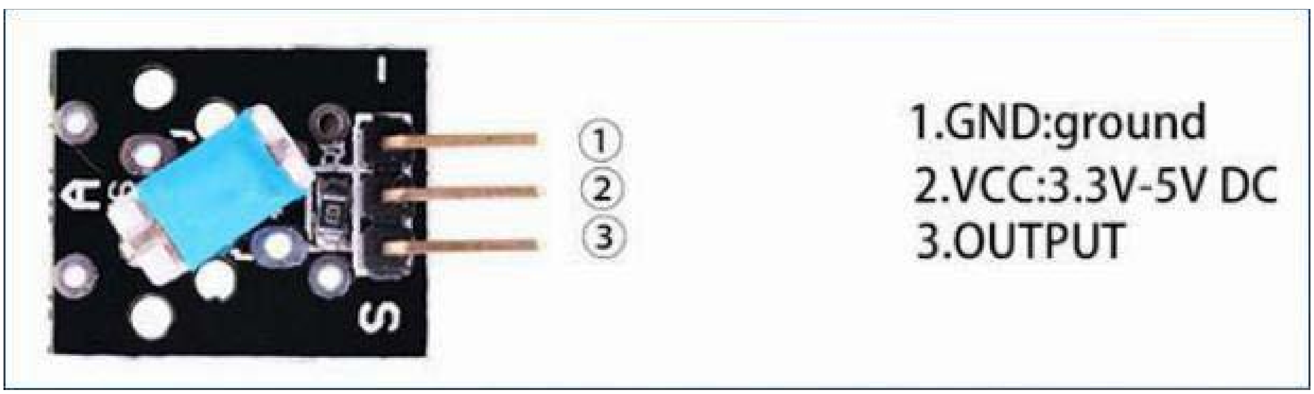
Including tilt switch module and mercury switch module.

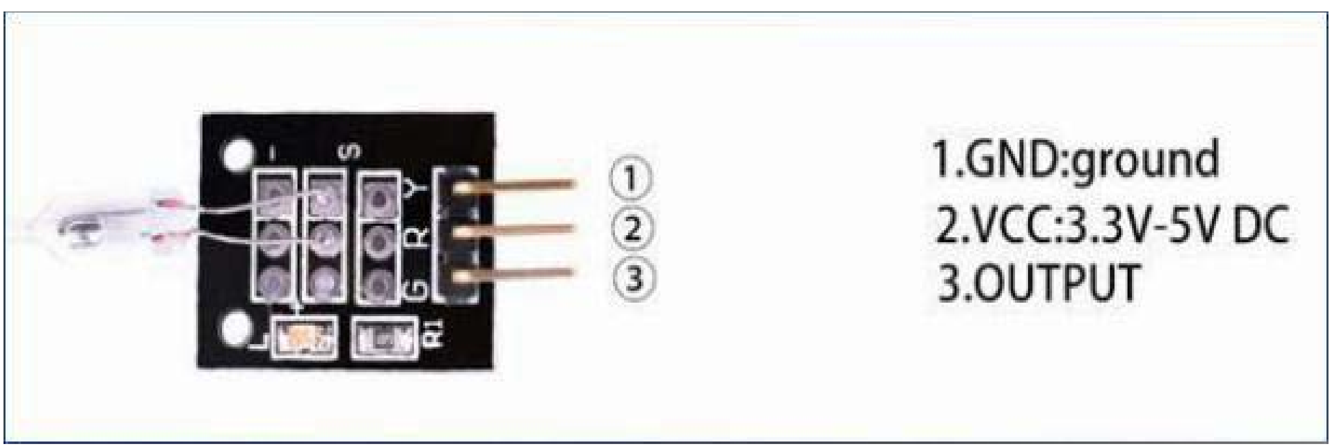
Ball Switch sensor

A built-in 10 K Ohm resistor connected between the middle and 'S' pin is available for pull up or pull down use. The switch contacts connect to the two outer pins .Load switching max: 12V DC 50mA.

Mercury Tilt Switch

Mercury tilt switch which makes or breaks depending on its attitude.





Component Required:

(1)x Elegoo Uno R3

(1) x USB cable

(1) x Mercury switch module

(1)x tilt switch module

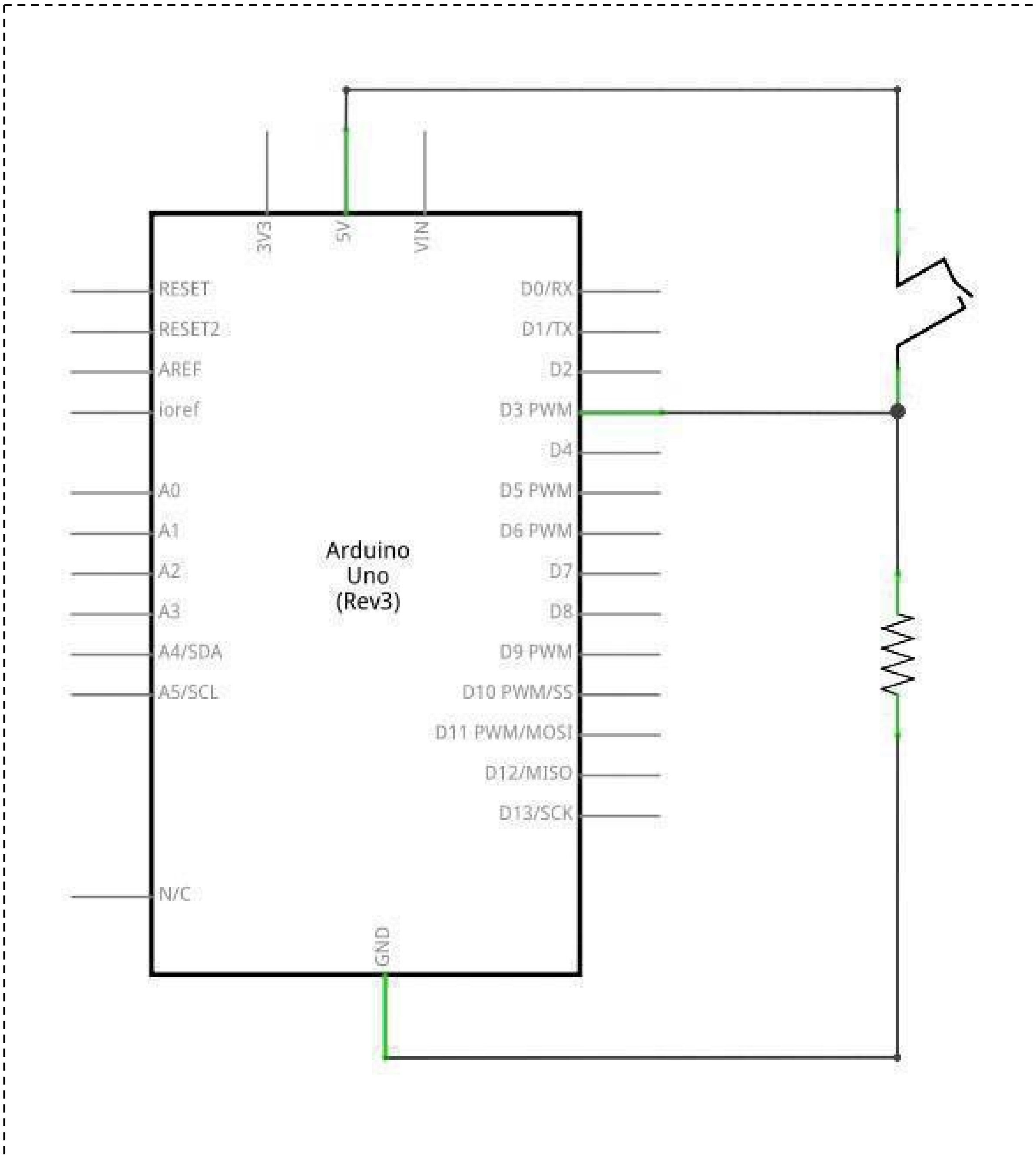
(x)x F-M wires

Principle:

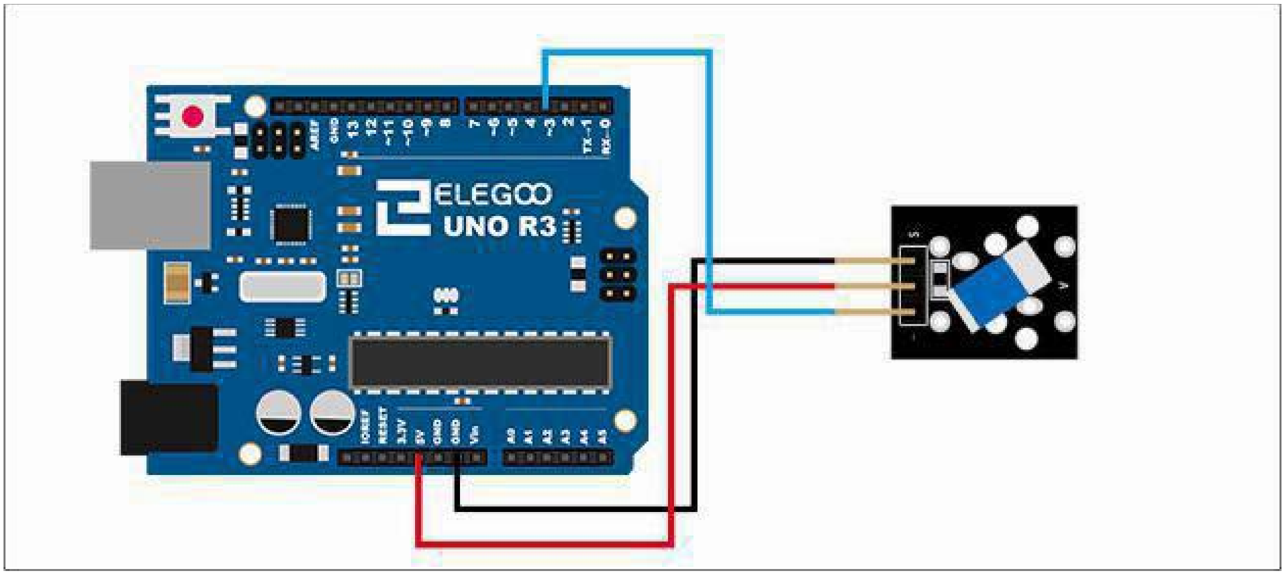
Switch and number 13 port have he built-in LED simple circuit .to produce a switch flasher, we can use connect the digital port 13 to the built-in LED and connect the switch Sport to number 3 port of Elegoo Uno board. When the SWITCH sensing, LED twinkle light to the switch signal.

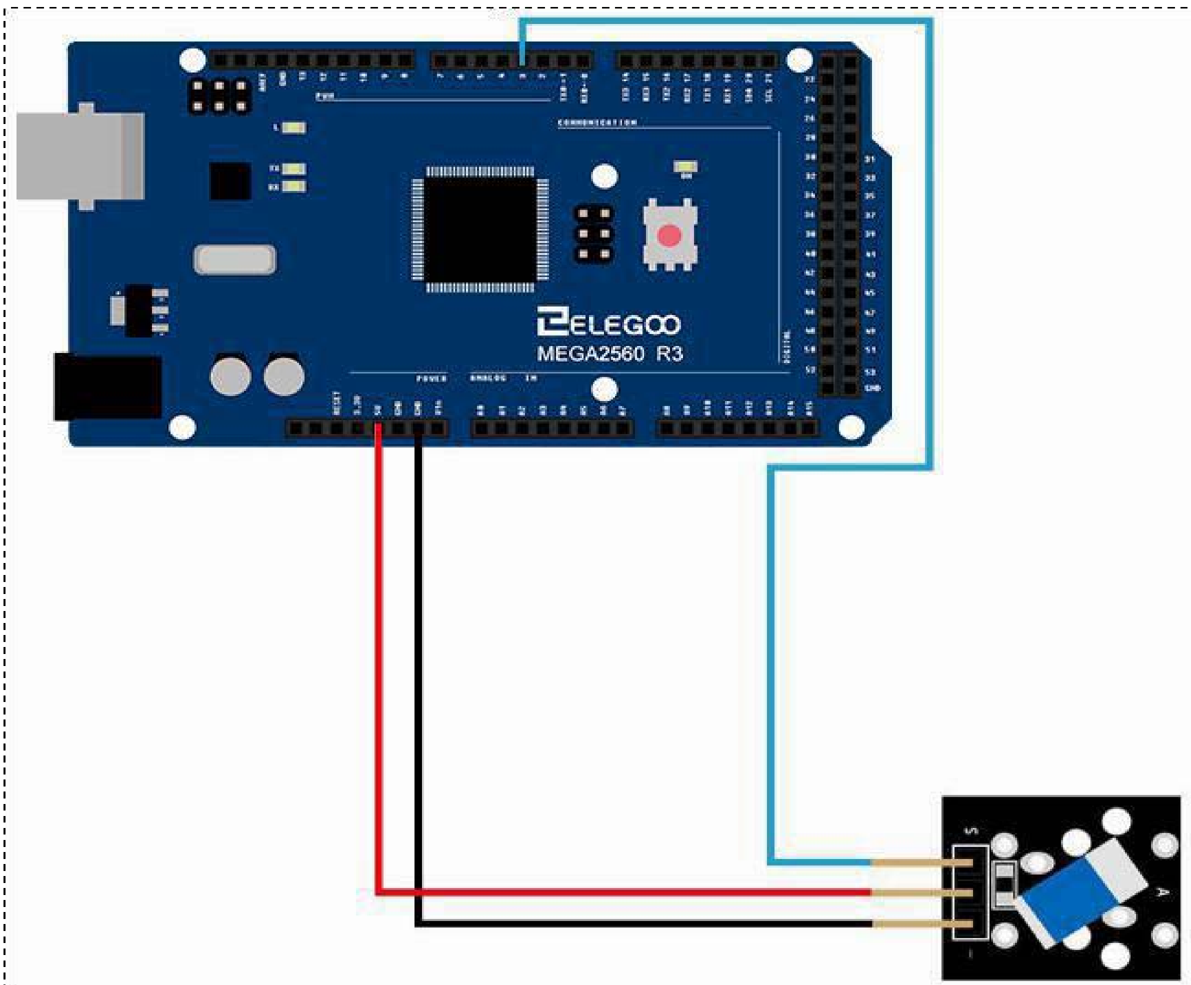
Connection

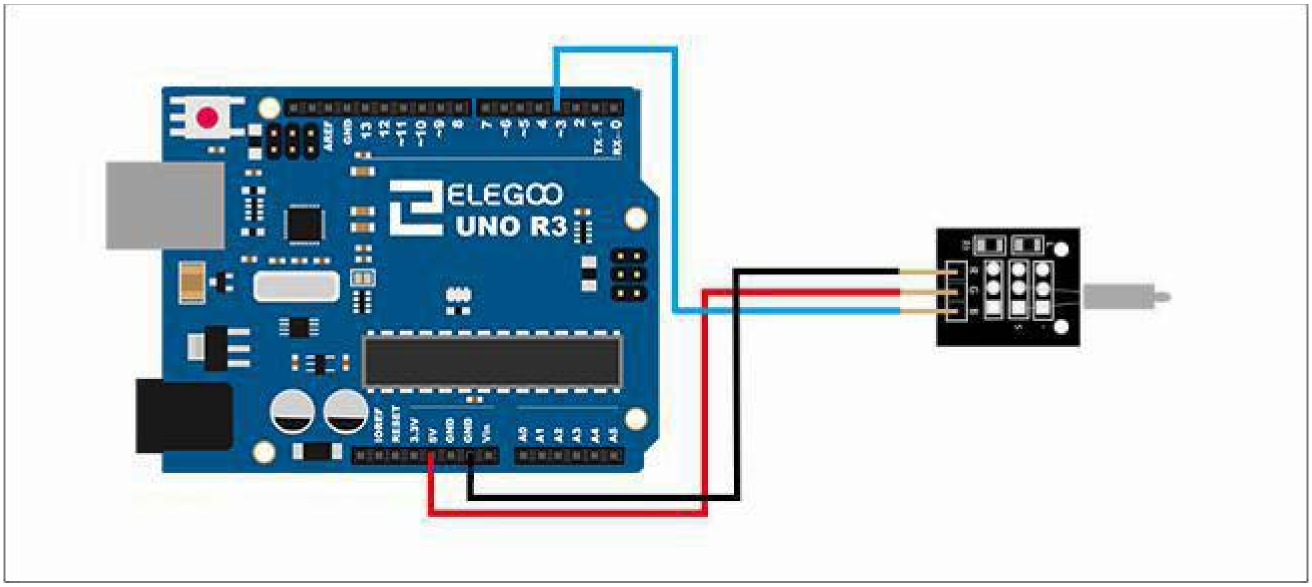
Schematic

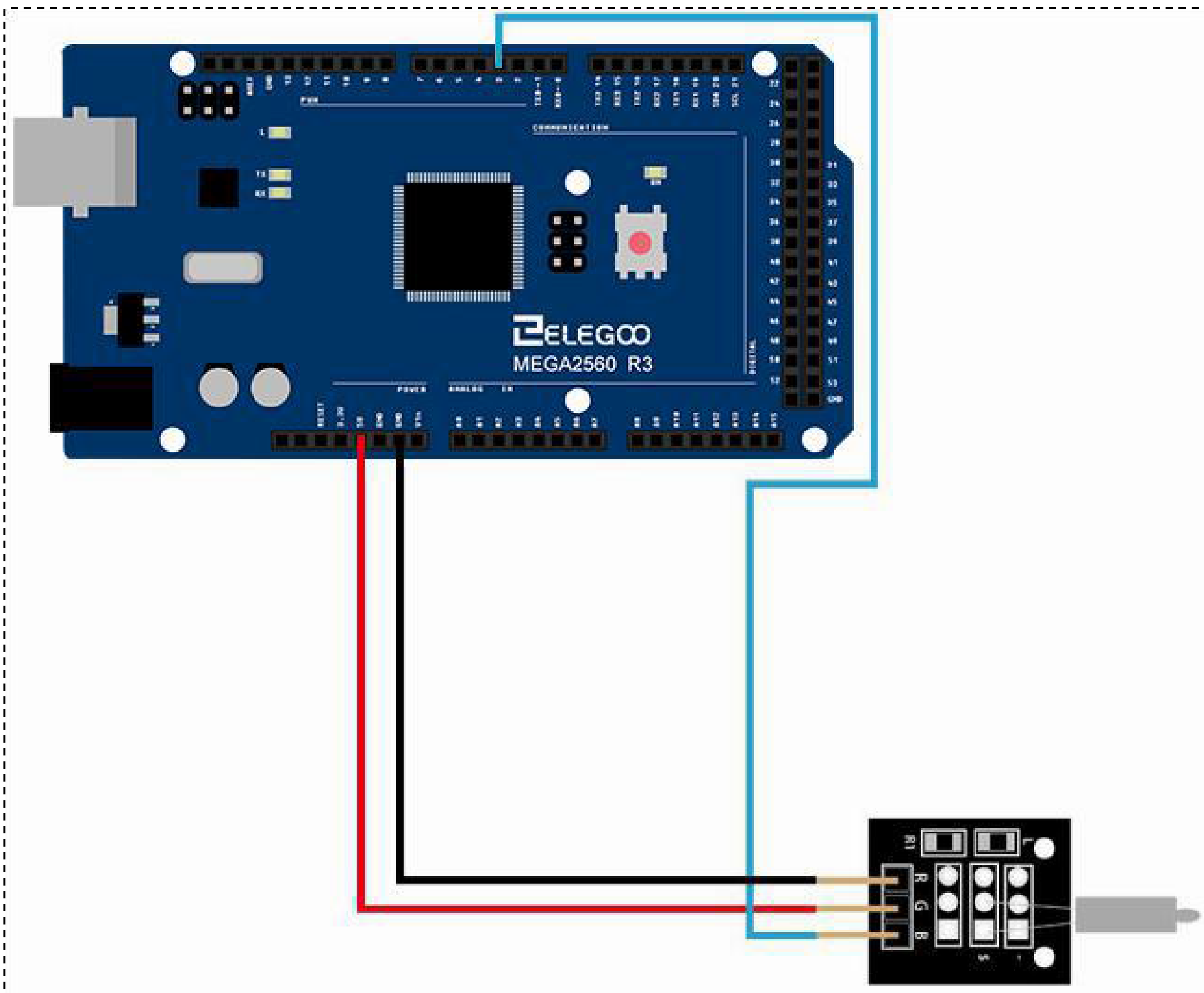


Wiring diagram

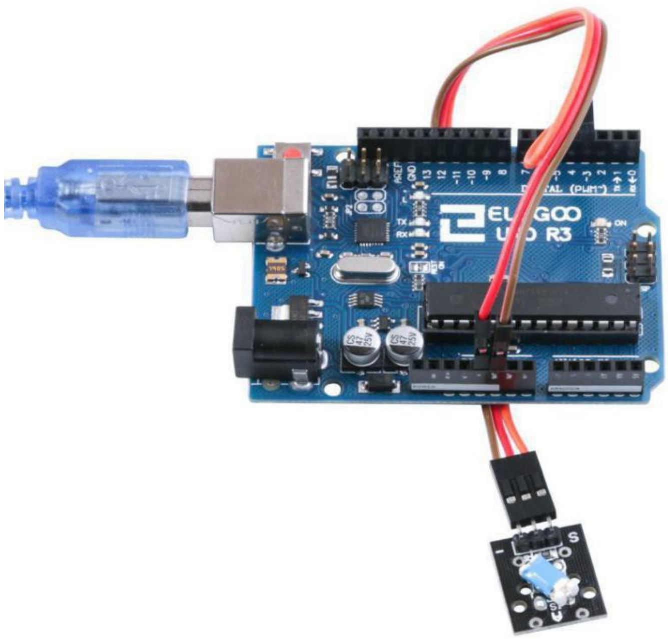


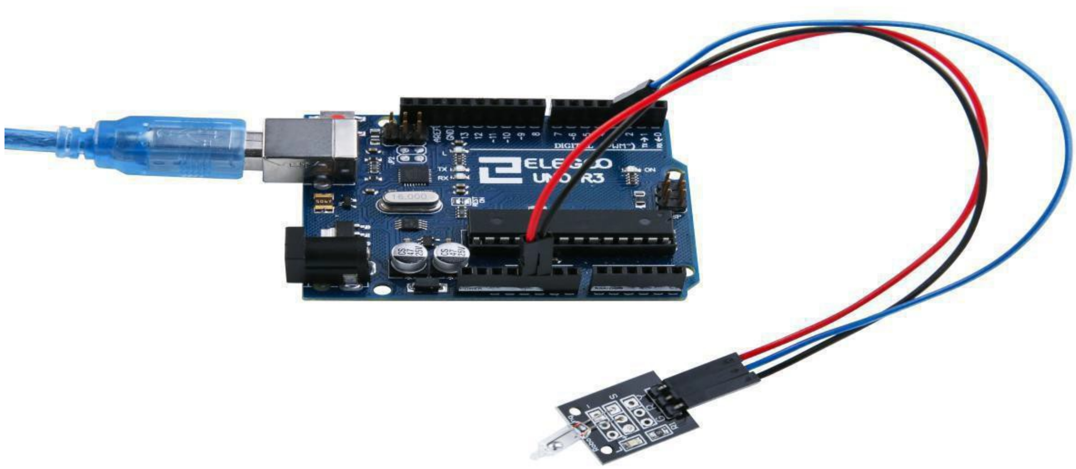



ELEGOO
UNO R3


ELEGOO
MEGA2560 R3

Result





Connect the circuit as above and upload the program, you can see led on and off.

Lesson 7 IR Receiver and IR Transmitter

Overview

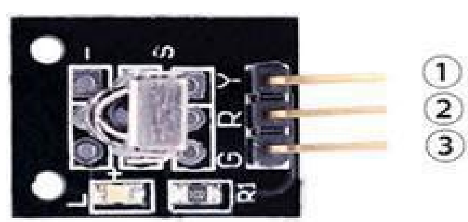
In this experiment, we will learn how to use Infrared Receiver and IR transmitter module.

In fact now in our daily life they play an important role, a lot of household electrical appliances are used to this kind of device, such as air conditioning, TV, DVD, etc. Actually it is based on its wire less remote sensing and it is very convenient by using them.

IR Receiver module

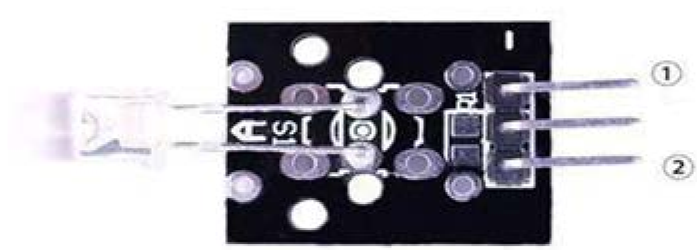
Infrared sensor type 1838 for use with 38 KHz IR signals.

- Supply voltage: 2.7 to 5.5 V
- Frequency: 37.9 KHz
- Receiver range: 18 m (typical)
- Receiving angle: 90°



IR Transmitter module

The IR-LED can be used to build a light barrier or an IR remote control signal transmitter.



Component Required:

(2)x Elegoo Uno R3

(1)x USB cable

(1)x IR Receiver module

(1)x IR Emission module

(x) x F-M wires

Component Introduction

IR RECEIVER SENSOR:

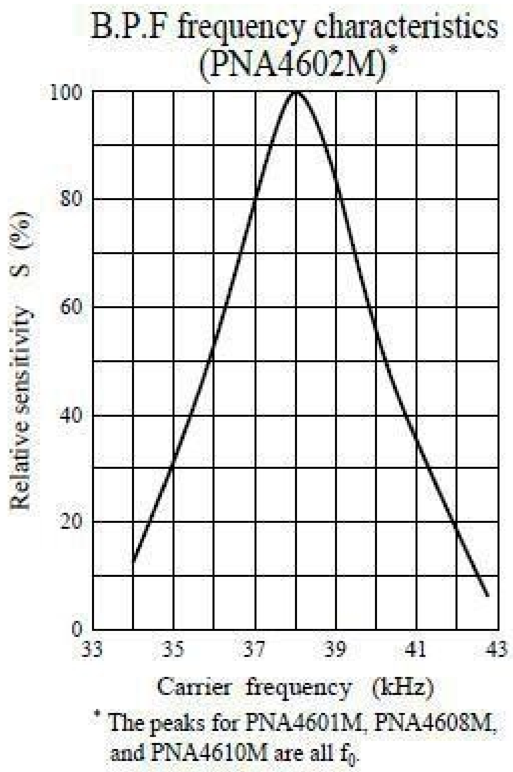
IR detectors are little microchips with a photocell that are tuned to listen to infrared light. They are almost always used for remote control detection - every TV and DVD player has one of these in the front to listen for the IR signal from the clicker. Inside the remote control is a matching IR LED, which emits IR pulses to tell the TV to turn on, off or change channels. IR light is not visible to the human eye, which means it takes a little more work to test a setup.

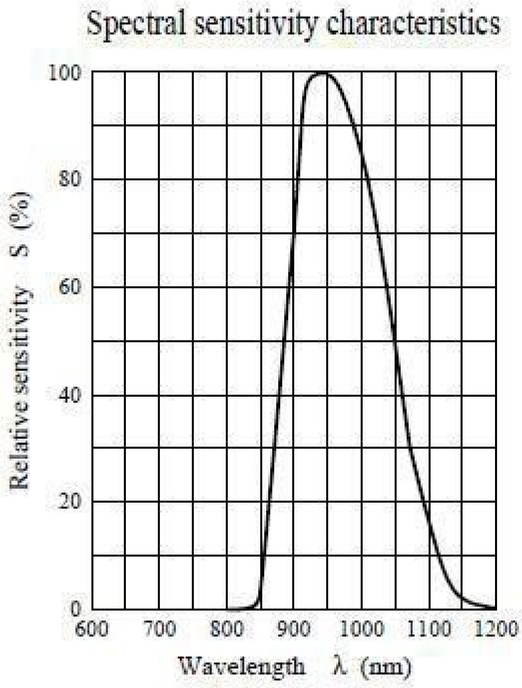
There are a few difference between these and say a CdS Photocells:

IR detectors are specially filtered for Infrared light, they are not good at detecting visible light. On the other hand, photocells are good at detecting yellow/green visible light, not good at IR light

- IR detectors have a demodulator inside that looks for modulated IR at 38 KHz. Just shining an IR LED won't be detected, it has to be PWM blinking at 38 KHz. Photocells do not have any sort of demodulator and can detect any frequency (including DC) within the response speed of the photocell (which is about 1 KHz)
- IR detectors are digital out - either they detect 38 KHz IR signal and output low (0 V) or they do not detect any and output high (5V). Photocells act like resistors, the resistance changes depending on how muchlight they are exposed to.

What You CanMeasure



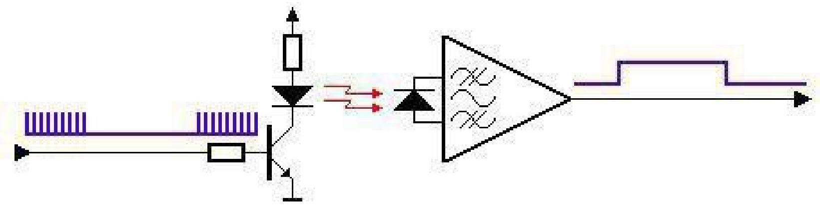


As you can see from these data sheet graphs, the peak frequency detection is at 38 KHz and the peak LEDcolor is 940 nm .You can use from about 35 KHz to 41 KHz but the sensitivity will drop off so that it wont detect as well from a far. Like wise, you can use 850 to 1100 nm LEDs but they won't work as well as 900 to 1000 nm so make sure to get matching LEDs! Check the datasheet for your IR LED to verify the wavelength try to get a 940 nm - remember that 940 nm is not visible light (its Infra Red)!

Principle

Firstly, let's know the structure of the infrared receiving head: there are two important elements inside the infrared receiving head, IC and PD. IC is receiving head processing components, mainly composed of silicon and circuit. It is a highly integrated device. The main function is filter, plastic, decoding, amplification, etc. PD is a photosensitive diode. The main function is to receive the light signal.

Below is a brief working principle diagram:



Infrared emitting diode launch out the modulation signal and infrared receiver head will receive, decode, filter and soon to regain the signal.

Infrared emitting diode: keep clean and in good condition. All the parameters in the process of working shall not exceed the limit value (positive To the current 30~60 mA, positive pulse current 0.3~1 A, reverse voltage 5V, dissipation power 90 mW, working temperature range -25~+80℃, and storage temperature range between 40~100℃, the welding temperature 260℃) infrared to be with a closed head should be matching use, other wise it will influence the sensitivity.

Connection

Firstly, take a UNO board and connect the Infrared Emission module according to the attached picture.

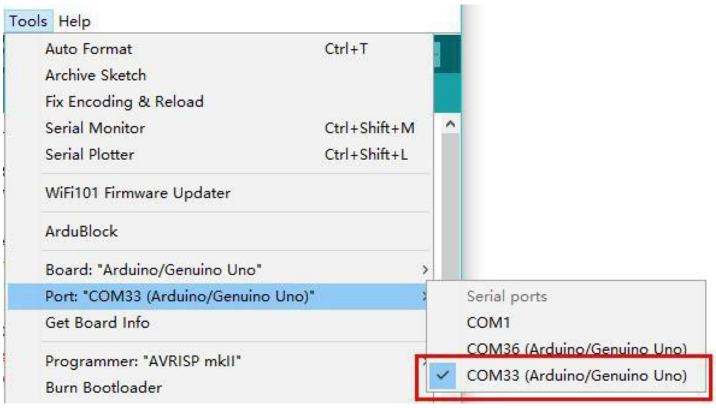
Then follow the previous procedure to write the Infrared Emission program.

IR_Emission

Next, take another UNO board, connect the Infrared Receiver module as shown below and write IR Receiver program.

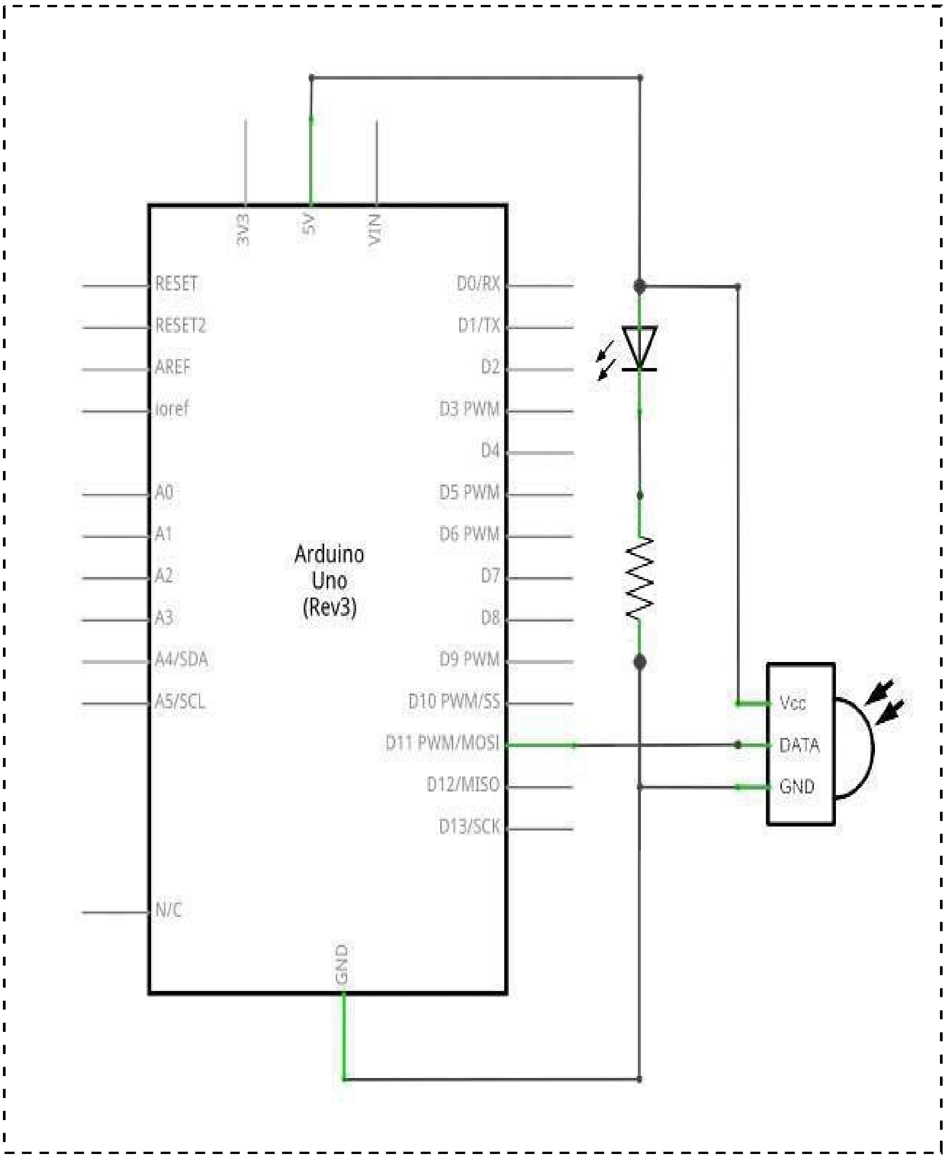
IR_Receiver

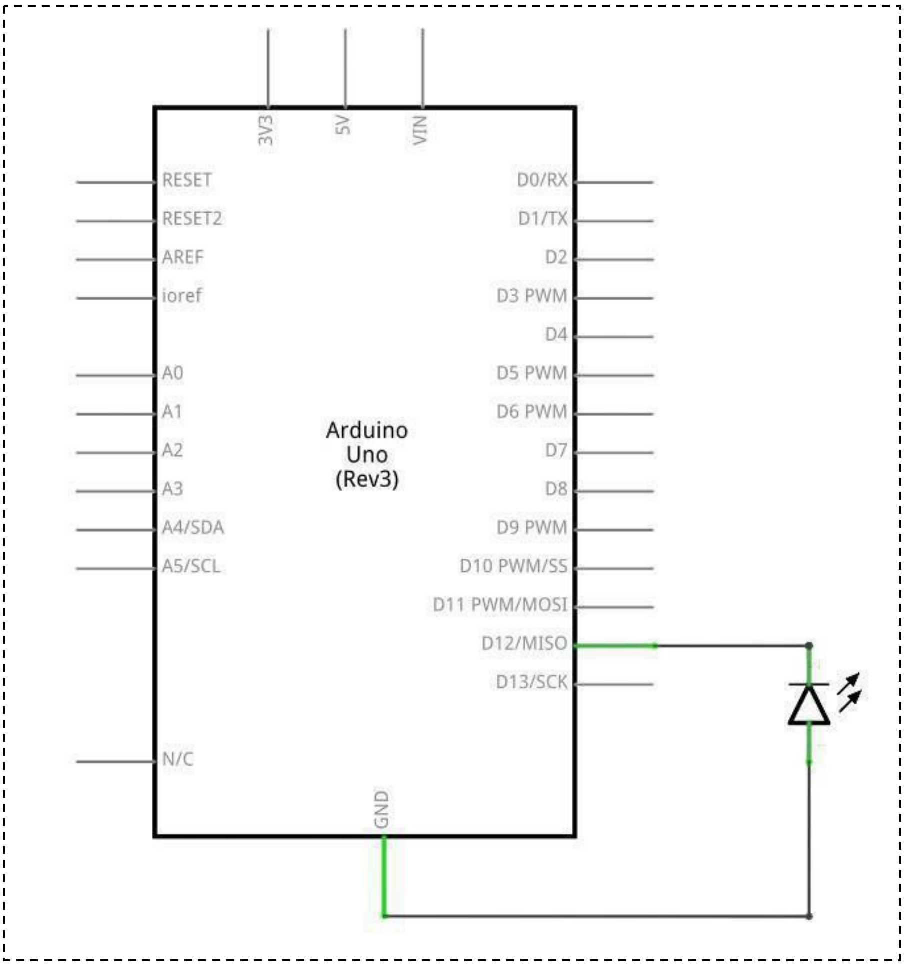
Note: Before downloading, be sure to reselect the download port. There will be two serial ports When you connect two UNO boards, we must choose the new one, as the picture shown.



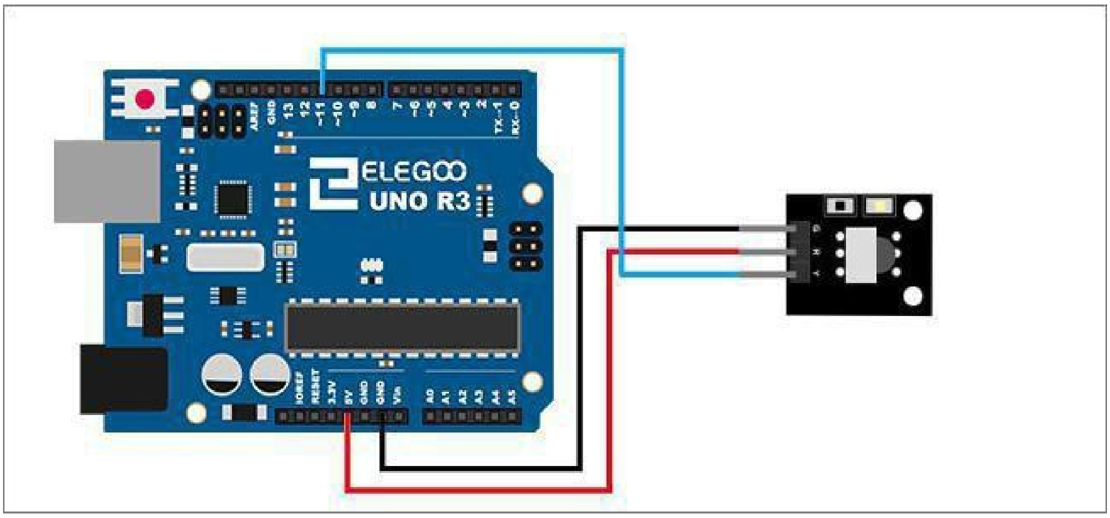
After wrote successfully, align the infrared emission module with the infrared receiver module. Observe the UNO of the Infrared receiver, if you can see the L light flashes, it says the experiment is successful. The principle of the experiment is that the infrared emission module transmits the specific coded signal .After the infrared receiver module received, the UNO will decode and operates the L lamp to blink. This experiment can be extended use: The IR signal control UNO to operate other modules such as robot, toys, appliances and etc.

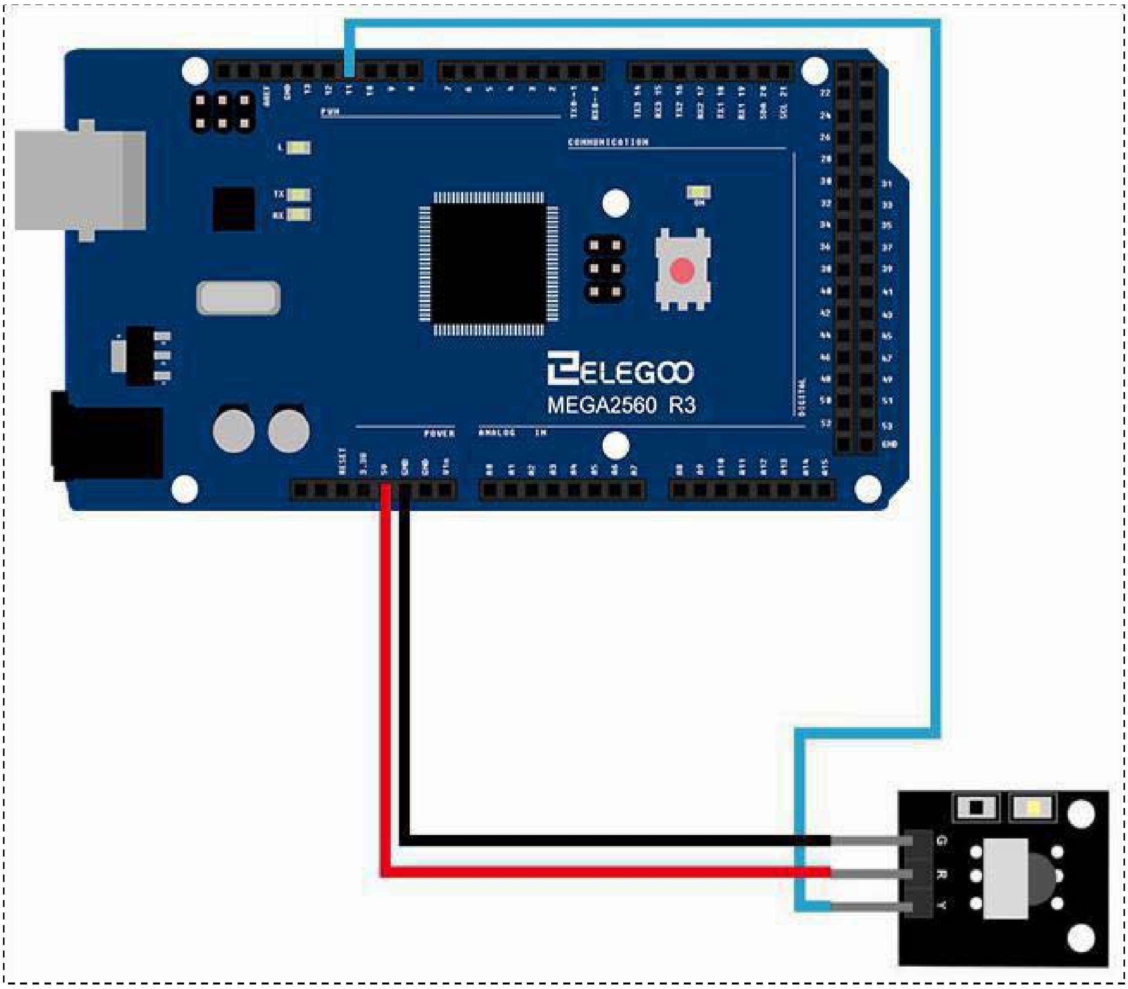
Schematic

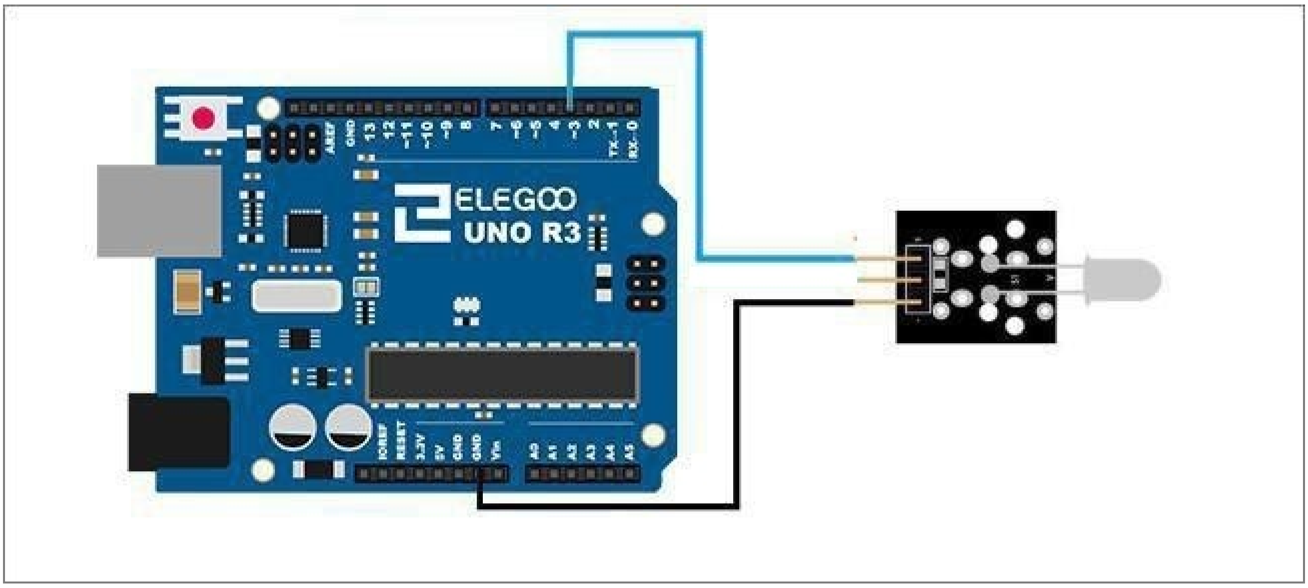


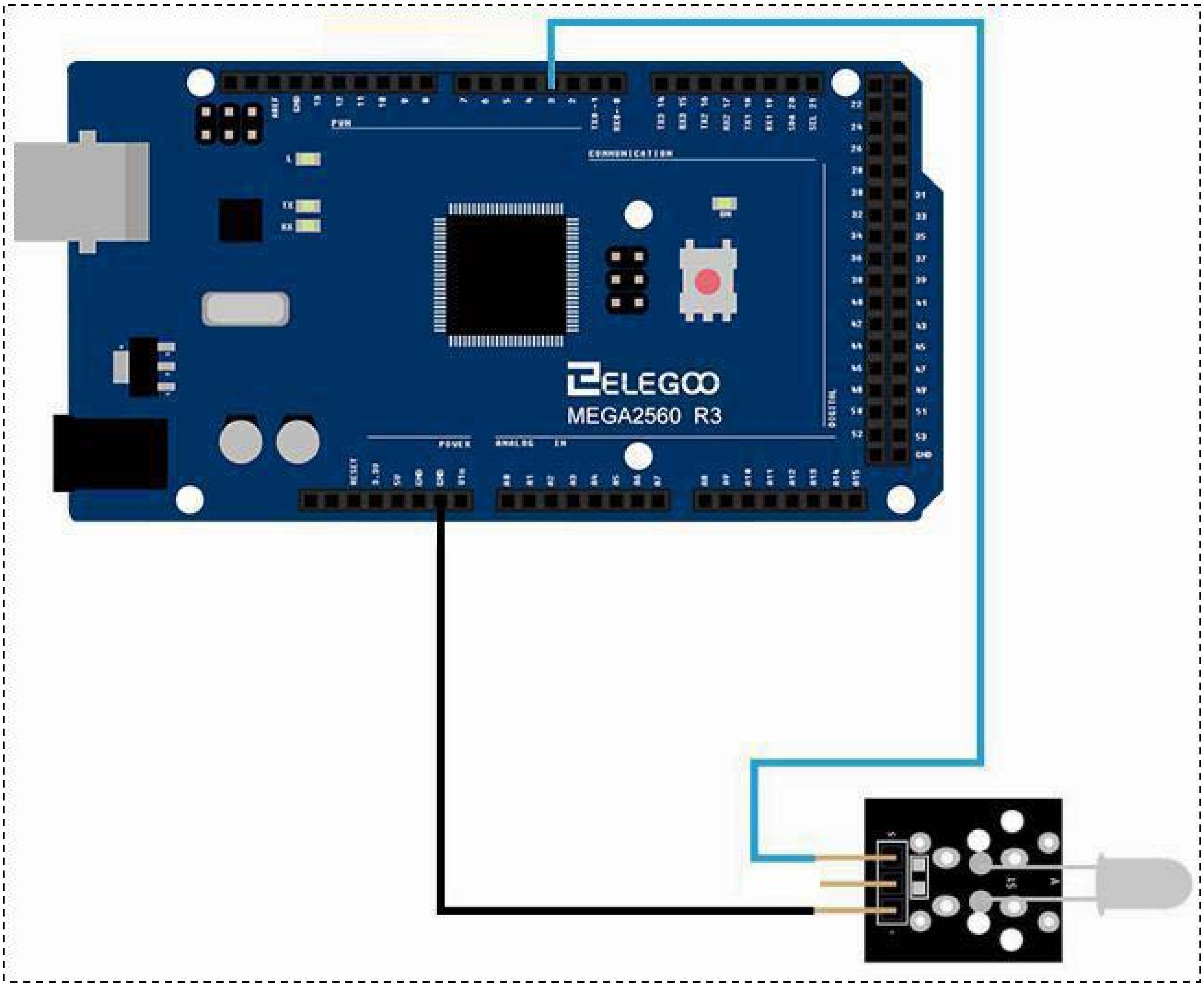


Wiring diagram

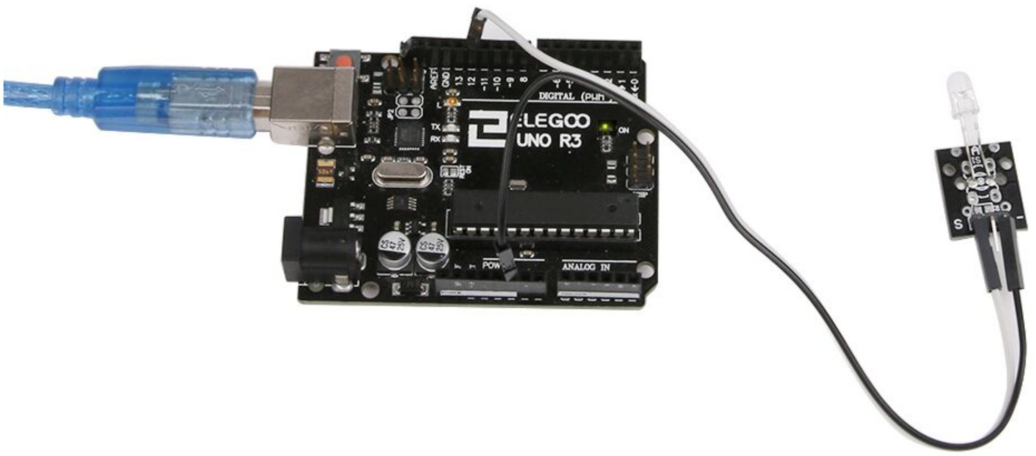



ELEGOO
MEGA2560 R3


ELEGOO
UNO R3


ELEGOO
MEGA2560 R3

Result:



Trouble shooting:

If the L lamp didn't flash, it may be the error of connection or the wrong operation when you wrotethe program .Please refer to the previous steps and modify the mistakes.

But if you are sure the connection and program are correct. There are two ways to help you determine whether the IR receiver module and the IR emission module are damaged.

The receiver module.

Please use your TV remote control to control the IR receiver module. If the IR receiver module flashes, it shows the IR receiver module is normal. If not, it couldn't be used.

The emission module.

Open your Android phone's camera, put in front of the IR emission's LED. If you could see thepurple light, it shows this module is correct, or else it'swrong.

If the modules came across defective, please contact us and we will send the replacement to yourplace.

Lesson 8 Active Buzzer Module

Overview

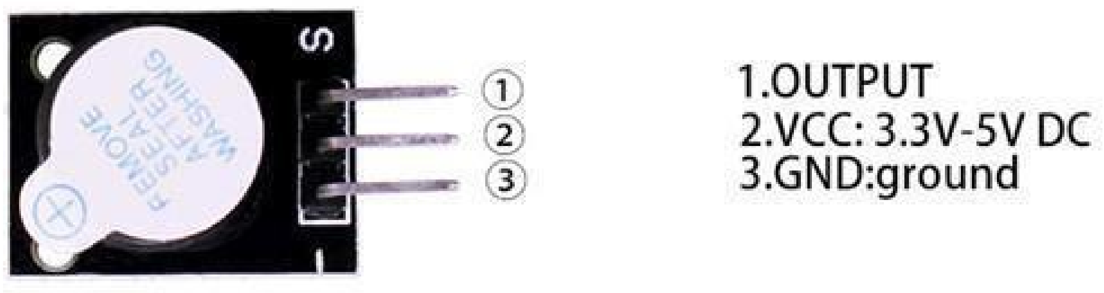
In this experiment, we will learn how to use buzzer module.

With the Arduino we can complete a lot of interactive work, commonly what we used is the light shows. And we has been use the LED small lights in the experiment before. In this experiment we will make the circuit having noise. The common components that can make sound are buzzer and speakers. Compared to the speaker, buzzer is more simple and easy to use so in this experiment we adopts the buzzer.

Active buzzer module

Electronic buzzer for 5 V operation. *Ensure correct polarity!!*
Positive supply to the '-' pin and ground to the 'S' pin!
Data: Typical operating frequency 4000 Hz at 80 dB min, 5V DC at 5 mA typical
TMB12A 05 or equivalent.

Tip to avoid mix up: The buzzer housing is slightly taller than the loudspeaker housing and has a label showing the + pinidentification.



Component Required:

(1) x Elegoo Uno R3

(1) x USB cable

(1) x Active buzzer module

(x) x F-M wires

Component Introduction

Active Buzzer:

As a type of electronic buzzer with integrated structure, buzzers, which are supplied by DC power, are widely used in computers, printers, photocopiers, alarms, electronic toys, automotive electronic devices, telephones, timers and other electronic products for voice devices.

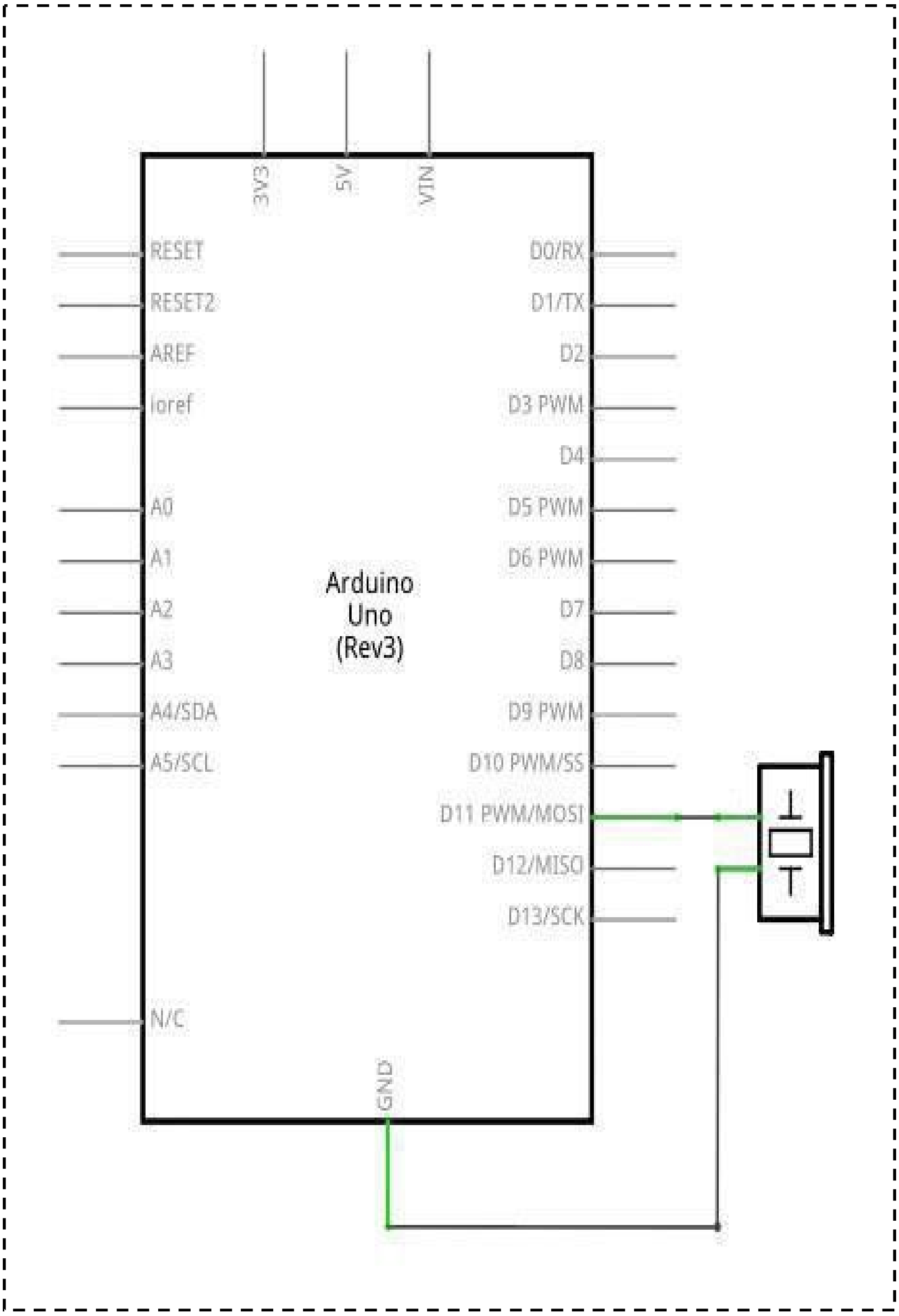
Turn the pins of two buzzers face up, and the one with a green circuit board is a passive buzzer, while the other enclosed with a black tape is an active one.

The difference between an active buzzer and a passive buzzer is:

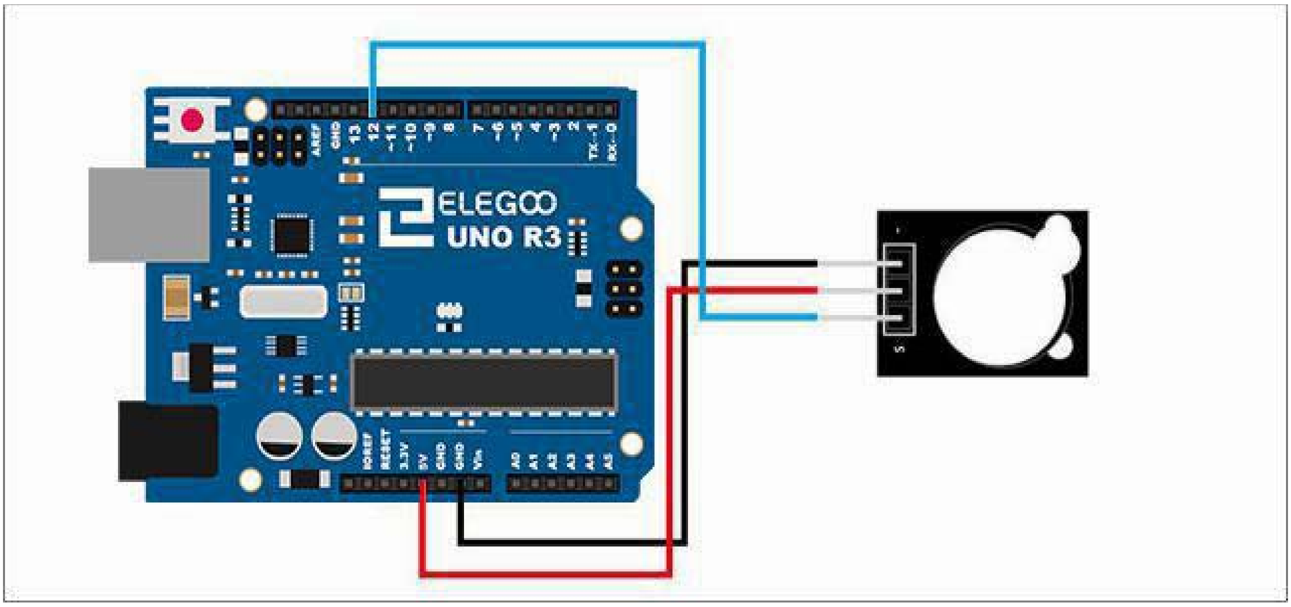
An active buzzer has a built-in oscillating source, so it will make sounds when electrified. But a passive buzzer does not have such source, so it will not tweet if DC signals are used; instead, you need to use square waves whose frequency is between 2K and 5K to drive it. The active buzzer is often more expensive than the passive one because of multiple built-in oscillating circuits.

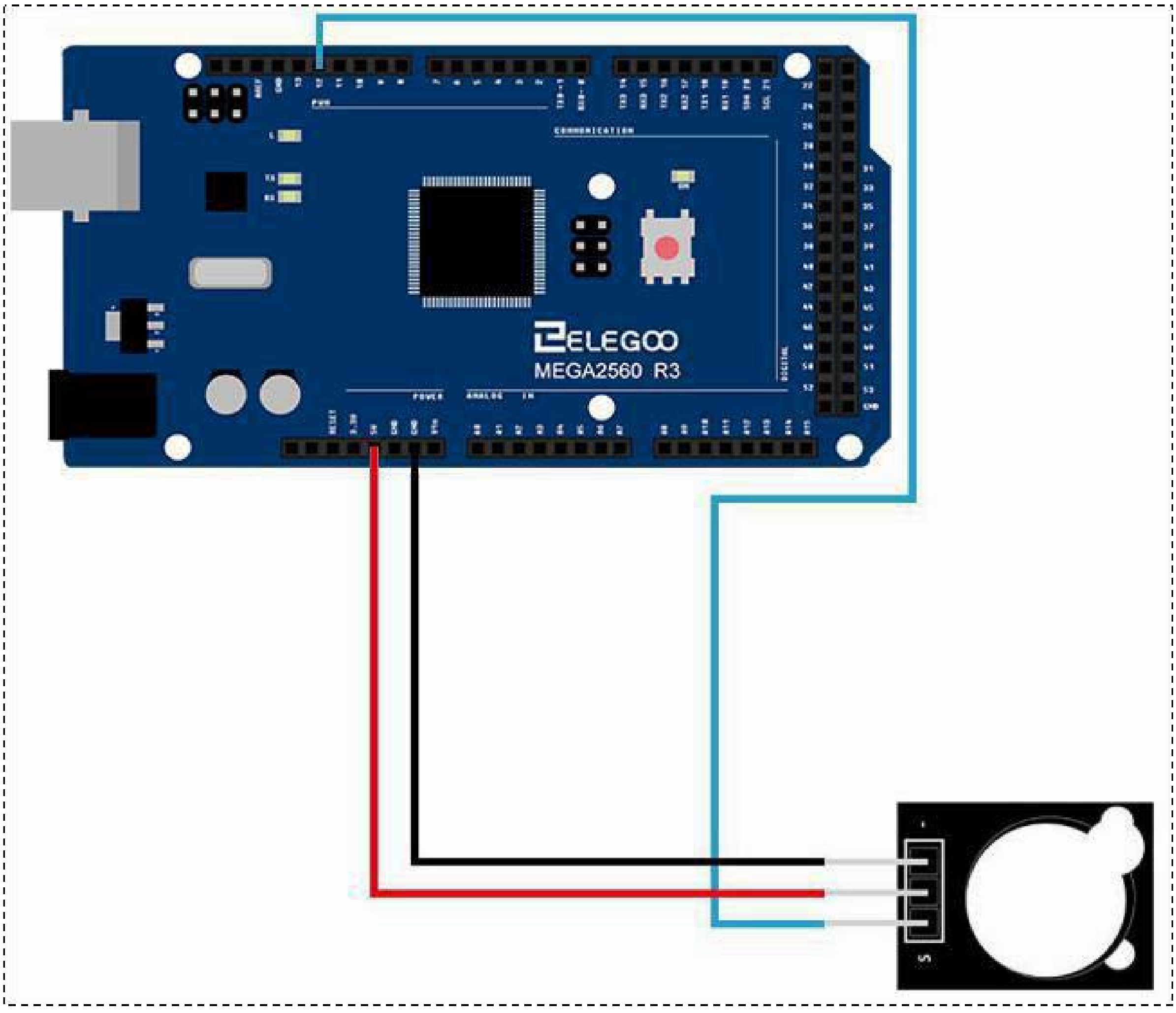
Connection

Schematic



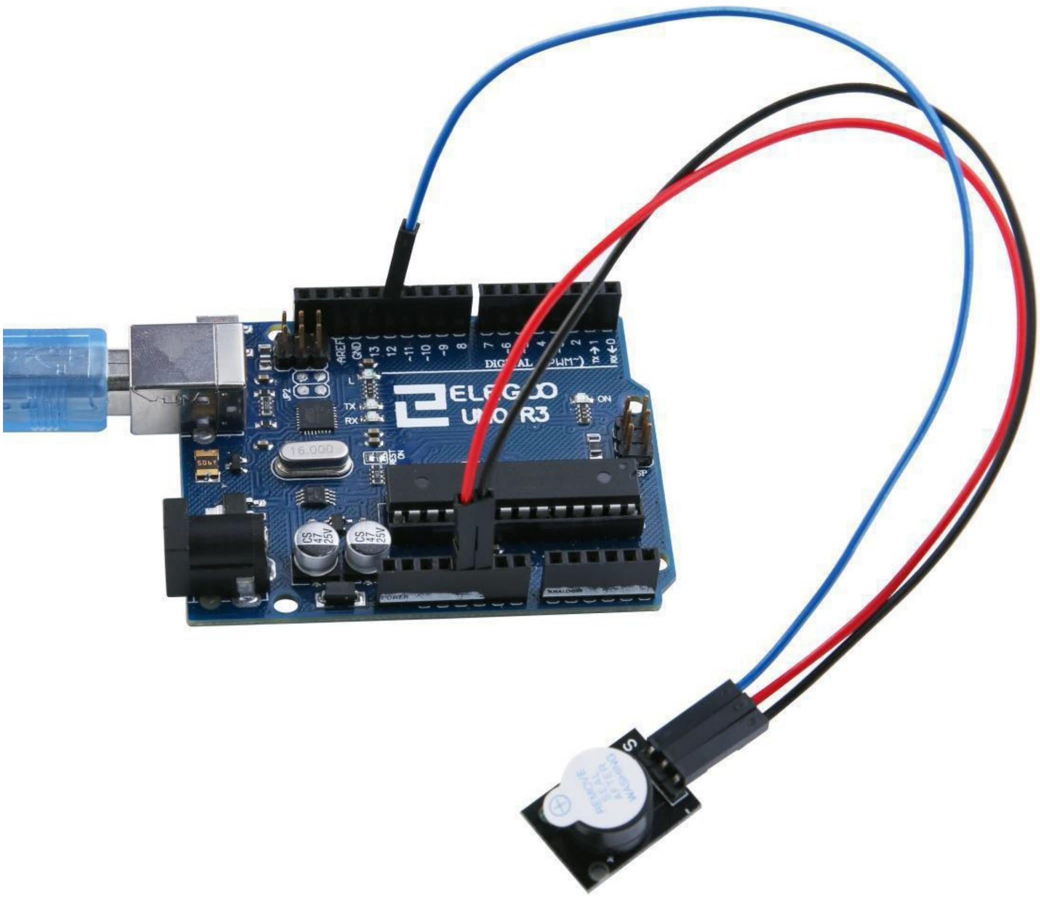
Wiring diagram





Result

After we connect the circuit as the picture, we upload the program of each module. We can hear that the active buzzer can make sound in one voice. And the passive can sing a song.



Lesson 9 Passive Buzzer Module

Overview

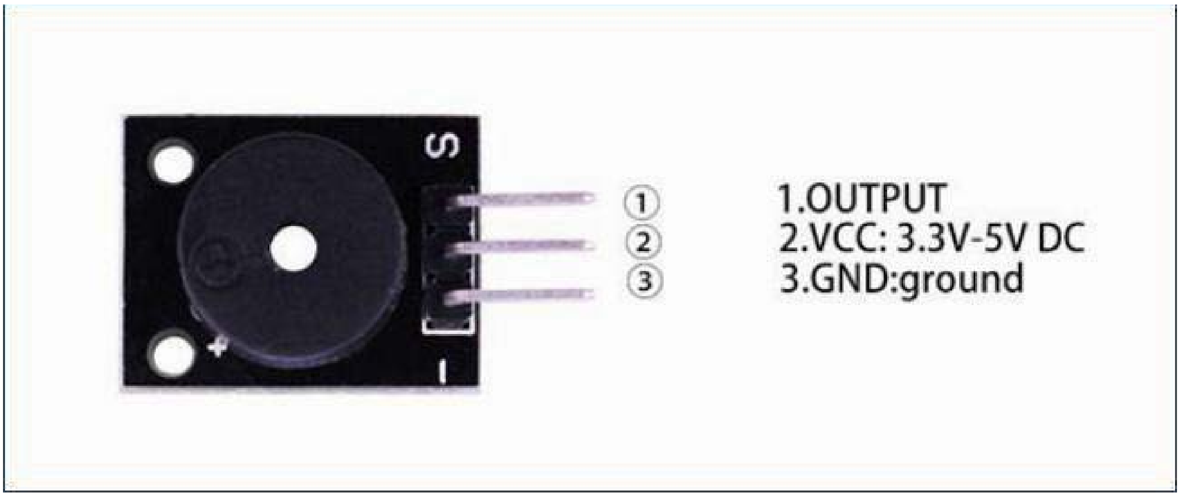
In this lesson, you will learn how to use a passivebuzzer.

The purpose of the experiment is to generate eight different sounds, each sound lasting 0.5 seconds: from Alto Do (523 Hz), Re (587 Hz), Mi (659 Hz), Fa (698 Hz), So (784 Hz), La (880 Hz), Si (988 Hz) to Treble Do (1047 Hz).

Passive Buzzer

Mini loudspeaker module ca. 16 Ohm Impedance, (maximum continuous current through the speaker coil is approximately 25 mA.) Don't mix this one up with the buzzer module! The outer two pins connect to the speaker. Polarity is unimportant.

Tip to avoid mix up: The loud speaker housing is not as tall as the buzzer housing.



Component Required:

(1)x Elegoo Uno R3

(1)x USB cable

(1)x Passive buzzer module

(x)x F-M wires

Component Introduction

Passive Buzzer:

Passive buzzer, in fact, just use PWM generating audio, drives the buzzer, all owing the air to vibrate, can sound. Appropriately changed as long as the vibration frequency, it can generate different sound scale. For example, sending a pulse wave can be generated 523 Hz Al to Do, pulse 587 Hz can produce midrange Re, 659 Hz can produce midrange Mi. If you then with a different beat, you can playa song. Here be careful not to use the
Arduino analog Write () function to generate a pulse wave, because the frequency analog Write () is fixed (500Hz), no way to scale the output of different sounds.

Distinguish between Active and Passive Buzzer Teach you to distinguish between active and passive buzzer

Now a small buzzer on the market because of its small size (diameter of only ll mm), light weight, low price, solid structure, and is widely used in a variety of needs audible electrical Equipment, electronic production and microcontroller circuits, etc...

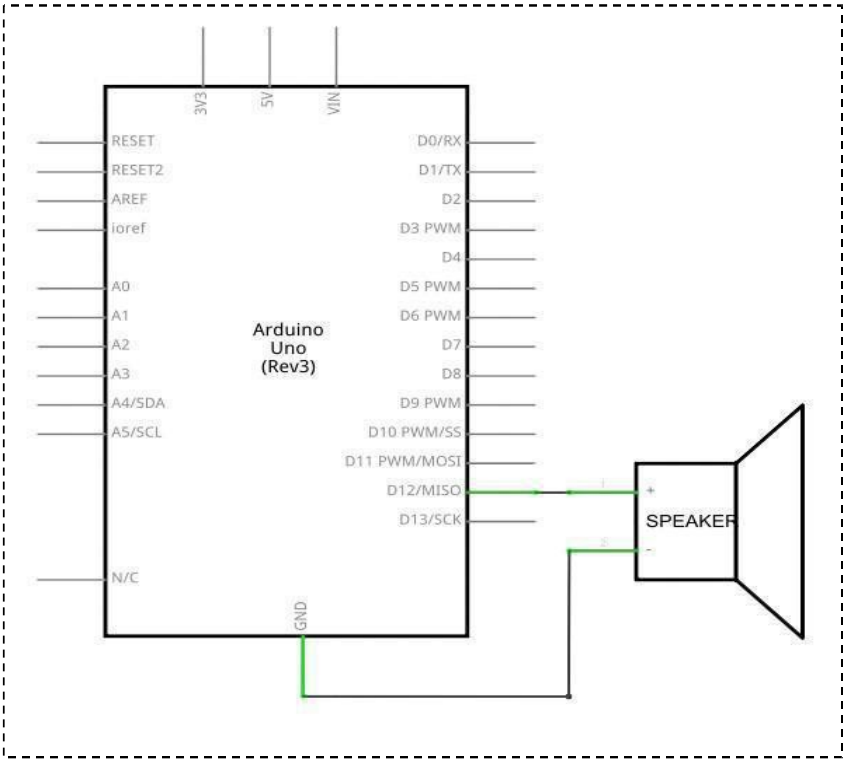
From the exterior, two kinds of buzzer seems the same, but acloser look, a slight difference between the height of the active buzzer a, height of 9 mm, and passive buzzer b height of 8 mm. When the buzzer as the two pins are facing up, you can see there is a green circuit board is passive buzzer, no closed circuit boards with a vinyl is active buzzer.

Further determine the active and passive buzzer, you can also use a multimeter to test the resistance profile Rxl file: buzzer with black pen then "+" pin, the red pen to touch on another pin back and forth, if trigger a cracking, cracking sound and the resistance only 8Ω (or 16Ω) is passive buzzer; If you can emit continuous sound, and the resistance in Europe and more than a few hundred, and is an active buzzer.

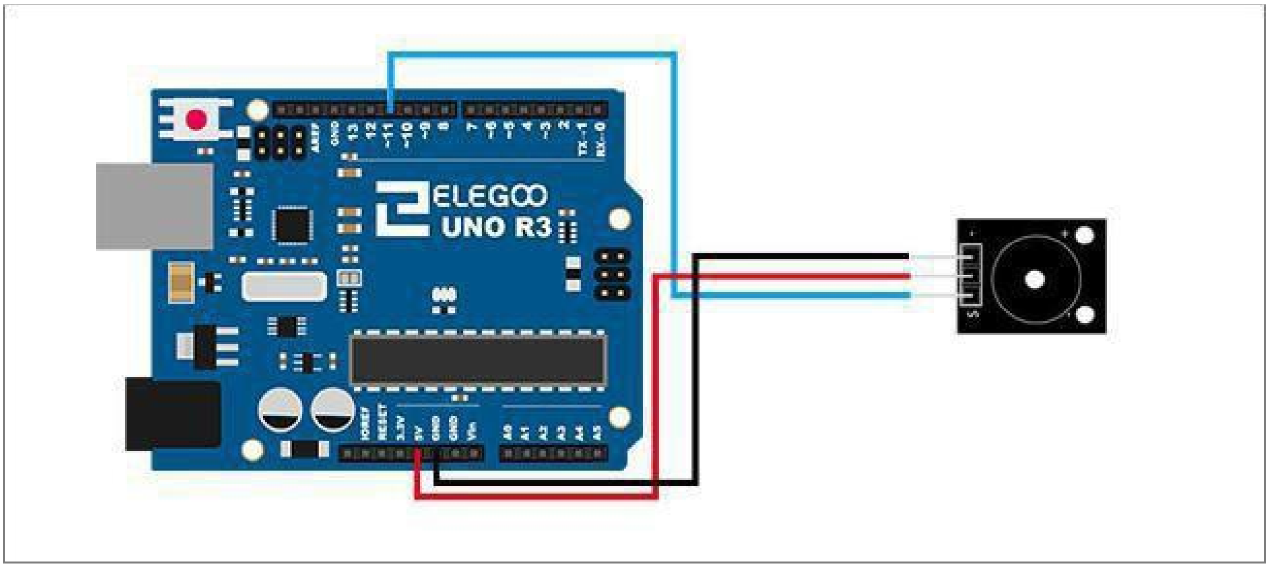
Active buzzer directly connected to the rated power (new buzzer has stated on the label) can be a continuous sound; rather passive electromagnetic buzzer and speaker are the same, you need to take in order to sound the audio output circuit.

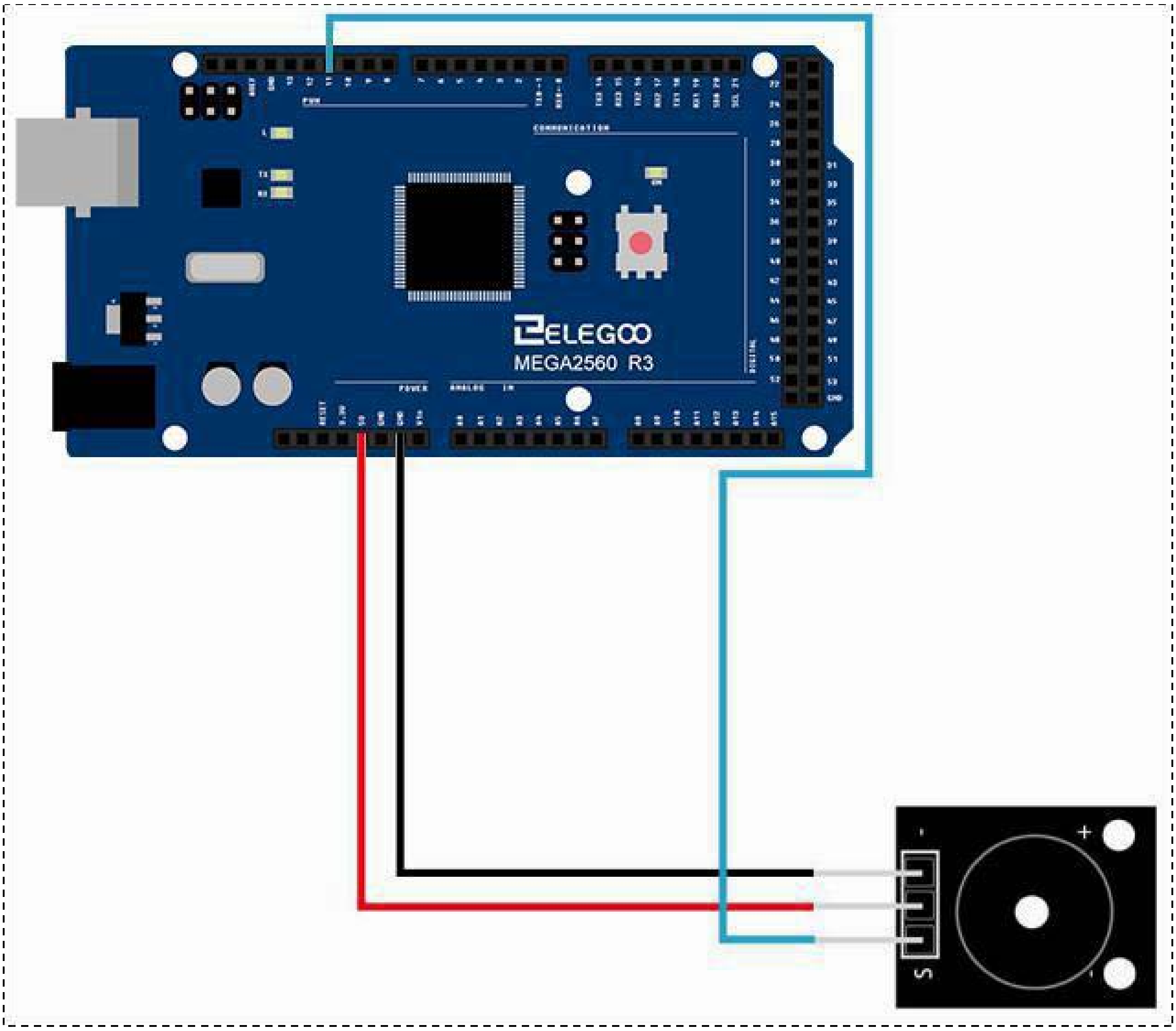
Connection

Schematic



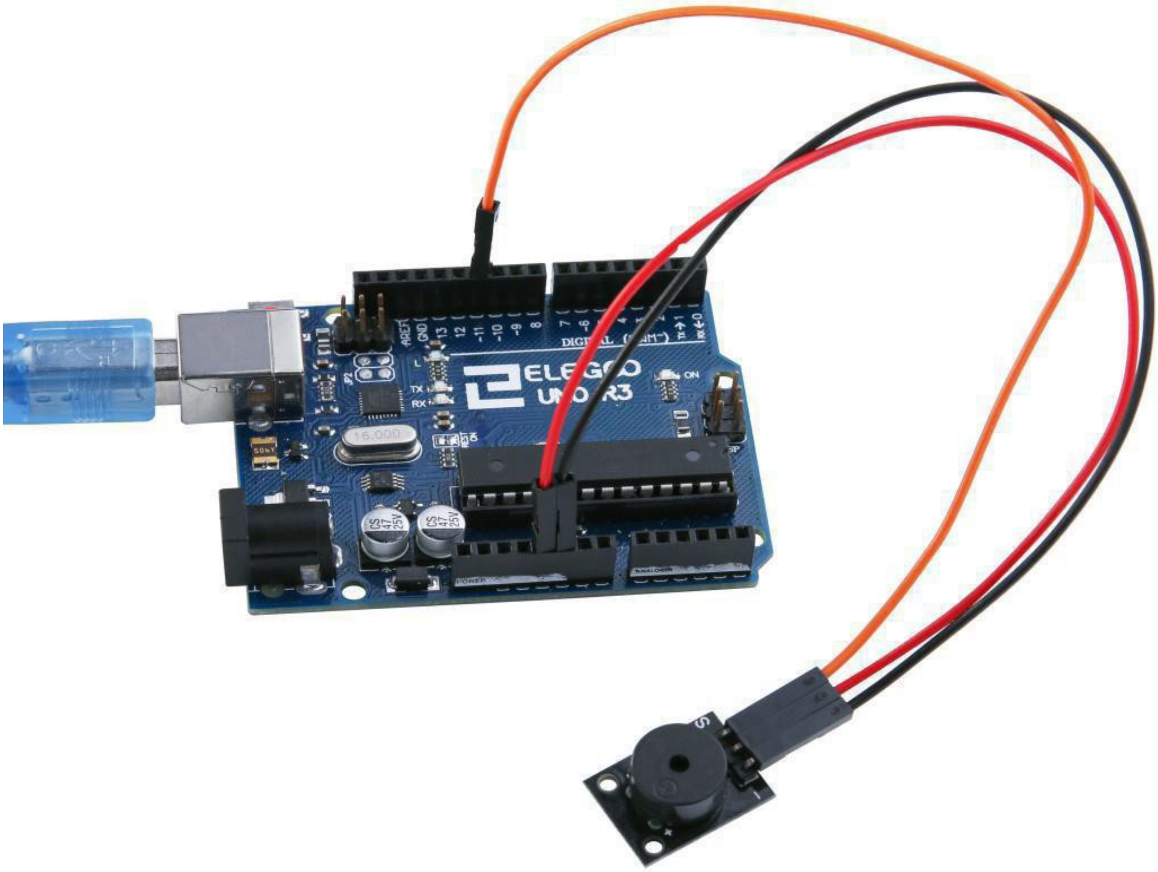
Wiring diagram





Result

After we connect the circuit as the picture, we upload the program of each module. We can hear that the active buzzer can make sound in one voice. And the passive can sing a song.



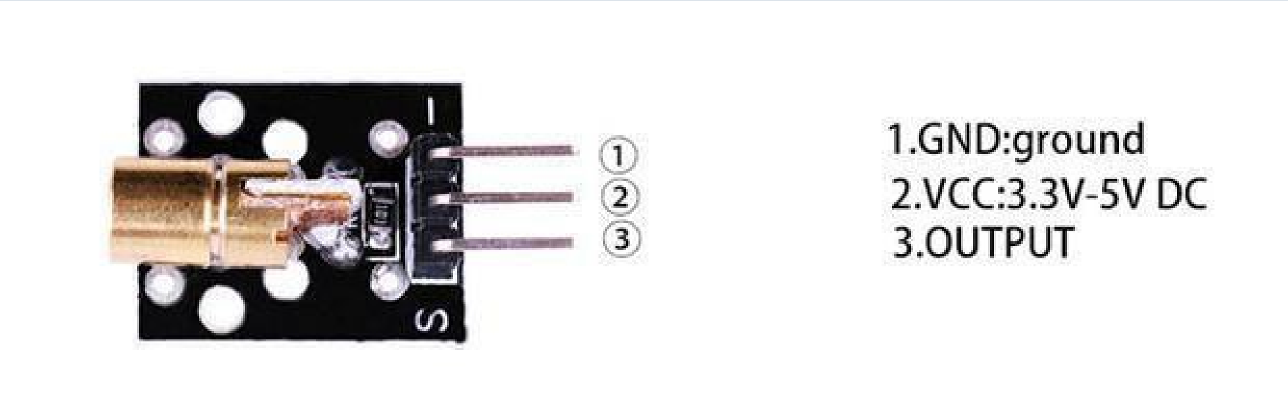
Lesson 10 Laser Module

Overview

In this experiment, we will learn how to use laser module.

Laser emitter

Red Laser module for direct connection to a 5 V supply. Connect the 5 V supply to the 'S' pin and ground to the '-' pin. Transmission wavelength: 650 nm



Component Required:

(1)x Elegoo Uno R3

(1)x USB cable

(1)x Laser module

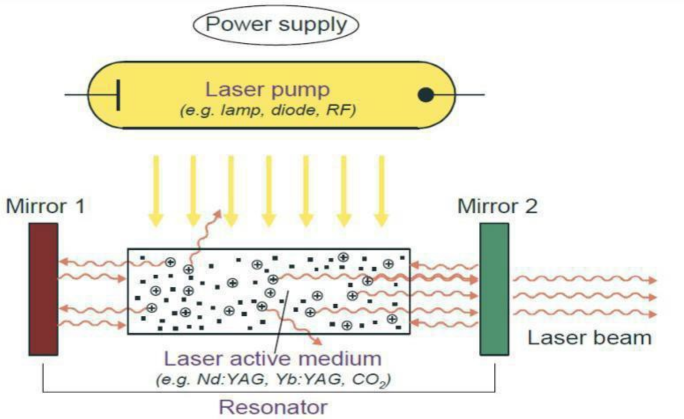
(X) x F-M wires

Component Introduction

Laser sensor:

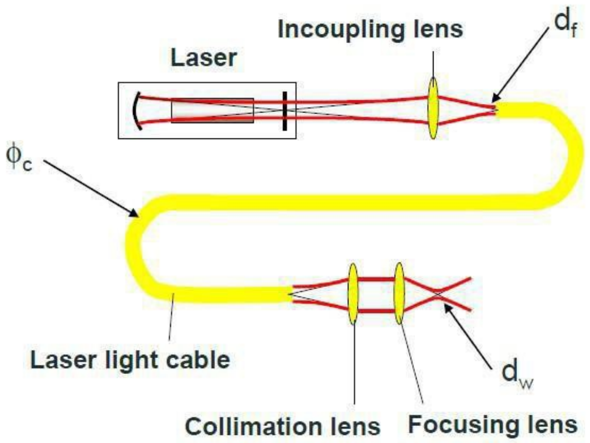
TRUMPF

Laser basics



TRUMPF

Spot size - YAG



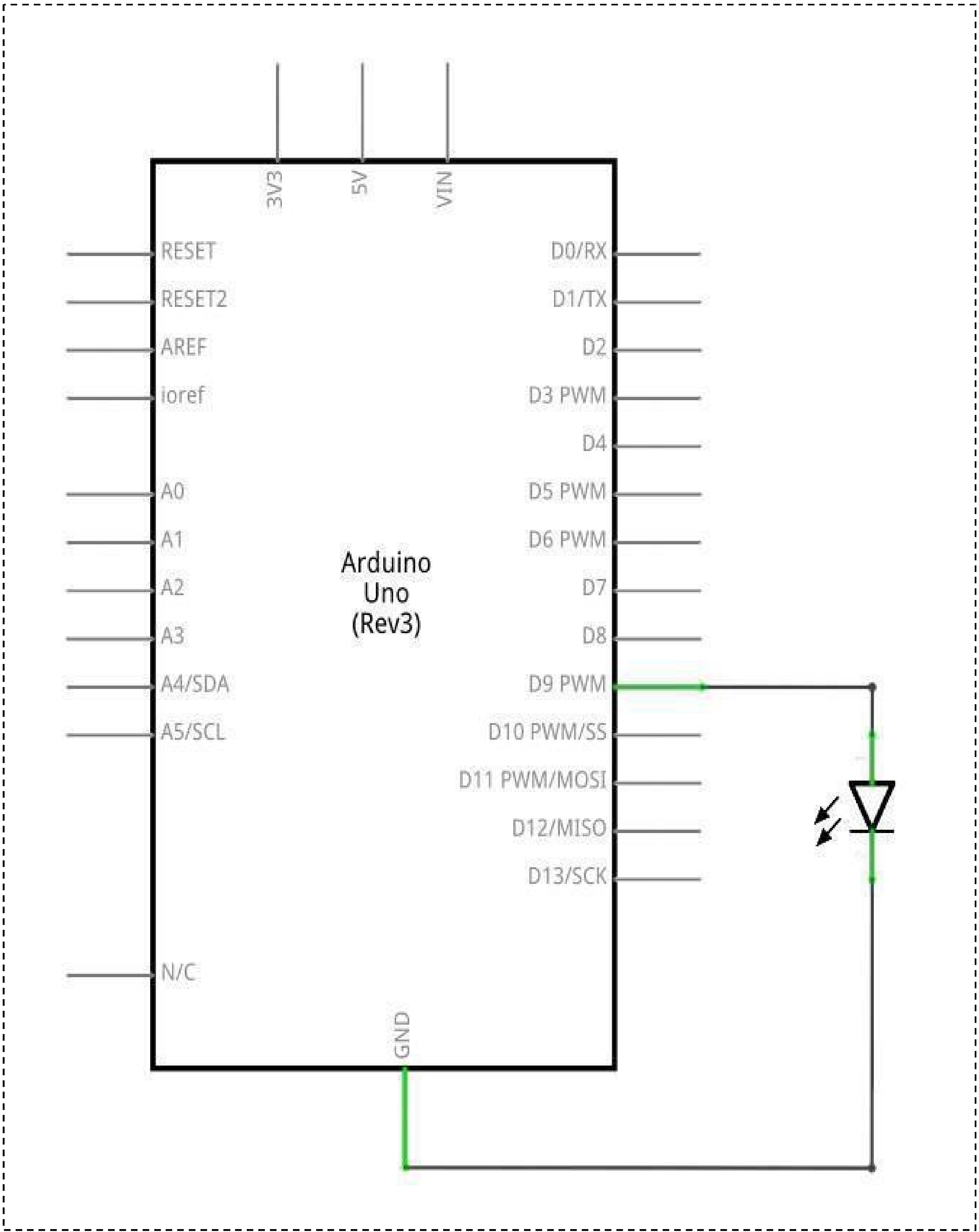
$$d_f = M^2(4\lambda f/\pi D)$$

$$d_f = 3BQ(4\lambda f/\pi D)$$

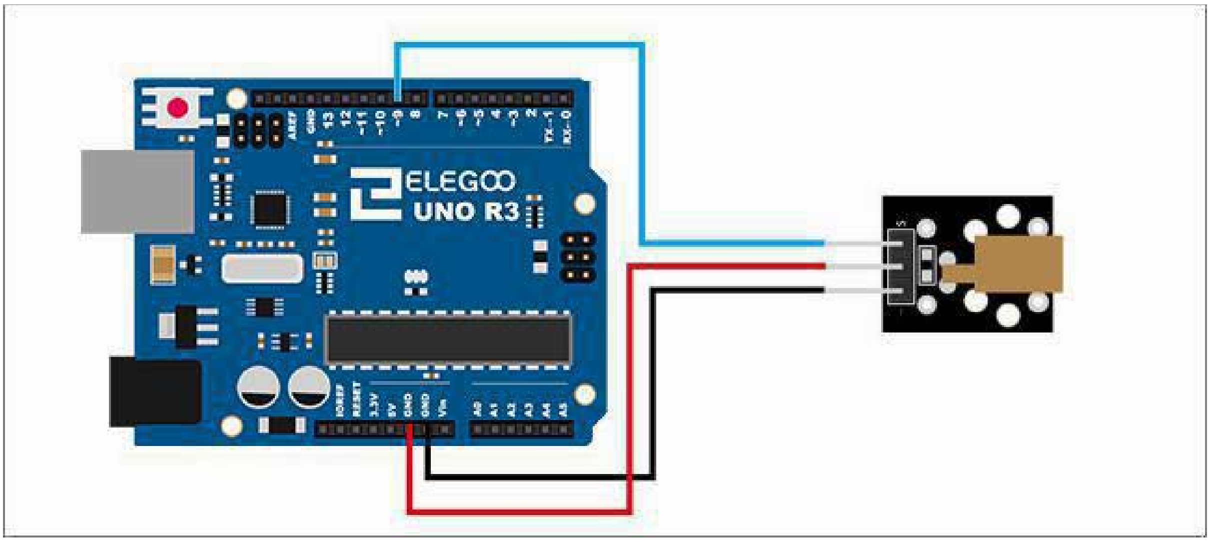
$$\boxed{d_w = \phi_c(f/ f_c)}$$

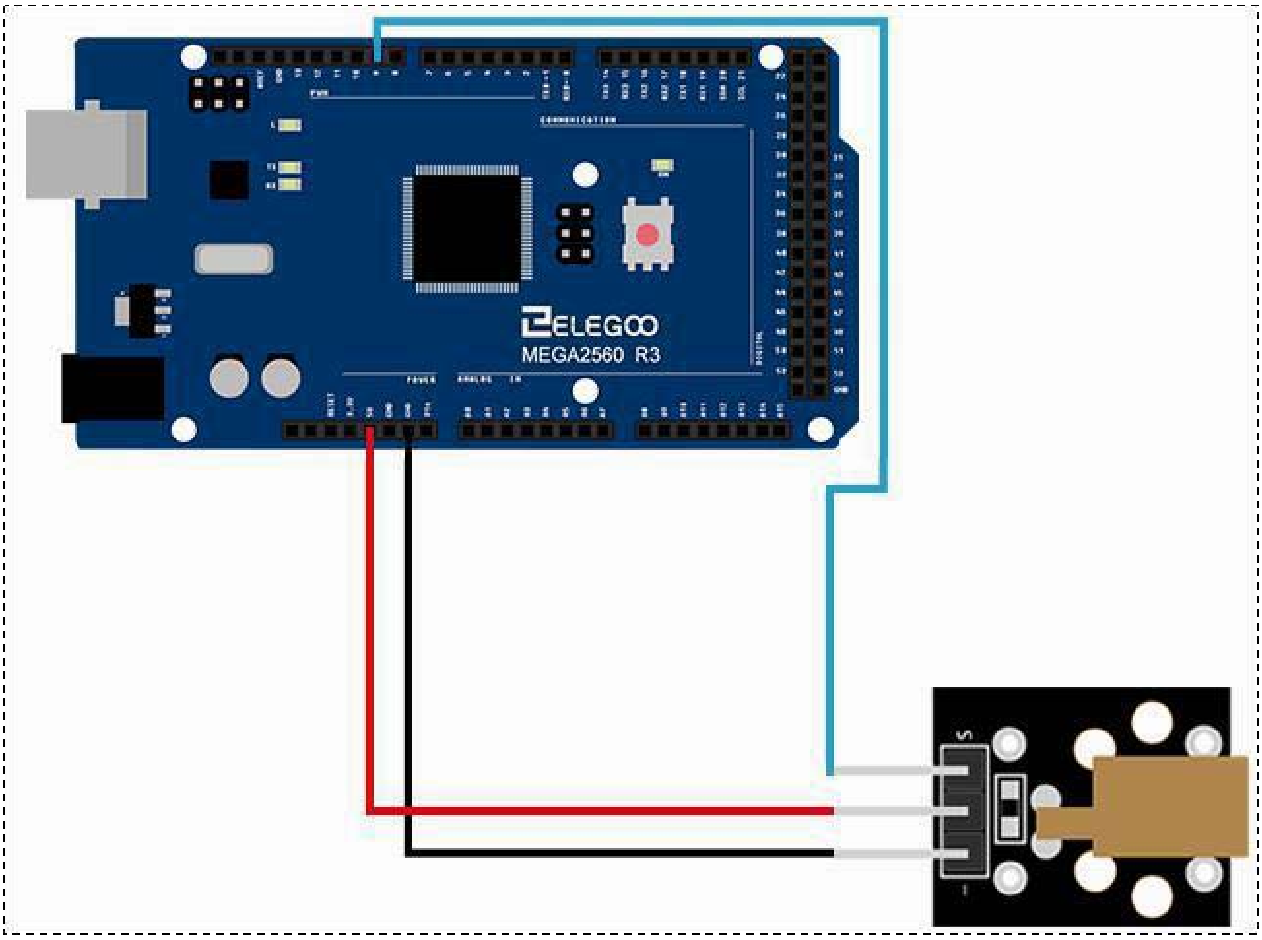
Connection

Schematic



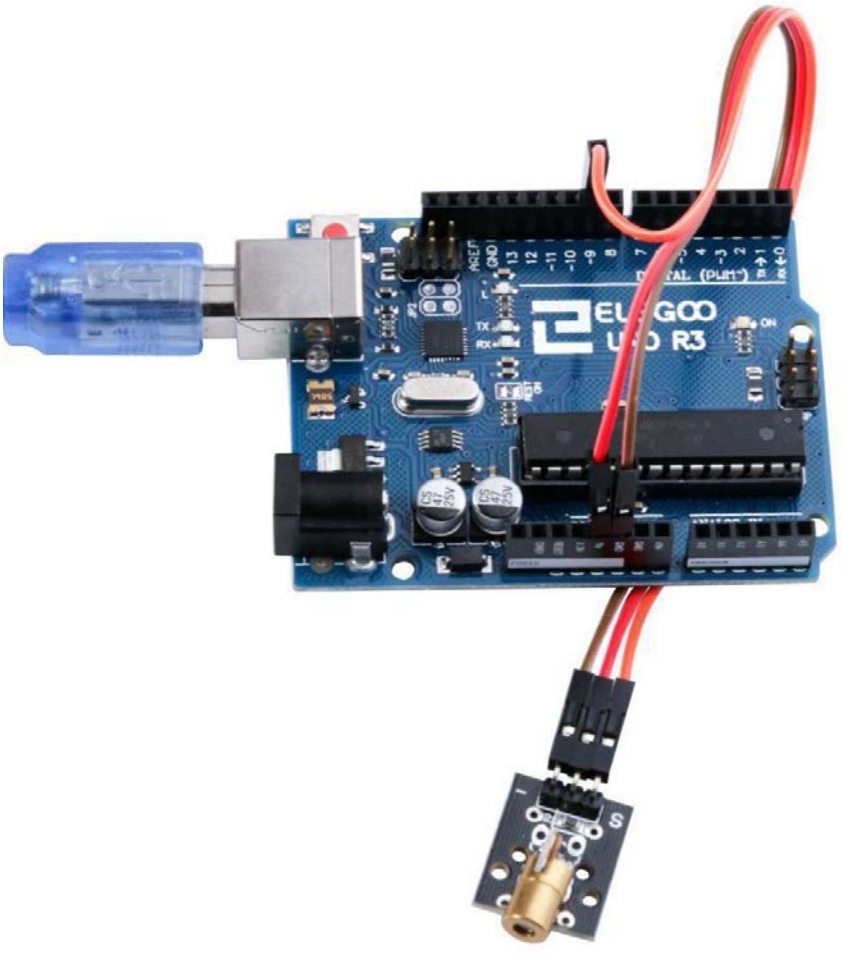
Wiring diagram





Result

After we connect the circuit as the picture, we upload the program. We can see the module can emission lasers.



Lesson 11 SMD RGB Module and RGB Module

Overview

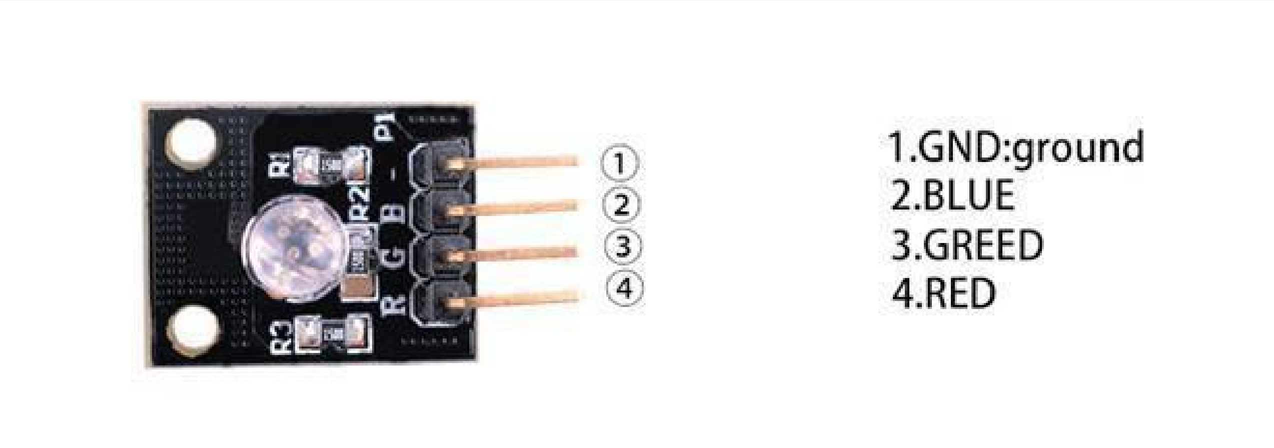
In this experiment, we will learn how to use SMD RGB LED module and RGB LED module.

Actually, the function of SMD RGB LED module and RGB LED module are almost the same. But we can choose the shape we like or we need.

SMD RGB LED module and RGB module are made from a patch of full-color LED. By adjusting the voltage input of R, G, B pins, we can adjust the strength of the three primary colors (red/blue/green) so as to implementation result of full color effect.

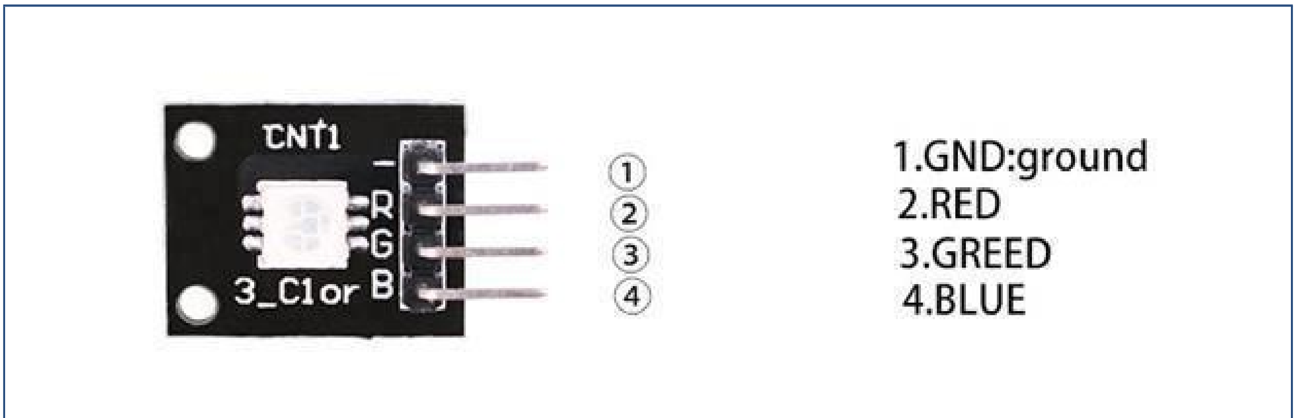
RGB LED Module

RGB-LED with clear lens and built-in 150 ohm series resistor for 5 V operation. The PCB printing is incorrect, it shows the blue and red connections switched. The LED has a common cathode (the '-'Pin).



SMD RGB LED Module

RGB-LED with an SMD housing and no series resistor. The PCB printing is incorrect, it shows the green and red connections switched. The LED has a common cathode (the '-'pin). A suitable resistor value would be 220 ohms.



Component Required:

(1)x Elegoo Uno R3

(1)x USB cable

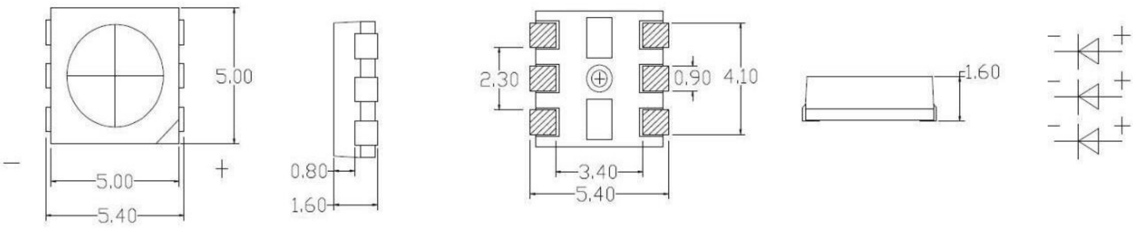
(1)x SMD RGB LED module

(1)x RGB LED module

(x)x F-M wires

Component Introduction

SMD RGB LED:

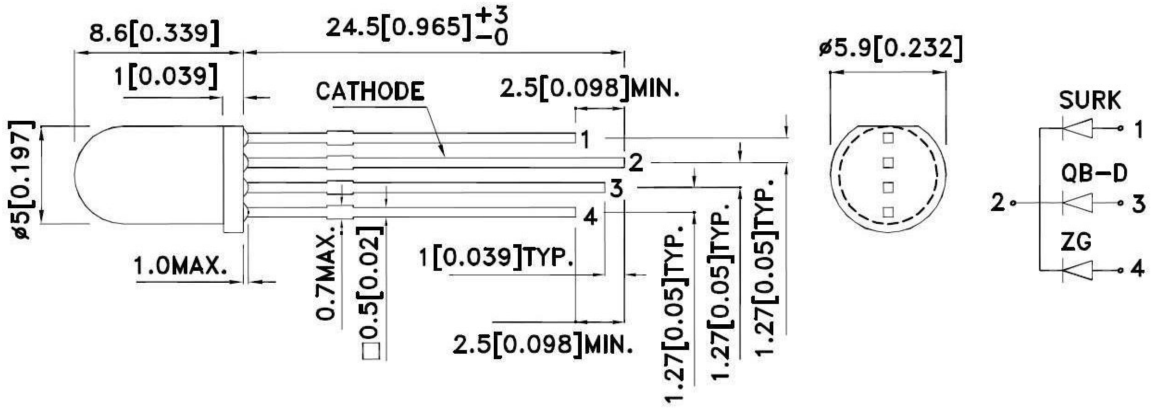


All dimensions are in millimeter

Tolerance is ±0. 25mm(0. 10″)unless otherwise noted

Parameter	Symbol	Value	Unit
Forward Current	If	20	mA
Reverse Voltage	Vr	5	V
Operating Temperature	Topr	-25～+85	℃
Storage Temperature	Tstg	-35～+85	℃
Soldering temperature	Tsol	260±5℃ (for4sec)	℃
Power Dissipation	Pd	R=40 G/B=60	mW
Pulse Current	I_{FP}	100	mA

RGB LED:



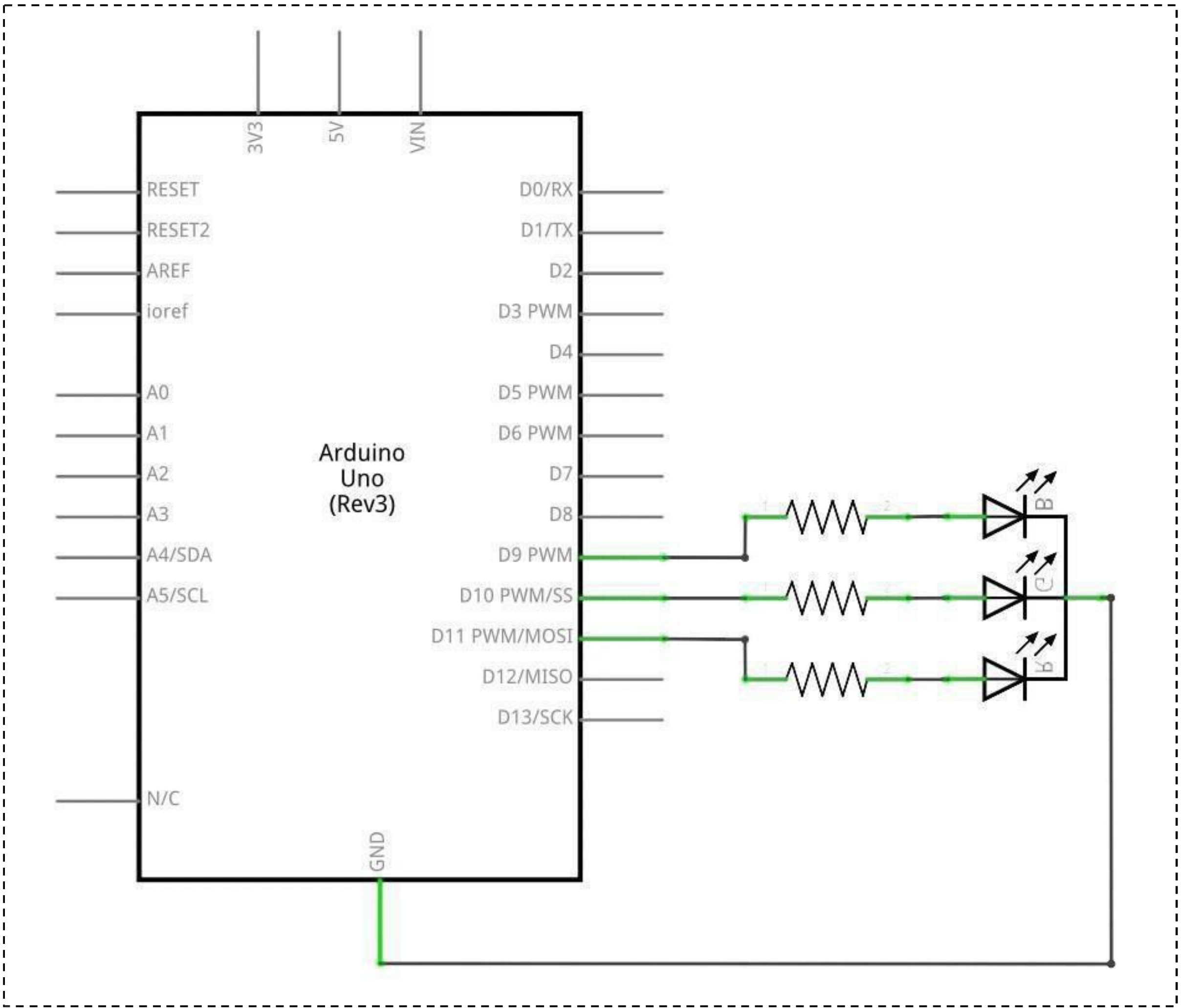
Electrical / Optical Characteristics at TA=25°C

Symbol	Parameter	Device	Typ.	Max.	Units	Test Conditions
λpeak	Peak Wavelength	Hyper Red Blue Green	650 468 515		nm	IF=20mA
λD [1]	Dominant Wavelength	Hyper Red Blue Green	630 470 525		nm	IF=20mA
Δλ1/2	Spectral Line Half-width	Hyper Red Blue Green	28 25 30		nm	IF=20mA
C	Capacitance	Hyper Red Blue Green	35 100 45		pF	VF=0V;f=1MHz
VF [2]	Forward Voltage	Hyper Red Blue Green	1.95 3.3 3.3	2.5 4 4.1	V	IF=20mA
IR	Reverse Current	Hyper Red Blue Green		10 50 50	uA	VR=5V

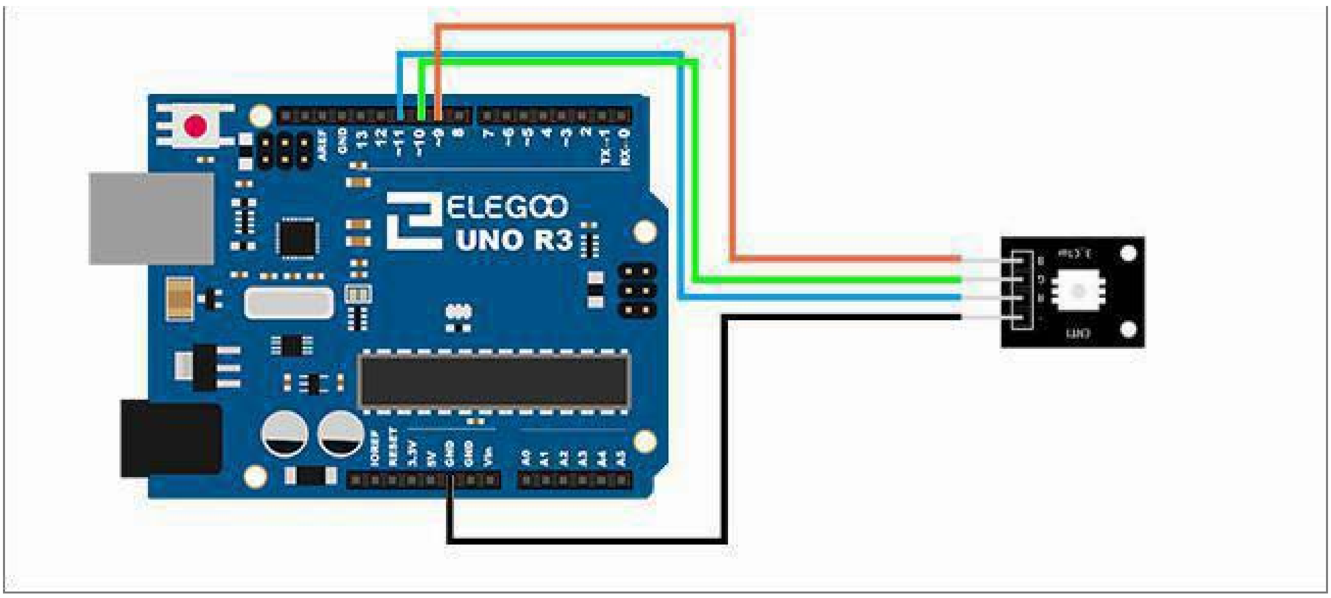
Notes:
1.Wavelength: +/-1nm.
2. Forward Voltage: +/-0.1V.

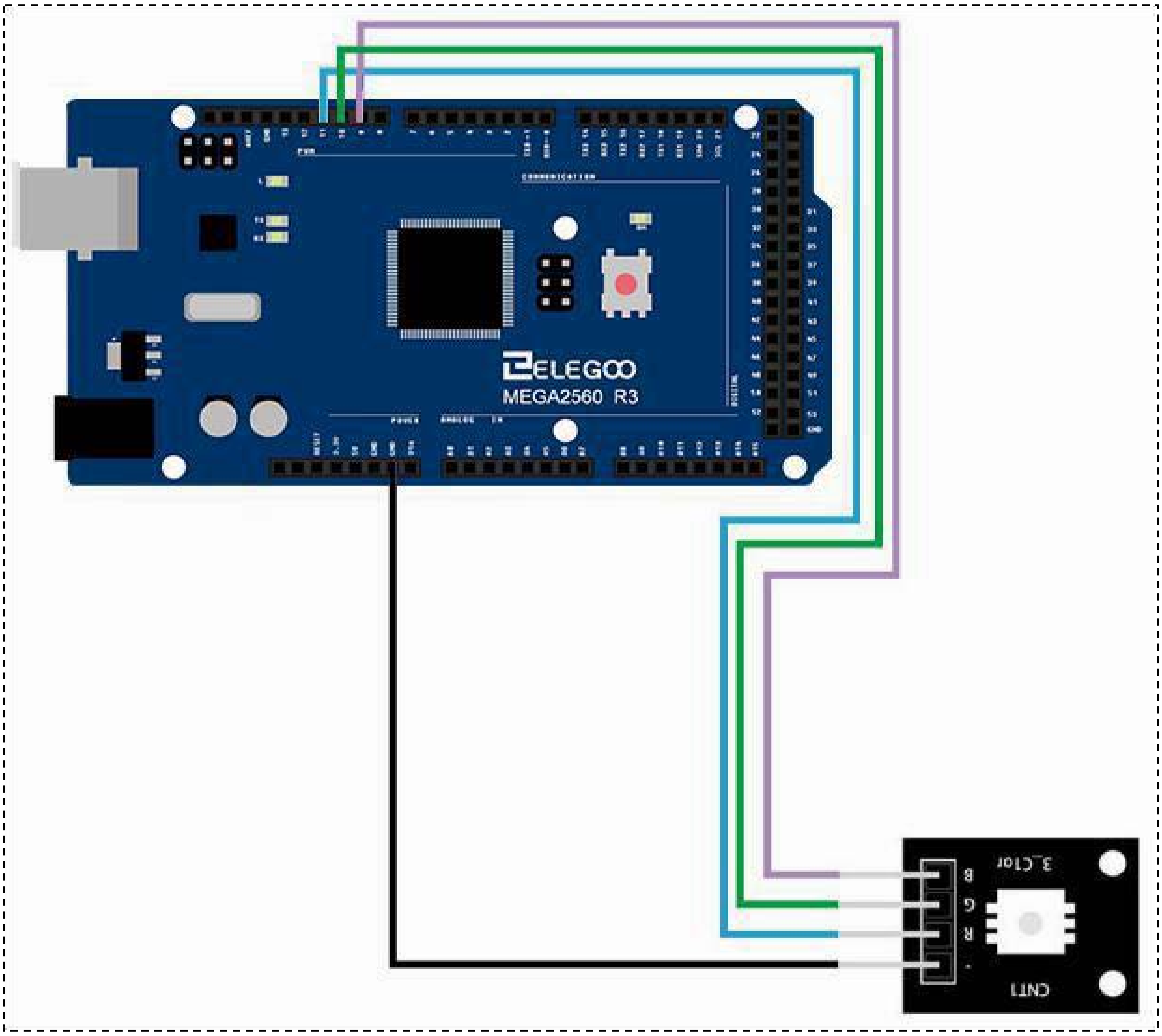
Connection

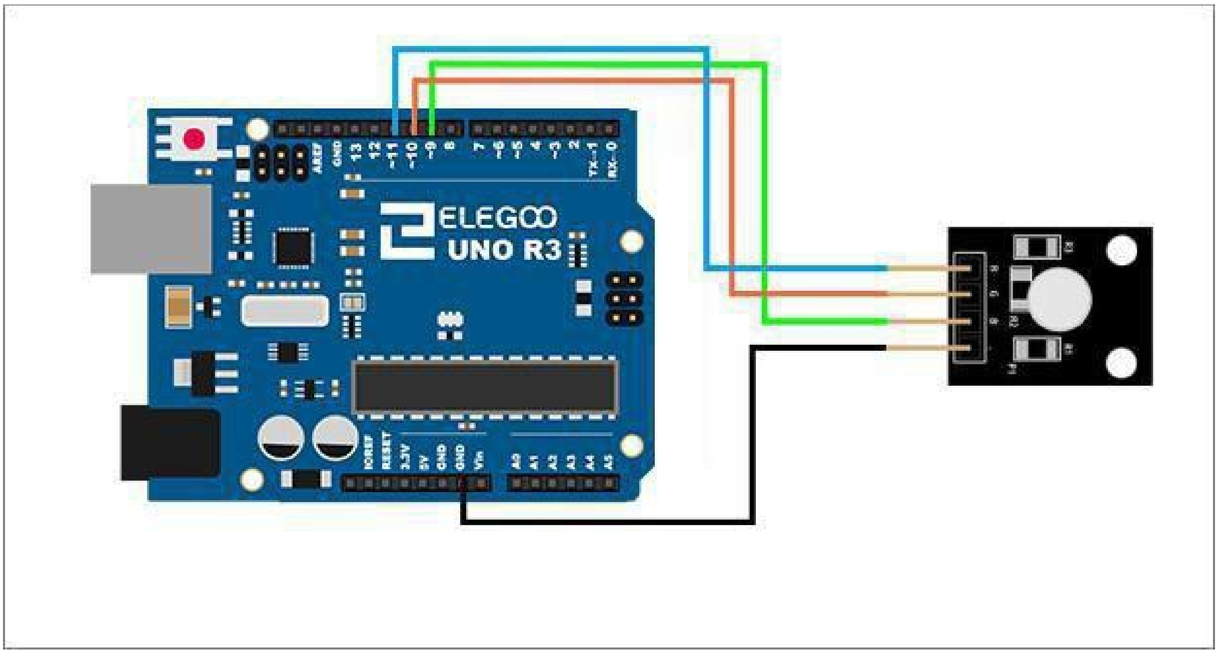
Schematic

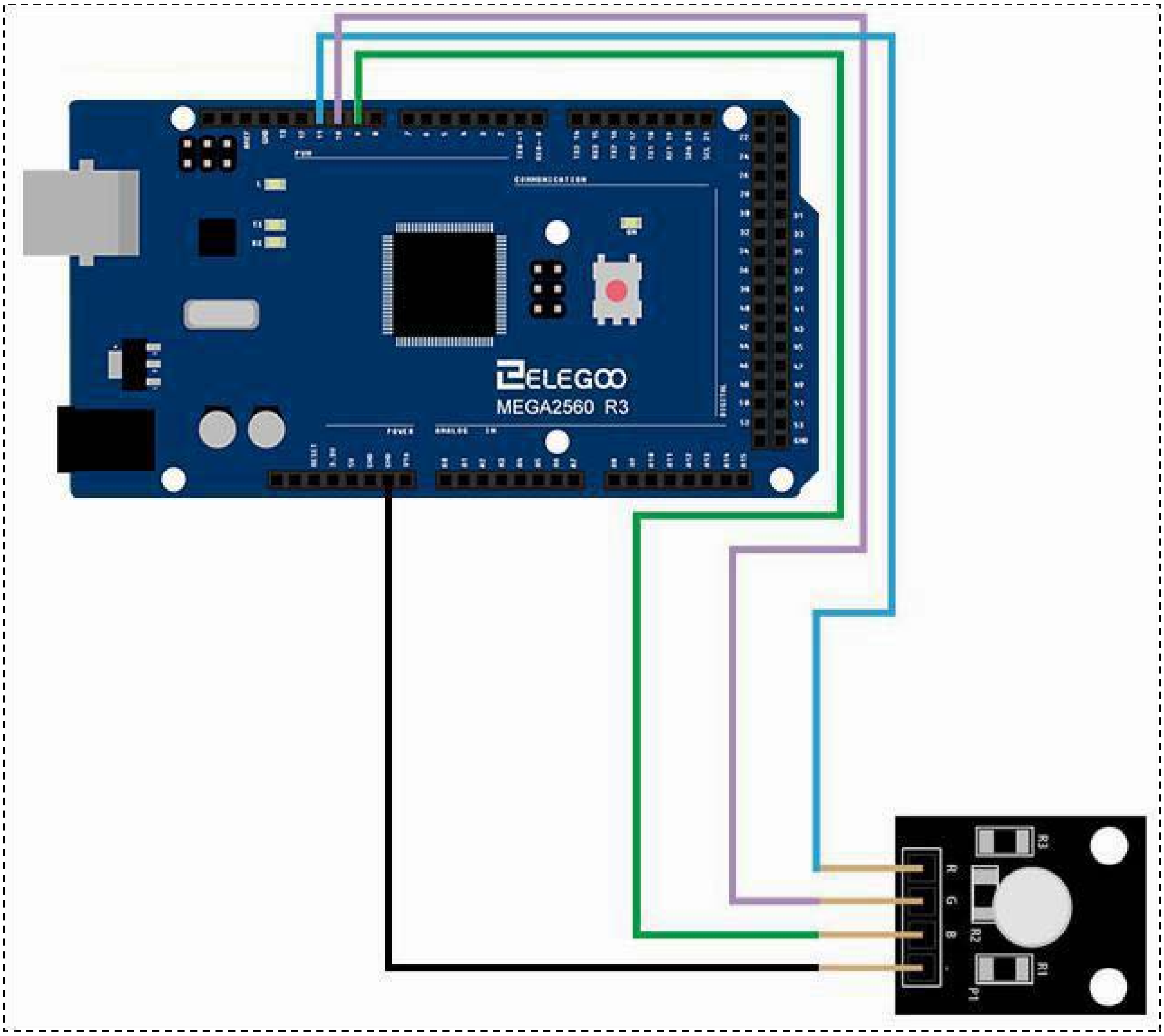


Wiring diagram



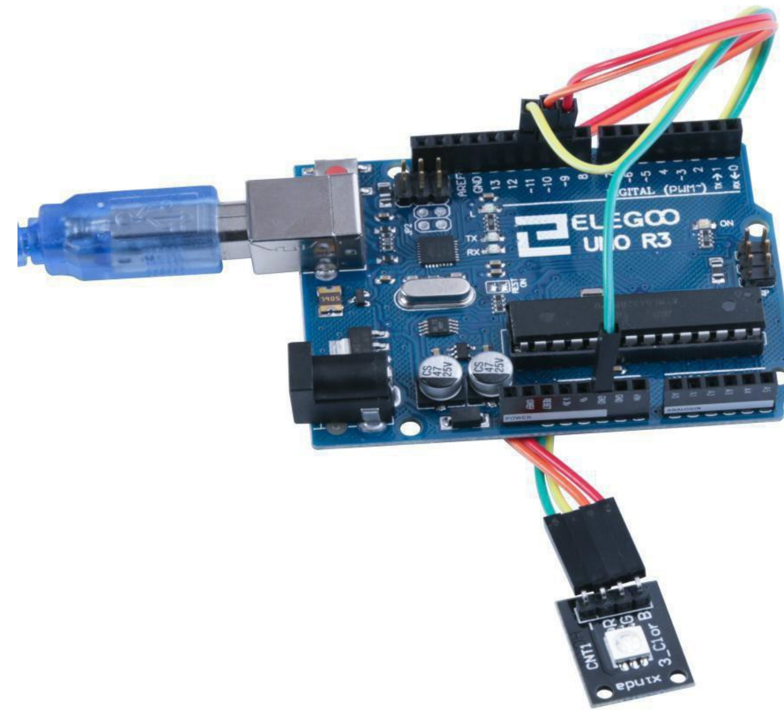

ELEGOO
MEGA2560 R3

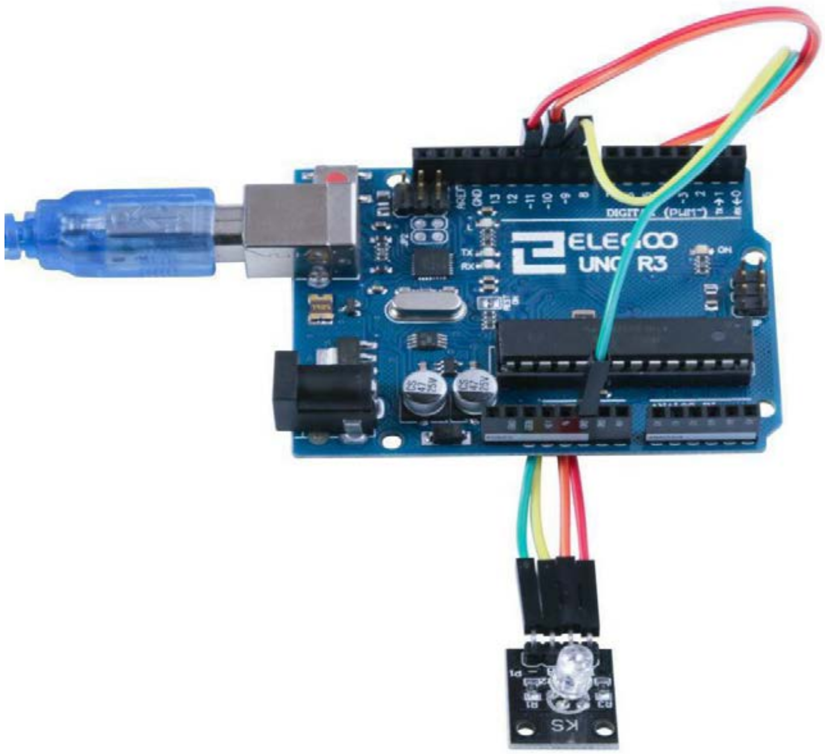

ELEGOO
UNO R3



Result

After we connect the circuit as the picture, we upload the program of each module. We can see the module changing their color as the code set. If you want to make it change the color in different way, you can revise the code.


ELEGOO


ELEGOO

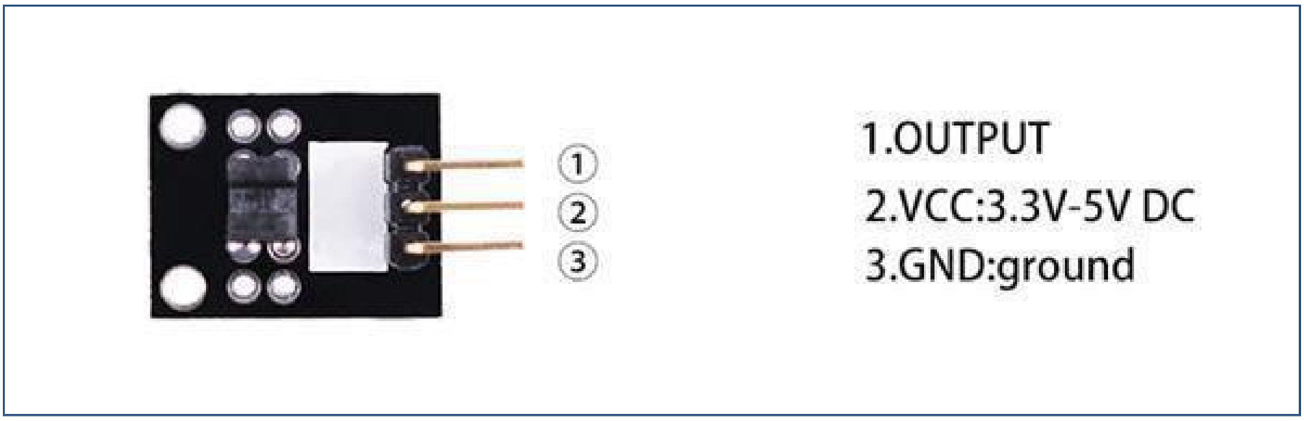
Lesson 12 Photo-Interrupter Module

Overview

In this experiment, we will learn how to use Photo-interrupter module.

Light blocking

Slotted light barrier. The middle pin connects to + 5 V supply and the pin marked'-' connects to ground. The output signal (with a 10 K ohm pull up to +5 V) is available on the pin on the right.



Component Required:

(1) x Elegoo Uno R3

(1) x USB cable

(1) x Photo-interrupter MODULE

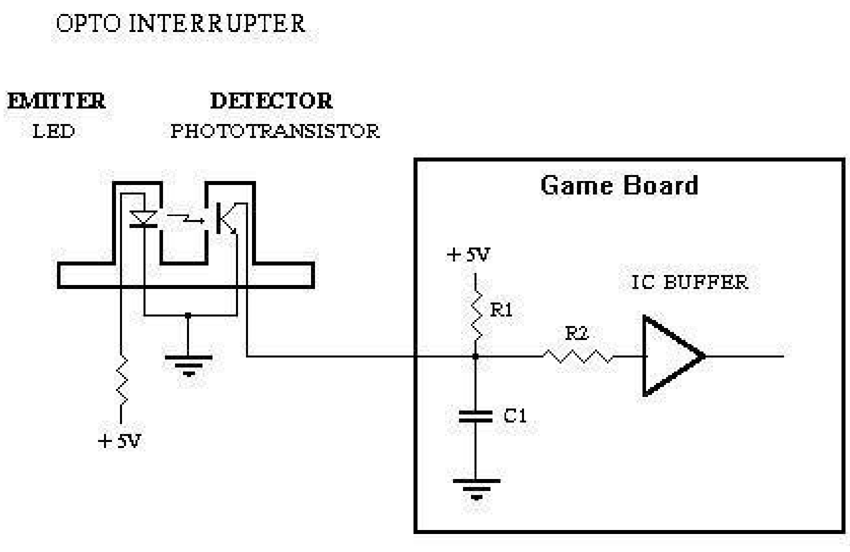
(x) x F-M wires

Component Introduction

Opto Interrupter Sensor:

Opto Interrupters are commonly used in many arcade games e.g. steering assembly in
Older driving games, scoring switches in Whack a Crock etc. Uninterrupted light beam will turn the phototransistor "ON" connecting the ground to the game board input. When the light beam is interrupted the phototransistor turns "OFF", the ground is disconnected from the input and the pull-up resistor R1 forces the input to go "HIGH"(5V level).

Principle



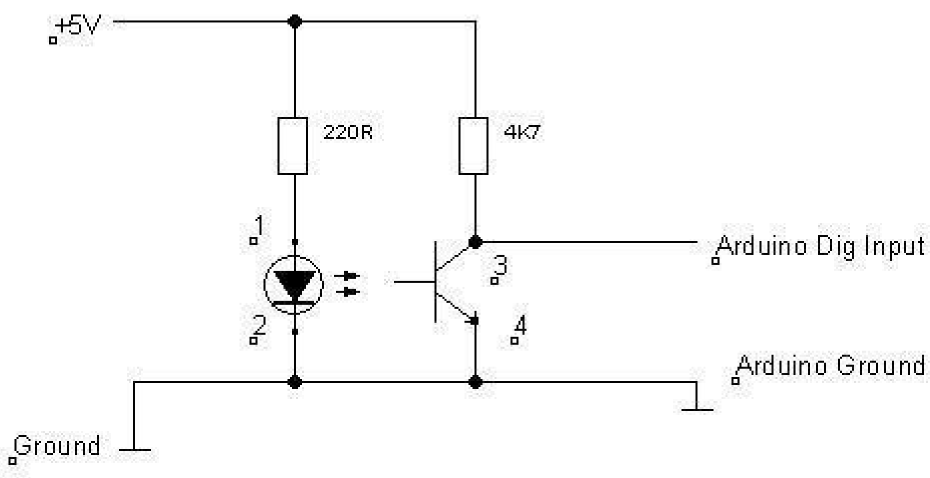
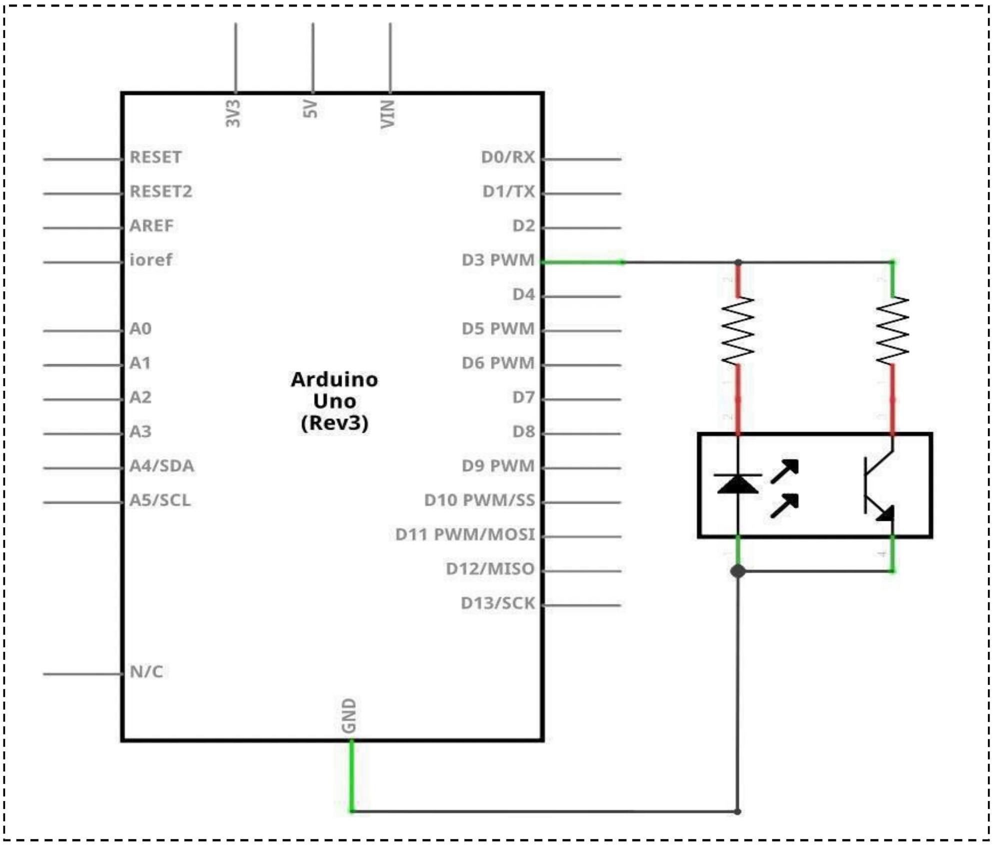


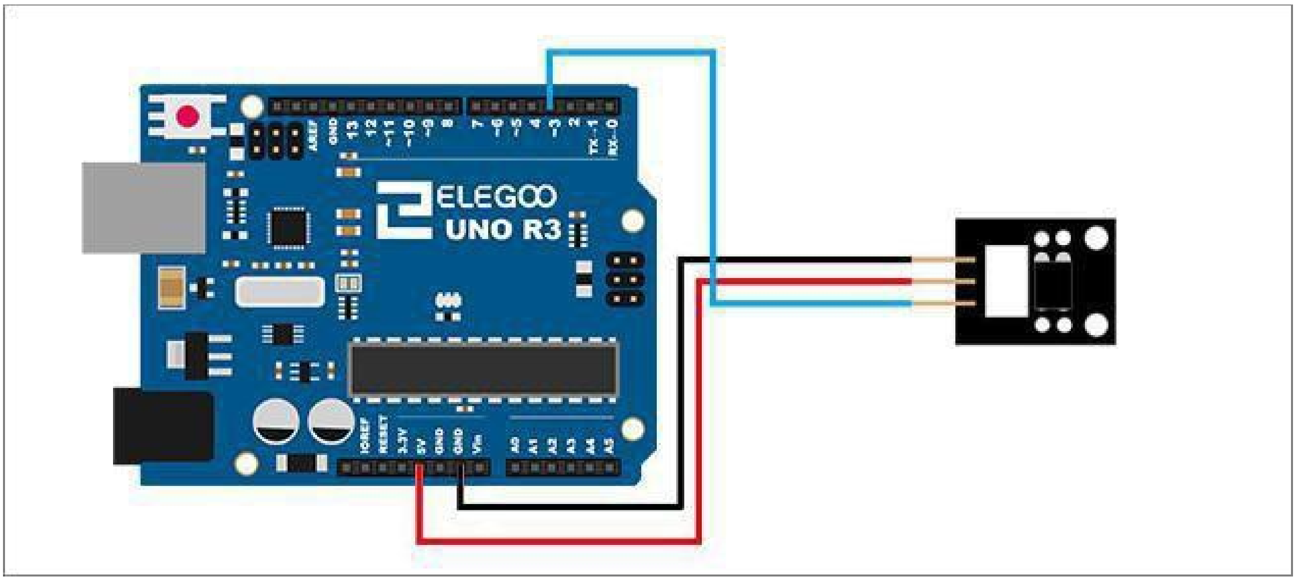
Photo-interrupter module and number 13 port have the built-in LED simple circuit. To produce a switch flasher, we can use connect the digital port 13 to the built-in LED and connect the Photo-interrupter MODULE S port to number 3 port of Elegoo Uno board. When the switch sensing, LED twinkle light to the switch signal.

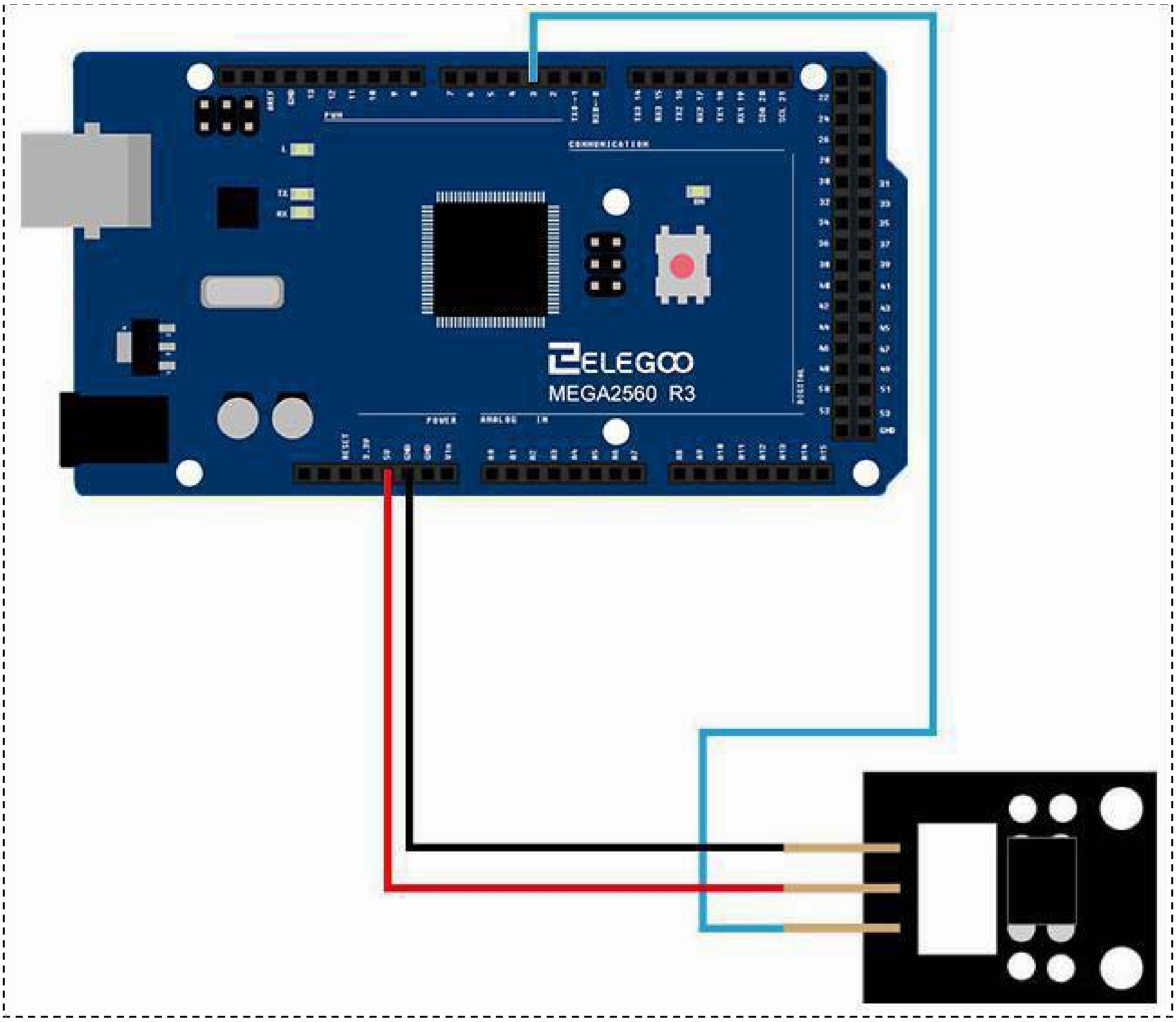
Connection

Schematic



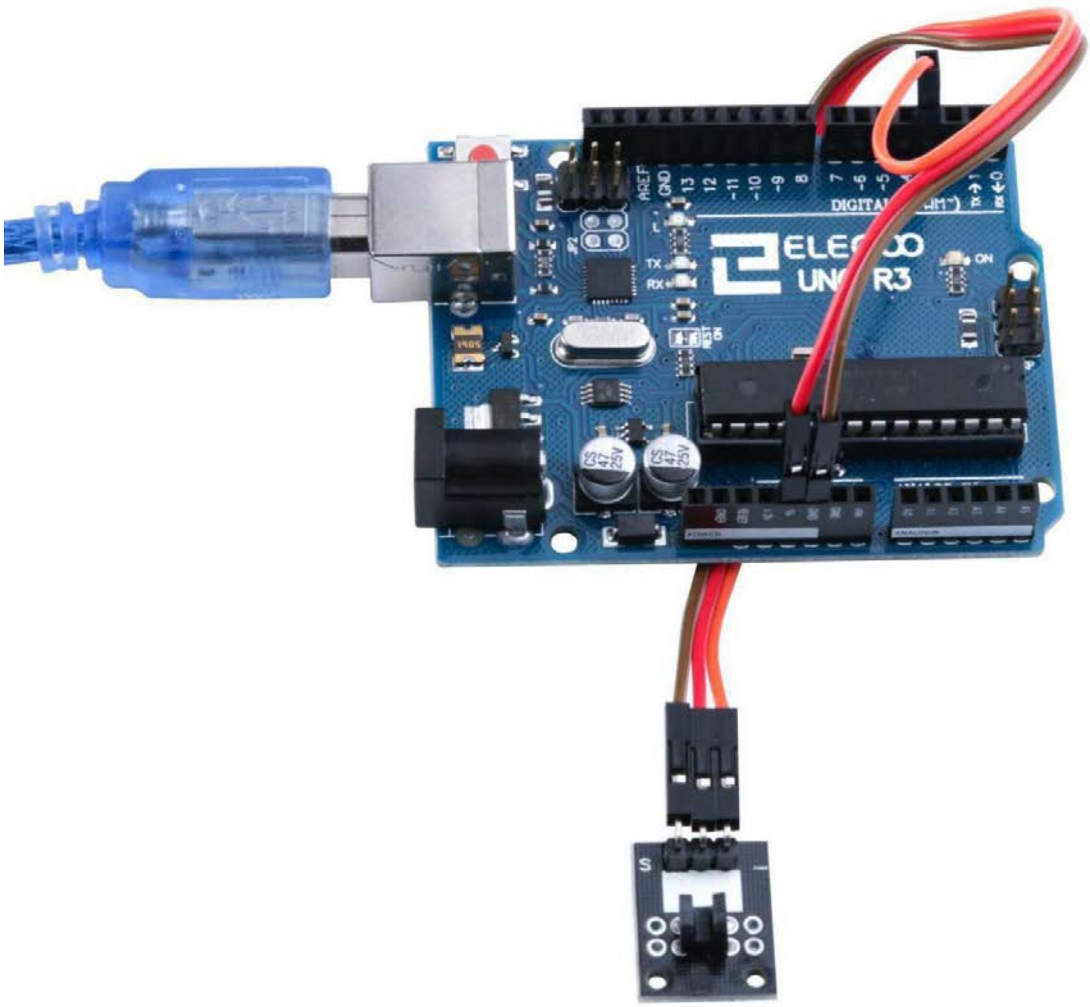
Wiring diagram





Result

After we connect the circuit as the picture, we upload the program, we sensing the opto Interrupter, the new can see the LED 13 light up and light off.


ELEGOO
UNO R3
DIGITAL
AREF
GND
13
12
~11
~10
~9
8
7
~6
~5
TX
RX
ON
S

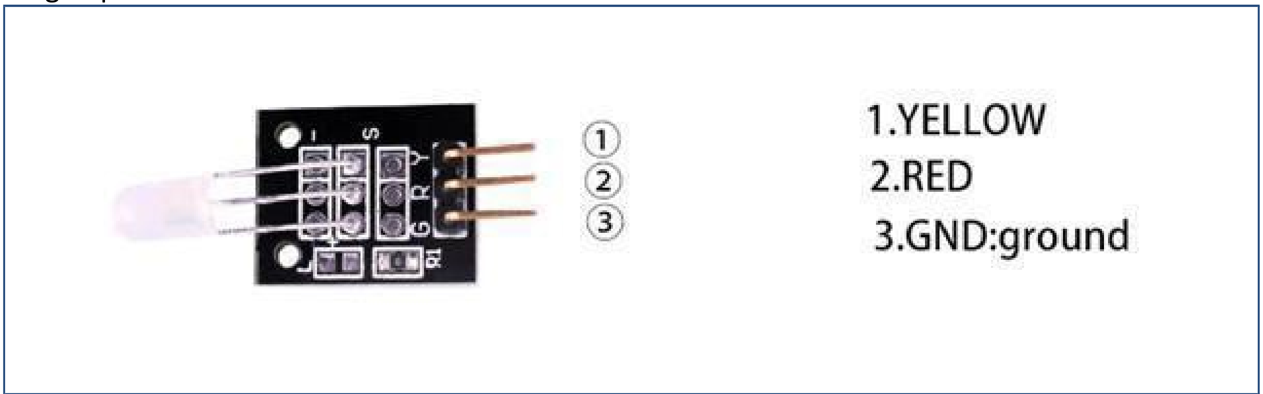
Lesson 13 Dual-Color Common-Cathode LED

Overview

In this experiment, we will learn how to use Dual-color Common-Cathode LED.

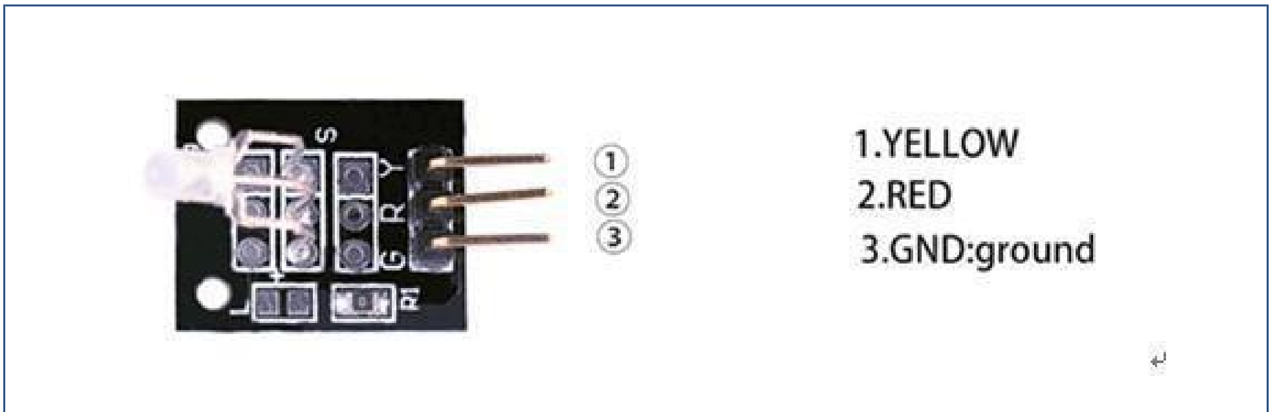
Two-Color LEDModule 5mm

The 5mm LED has a common cathode connected to the '-'pin on the PCB.The center pin connects to the red anode and the 'S' pin connects to the green anode. No series resistor is included in the circuit. A suitable value for low voltage operation would be 220ohms.



Two Color LED Module 3mm

The 3mm bi-color LED has a common cathode (- pin), connected with the '-' pin on the PCB. The center pin activates the red light and the 'S' pin the green light .No series resistor is included in the circuit. A suitable value for low voltage operation would be 220 ohms.



Component Required:

(1) x Elegoo Uno R3

(1) x USB cable

(2) x Dual-color Common-Cathode LED

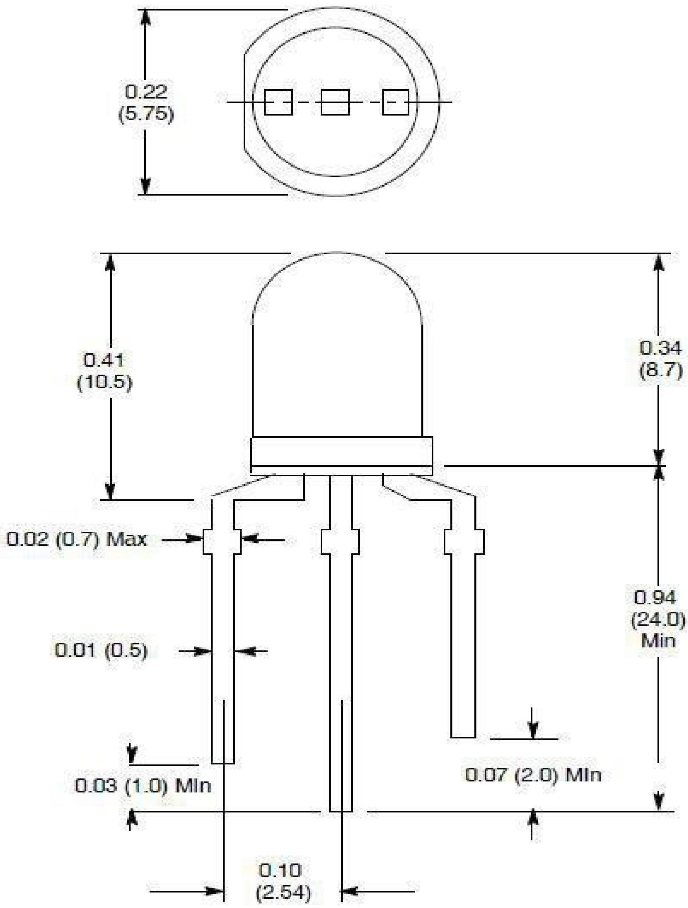
(x) x F-M wires

Component Introduction

Dual-color Common-Cathode led:

Electro–Optical Characteristics: (T_A = +25°C unless otherwise specified)

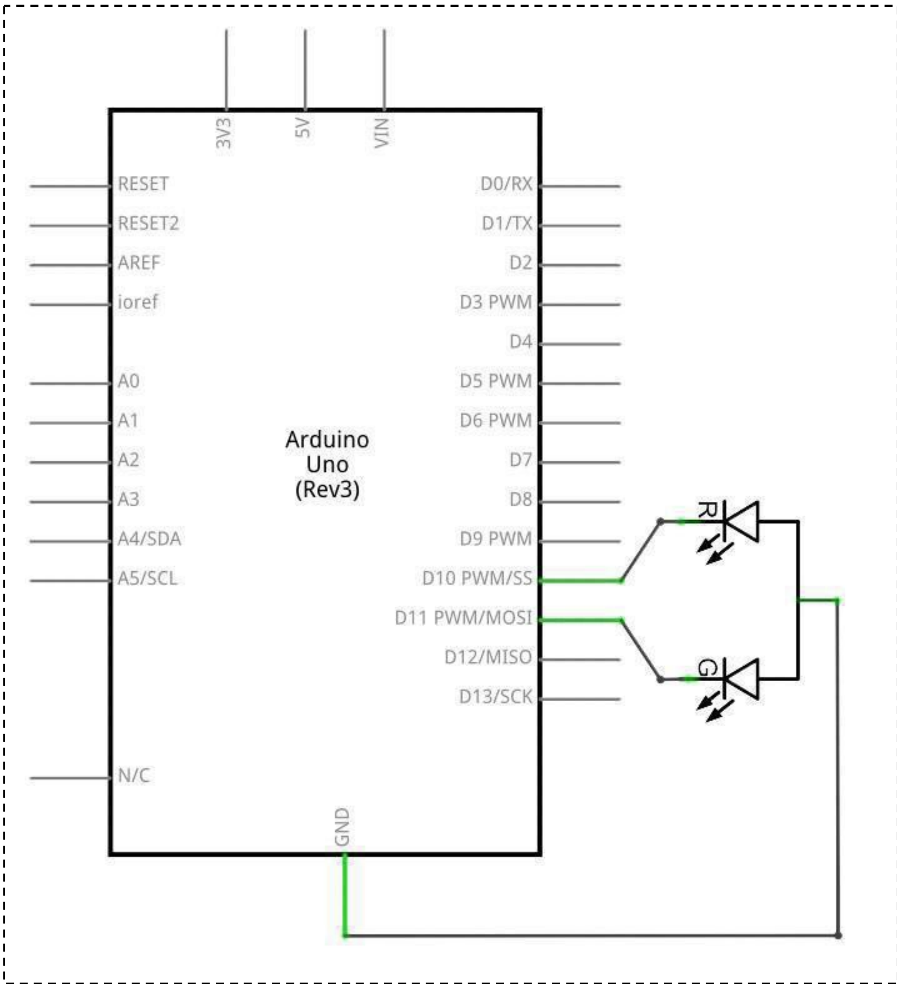
Parameter	Symbol	Test Conditions	Min	Typ	Max	Unit
View Angle of Half Power	2Θ 1/2	IF = 20mA	–	40	–	deg
Forward Voltage High Efficiency Red	VF	IF = 20mA	–	2.05	2.80	V
Yellow–Green			–	2.15	2.80	V
Luminous Intensity (Note 1)	IV	IF = 20mA	35	60	–	mcd
Peak Emission Wavelength High Efficiency Red	λp	IF = 20mA	–	625	–	nm
Yellow–Green			–	570	–	nm
Dominate Wave Length (Note 2) High Efficiency Red	λd(HUE)	IF = 20mA	–	618	–	nm
Yellow–Green			–	567	–	nm



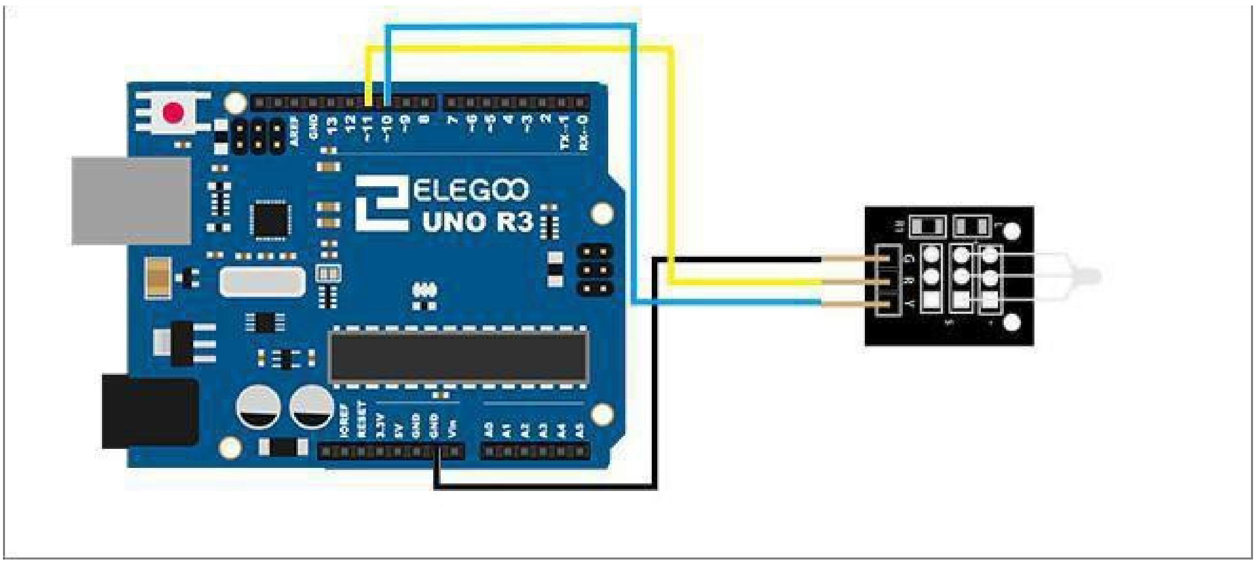
1. Red +
2. Common Lead –
3. Green +

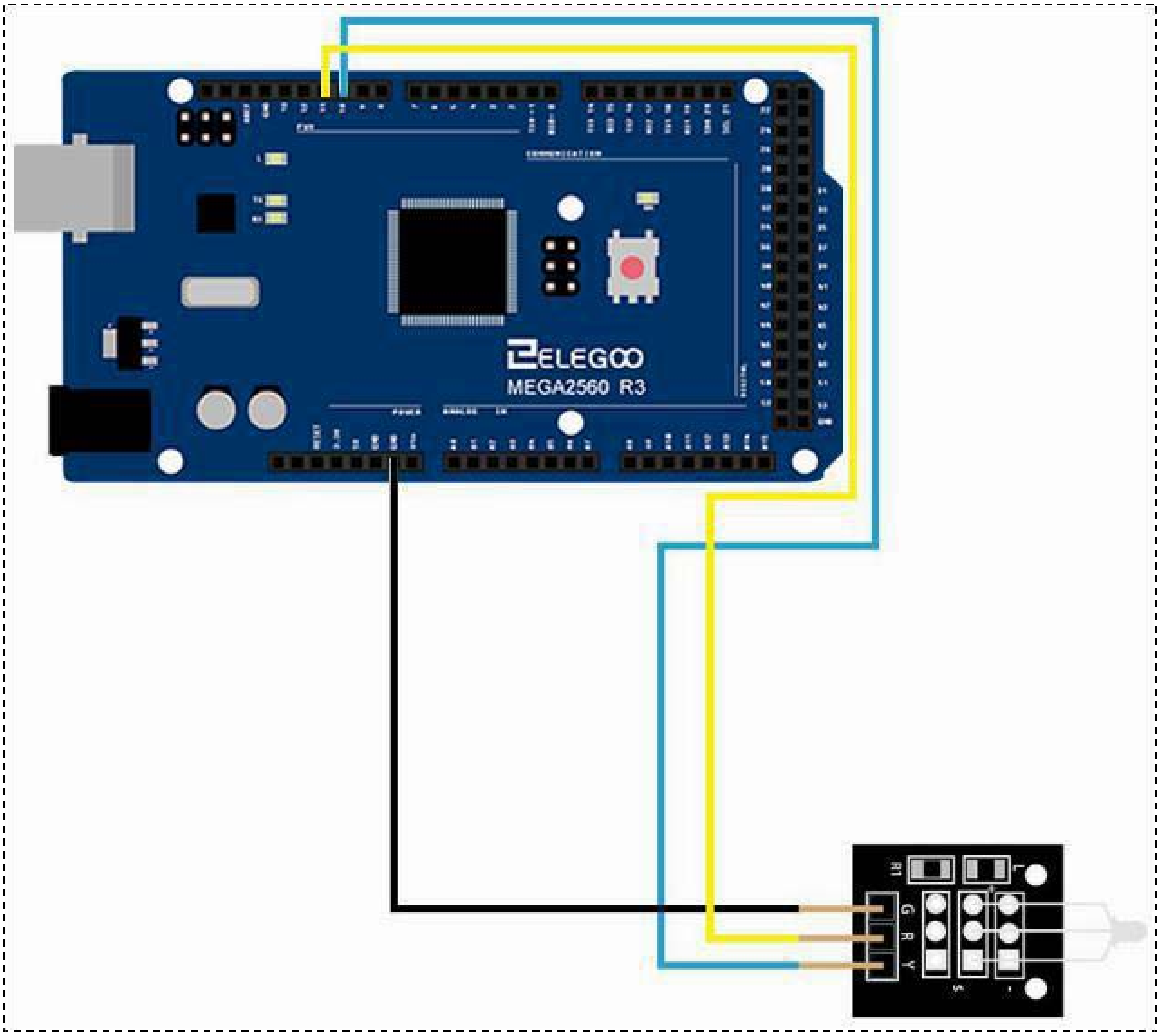
Connection

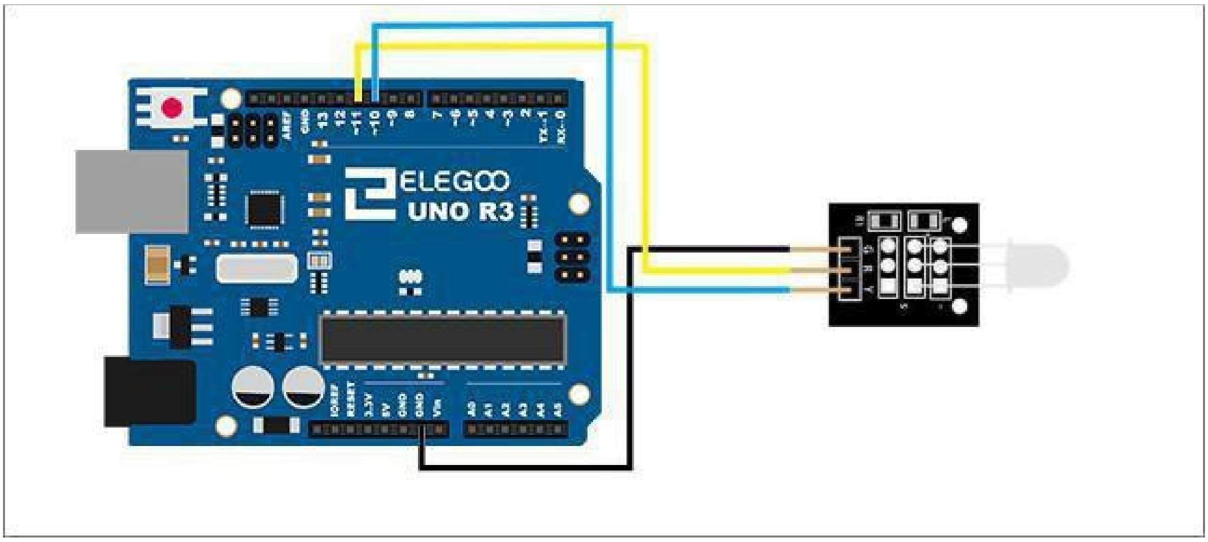
Schematic

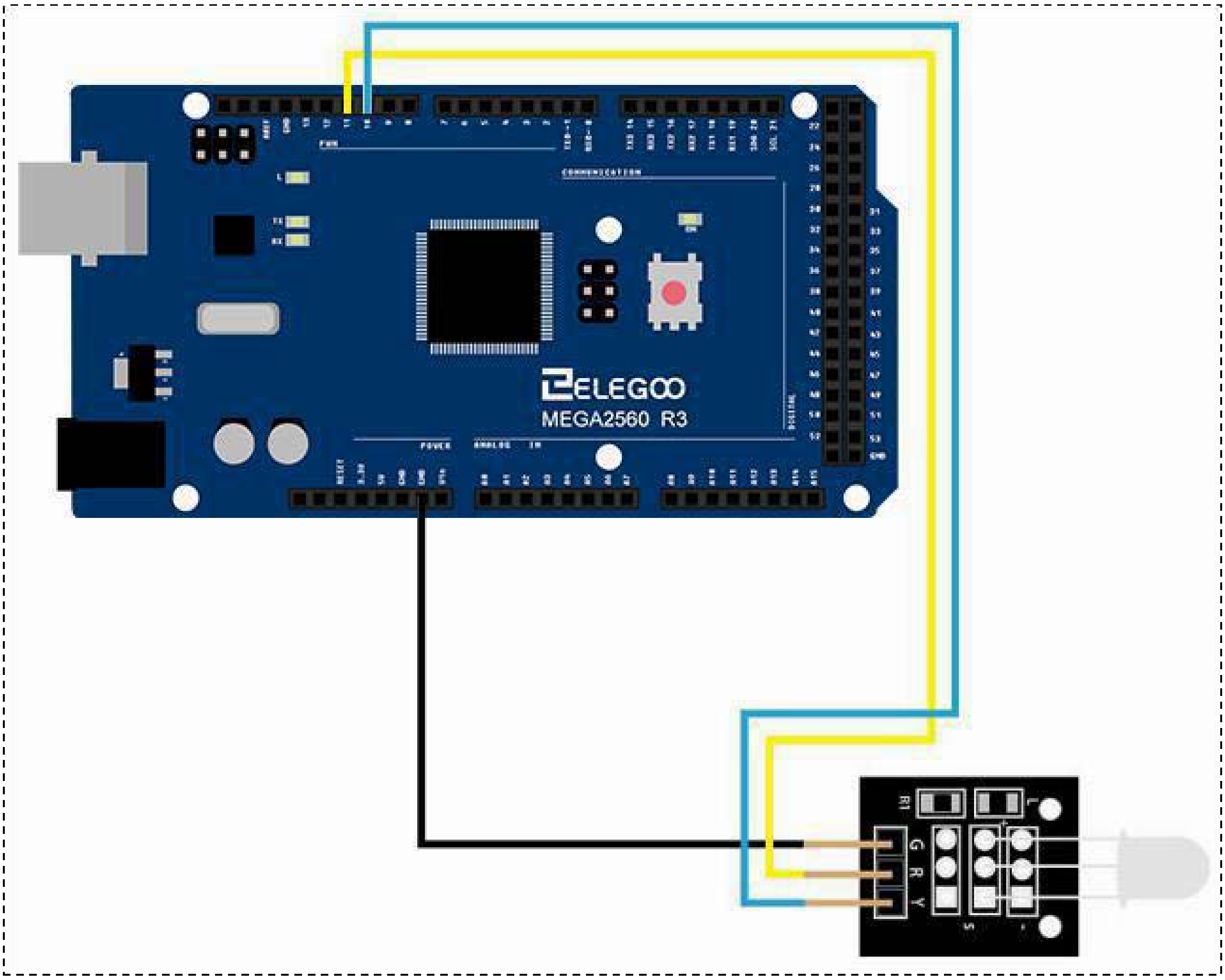


Wiring diagram



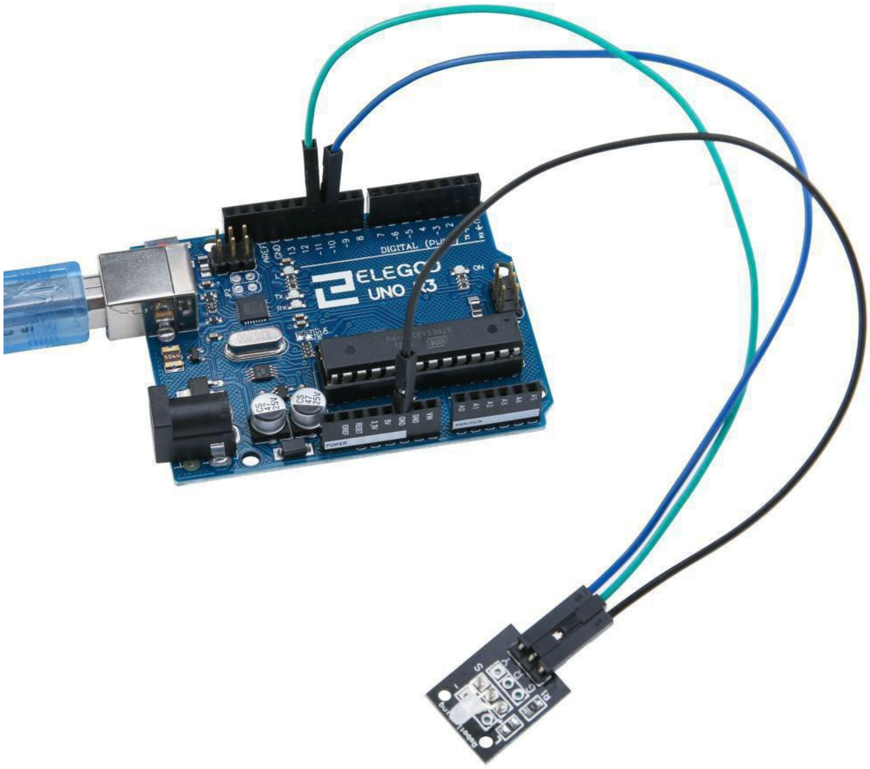

ELEGOO
MEGA2560 R3

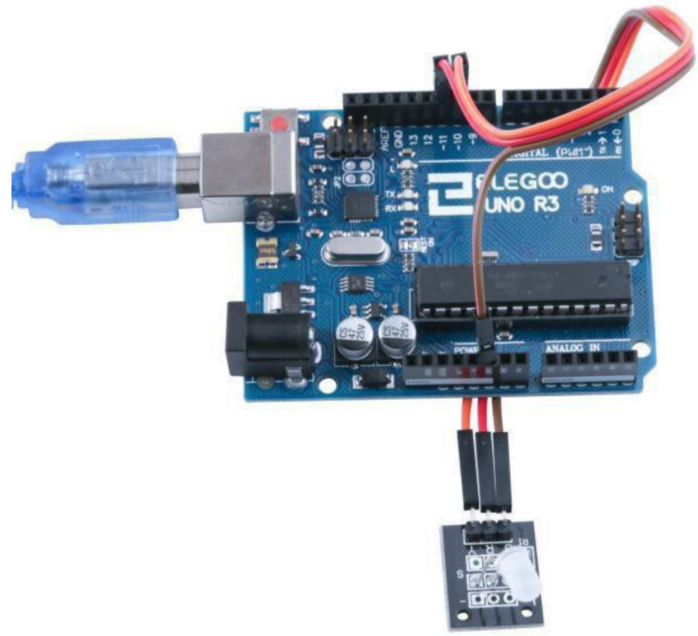

ELEGOO
UNO R3



Result

After we connect the circuit as the picture, we upload the program of each module. We can see the module changing their color as the code set. If you want to make it change the color in different way, you can revise the code.





Lesson 14 Light Dependent Resistor Module

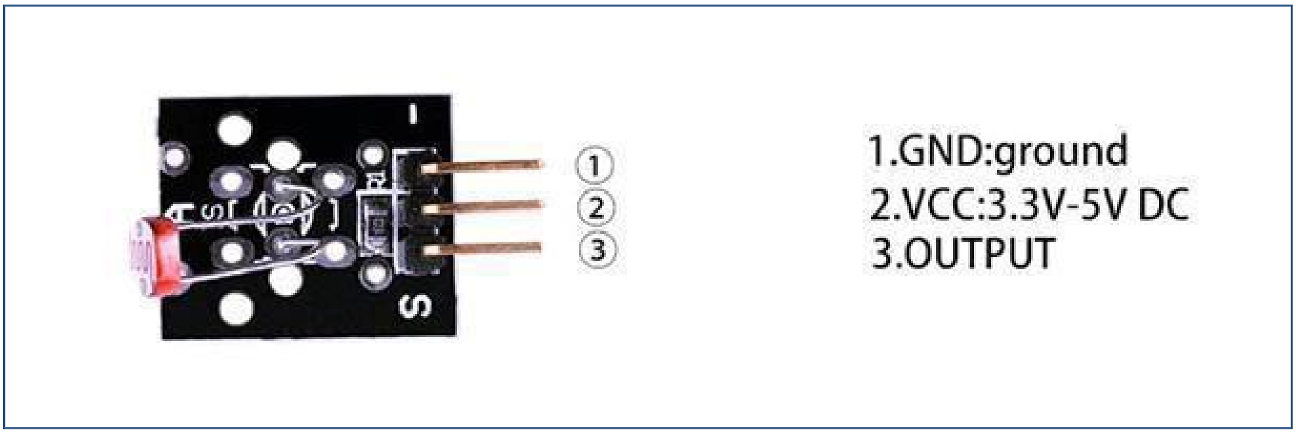
Overview

In this experiment, we will learn how to use the Light Dependent Resistor Module.

Light Dependent Resistor Module is very common in our daily life. It is mainly used in intelligent switch so as to bring convenience to our life. At the same time, in our daily life, we also use it in electronic design. So in order to use it in a better, we provide the corresponding modules to help us to use it moreconvenientlyandefficiently.

Photo Resistor Module

LDR (Light Dependent Resistor). Dark resistance >20M Ohm, light <80 Ohm. The two outer pins connect to the LDR. A fixed 10 K ohm resistor connected between the middle pin and the 'S' pin is included on the module. This simplifies the building of a measurement bridgecircuit.



Component Required:

(1) x Elegoo Uno R3

(1) x USB cable

(1) x photo resistor module

(x) x F-M wires

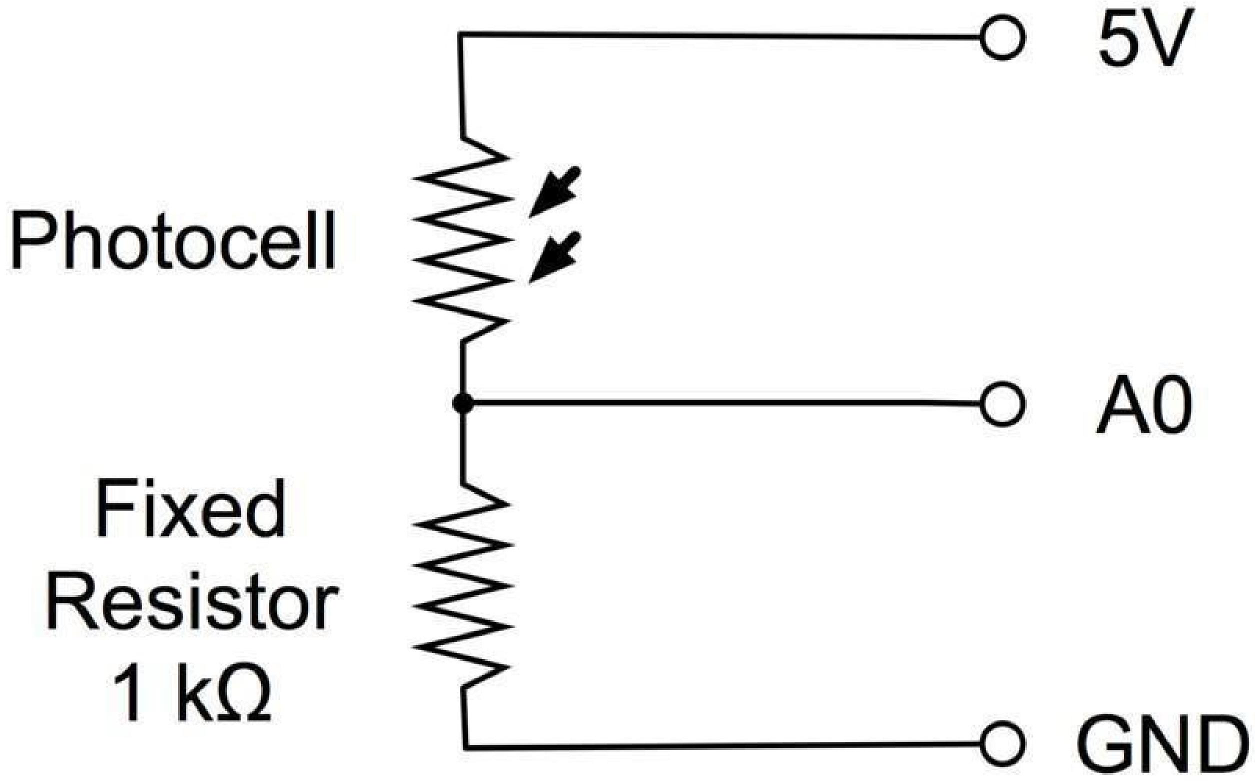
Component Introduction

PHOTOCELL:

The photocell used is of a type called a light dependent resistor, sometimes called an LDR. As the name suggests, these components act just like a resistor, except that the resistance changes in response to how much light is falling on them.

This one has a resistance of about 50 kΩ in near darkness and 500 Ω in bright light. To convert this varying value of resistance into something we can measure on an Arduino's analog input, it need to be converted into a voltage.

The simplest way to do that is to com bine it with a fixed resistor.



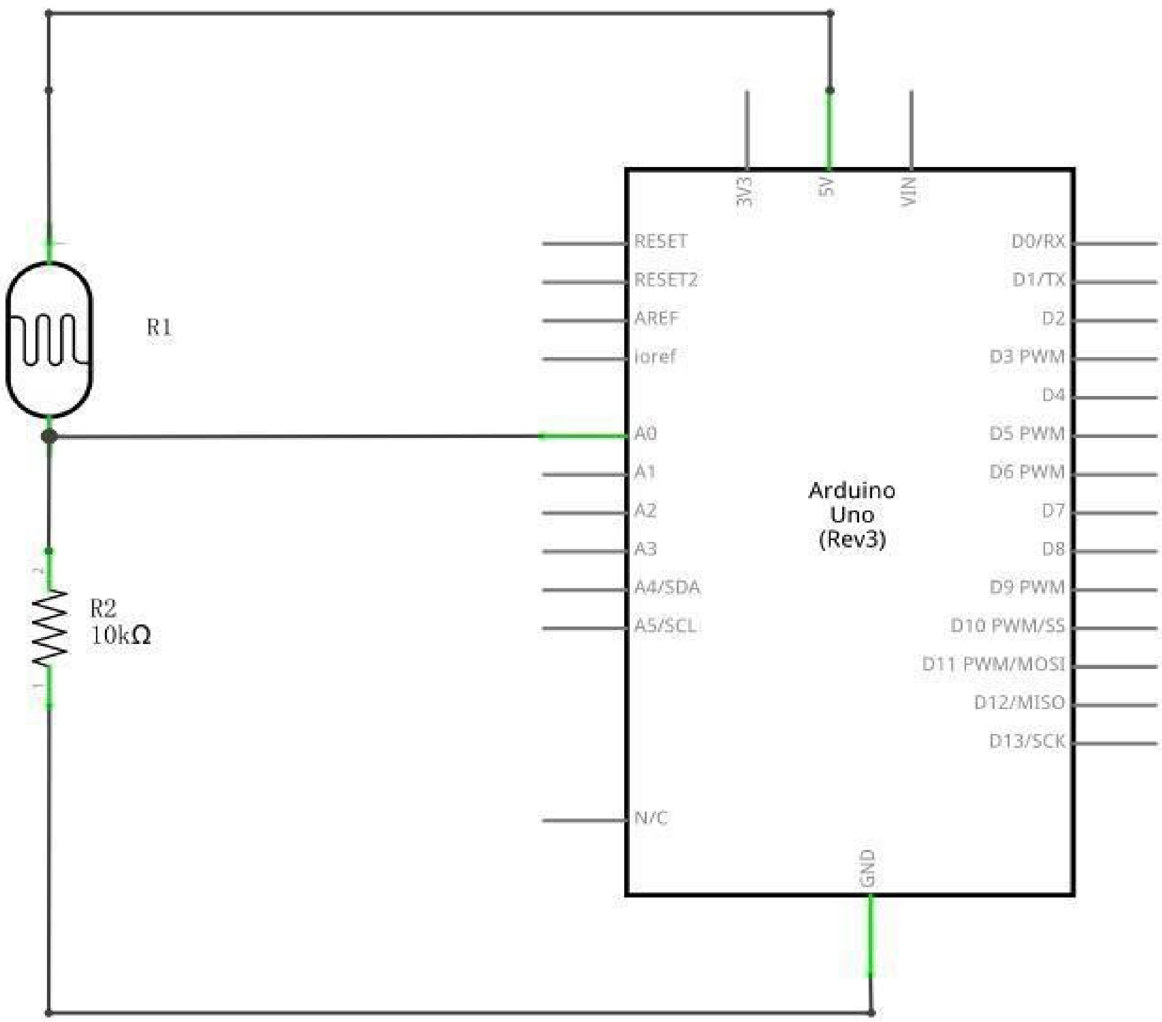
The resistor and photocell together behave rather like a pot. When the light is very bright, then the resistance of the photocell is very low compared with the fixed value resistor, and so it is as if the pot were turned to maximum.

Whenthephotocellisindulllighttheresistancebecomesgreaterthanthefixed1kΩresistor and it is as if the pot were being turned to wards GND.

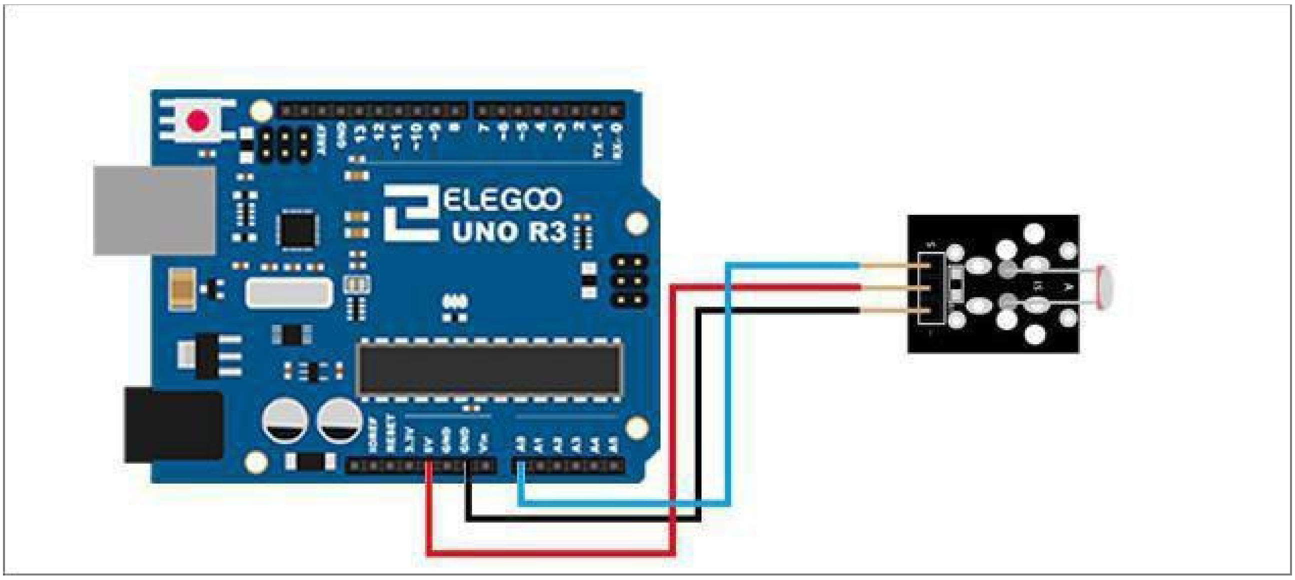
Load up the sketch given in the next section and try covering the photo cell with your finger, and holding it near alight source.

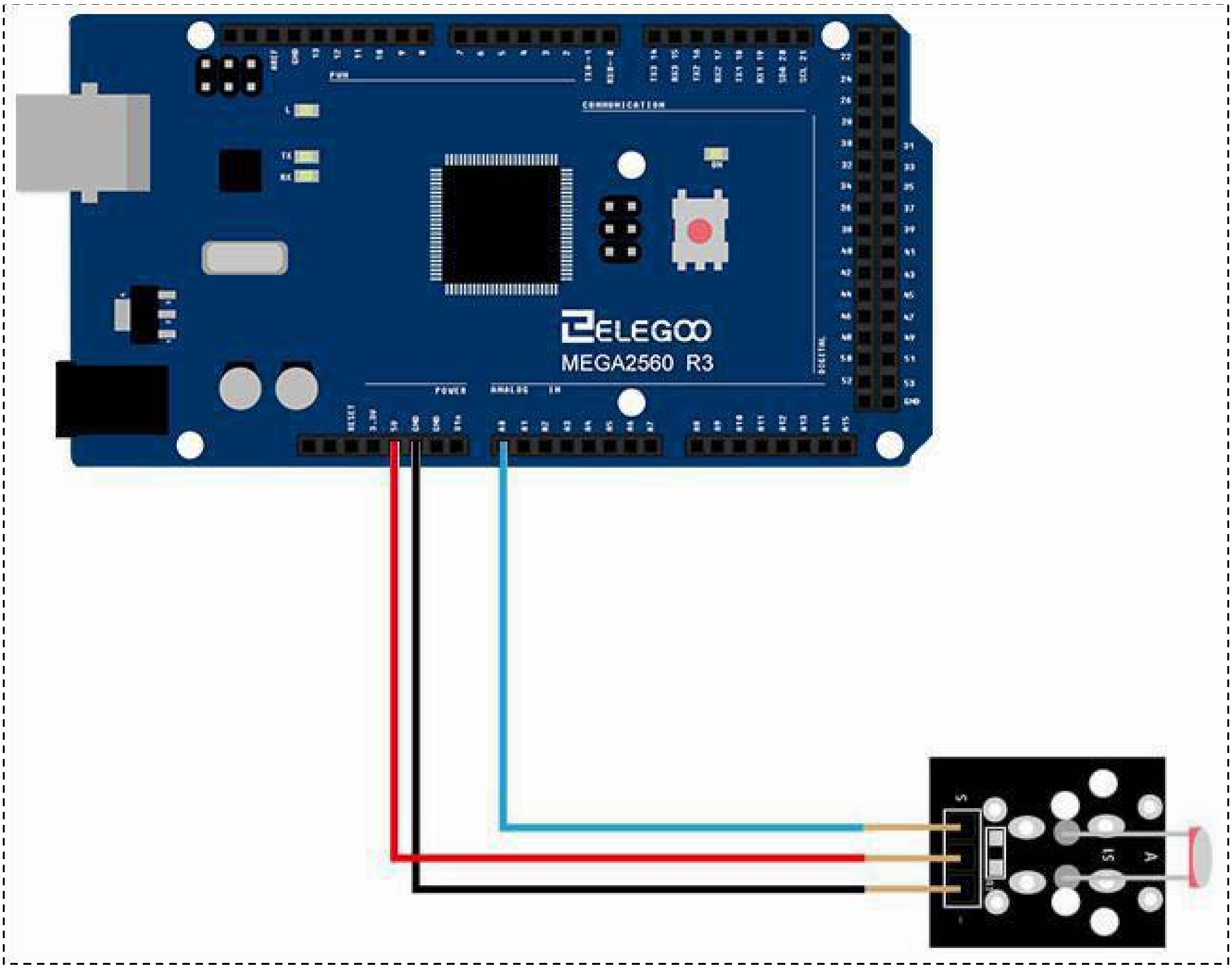
Connection

Schematic

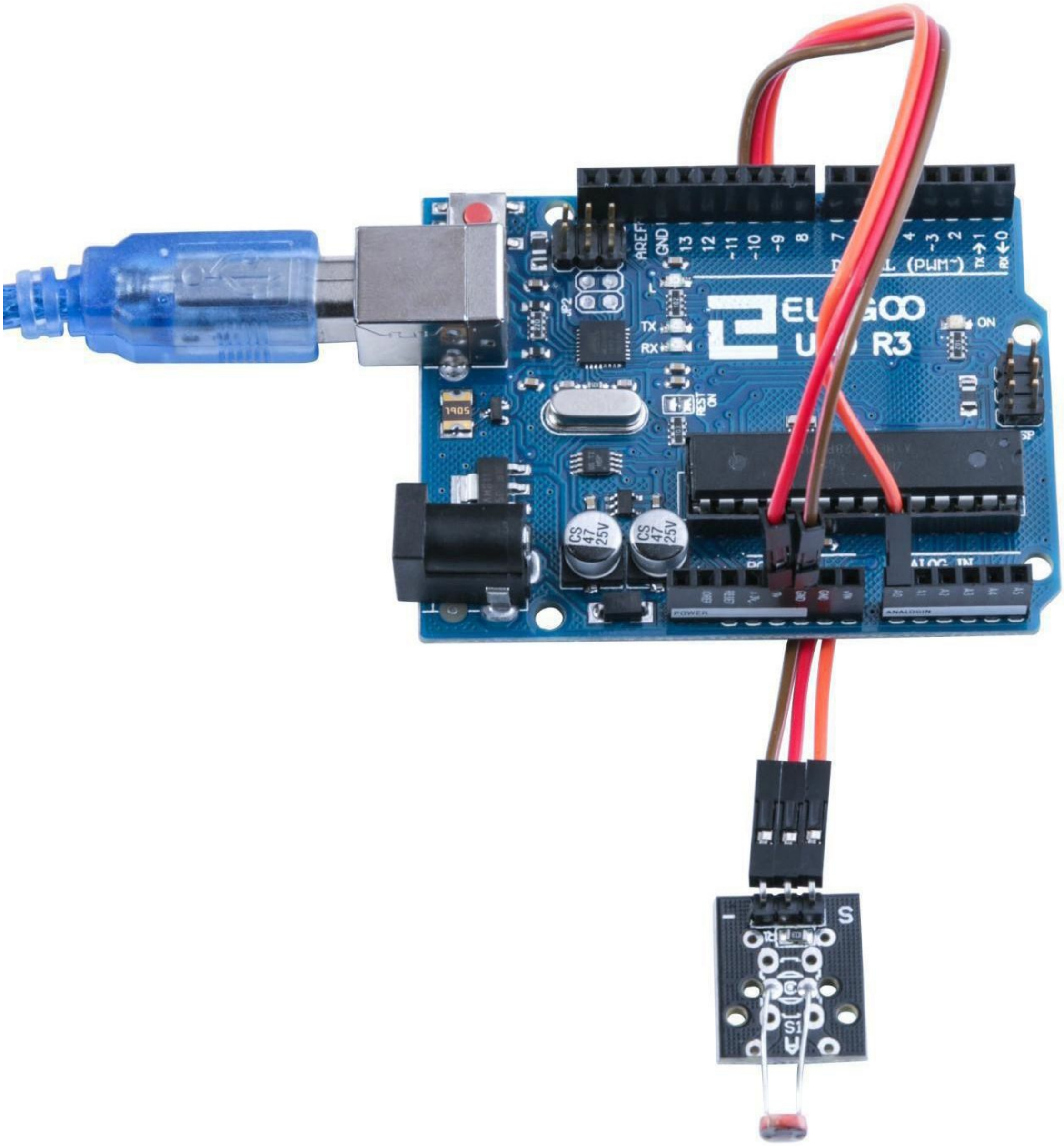


Wiring diagram

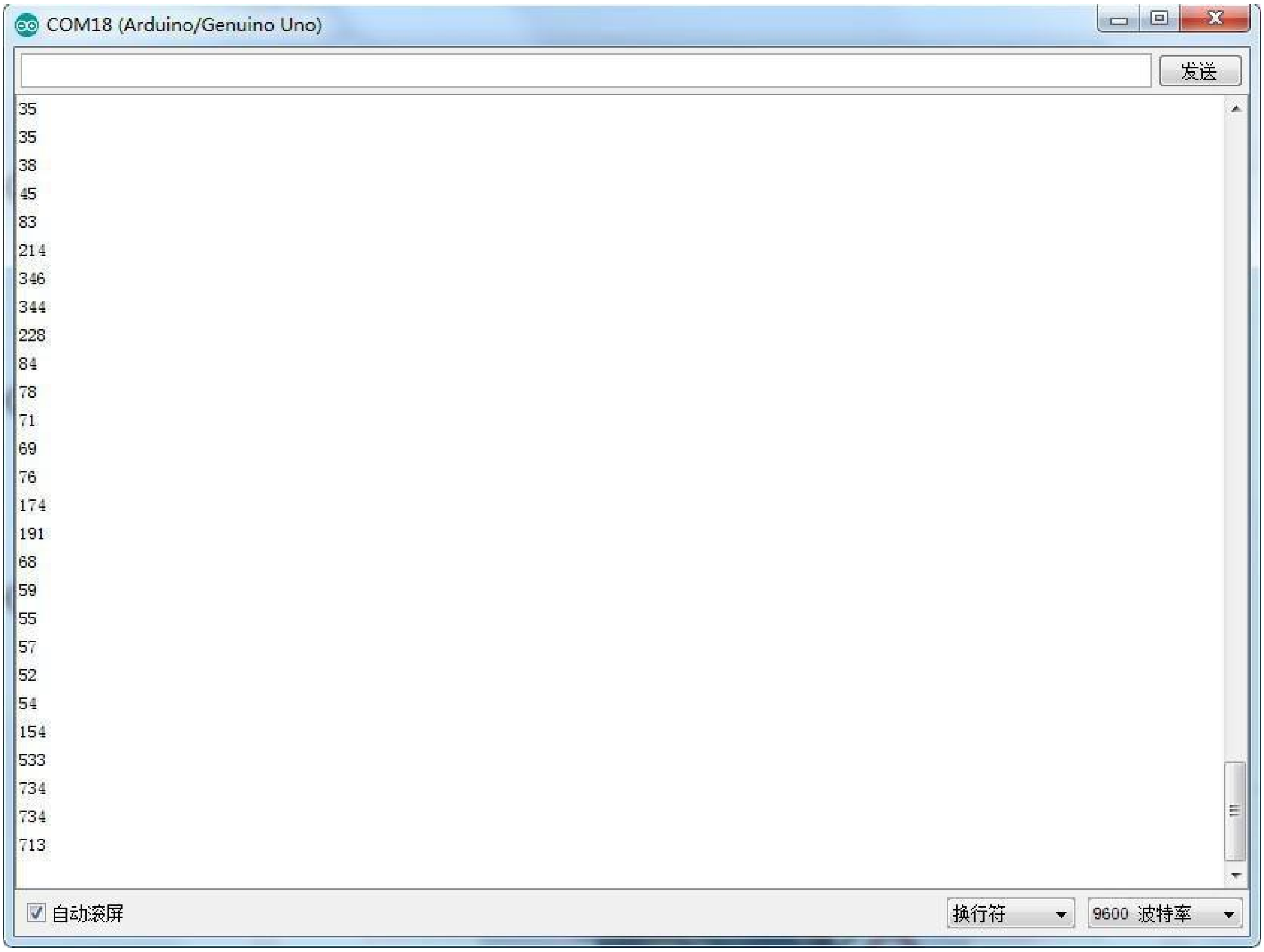




Result



Uploadthe programthen openthe monitor, we can see the data as below:



In the test, we only read the output analog voltage value of photo-resistor module. In the test results, we will find that when there is lighting, high voltage out put equivalently of switch on, when there is no light, low voltage equivalently of switch off. This is what we can use this in practice.

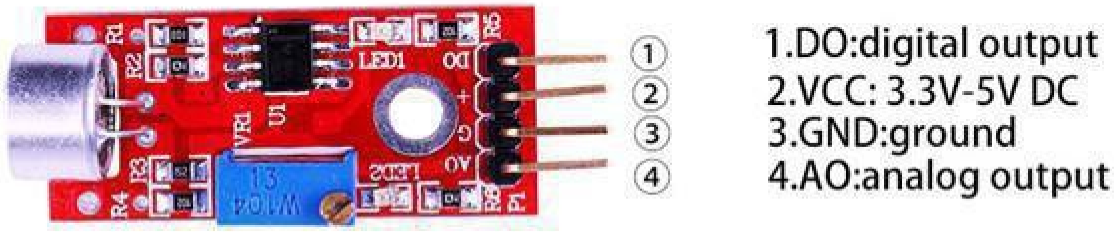
Lesson 15 Large Microphone Module and Small Microphone module

Overview

In this experiment, we will learn how to use the High-sensitive Voice Sensor.

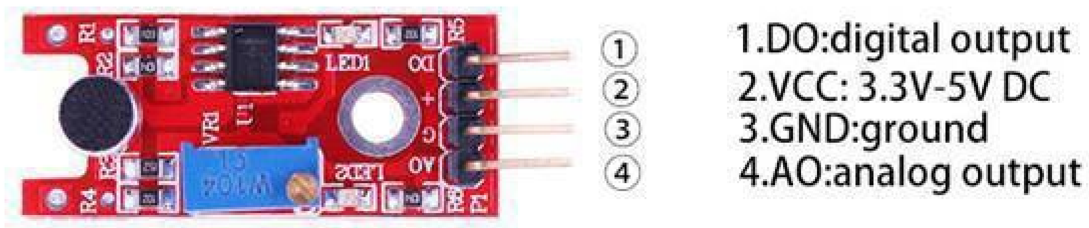
Large Microphone Module

A microphone module featuring a high-sensitivity large-for mat electret capsule. Output ‘DO’ (active high) is switched when the sound level exceeds a preset level. A pot allows adjustment of the level.



Small Microphone module

A microphone module with a small electret capsule. Output ‘DO’ (active high) is switched when the sound level exceeds a preset level. A pot allows adjustment of the level. Except for the smaller size of the capsule and its lower sensitivity the module is identical to the ‘Big Sound’ module.



Component Required:

(1) x Elegoo Uno R3

(1) x USB cable

(1) x Large Microphone Module

(1)xSmall Microphone Module

(x) x F-M wires

Component Introduction

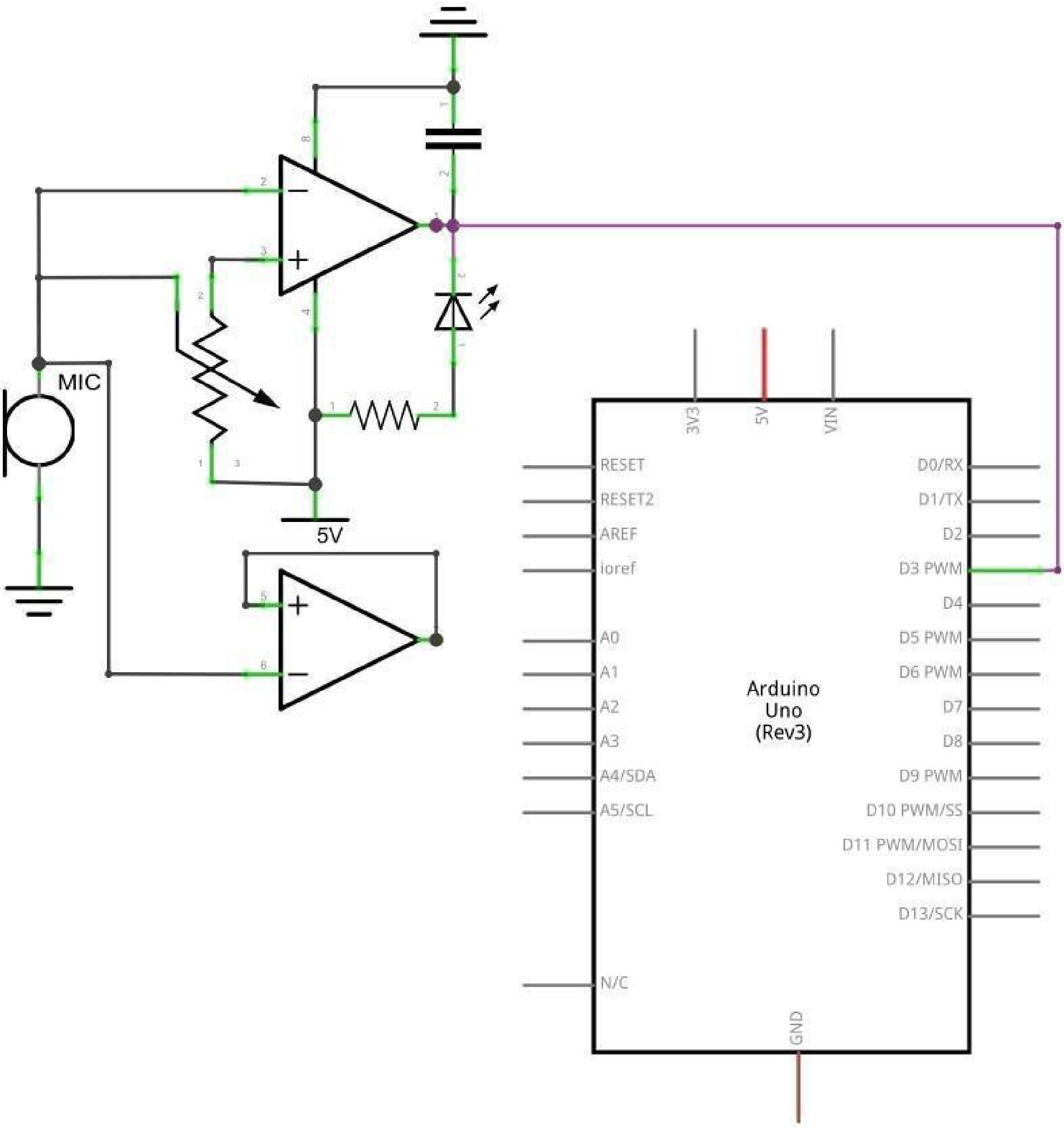
Sound sensor:

The sound sensor module provides an easy way to detect sound and is generally used for detecting sound intensity. This module can be used for security, switch, and monitoring applications. Its accuracy can be easily adjusted for the convenience of usage.

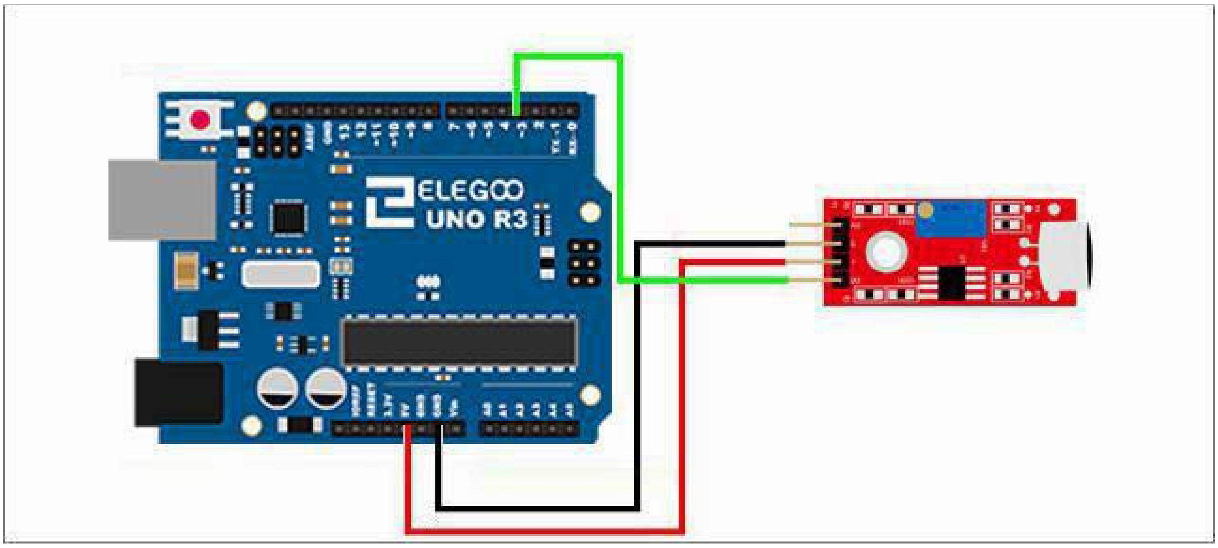
It uses a microphone which supplies the input to an amplifier, peak detector and buffer. When the sensor detects a sound, it processes an output signal voltage which is sent to a microcontroller then performs necessary processing.

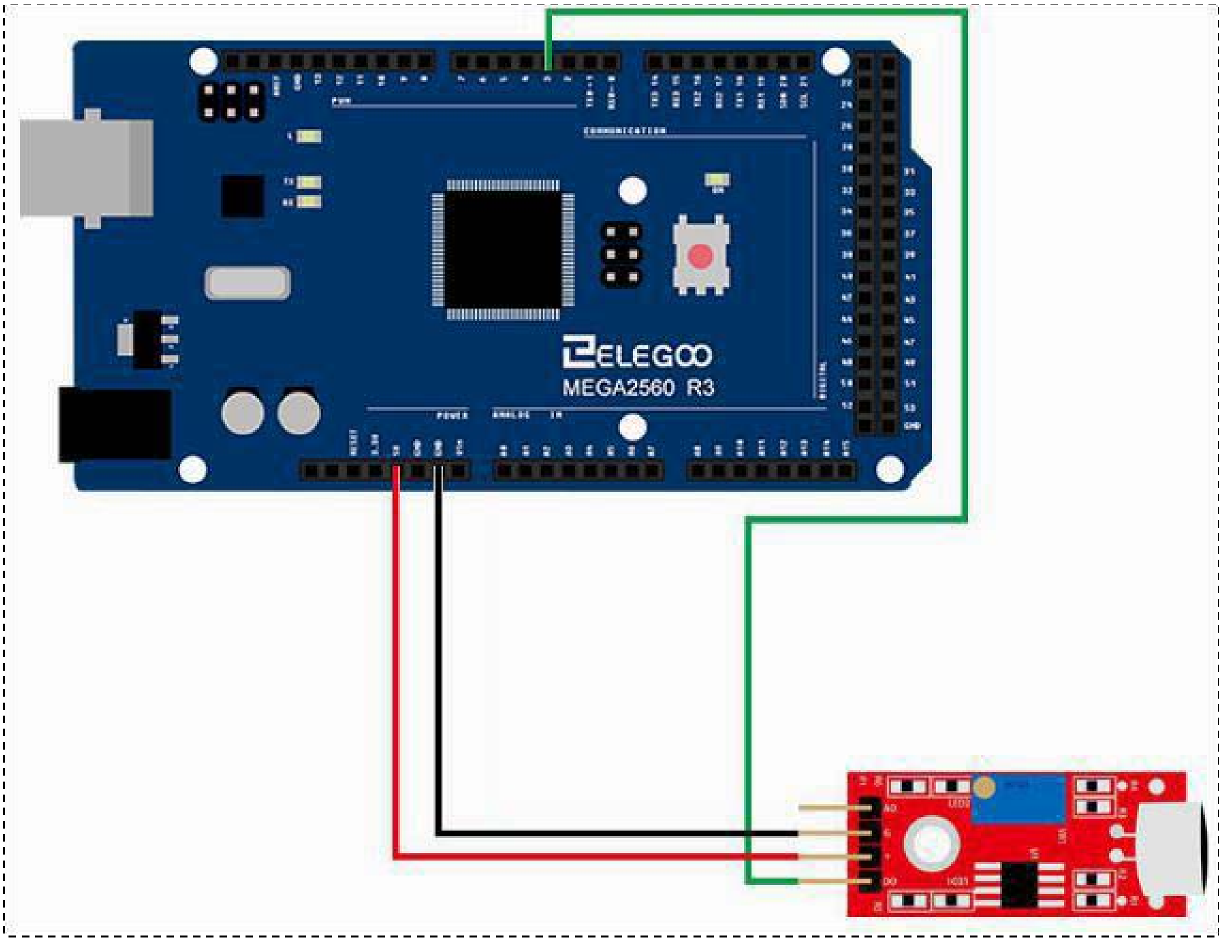
Connection

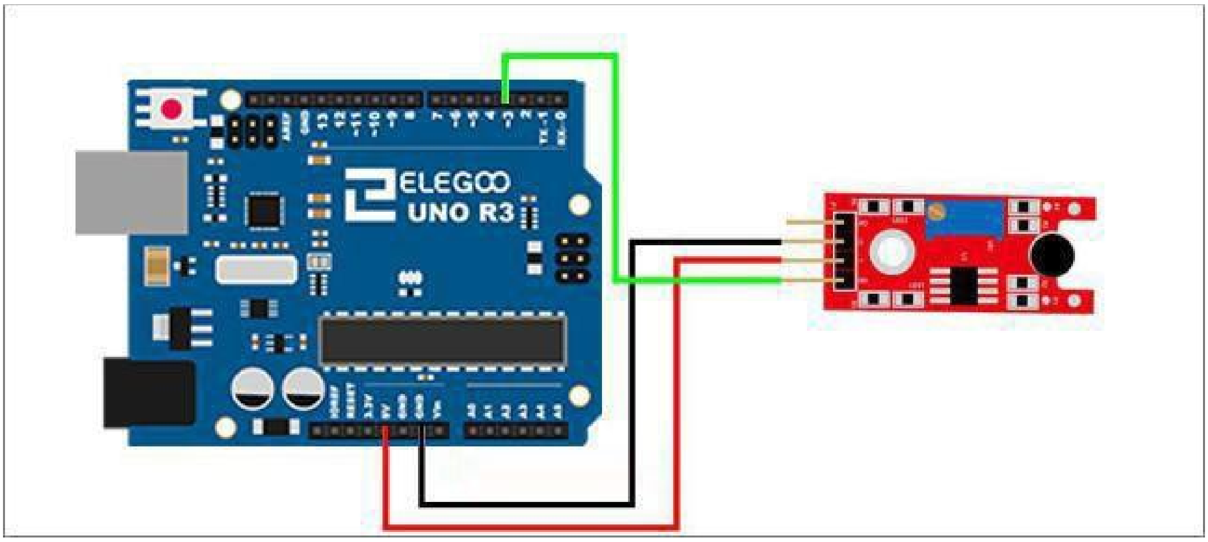
Schematic

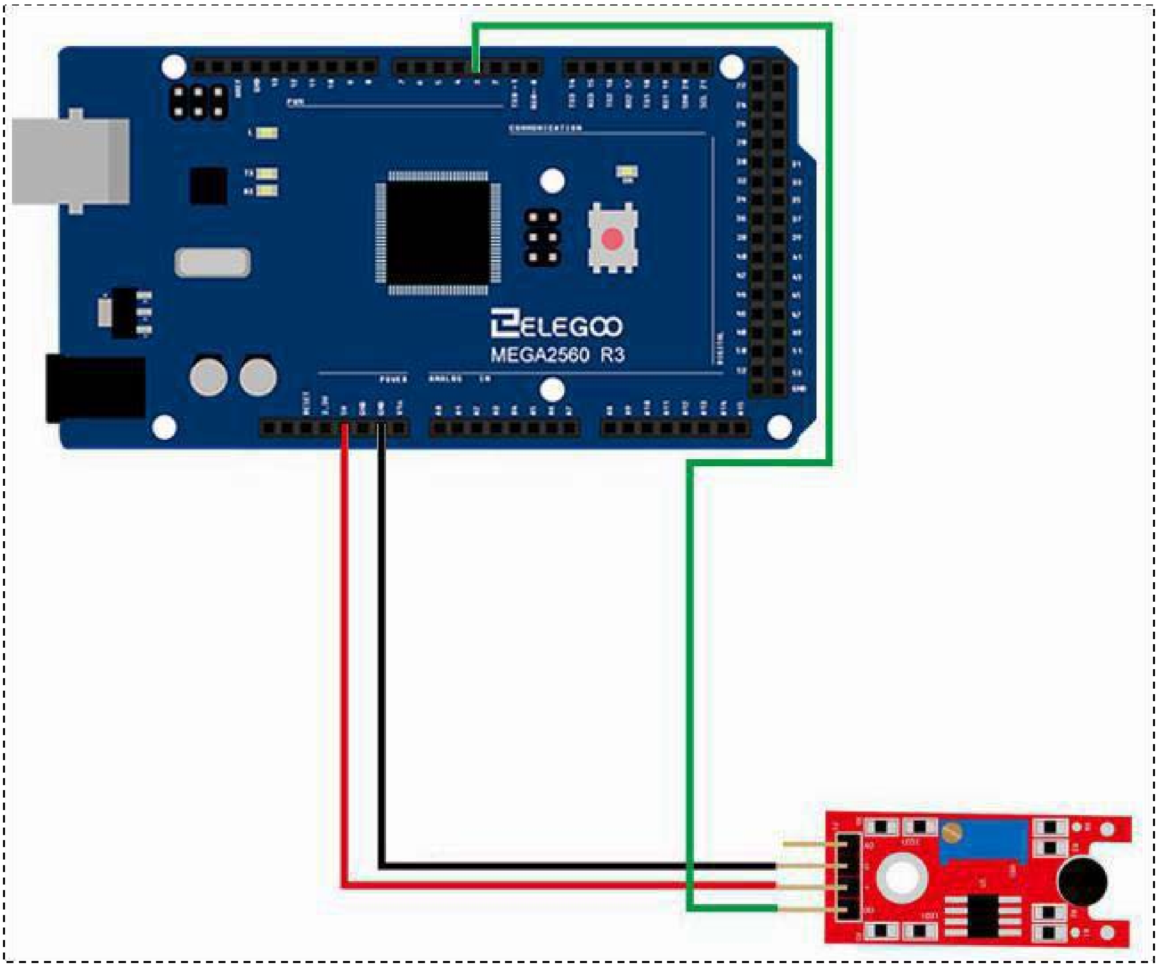


Wiring diagram









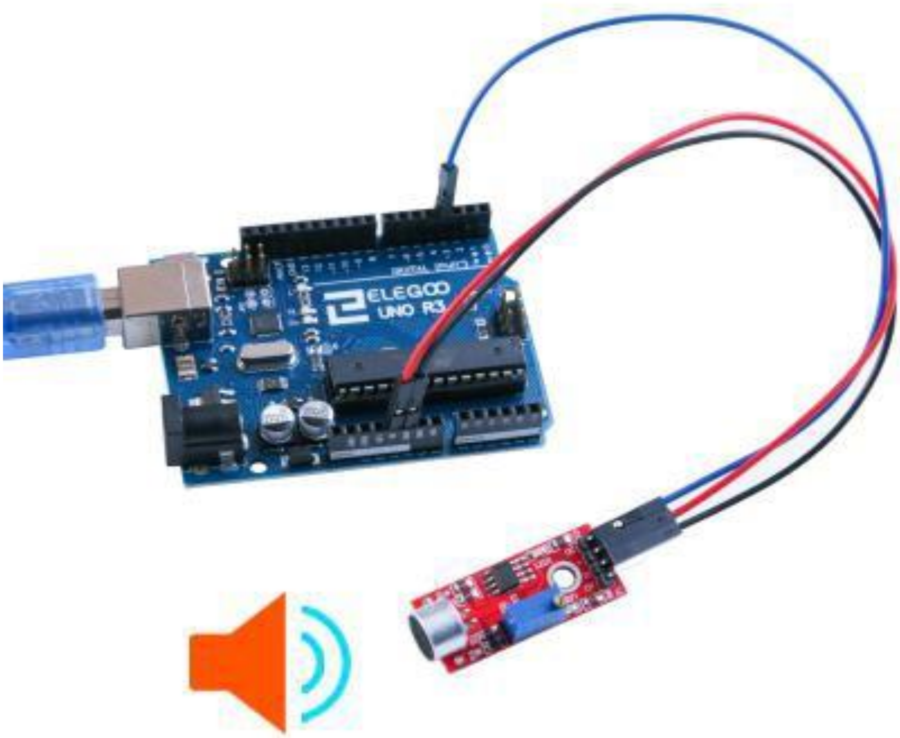
Result

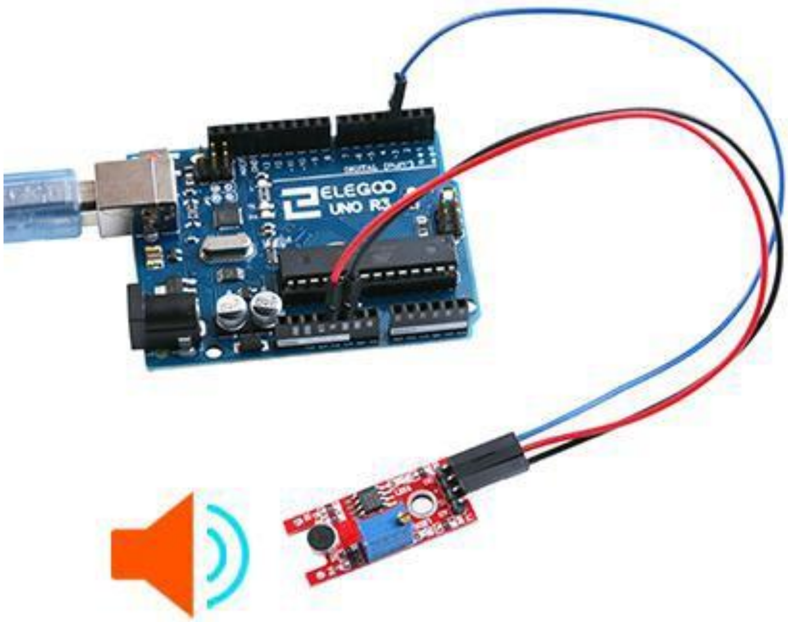
LED1 should be lit, showing power. LED2 indicates sound input, and the sensitivity is adjusted by potentiometer.

There is a tiny screw on the blue potentiometer block that you can use for adjustment.

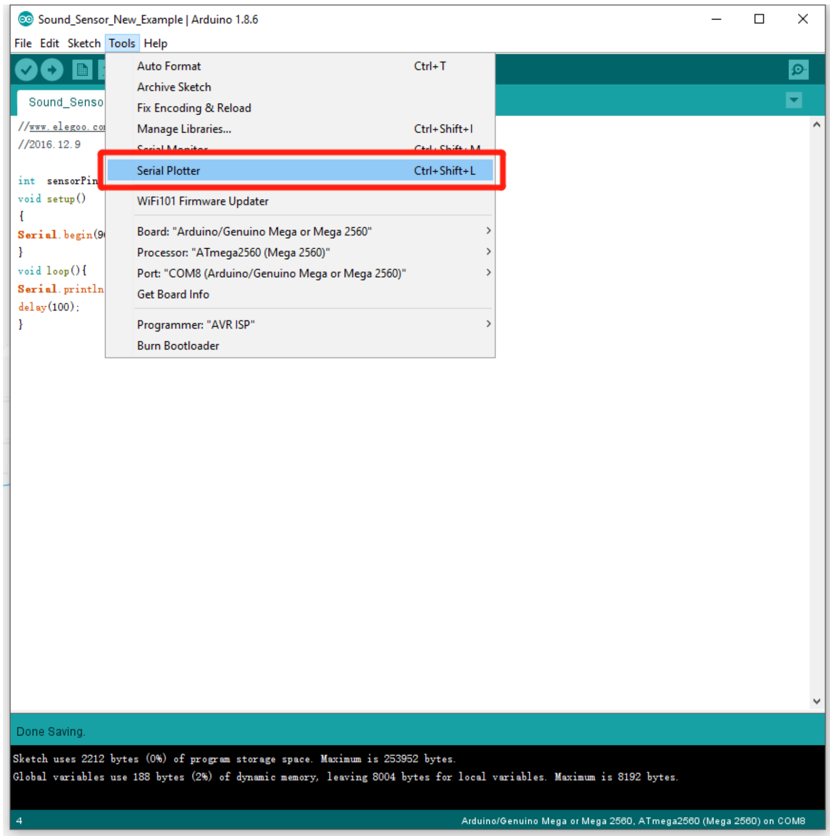
Turning that clockwise lowers the potentiometer value, while counter-clockwise raises the potentiometer value.

Use the potentiometer to adjust the Sound Sensor sensitivity.

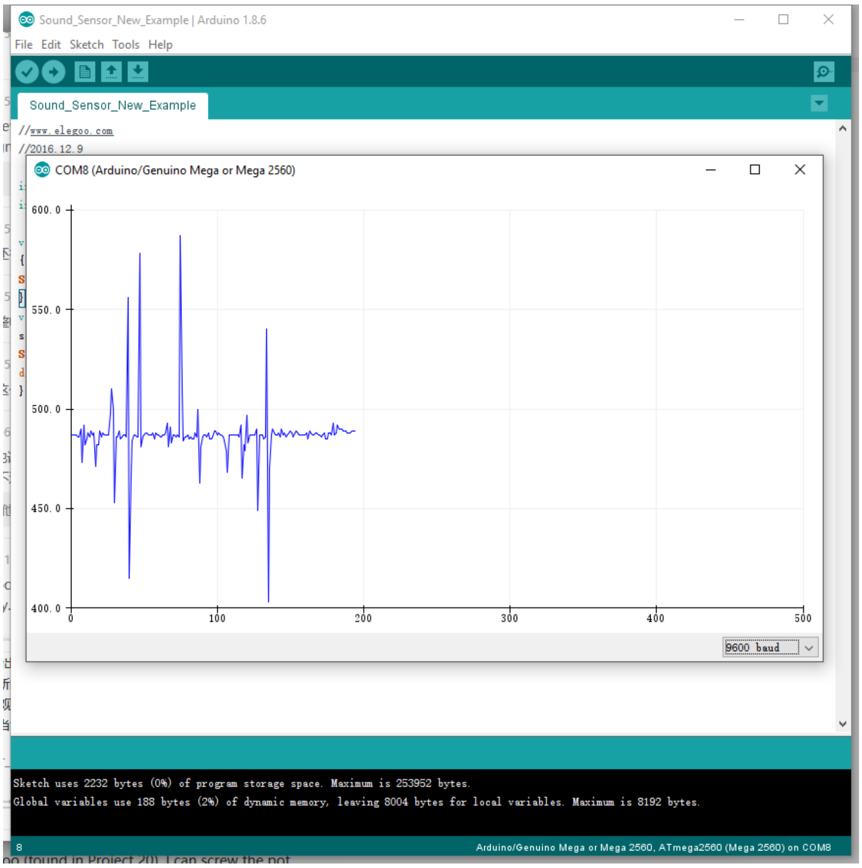




Open Serial Plotter:



Serial Plotter Example:



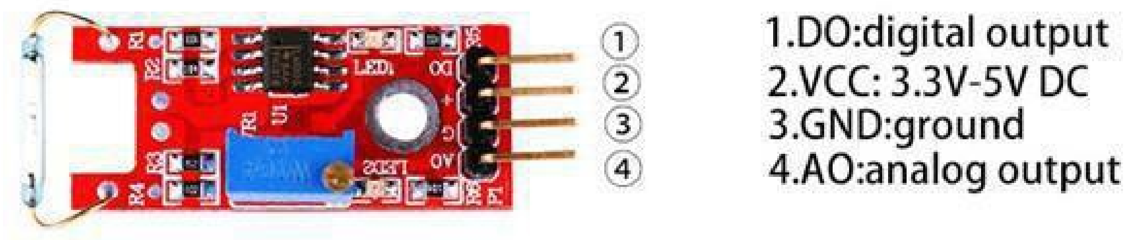
Lesson 16 Reed Switch Module and MiniReed Switch Module

Overview

In this experiment, we will learn how to use reed switch and minireed switch module.

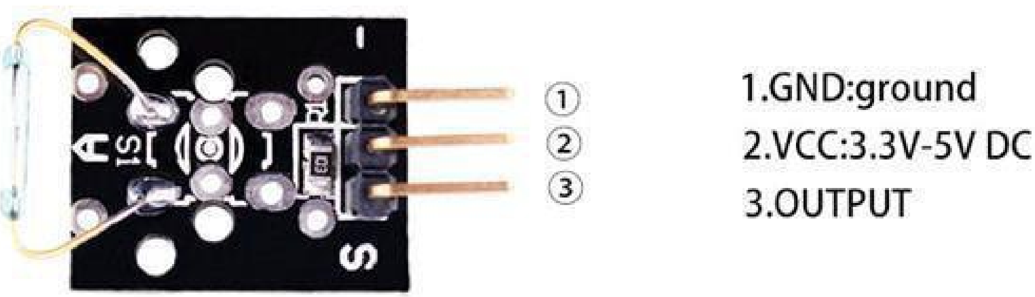
Reed Switch Module

This reed switch offers an analog as well as a digital interface. The ‘G’ pin connected to GND, the ‘+’ pin to 5V DC, the ‘AO’ pin offers the analog output while the ‘DO’ offers the digital output. A potentiometer is used as a pull up resistor.



Mini Reed Module

The reed switch is connected between the two outer pins on the PCB. Without a magnetic field the contacts remain open.
A built-in 10 K ohm resistor is connected between the center pin and the ‘S’ pin.
It can be used as a pull up or pull down resistor.



Component Required:

(1)x Elegoo Uno R3

(1) x usb cable

(1)x reed switch module

(1) Mini Reed switch module

(x)x F-M wires

Component Introduction

Reed Switch and Reed Sensor Activation:

Although a reed switch can be activated by placing it inside an electrical coil, many reed switches and reed sensors are used for proximity sensing and are activated by a magnet. As the magnet is brought in to the proximity of the reed sensor/switch, the device deactivates. The magnet is removed from the proximity of the reed sensor/switch, the device deactivates. However, the magnetic interaction involved in activating the reed switch contacts is not necessarily obvious. One way of thinking about the interaction is that the magnet induces magnetic poles in to the metal parts of the reed switch and the resulting attraction between theelectrical contacts causes the reed s

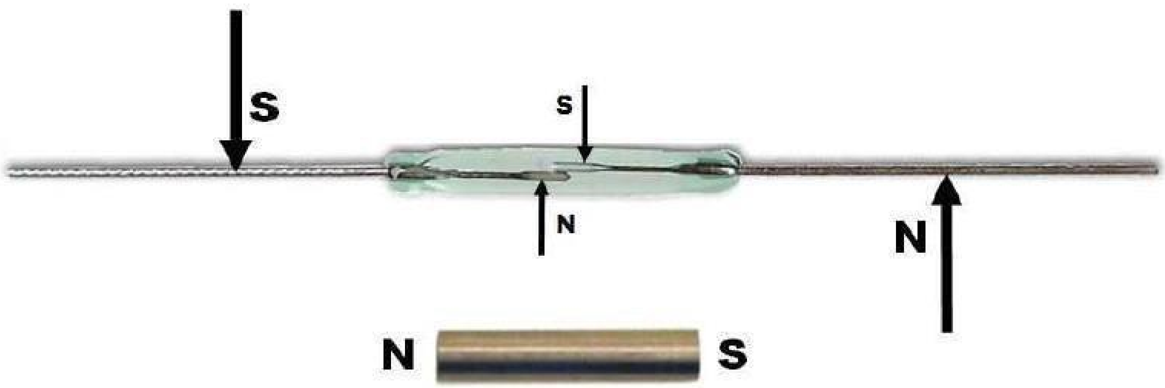


Figure 1 – Magnetic Induction

Another equally valid way of thinking about the interaction between a magnet and a reed switch is that the magnet induces magnetic flux through the electrical contacts. When the magnetic flux is high enough, the magnetic attraction between the contacts causes the reed switch to close.

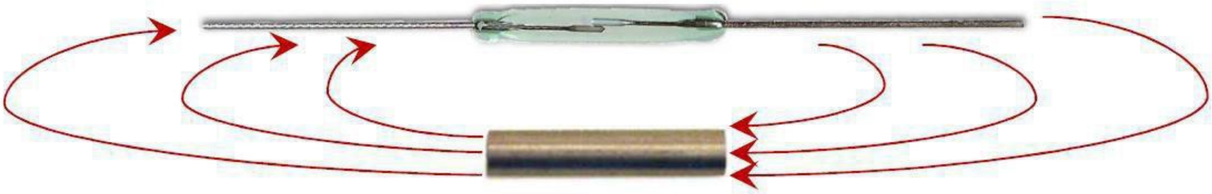


Figure 2 – Magnetic Flux

The following are examples of typical reed switch and reed sensor activate distances.

Difference between the reed switch module and mini reed switch module

As we can see, the reed switch module is bigger than the mini reed switch module. So the bigger one may

have more function than the mini one. The reed switch can output in two ways: digital and analog. The mini reed can only out put in digital.

In the 37 sensor kit, there have 7 red pcb modules. The difference between the red and small pcb is same as above.

Principle

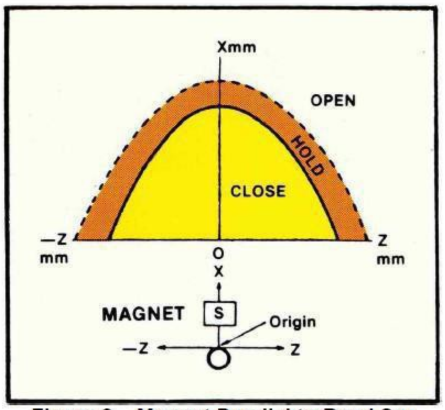


Figure 3 – Magnet Parallel to Reed Sw.

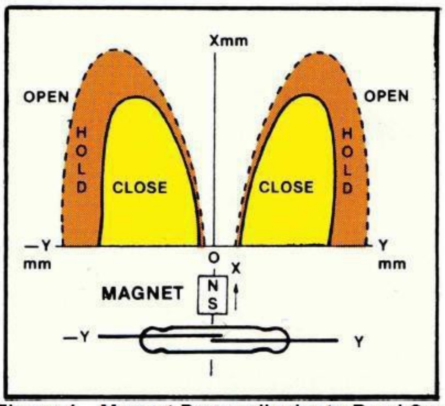


Figure 4 – Magnet Perpendicular to Reed Sw.

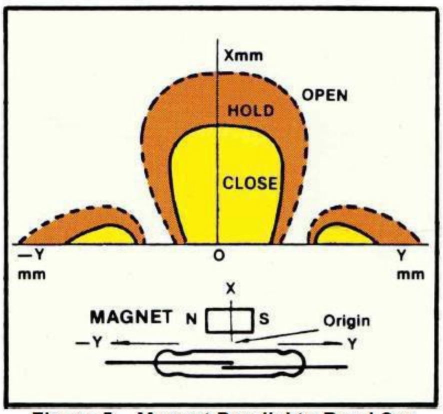
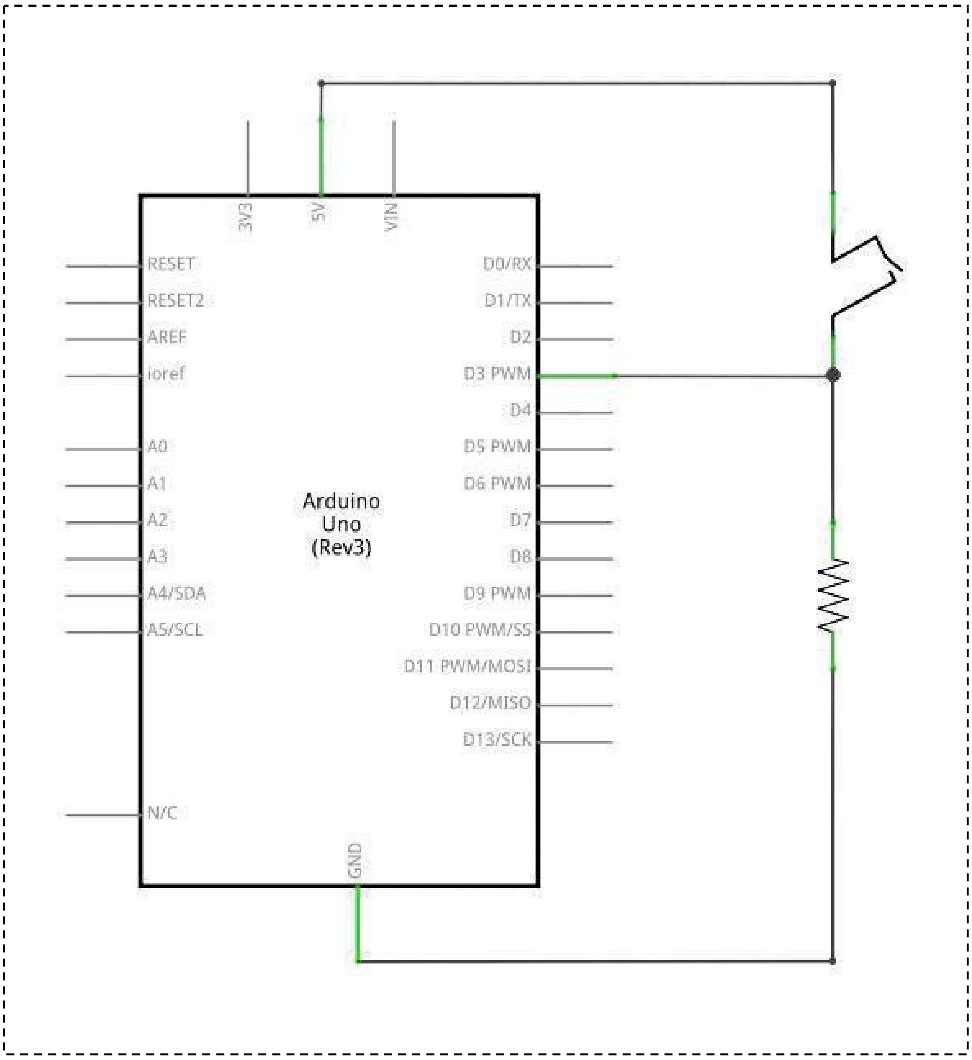


Figure 5 – Magnet Parallel to Reed Sw.

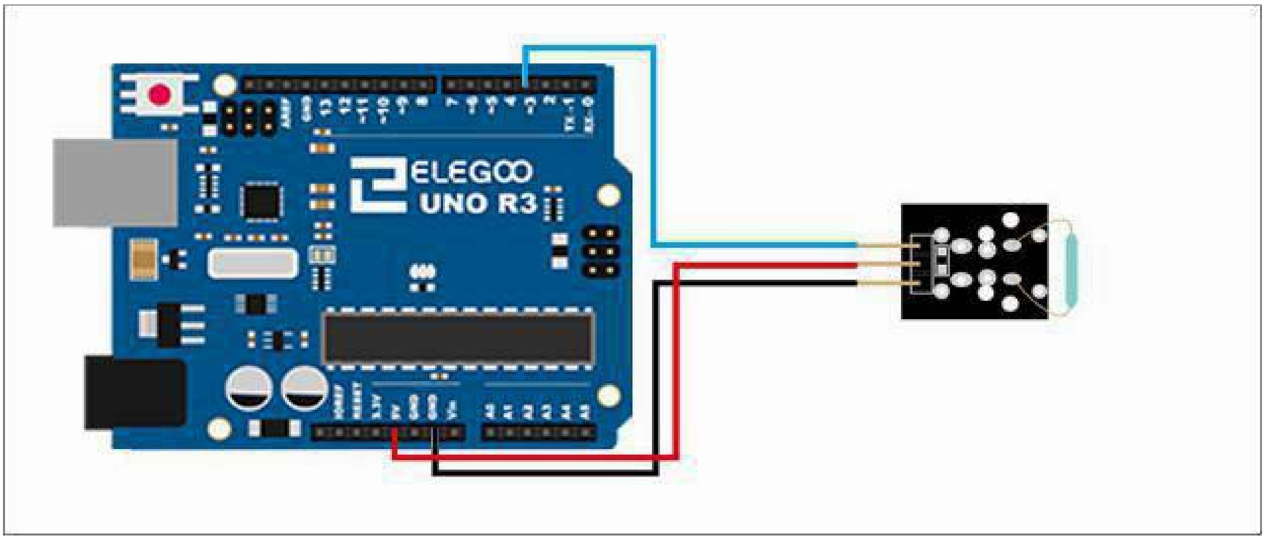
As can be seen, the magnetic orientation and location relative to the reed switch play important role sin the activation distances. In addition, the size of the activate regions (lobes) will vary depending on the strength of the magnet and the sensitivity of the reed switch. Proper orientation of the magnet with respect to the reed sensor/switch is an important consideration in meeting the application’s requirements across the tolerance range for mechanical systems, magnetic strength and reed sensor or reed switch sensitivity.

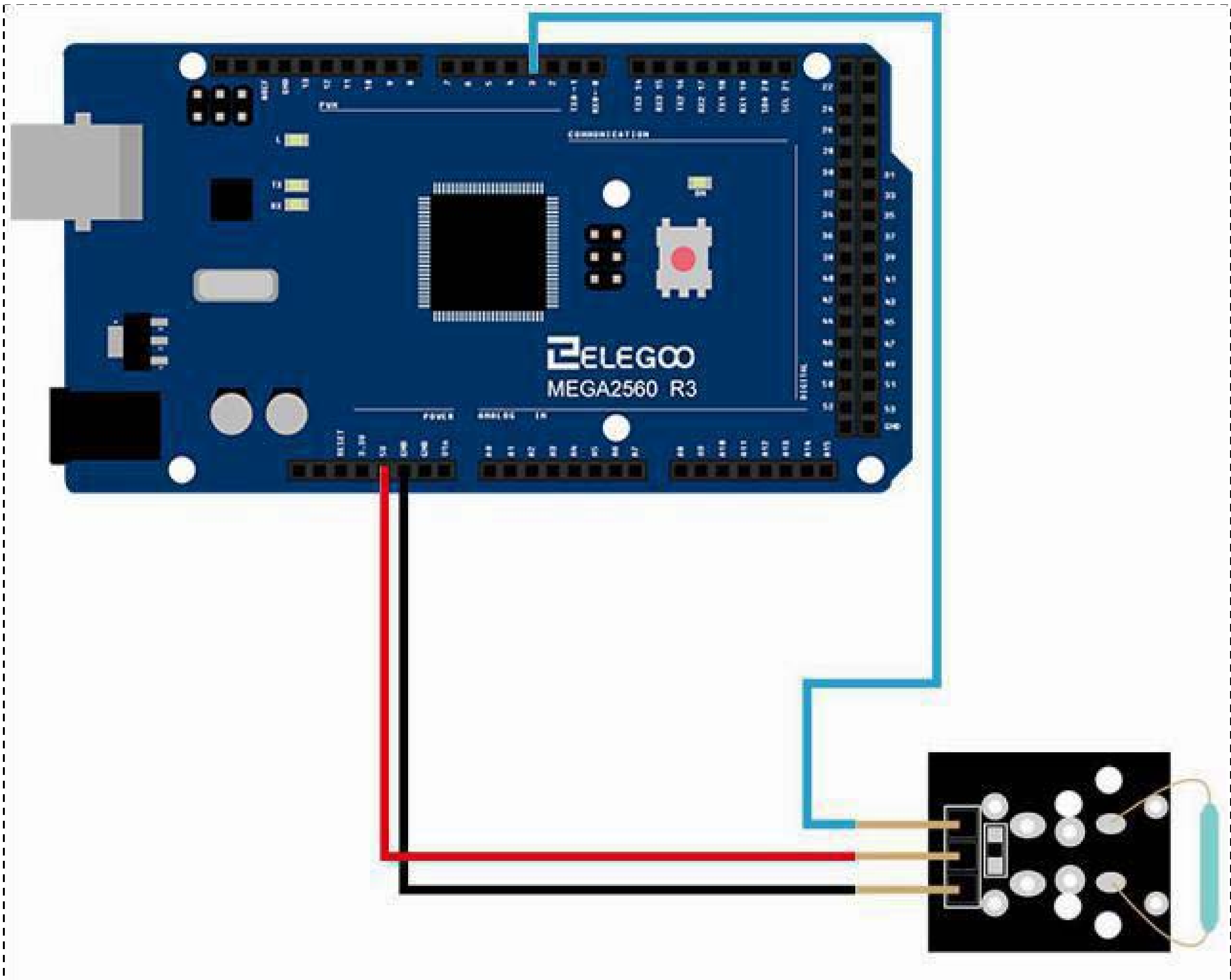
Connection of mini reed switch module

Schematic

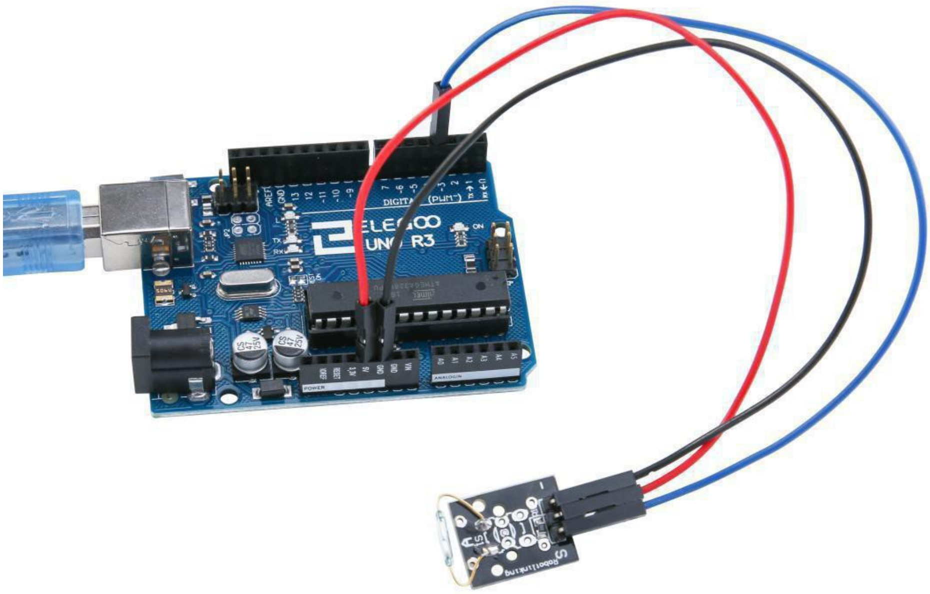


Wiring diagram





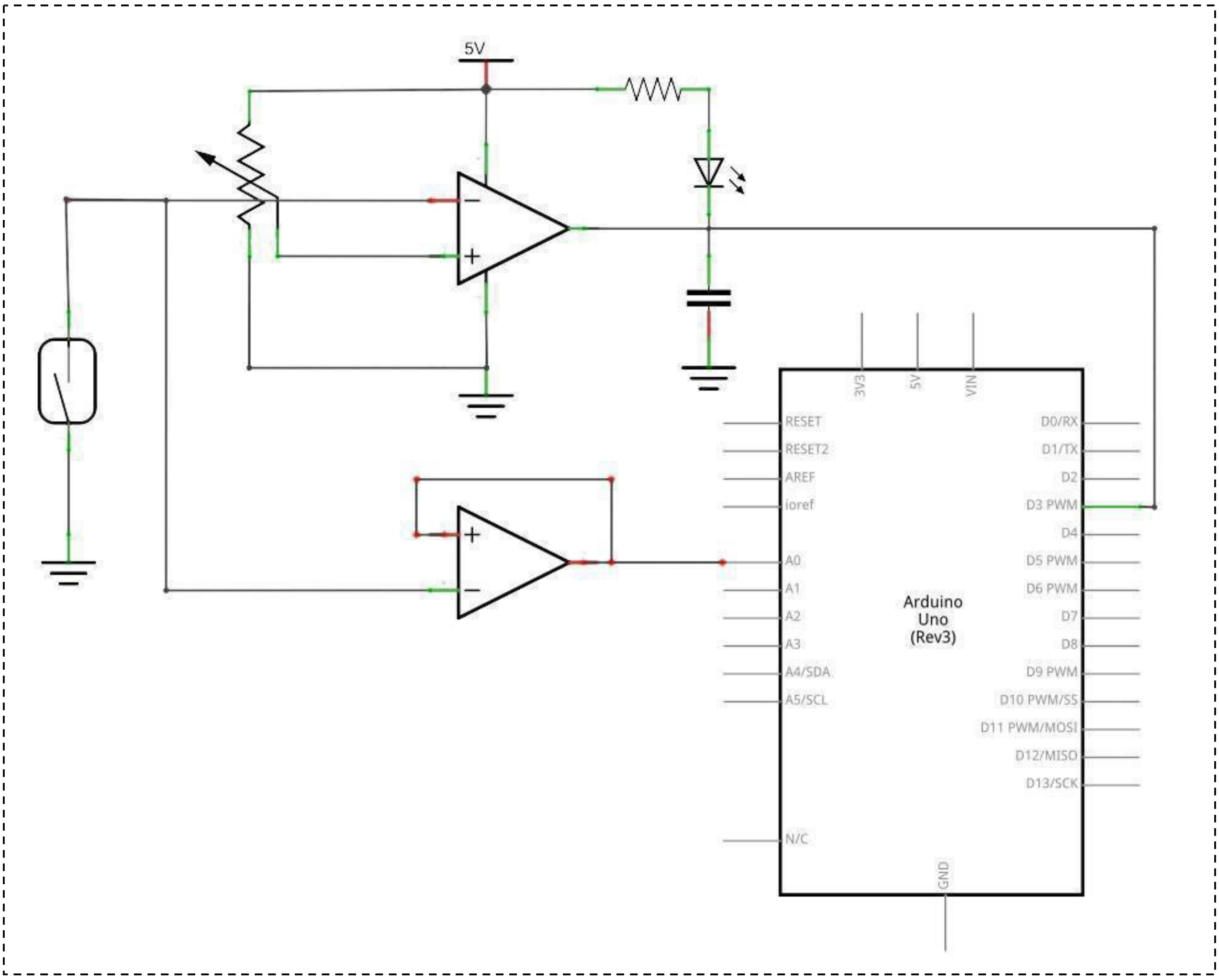
Result



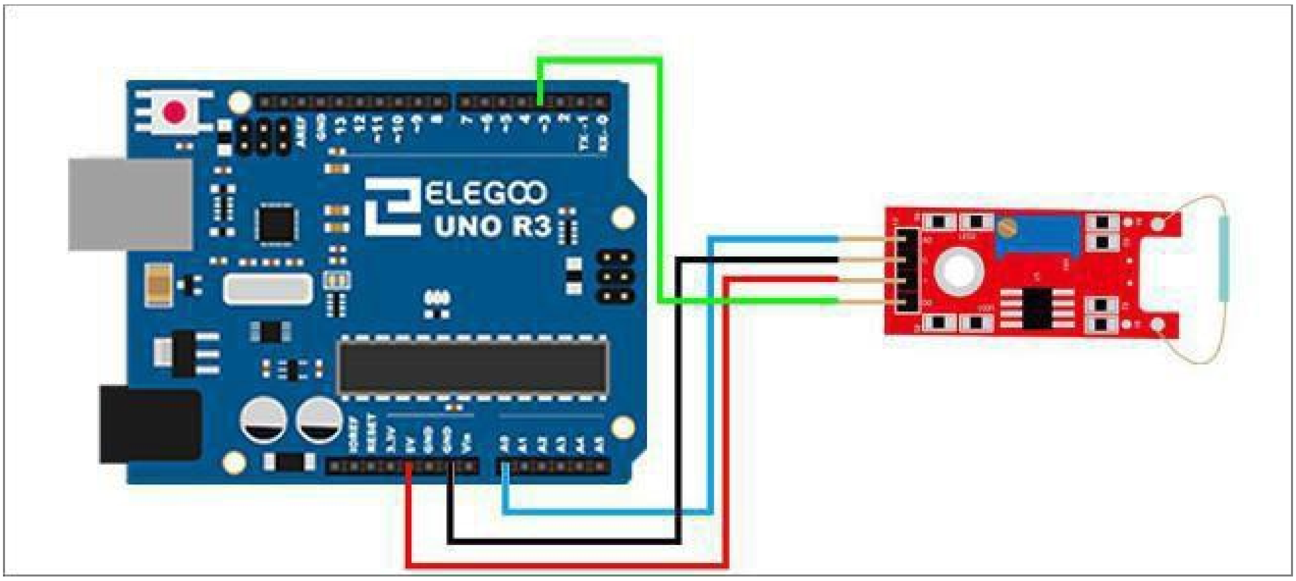
Upload the program, When the module access to magnetic, you can see the led on or off. Mini switch module and number 13 port have the built-in LED simple circuit. To produce a switch flasher, we can use connect the digital port 13 to the built-in LED and connect the switch module S port to number 3 port of Elegoo Uno board. When the switch sensing, LED twinkle light to the switch signal.

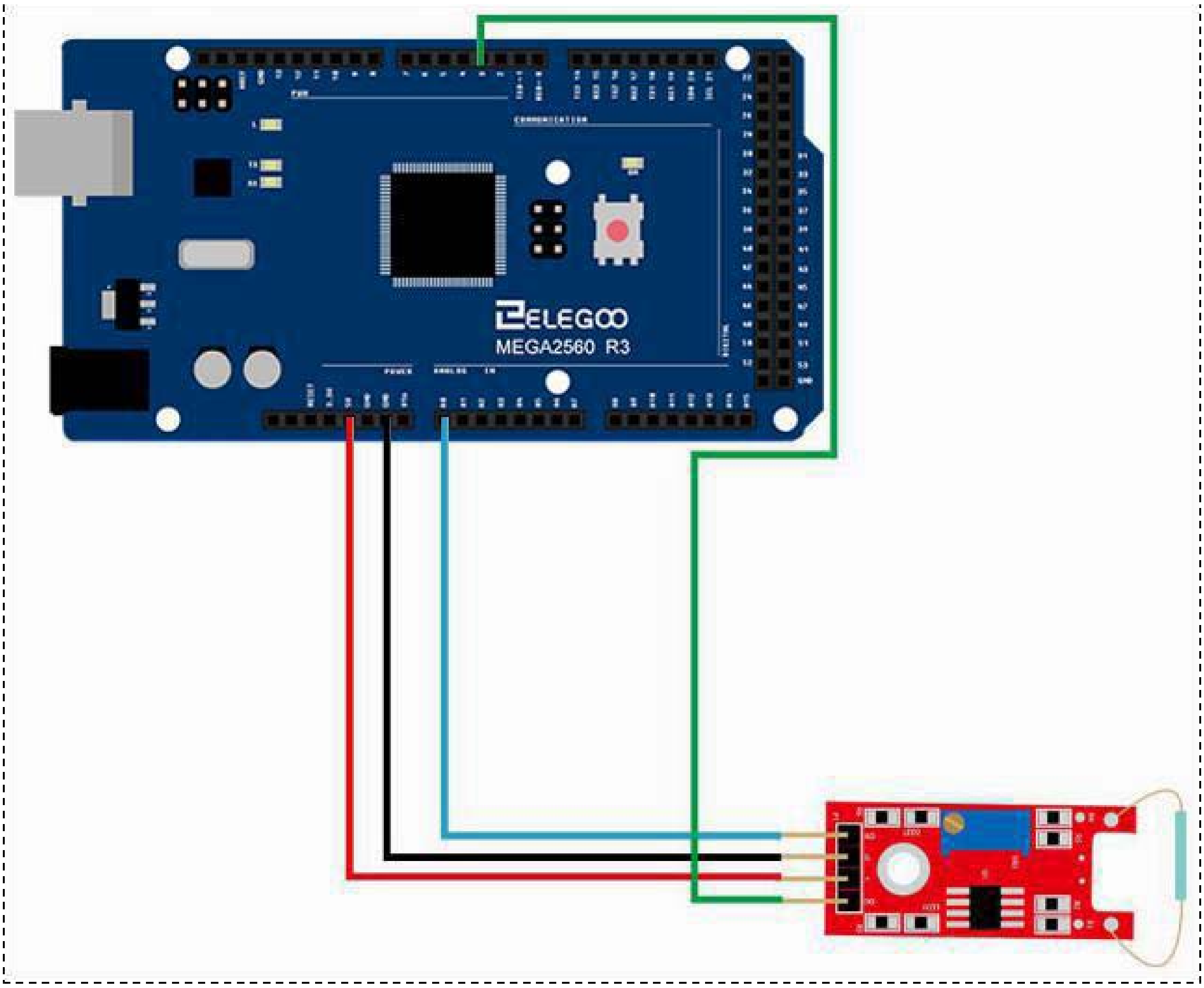
Connection of reed switch module

Schematic

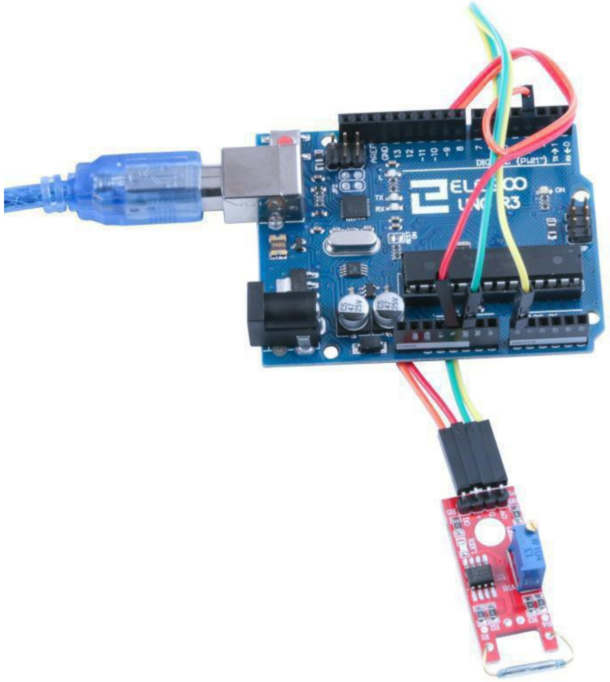


Wiring diagram



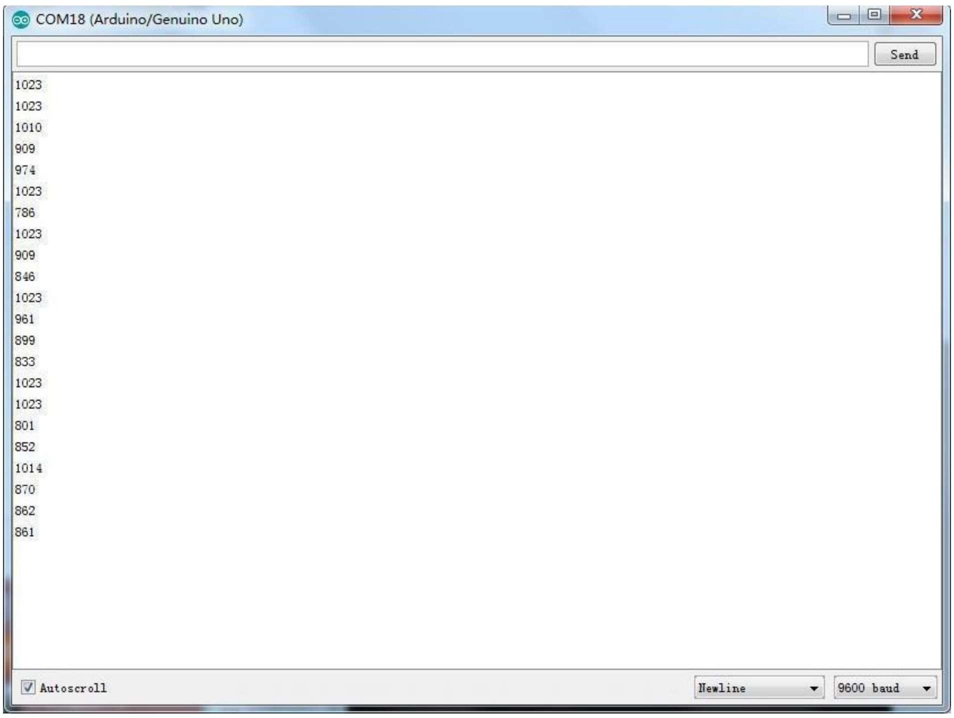


Result



In this experiment, we use the AO pin of reed switch module. When the sensor sensing magnetism, the module will out put a data which reflect the strength of the magnetism. The number is from 0 to 1023.

Uploadthe programthen openthe monitor, we can see the data as below:



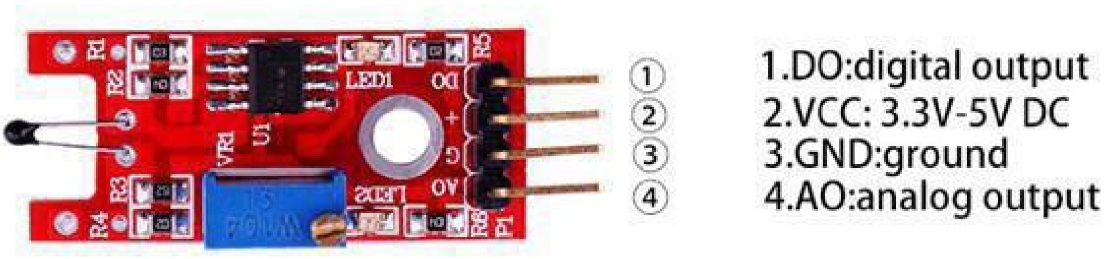
Lesson 17 Digital Temperature Sensor Module

Overview

In this experiment, we will learn how to use digital temperature module and analog temp module.

Digital Temp

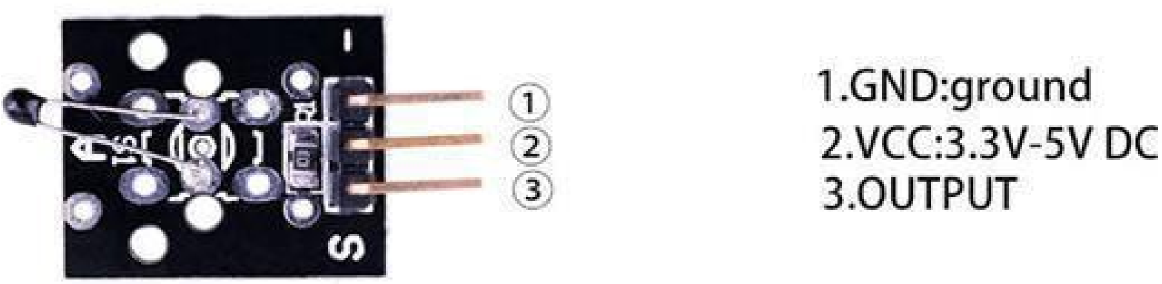
Temperature sensing module using an NTC thermistor. The output signal at 'DO' switches high when the preset (adjustable) temperature is reached. An analog output signal from the sensor is available at pin 'AO'.



Analog Temp

NTC Temperature sensor module. The sensor resistance is approximately 10 kohm at room temperature. The NTC sensor is connected between the two outer pins .A fixed 10 K ohm resistor connected between the middle pin and the 'S' pin is included on the module. This simplifies the building of a measurement bridge circuit.

- Temperature range: -55°C to +125°C
- Accuracy: +/- 0.5°C



Component Required:

(1) x Elegoo Uno R3

(1)x usb cable

(1)x digital temperature module

(1) x temperature module

(x) x F-M wires

Component Introduction

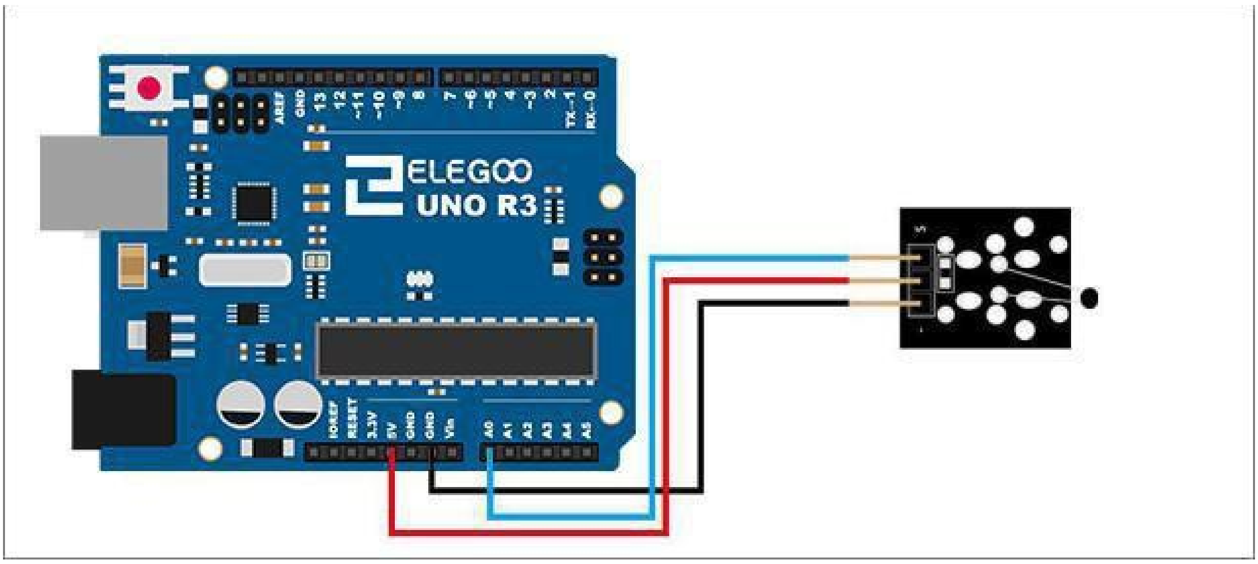
Thermistor:

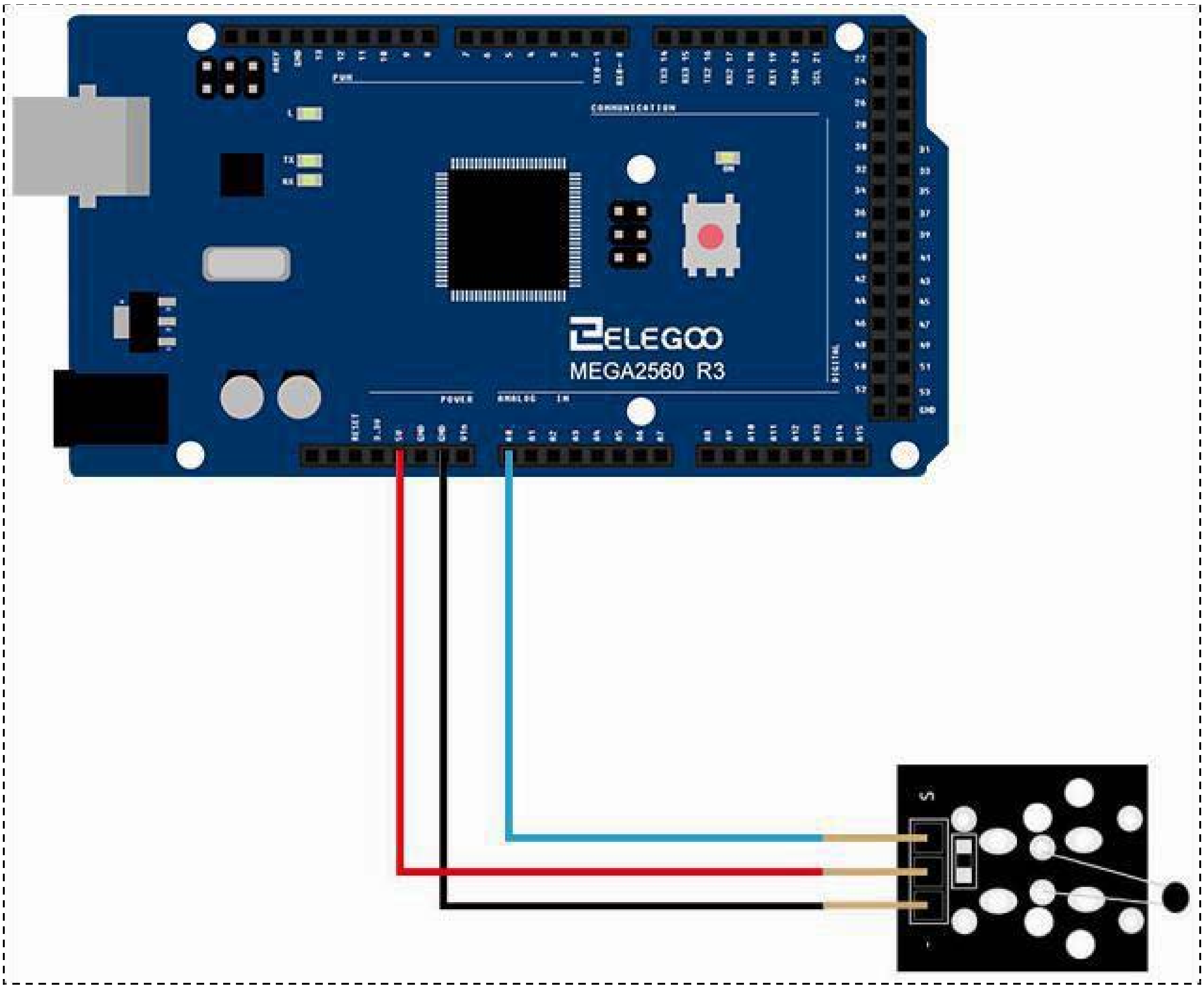
These thermistors have a narrow tolerance on the B-value, the result of which provides a very small tolerance on the nominal resistance value over a wide temperature range. For this reason the usual graphs of R = f(T) are replaced by Resistance Values at Intermediate Temperatures Tables, together with a formula to calculate the characteristics with a high precision.

RS stock no.	151-215	151-221	151-237	151-243
Resistance at +25°C	3kΩ	5kΩ	10kΩ	100kΩ
Temperature range	-55°C to +150°C			
Tolerance (0 to +70°C)	±0.2°C			
Dissipation constant	1mW			
Time constant	10s			

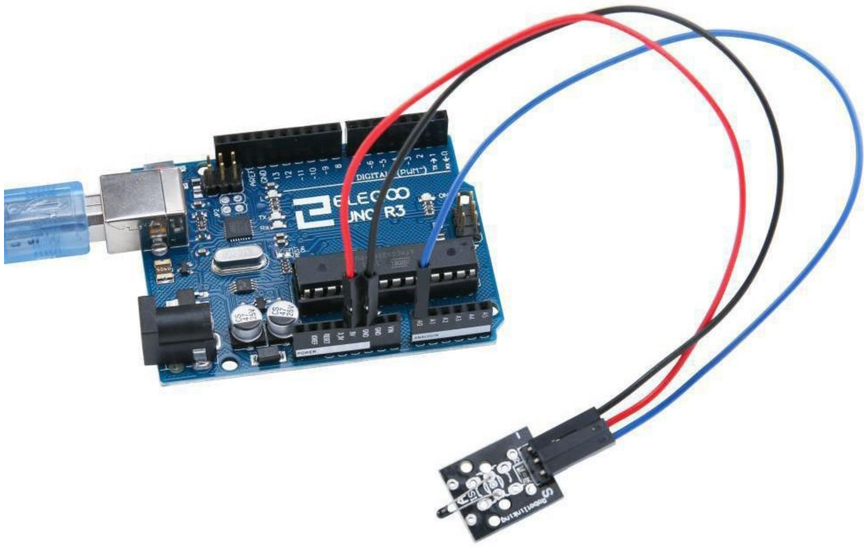
Connection of digital temperature module

Wiring diagram

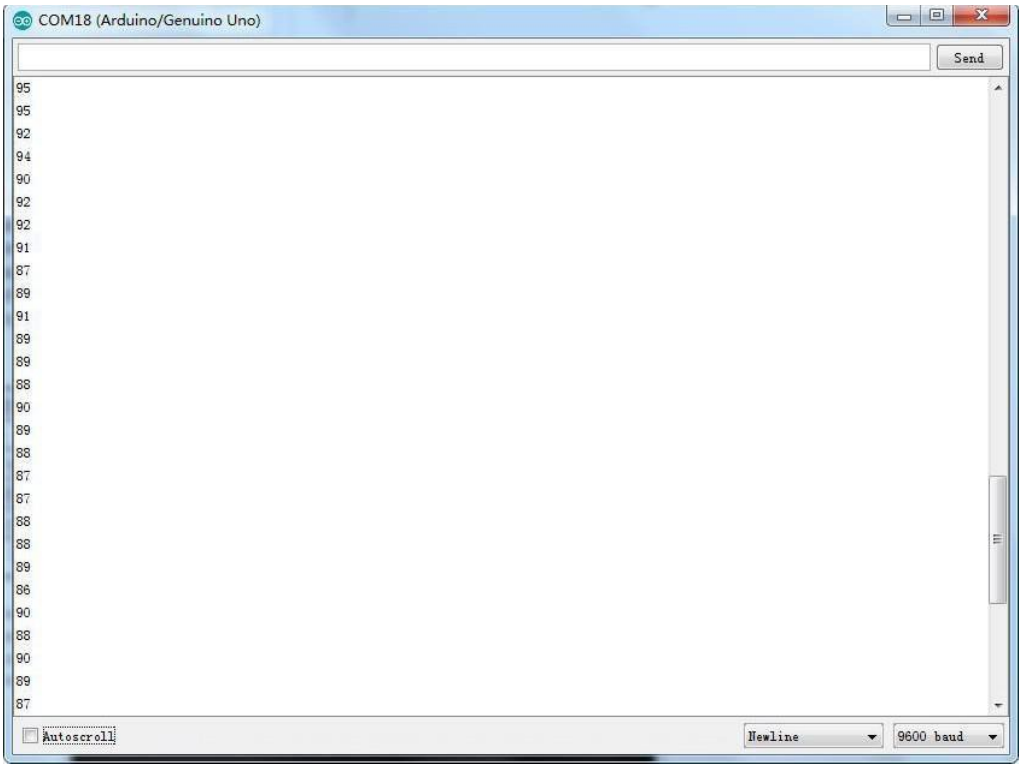




Result

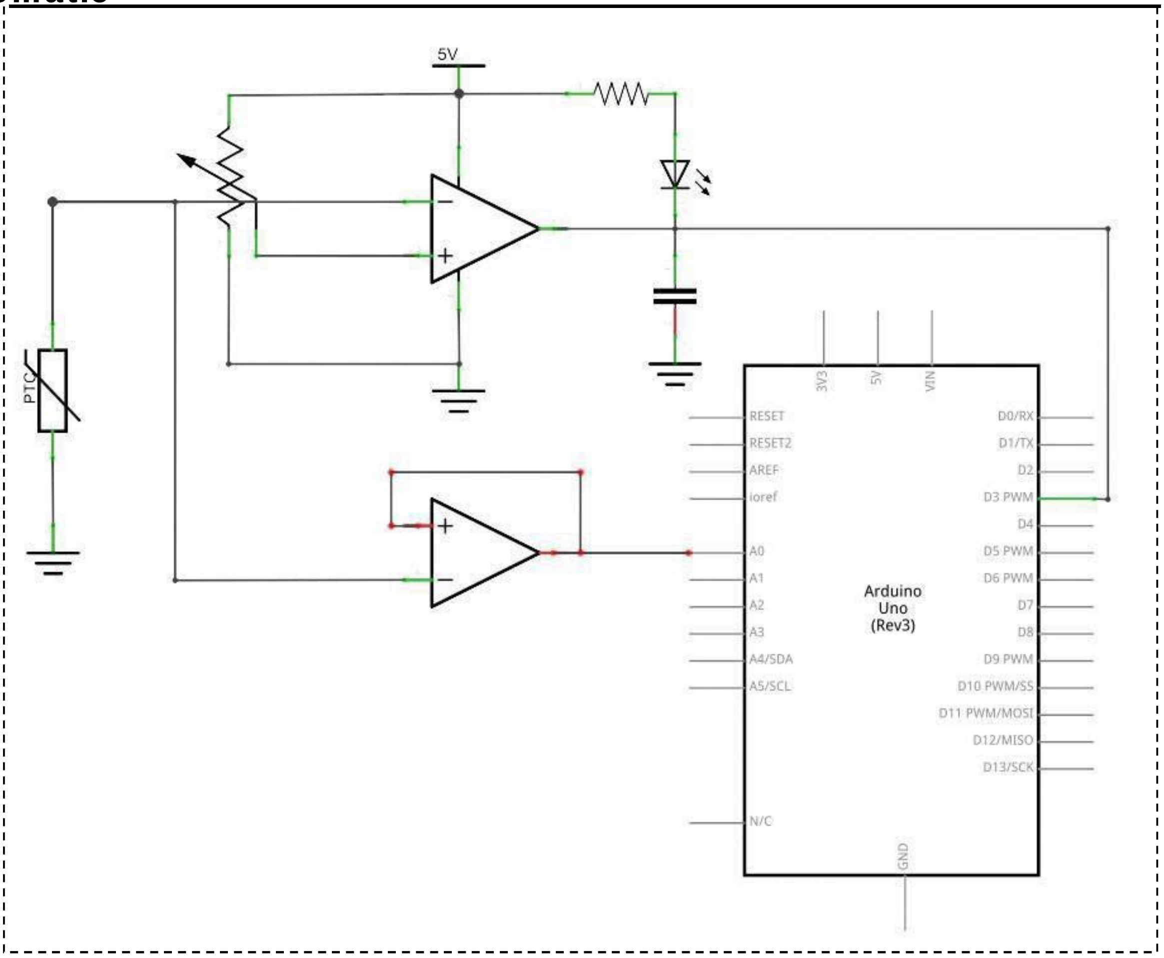


Uploadthe program, open the monitor then you can seethe data as below:

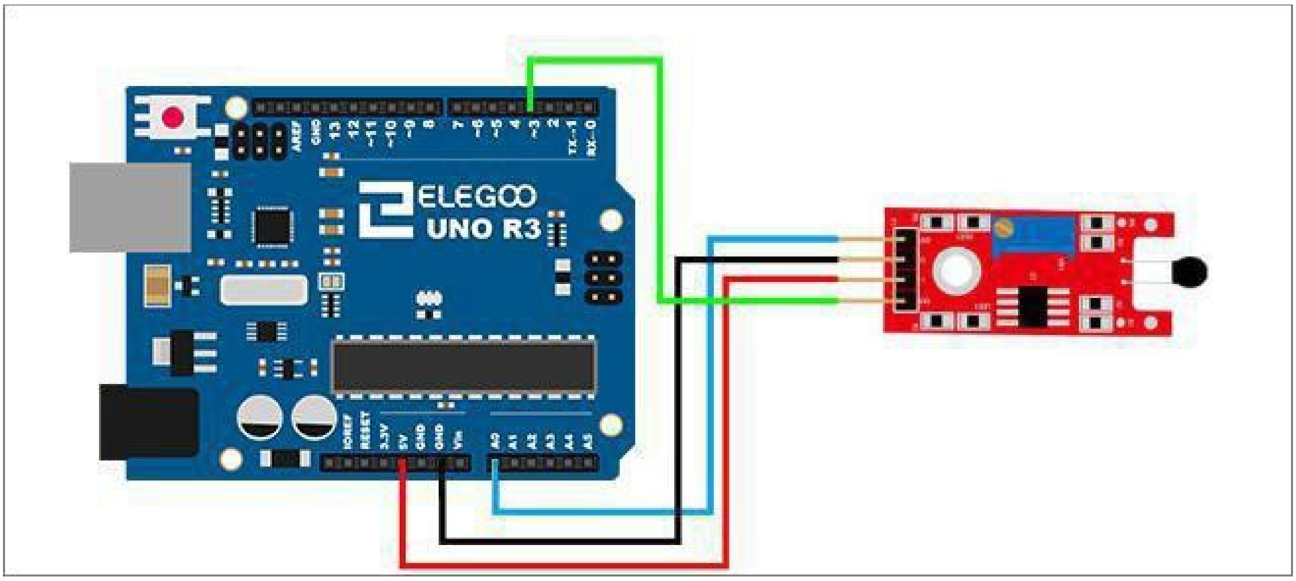


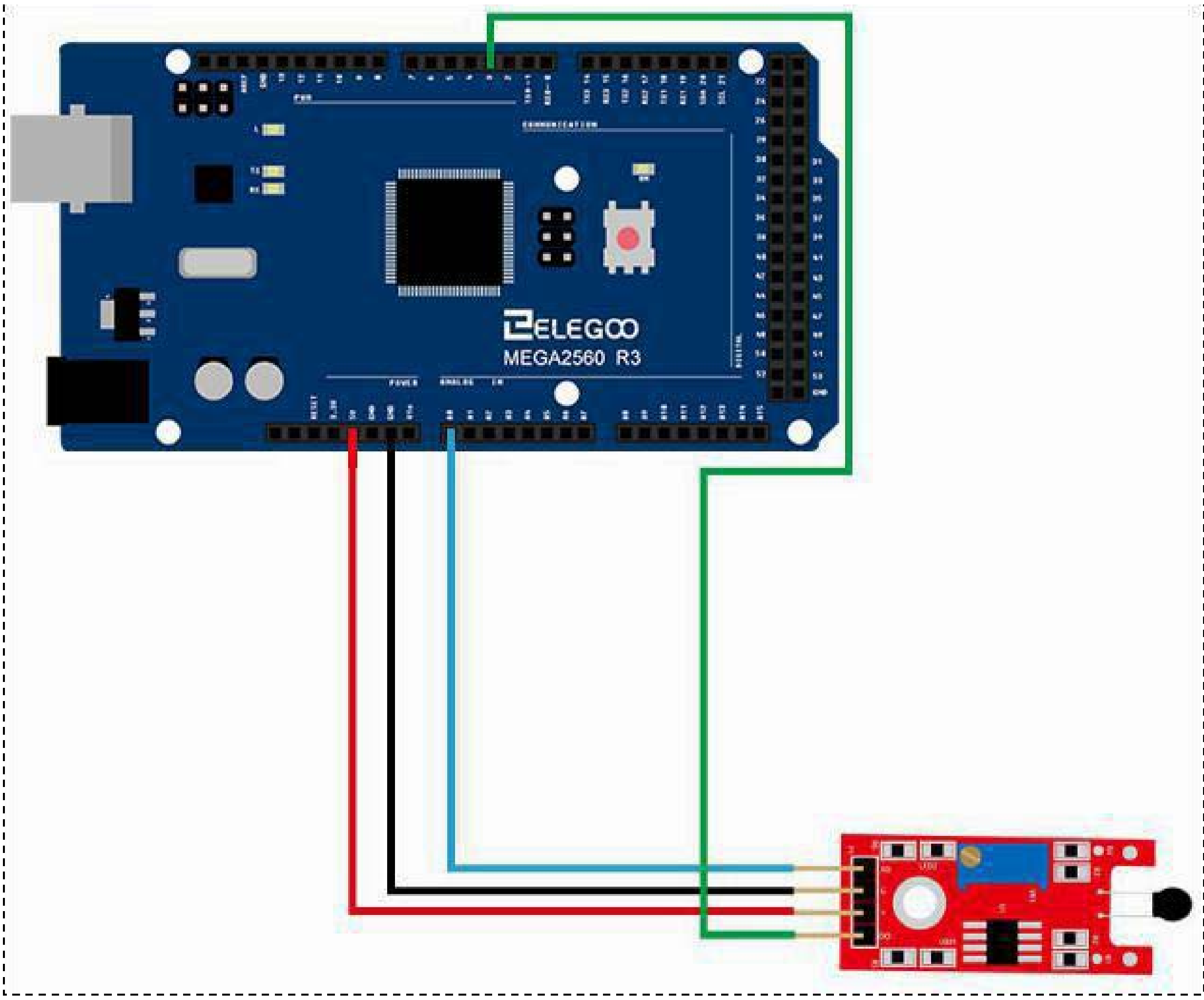
Connection of temperature module

Schematic

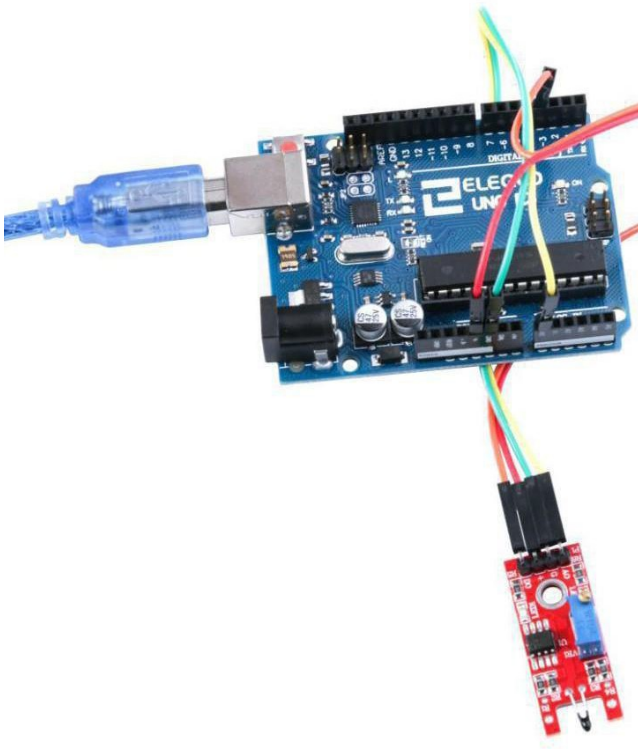


Wiring diagram

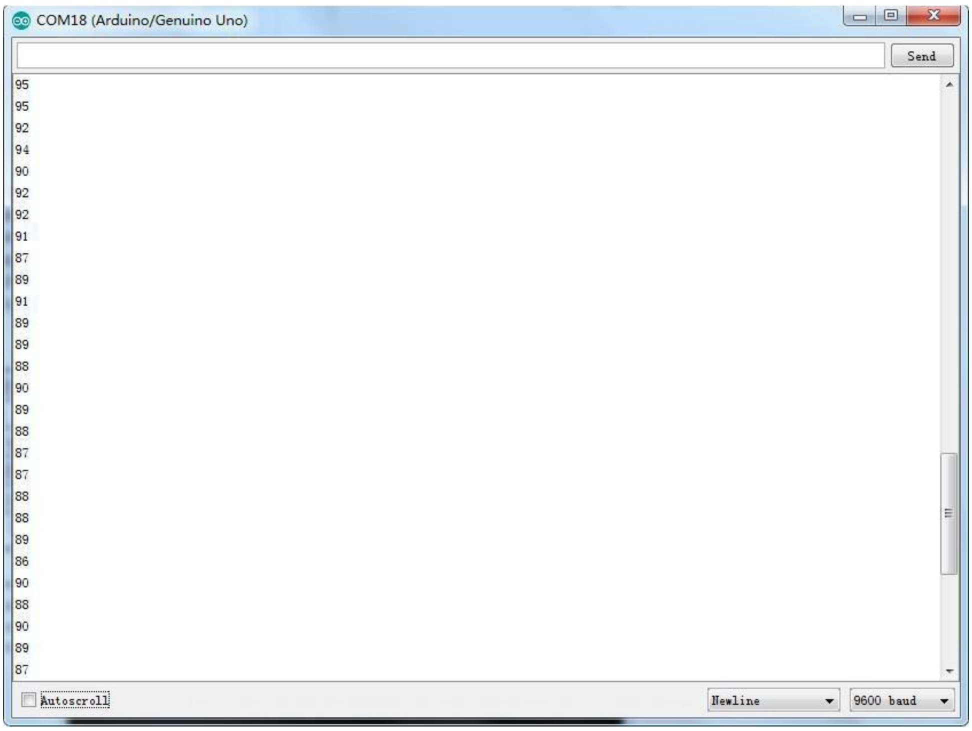




Result



Upload the program then open the monitor, we can see the data as below:



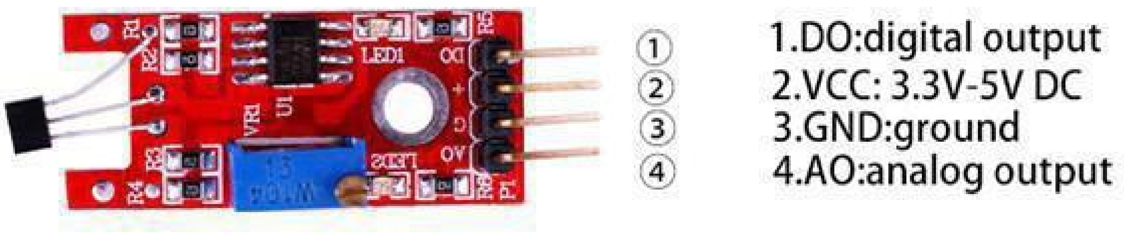
Lesson 18 Linear Hall and analog Hall Module

Overview

In this experiment, we will learn how to use the linear hall and analog hall module.

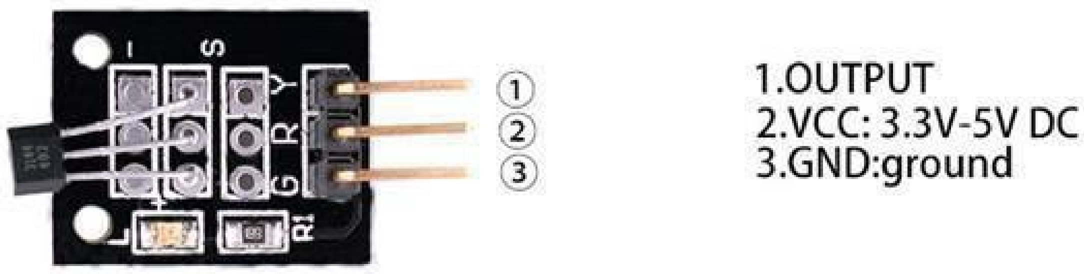
Linear Magnetic Hall Sensor

Linear Hall Sensor module to detect the presence of a magnetic field near the sensor. Variables such as field strength, polarity and position of the magnet relative to the sensor will affect point at which the 'DO' output switches to a high level (i.e. active high). The circuit sensitivity canbe adjusted with a pot. An analog output signal from the sensor is availableat pin 'AO'.



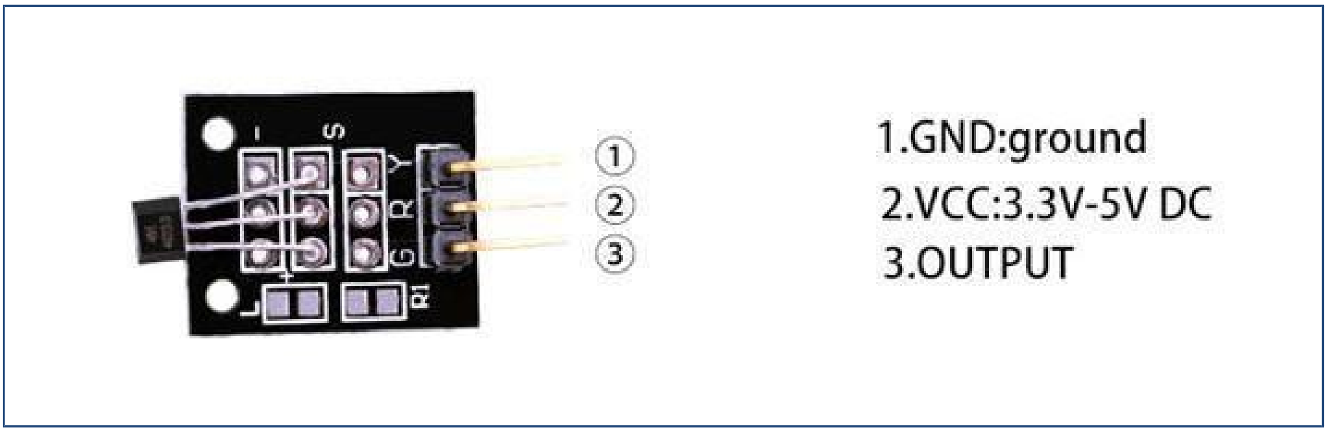
Hall Effect Sensor

A Hall sensor module with analog output signal. The ground pin ismarked'-', center pin is +5 V supply (Vs) and the output signal is on the'S'pin.



Class Hall Magnetic Sensor

The Hall-Sensor-Switch (bipolar) module features a 44E311, 3144EUA-Sor 3144LUA-S sensor together with an LED and resistor. The LED switches on when a magnetic field is detected. The ground pin is marked '-', center pin is +5 V supply (Vs) and the output signal is on the 'S'pin.



Component Required:

(1)x Elegoo Uno R3

(1)x USB cable

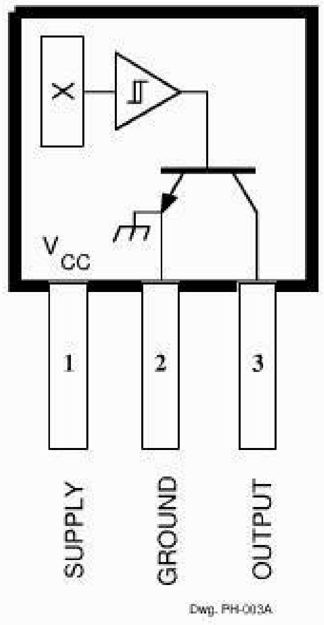
(1)x Linear Hall module

(1)x Analog Hall module

(1)x F-M wires

Component Introduction

Hall Sensor:

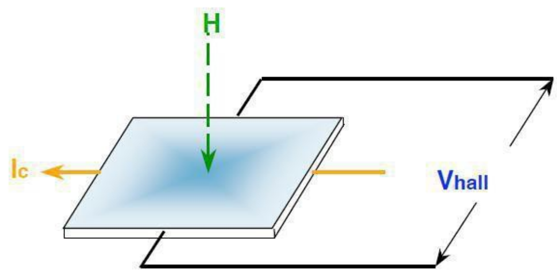


Pinning is shown viewed from branded side.

ABSOLUTE MAXIMUM RATINGS at T_A = +25°C

Supply Voltage, V_{CC}	28 V
Reverse Battery Voltage, V_{RCC}	-35 V
Magnetic Flux Density, B	Unlimited
Output OFF Voltage, V_{OUT}	28 V
Reverse Output Voltage, V_{OUT}	-0.5 V
Continuous Output Current, I_{OUT}	25 mA
Operating Temperature Range, T_A	
Suffix 'E–'	-40°C to +85°C
Suffix 'L–'	-40°C to +150°C
Storage Temperature Range, T_S	-65°C to +170°C

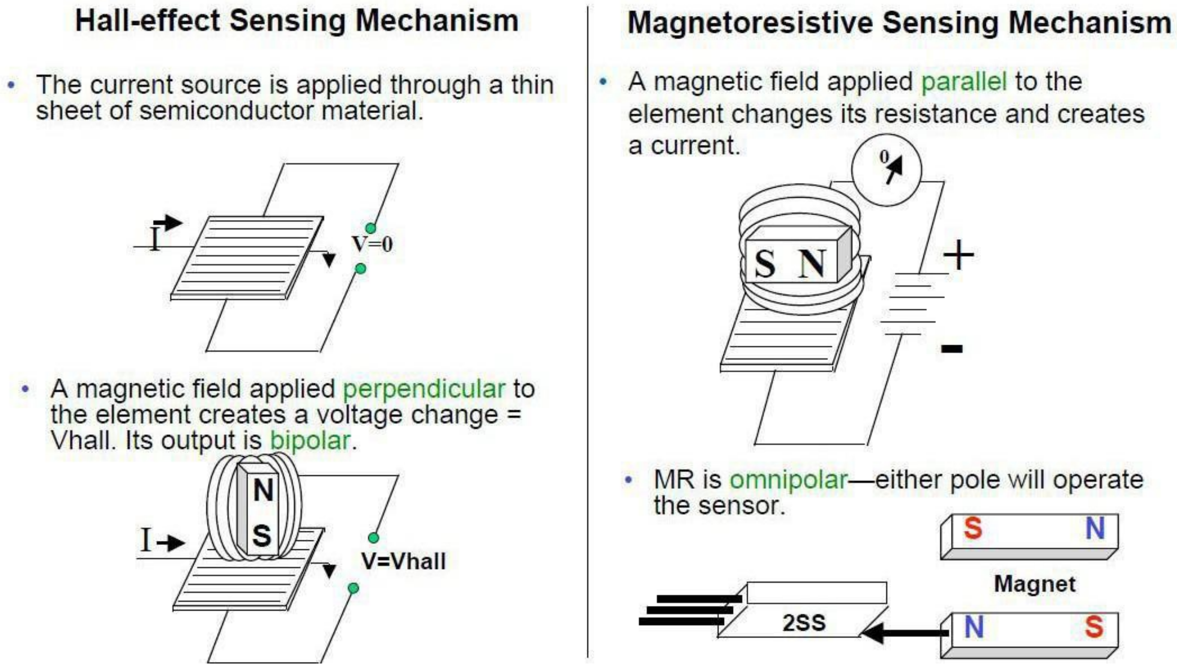
$\boxed{V_{hall} \propto I_c \times H}$



Vhall = Output Hall-effect voltage

H = Magnetic Flux created by magnet or current-carrying conductor

Ic = Constant supply current



Design Factors–Magnetic TypesUnipolar: Only as out pole will operate the

sensor. The sensor turns on with thes out pole (+) and off when the

South Pole is removed.

• Bipolar: Sensor out put is pole-dependent. As out pole(+) is designed to

Activate the sensor; a north pole (-) is designed to deactivate. It's possible that the sensor could turn

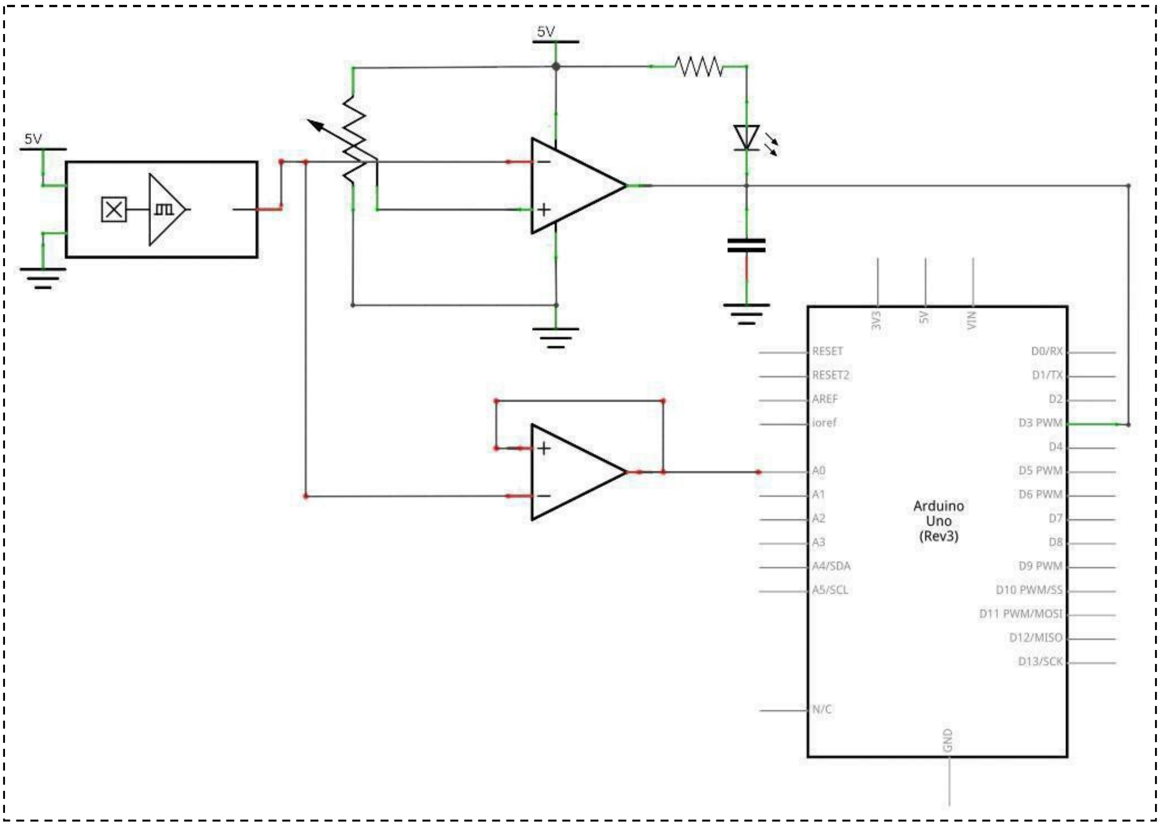
off and still be with in a positive Gauss level.

• Latching: Specifications are tighter on latching. Some times it is designed to make certain that when the south pole (+) is removed from the sensor, it will stay on until it sees the opposite pole (-).
• Omni polar: The sensor is designed to operate with either magnetic pole (+or-).
• Ratio metric linear: Out put is proportional to magnetic field strength.

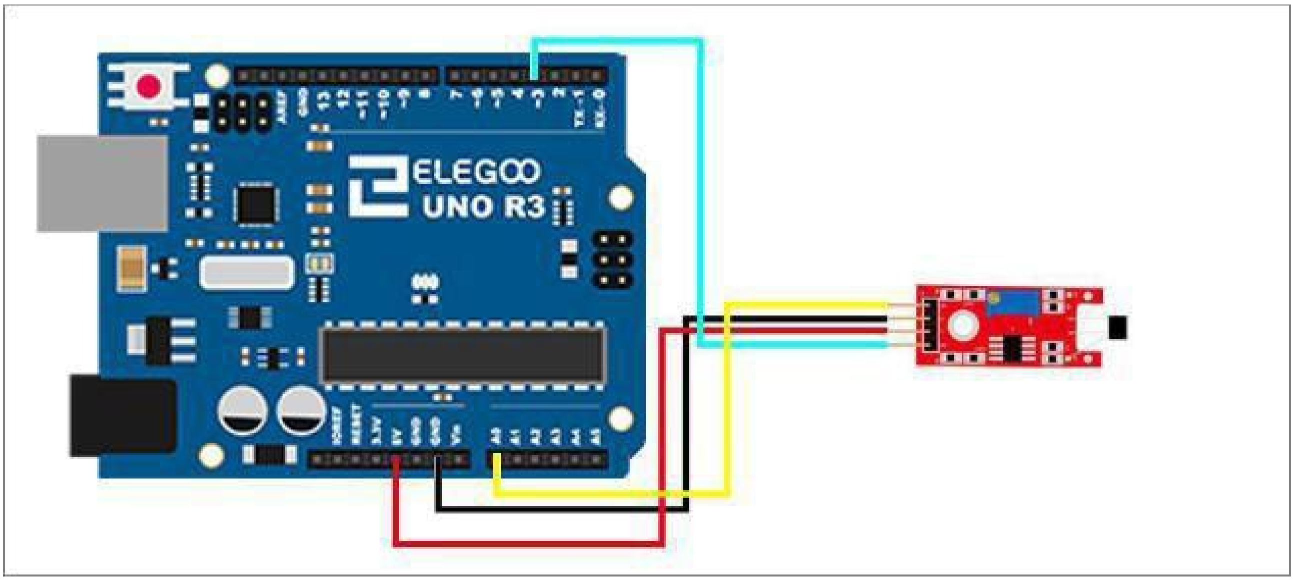
Output sensitivity range is 2.5 – 3.75 mV per unit of Gauss.

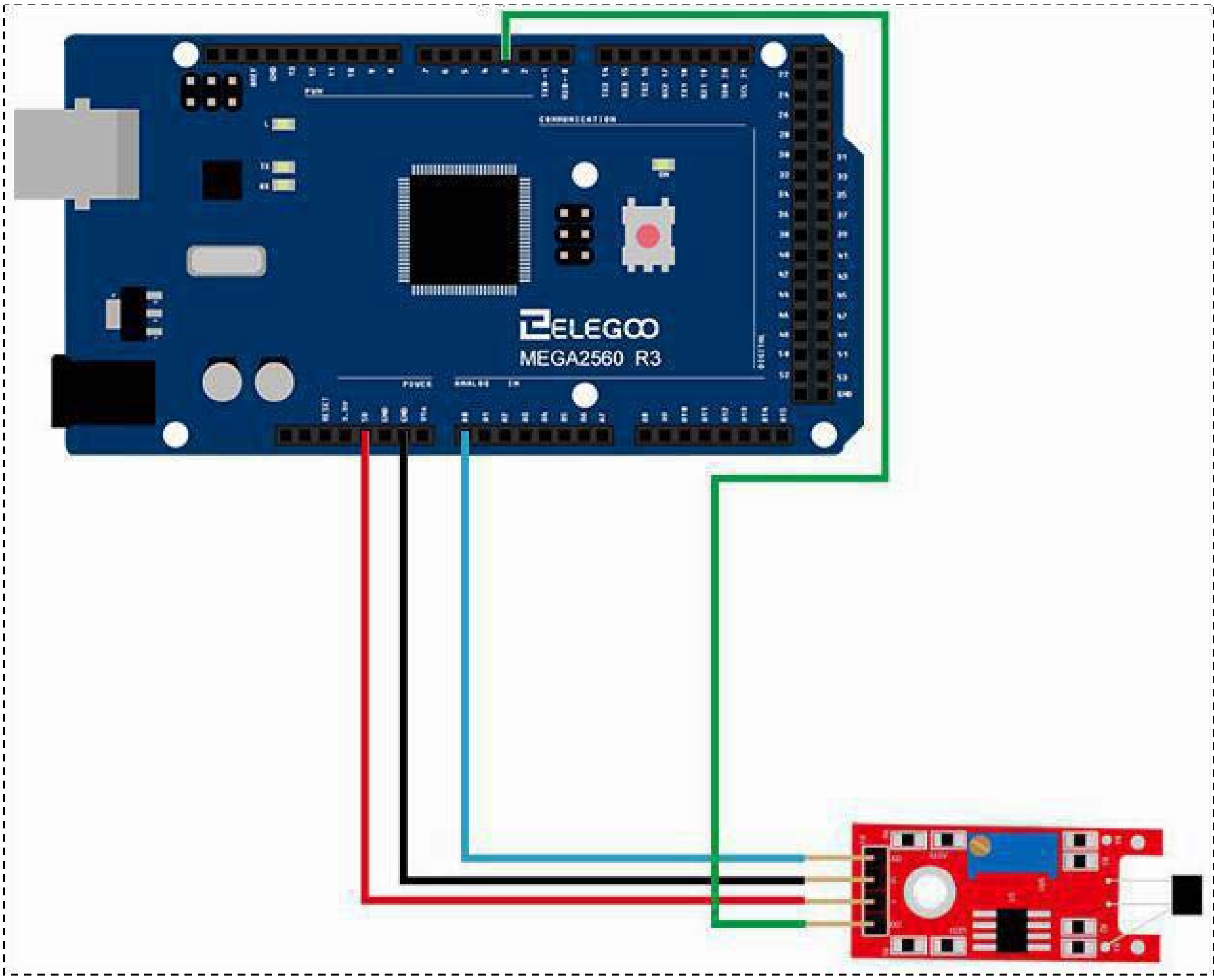
Connection

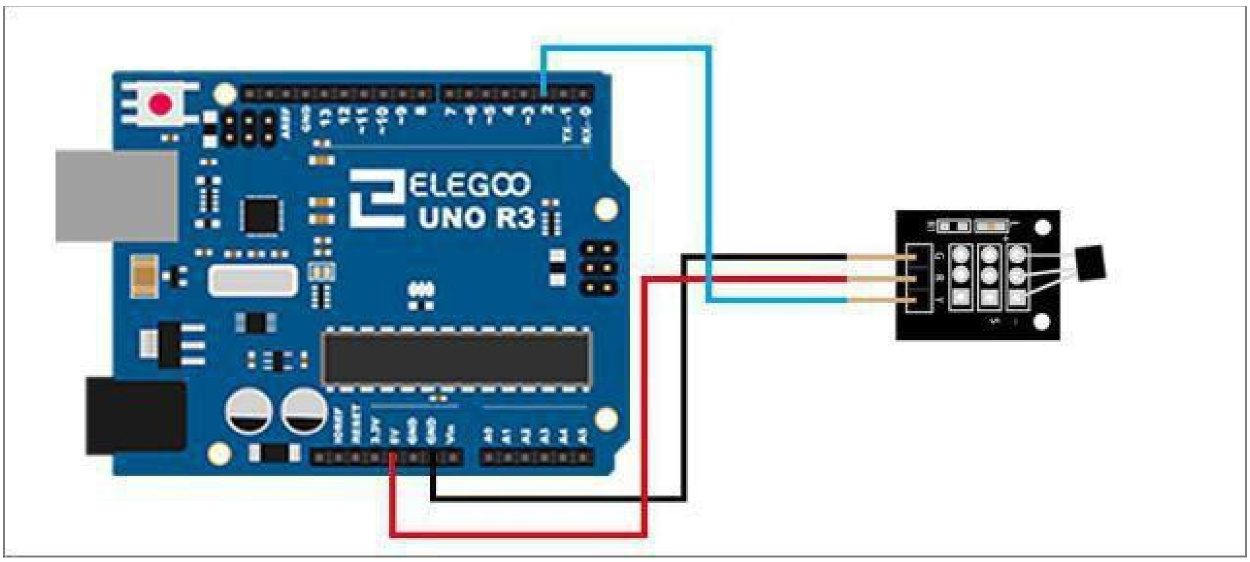
Schematic

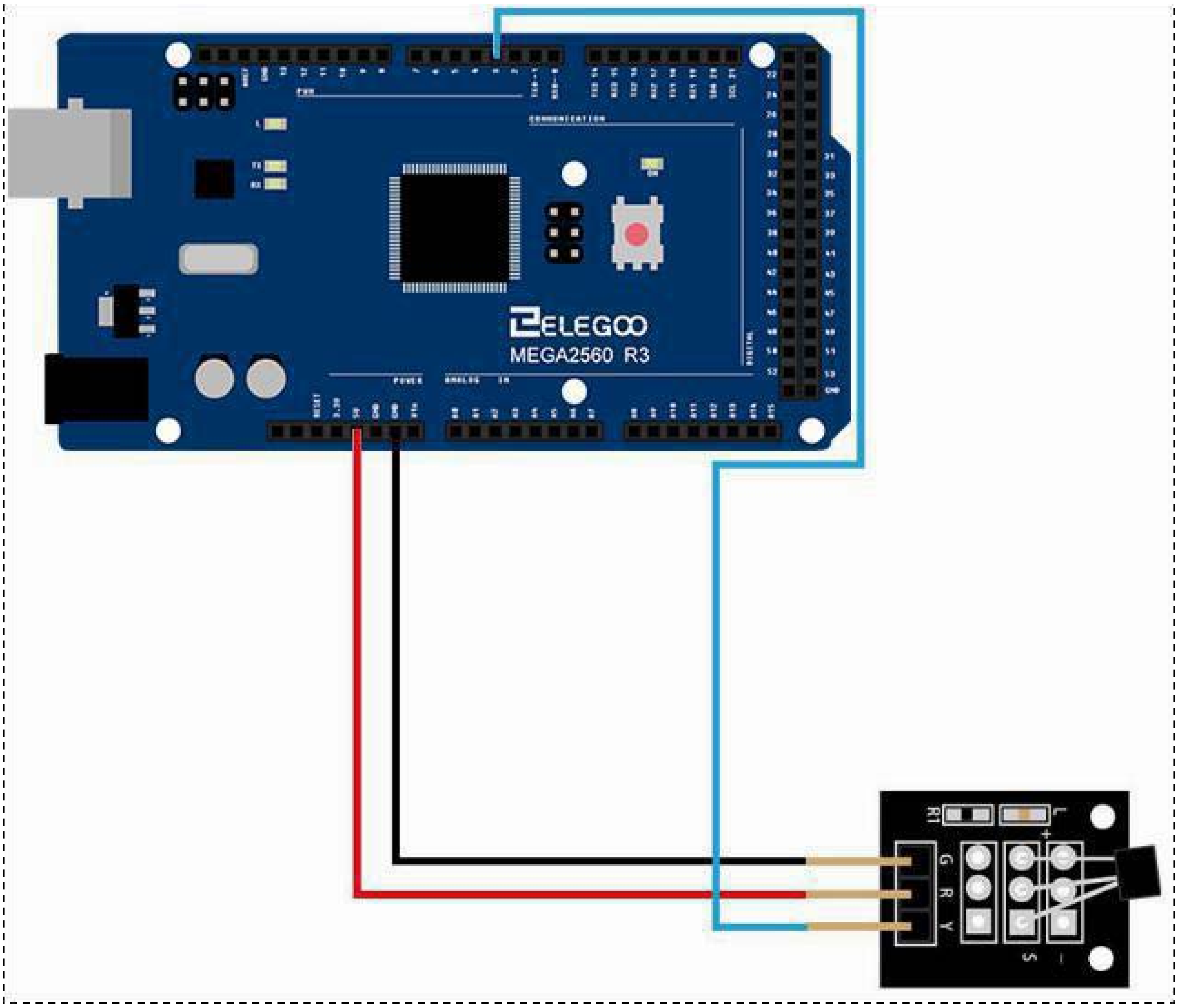


Wiring diagram



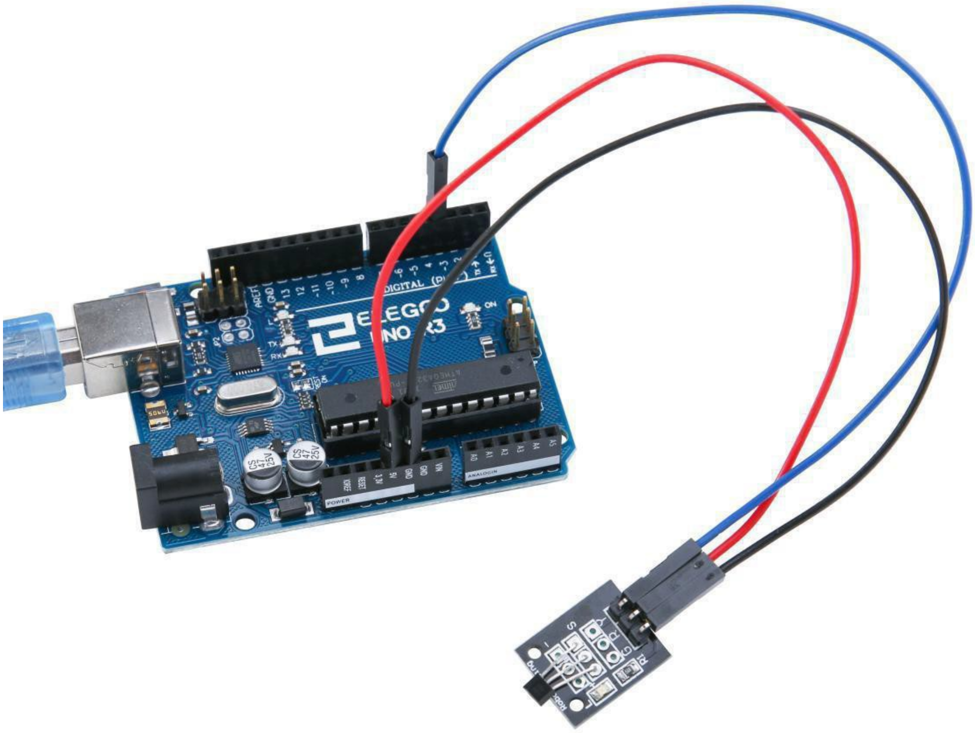

ELEGOO
MEGA2560 R3

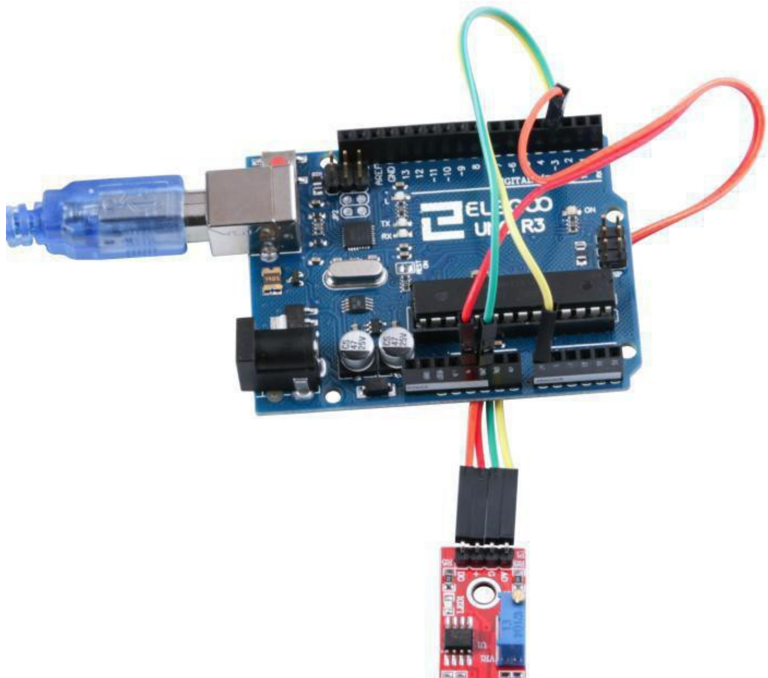

ELEGOO
UNO R3



Result

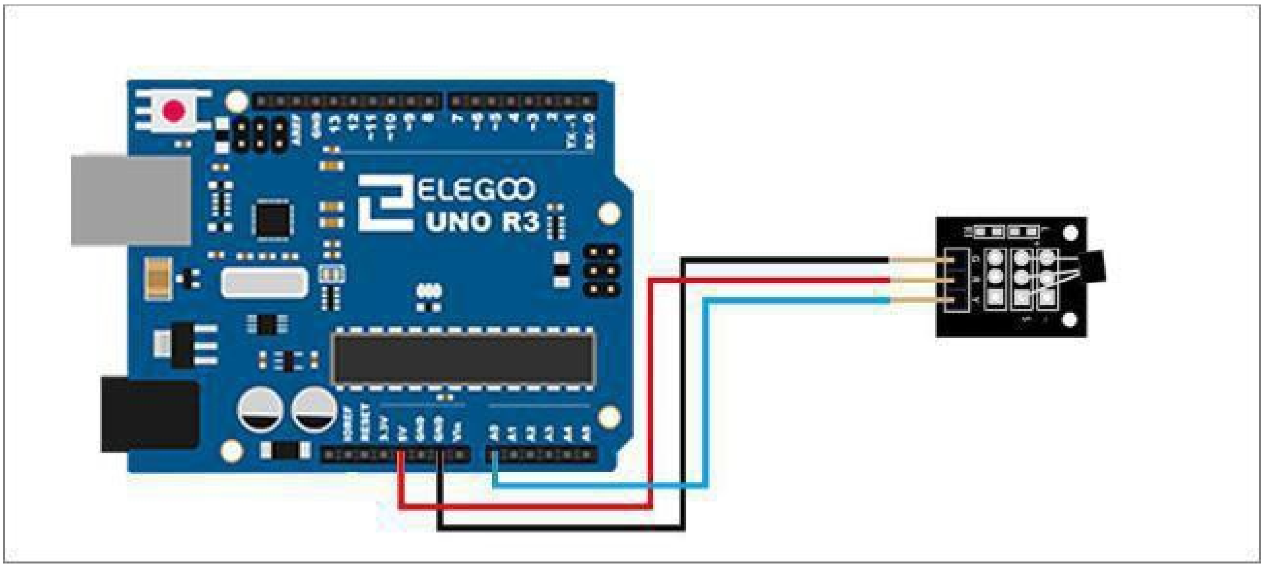
For the hall sensor module, we can choose the output: digital output or analog output. In the following picture, we use the DO port to output. So we can see that if the hall sensor sensing the magnetic force, the light will turn on.

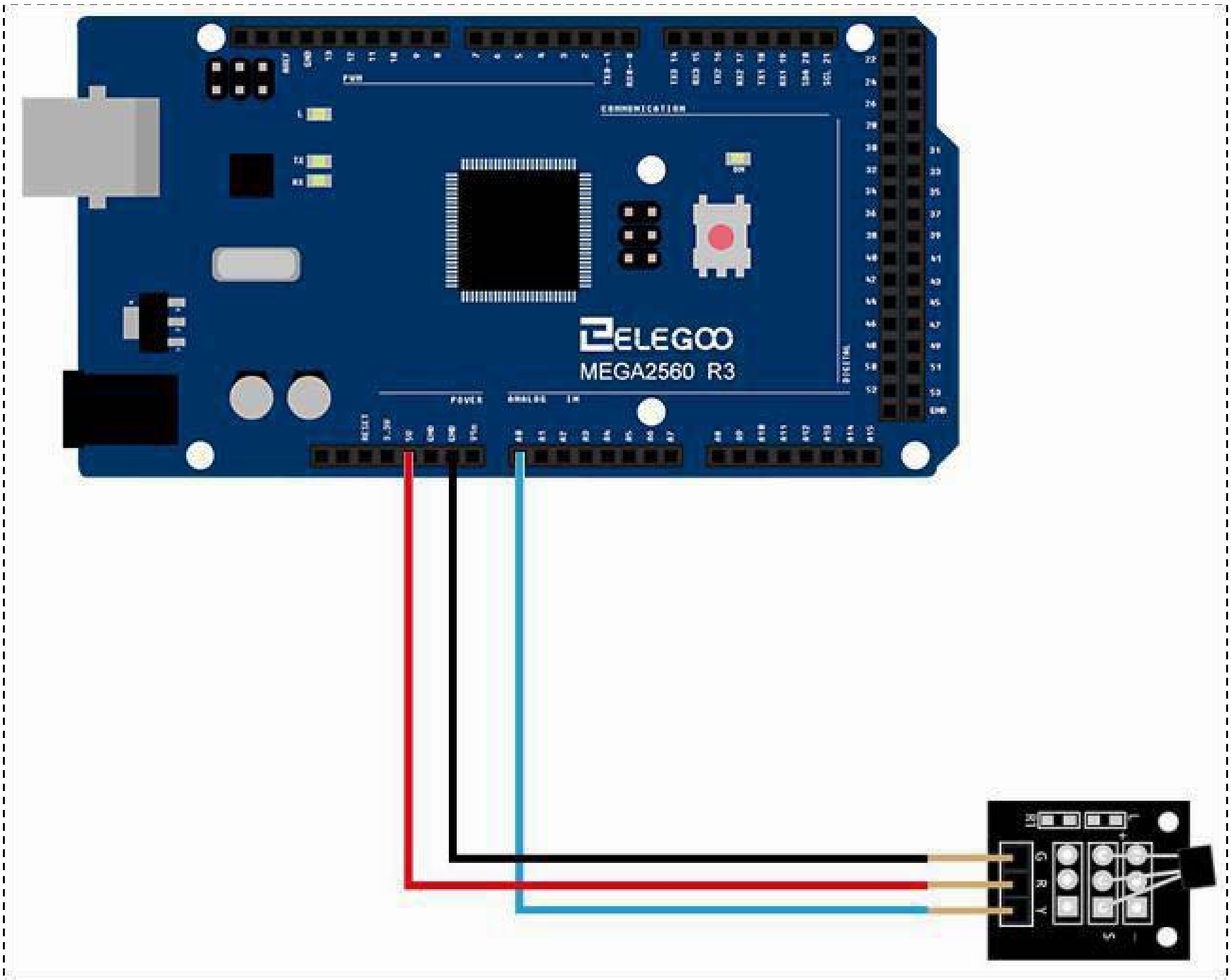




Connection

Wiring diagram

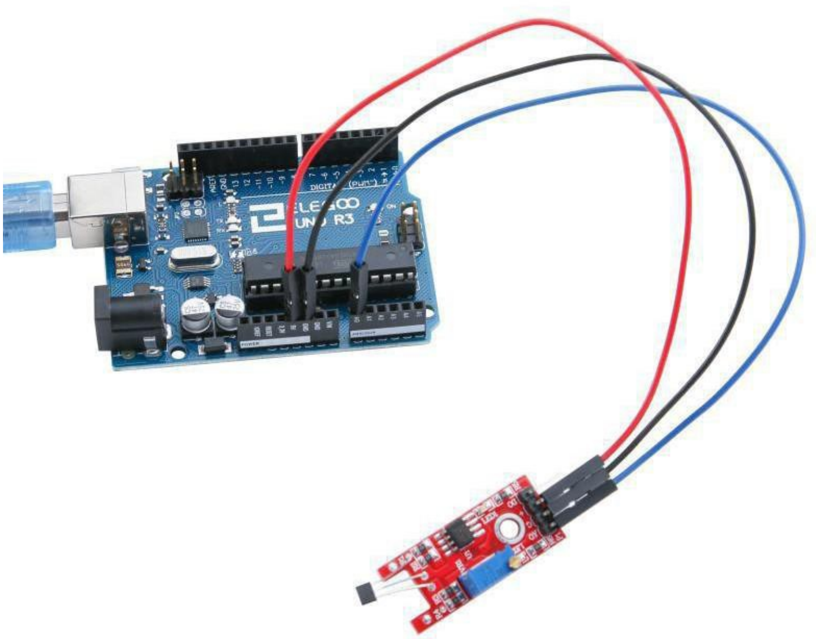




Result

In this example, If no magnetic field is present, the signal line of the sensor is HIGH (3.5 V). If a magnetic field is presented to the sensor, the signal line goes LOW, at the same time the LED on the sensor lights up.

The connection as below:



Lesson 19 Flame Sensor Module

Overview

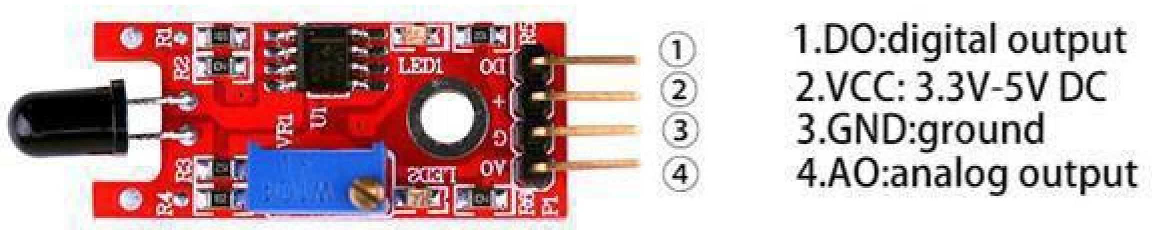
In this experiment, we will learn how to use the flame sensor module.

This module is sensitive to the flame and radiation. It can detect infrared light source in the range of a wavelength 760 nm-1100 nm. The detection distance is up to 100 cm. The Flame sensor can output digital or analog signal. It can be used as a flame alarm or in fire fighting robots.

Flame

A sensor module to detect flames. The spectral sensitivity of the sensor is optimized to detect emissions from naked flames. The output signal 'DO' is pulled high (active high) when a flame is detected. The switching threshold is adjustable via a preset pot. An analog output signal from the sensor is available at pin 'AO'.

- Typical spectral sensitivity: 720-1100 nm
- Typical detection angle: 60°



Component Required:

(1)x Elegoo Uno R3

(1)x USB cable

(1)x Flame sensor module

(x) x F-M wires

Component Introduction

Flame sensor:

•Detectsaflameoralightsourceofawavelengthintherangeof760nm-1100nm

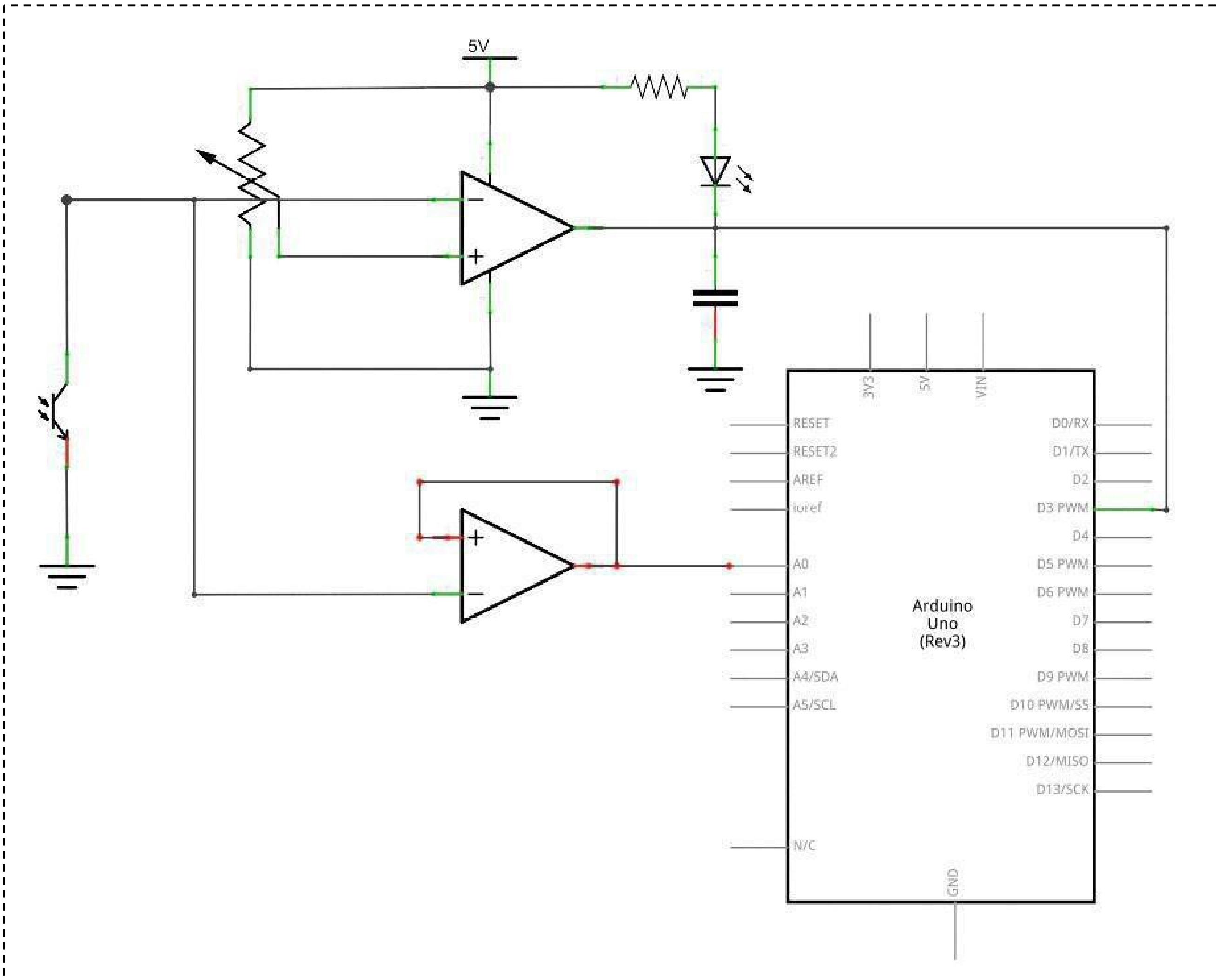
- Detection distance: 20 cm (4.8V) ~ 100 cm (1V)
- Detectionangle about 60 degrees, it is sensitive to the flame spectrum.
- Comparator chip LM393 makes module readings stable.
- Adjustable detection range.
- Operating voltage 3.3V-5V
- Digital and Analog Output

analog voltage output

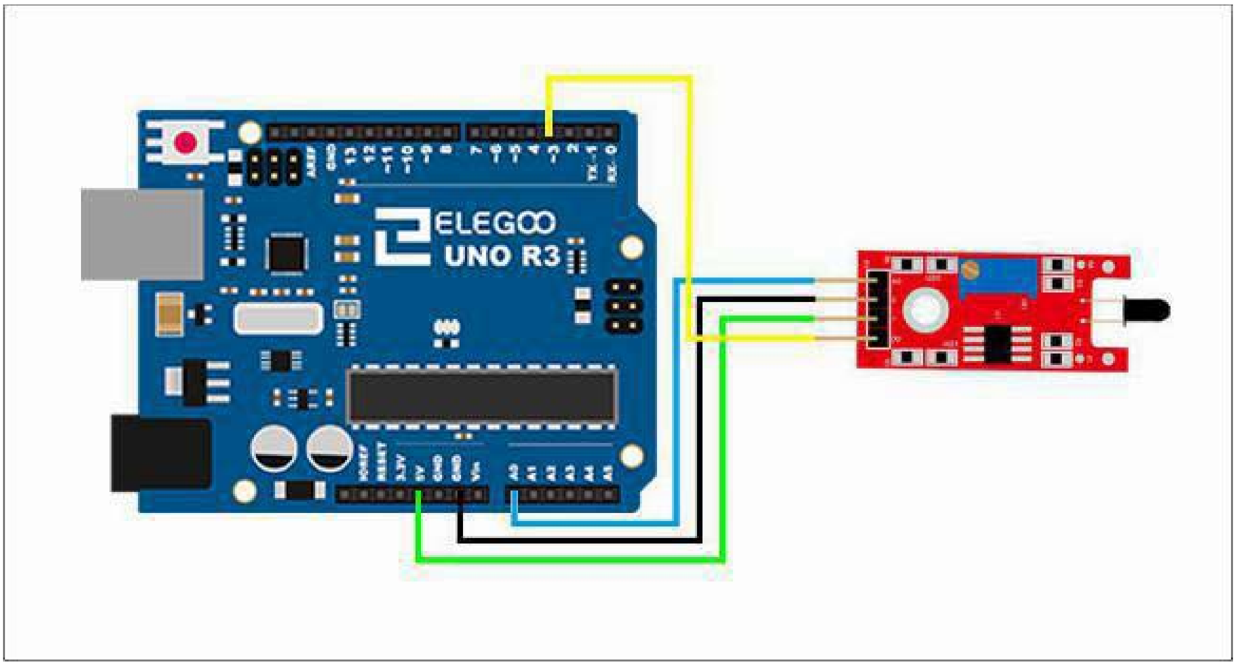
- Power indicator and digital switch output indicator

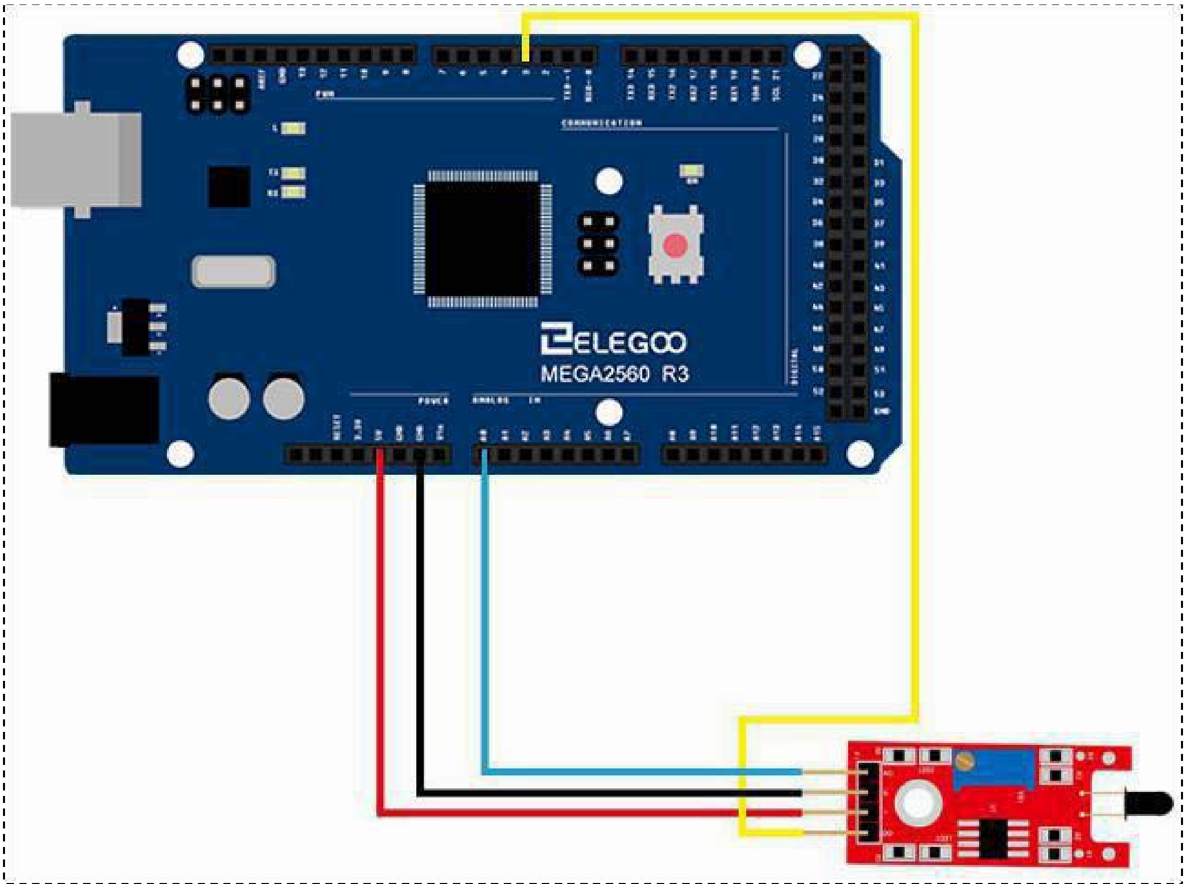
Connection

Schematic



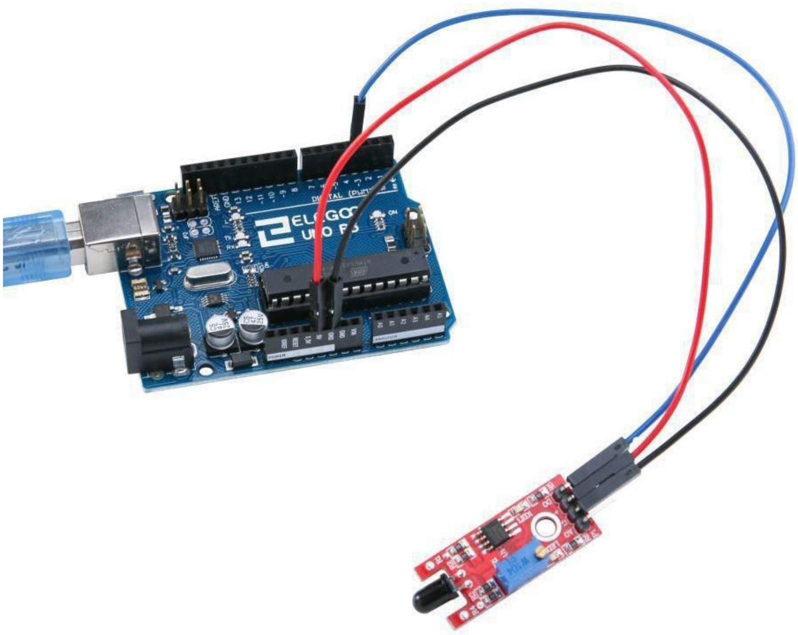
Wiring diagram



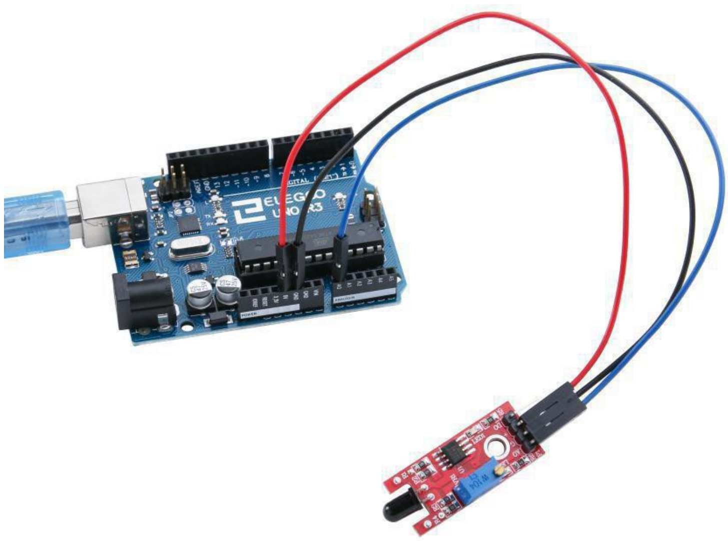


Result

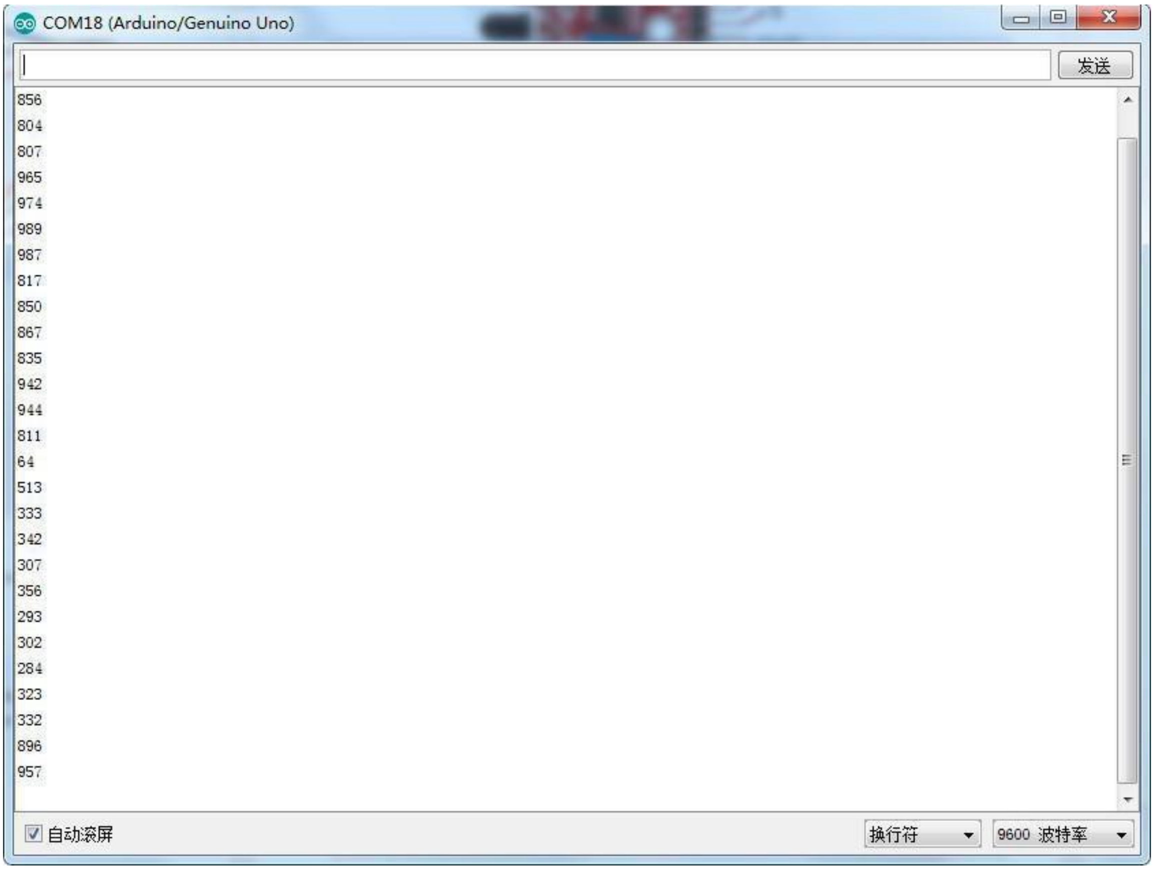
For the flame sensor module, we can choose the output: digital output or analog output. In the following picture, we use the DO port to output. so we can see that if the flame sensor sensing the flame, the light will turn on.



In the following picture, we use the AO port to output. so we can see that if the flame sensor sensing the flame,themodulewilloutputadatawhichreflectthestrengthoftheflame.The number is from 0 to 1023.



Uploadthe programthen openthe monitor, we can see the data as below:



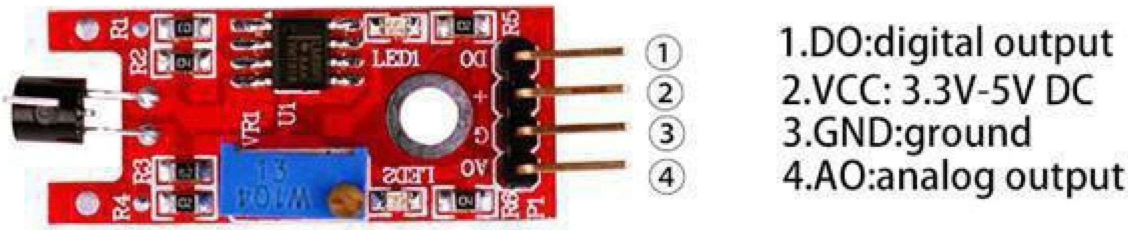
Lesson 20 Touch Sensor Module

Overview

In this experiment, we will learn how to use the mental touch module.

Touch

Touch sensitive switch. Touching the sensor pin produces an output at the 'DO' pin. The output is not a clean signal but includes 50 Hz mains induced signals ('mains hum'). The output signal is 'active high' and the circuit sensitivity can be adjusted with a pot. An analog output signal from the sensor is available at pin 'AO'.



Component Required:

(1) x Elegoo Uno R3

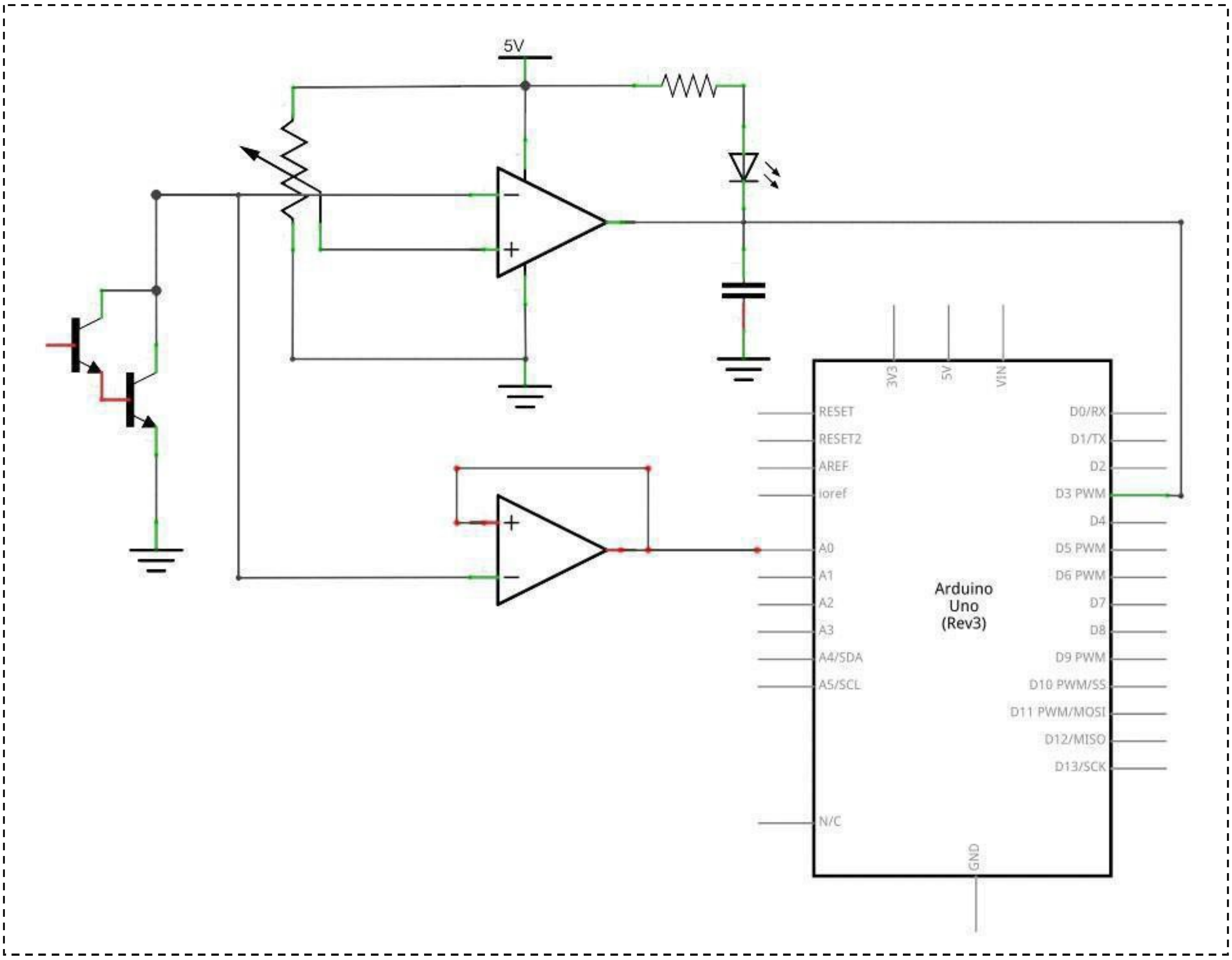
(1) x USB cable

(1) x Mental touch module

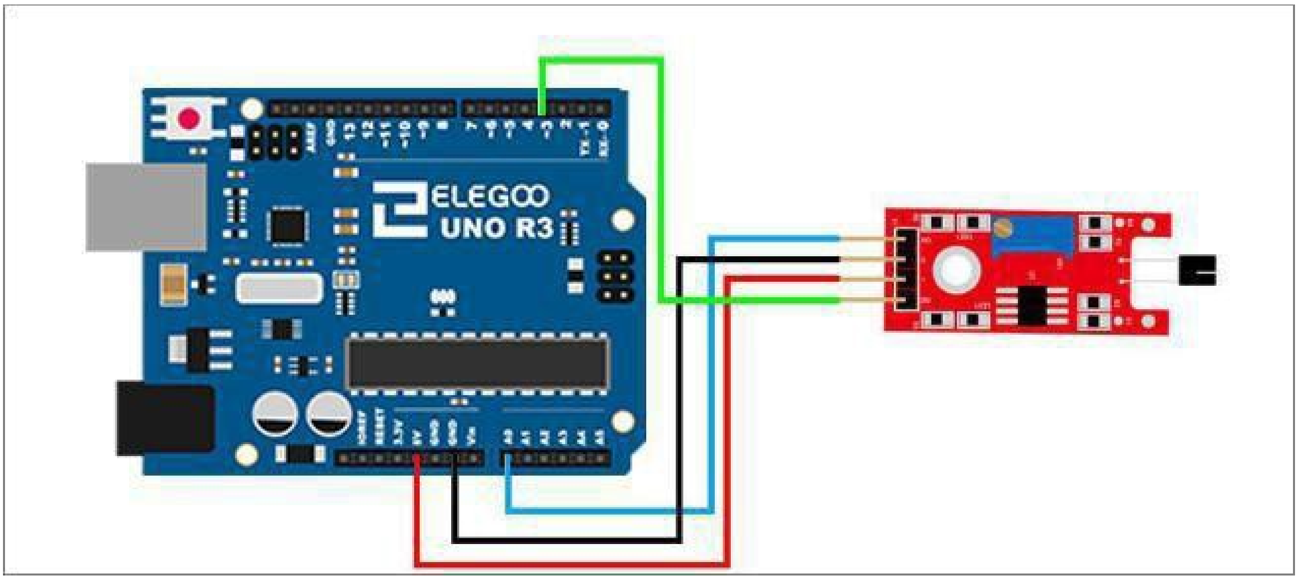
(x) x F-M wires

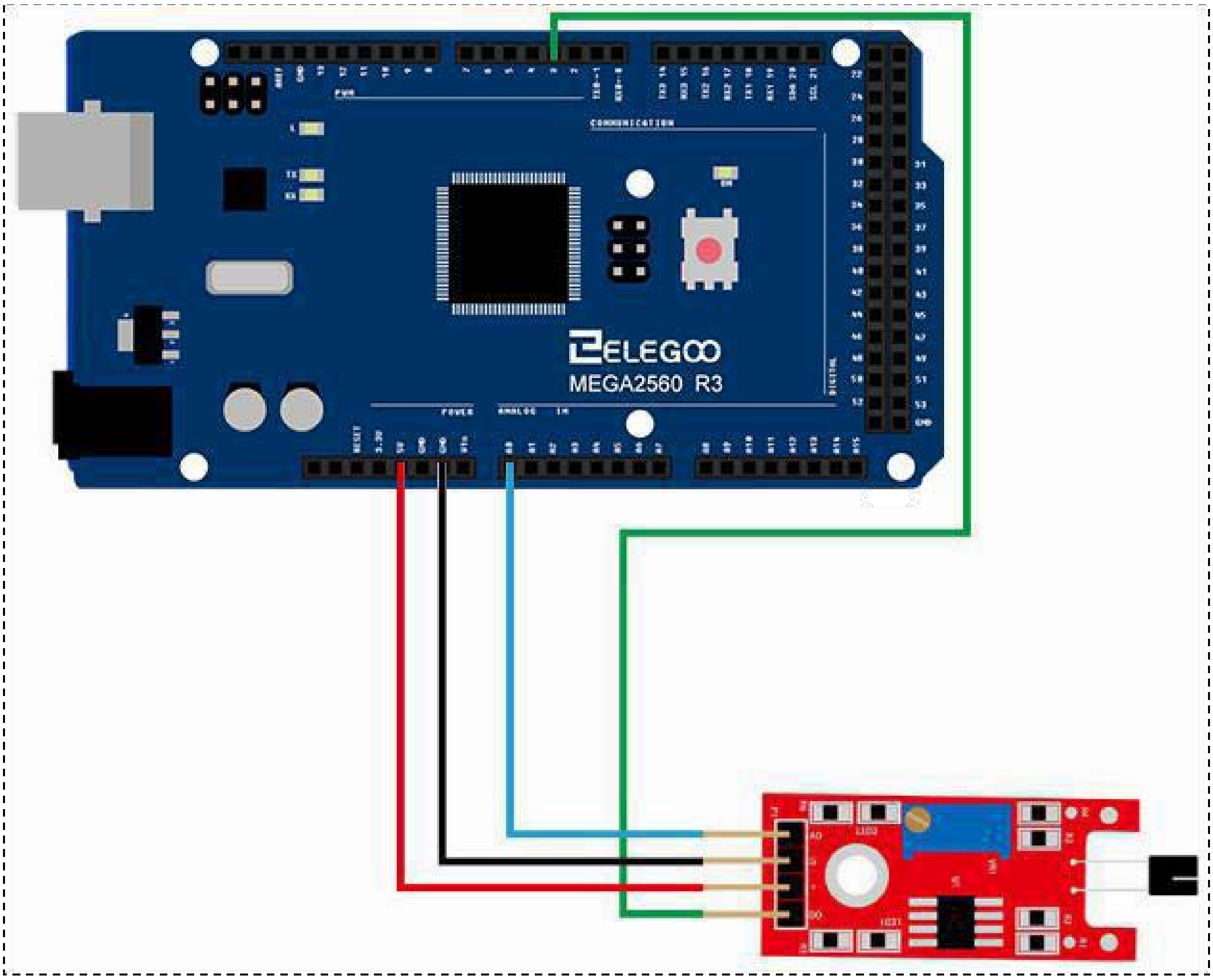
Connection

Schematic



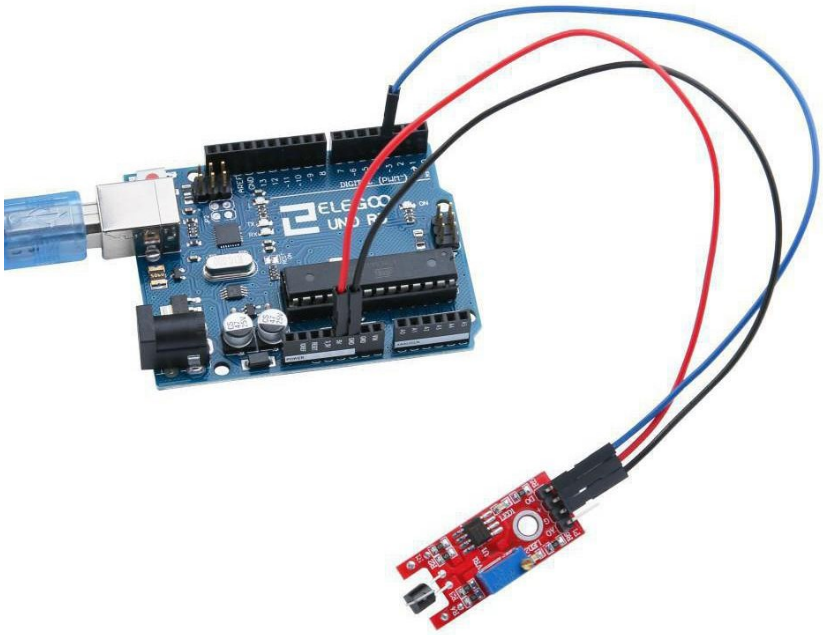
Wiring diagram



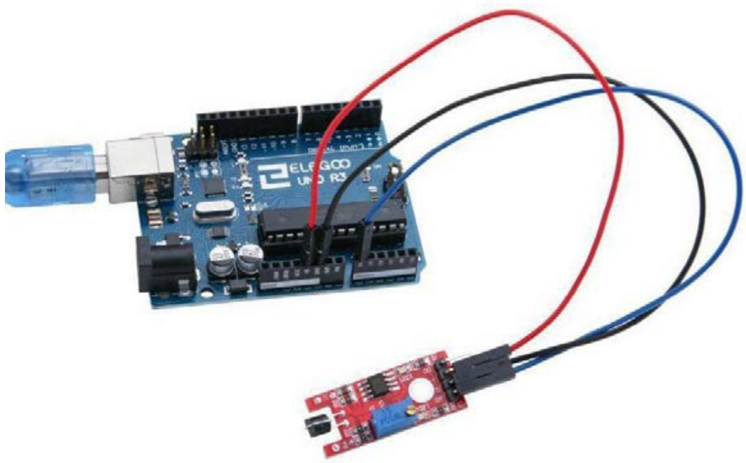


Result

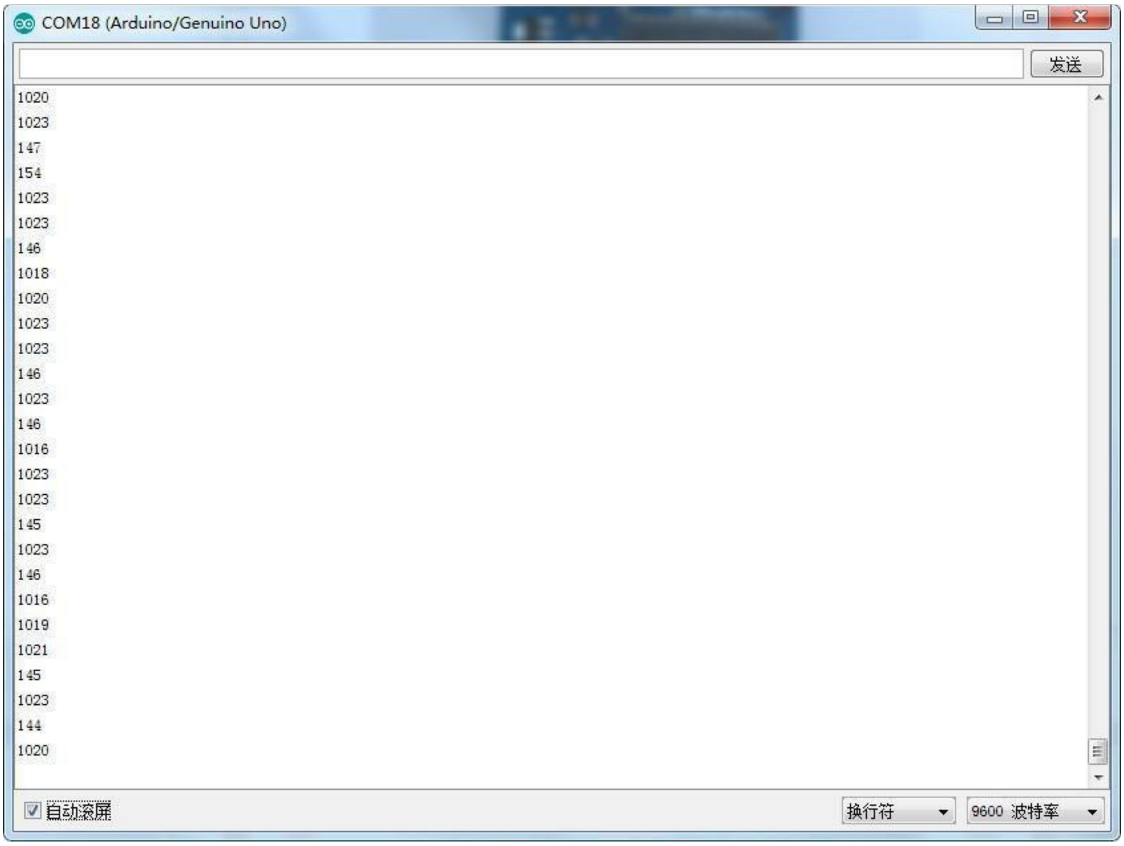
For the mental touch module, we can choose the output: digital output or analog output. In the following picture, we use the DO port to output. so we can see that if the sensor sensing, the light will turn on.



In the following picture, we use the AO port to output. so we can see that if the sensor sensing ,the module will output data. The number is from 0 to 1023.



Upload the program then open the monitor, we can see the data as below:



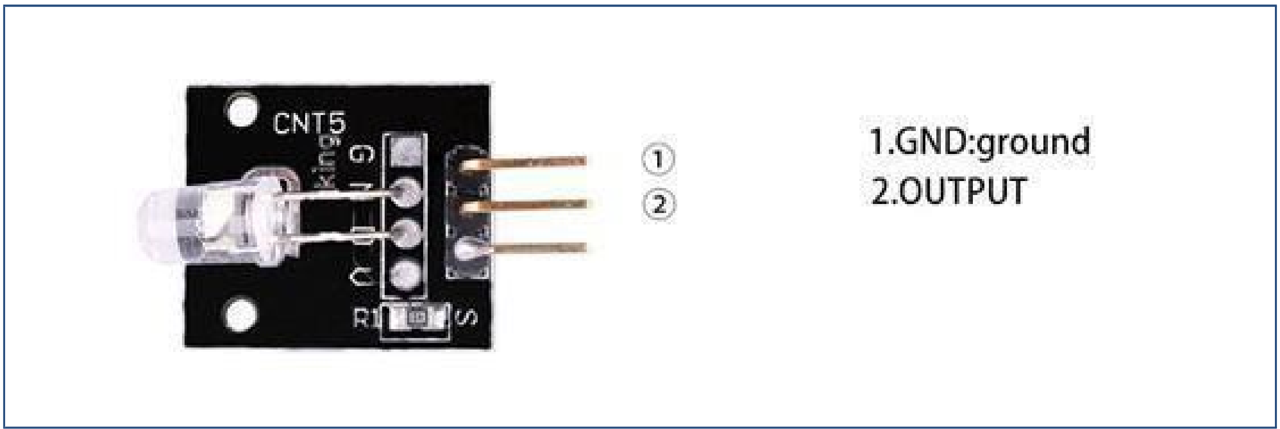
Lesson 21 Seven-Color Flash Module

Overview

In this experiment, we will learn how to use the Seven-Color flash Module

Seven-Color flash Module

Clear 5 mm LED for direct operation from 5V. The LED color automatically Cycles through a seven-color sequence. The 5 V supply connects to the 'S' pin and ground on the centerpin.



Component Required:

(1)x Elegoo Uno R3

(1)x USB cable

(1)x Seven-Color flash Module

(x)x F-M wires

Component Introduction

Seven-Color flash Module:

7 color flashing LED module automatically uses 5mm round high-brightness light-emitting diode which has the following characteristics:

Product Type: LED

Product Model: YB-3120B4PnYG-PM Shape: Round

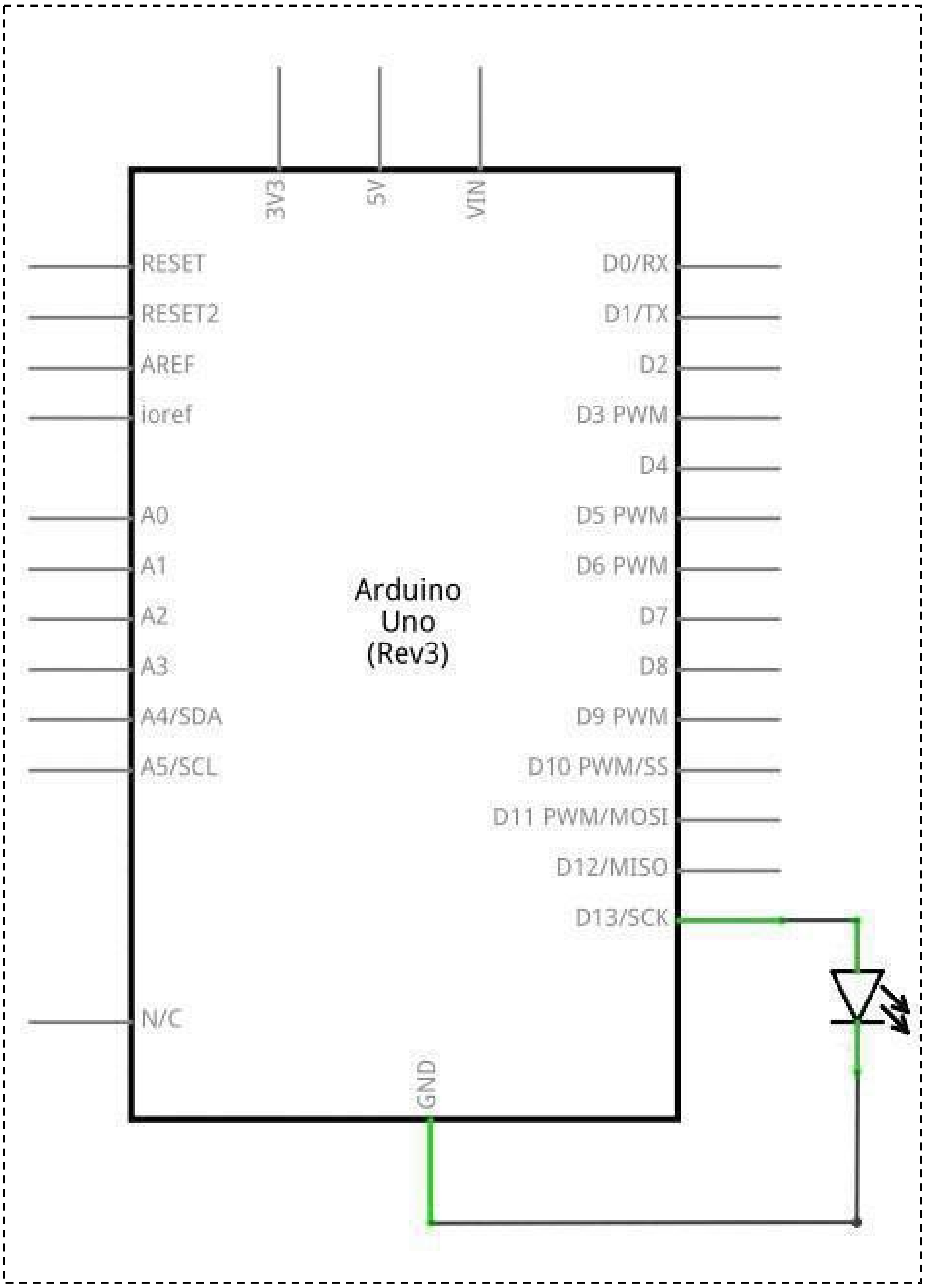
LED 5 mm DIP type
Color: pink yellow green (high brightness) Lens type: white

mist

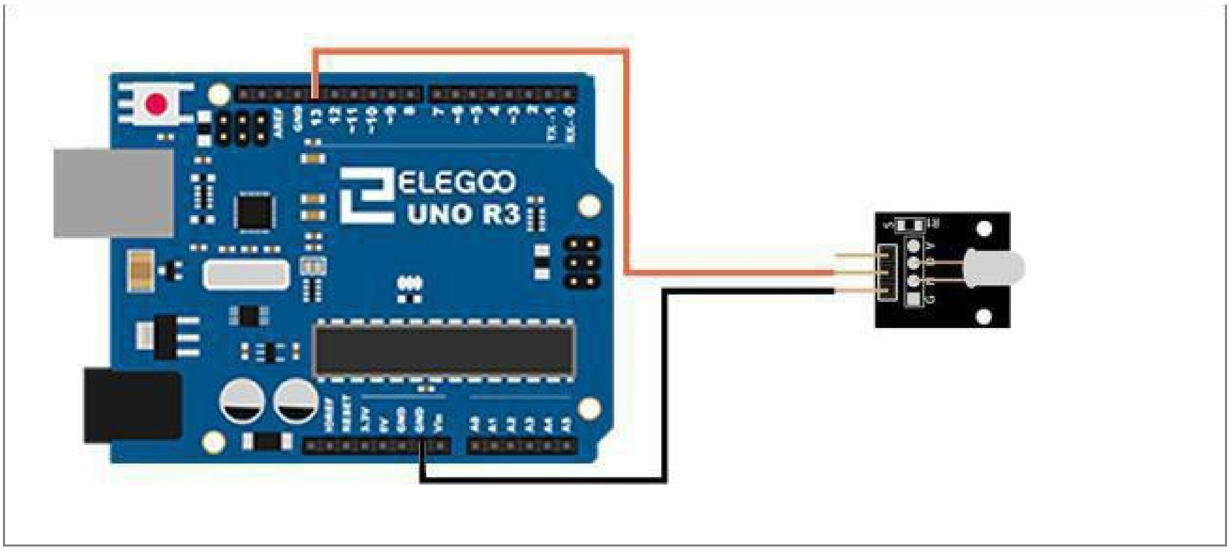
Standard Forward Voltage:3.0-4.5 V

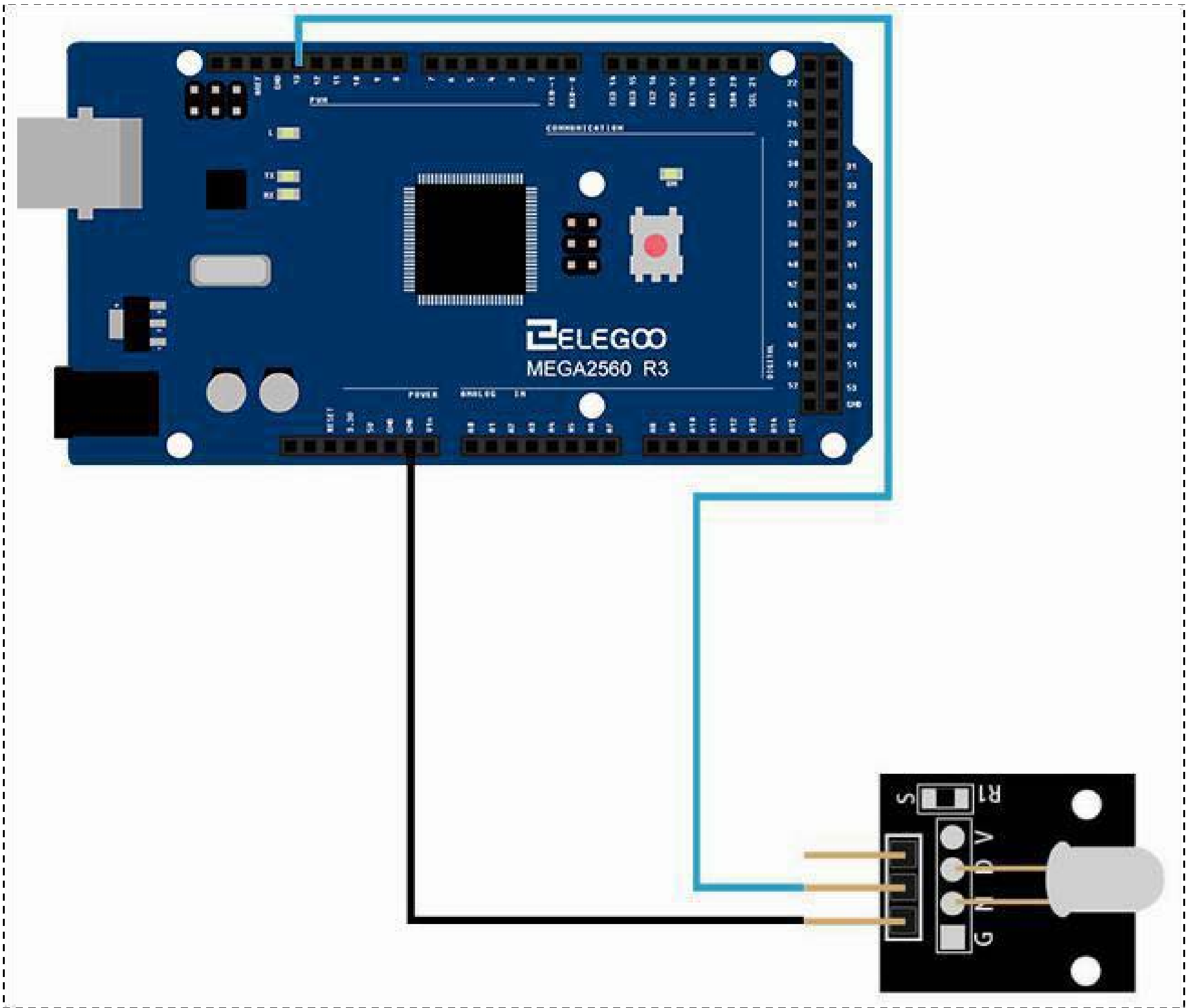
Connection

Schematic

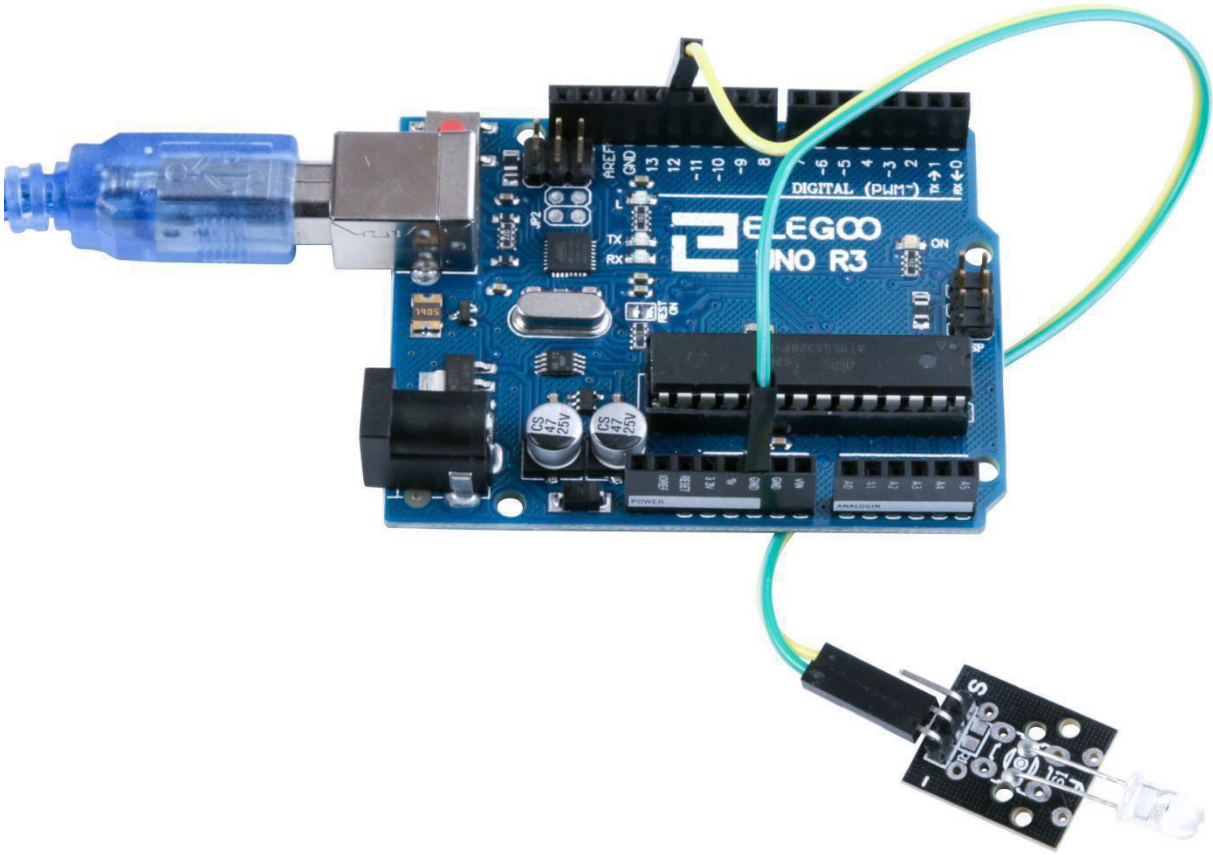


Wiring diagram


ELEGOO
UNO R3


ELEGOO
MEGA2560 R3

Result



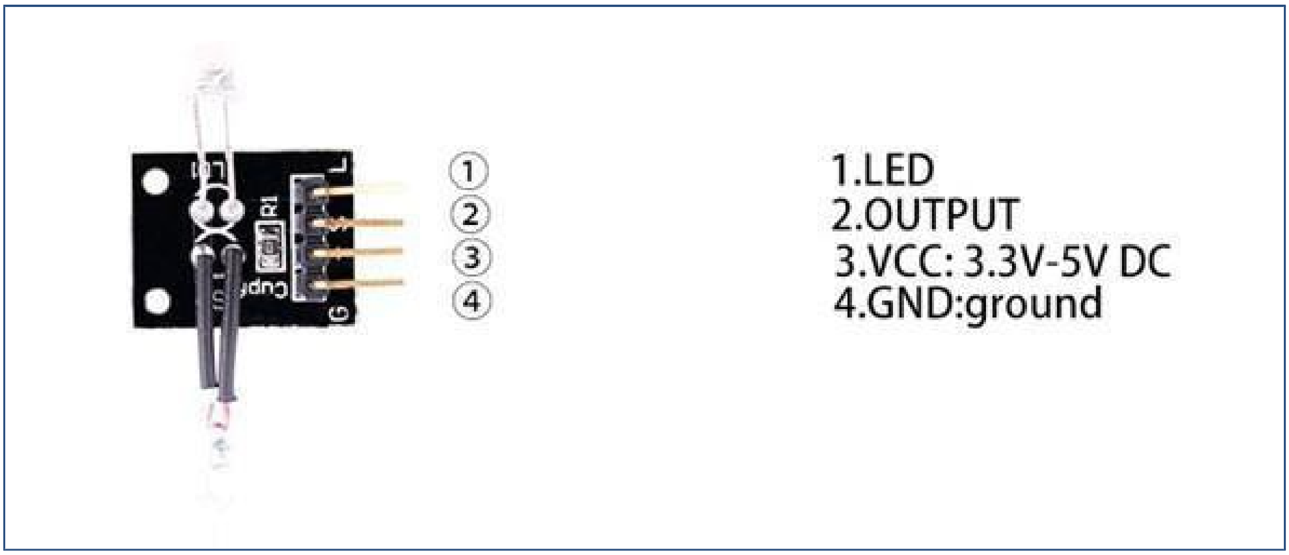
Lesson 22 Switch Light Module

Overview

In this experiment, we will learn how to use the switch light module.

2 x switch light module

This module incorporates a mercury tilt switch and clear red LED. ‘G’ is the common connection to the LED cathode and one terminal of the switch. ‘S’ is the other switch contact and ‘L’ connects to the LED anode (a series resister is required for the LED, 220 ohms for example). The ‘+’ pin connects to a 10 K ohm pull up resistor connected to ‘S’ ofthe switch.



Component Required:

(1)x Elegoo Uno R3

(1) x USB cable

(1)x Switch light module

(x) x F-M wires

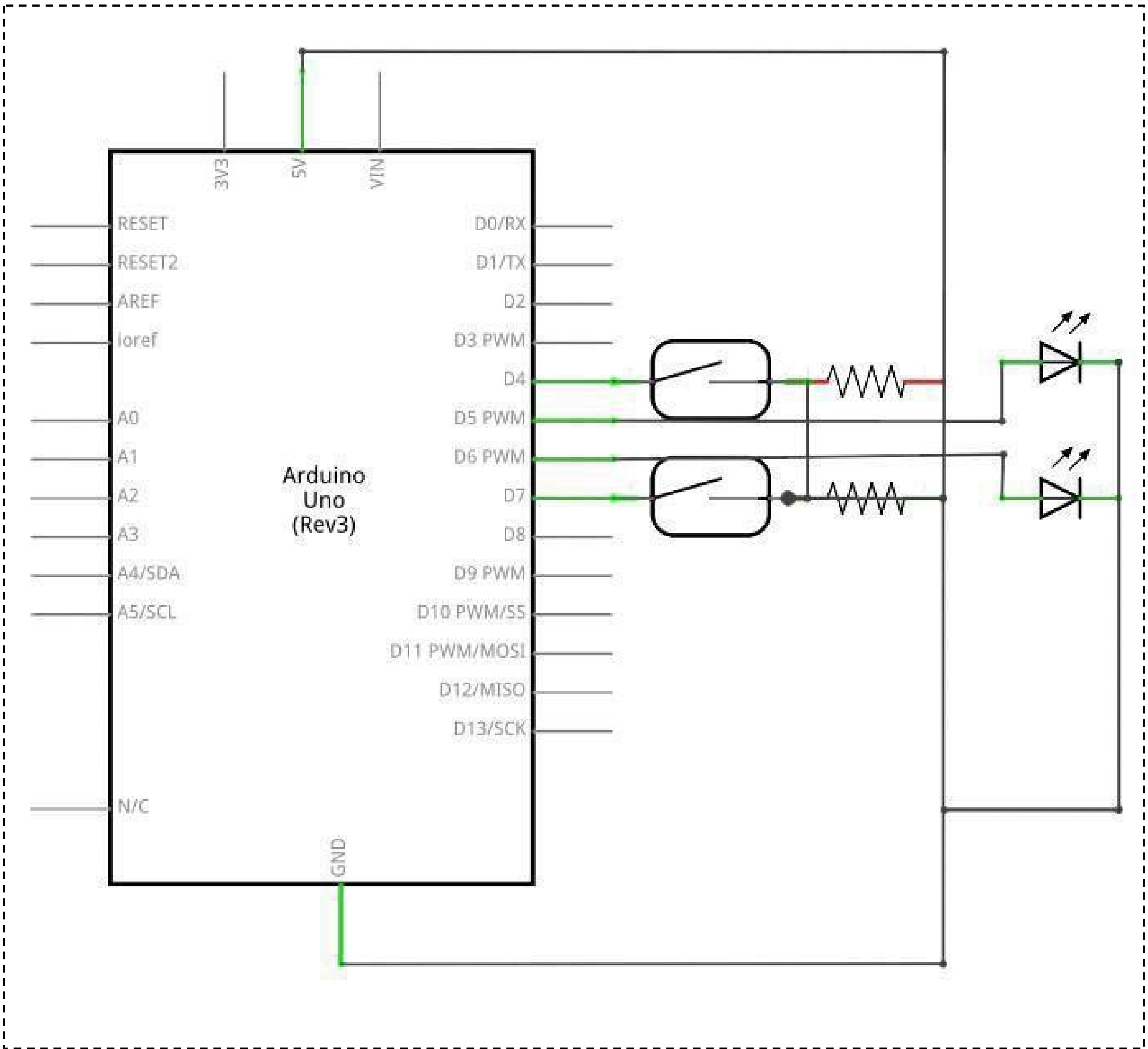
Component Introduction

Magic lightcup module:

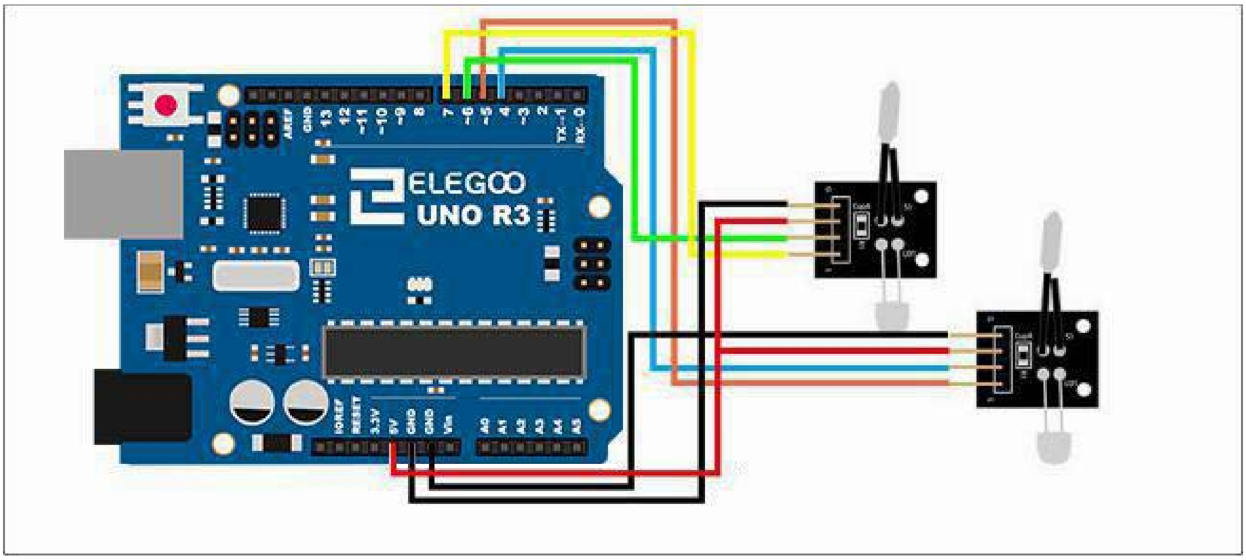
Magic light cup module is a product which can easily interact with the ARDUINO module, the principle is to use the PWM dimming, change the brightness of the two modules. Mercury switch provides digital signal, trigger PWM adjustment. Through the design program, we can see the result which is similar to the effect of two cup full of light pouring to each other.

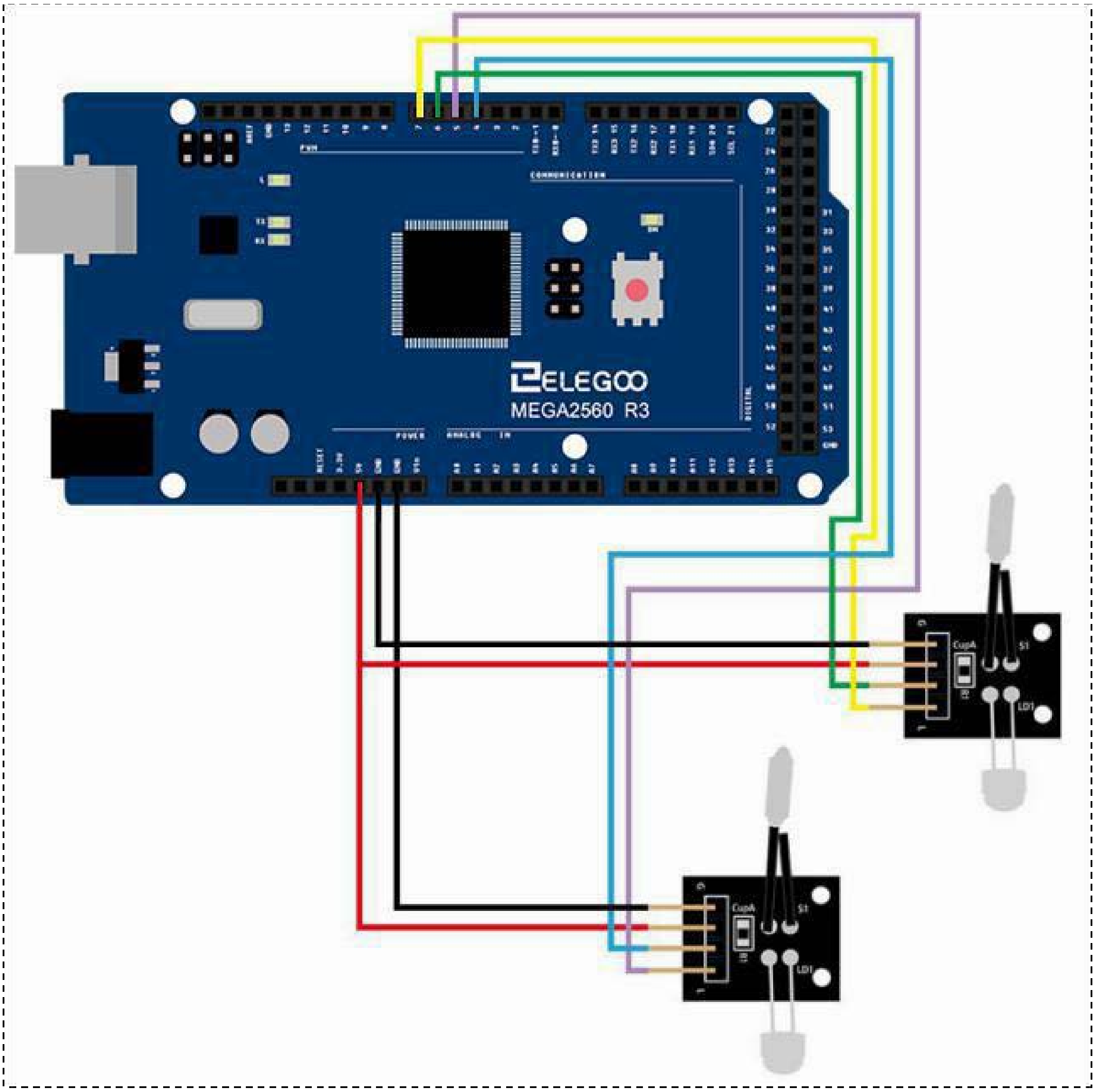
Connection

Schematic

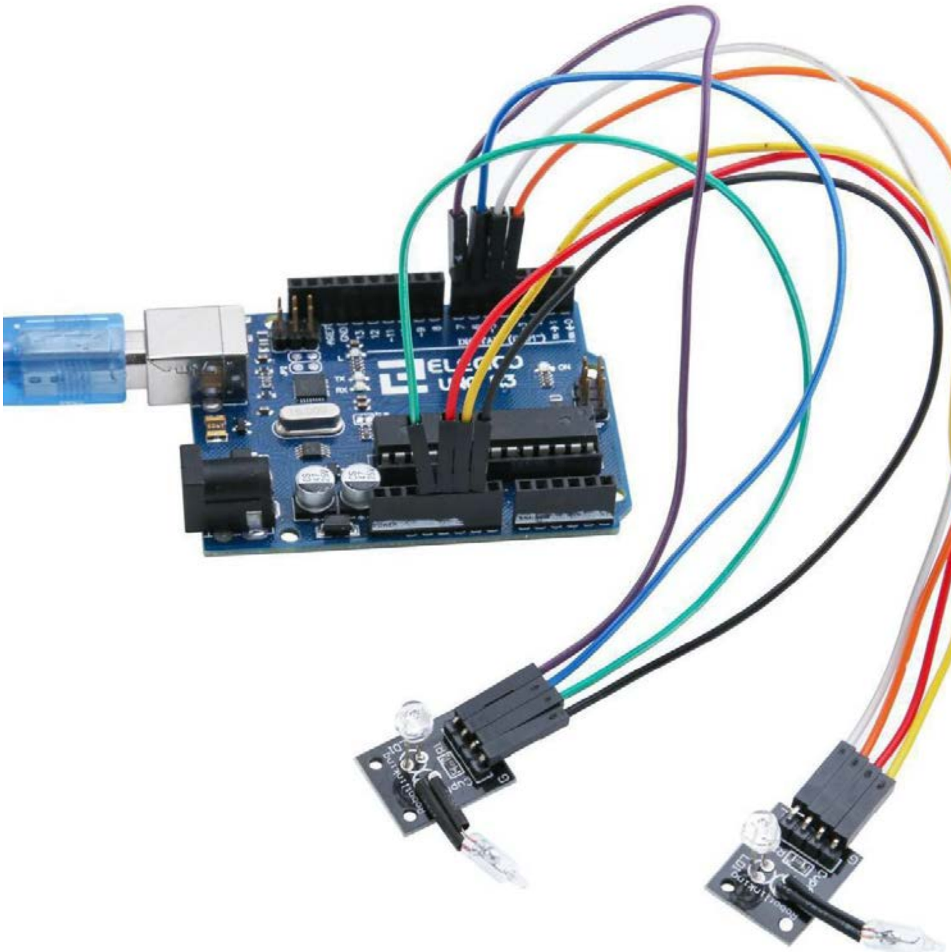


Wiring diagram





Result



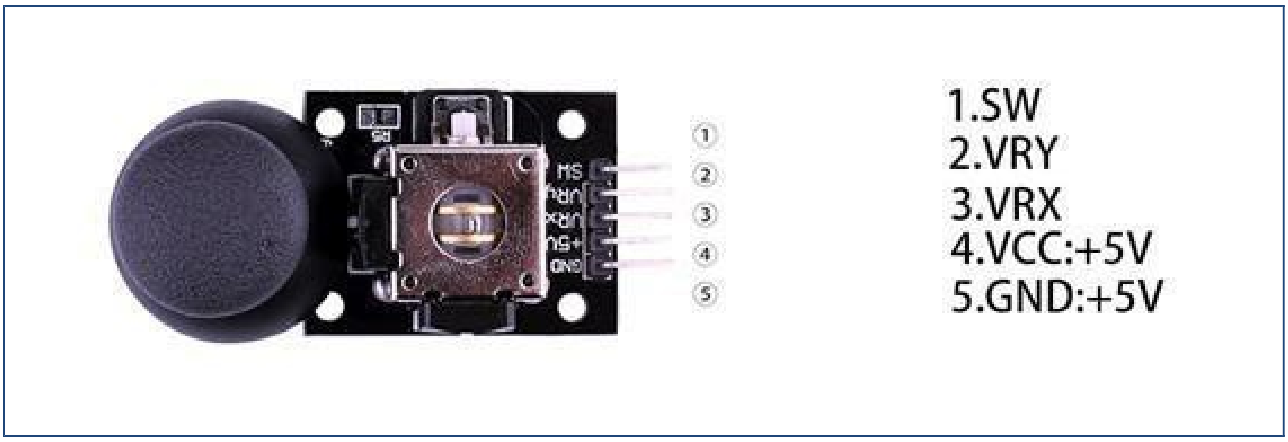
Lesson 23 Joystick Module

Overview

Just like a joystick on game console. you can control x, y and z dimensions input by this joystick module. It can be considered as combination of potentiometers and one button. Data type of the x, y dimensions are analog input signals and z dimension is digital input signal. Thus the x and y ports connect to analog pins of Sensor Shield, while z port connects to digital pin.

Joystick

An analog 2-axis joystick with 2x 10K ohm pots and push button function.
Connector pin descriptions are printed on the PCB. A push-on operating knob is included with themodule.



Component Required:

(1)x Elegoo Uno R3

(1)x USB cable

(1)xJoystick module

(x)x F-M wires

Component Introduction

Joystick sensor:

Lots of robot projects need joystick. This module provides an affordable solution. By simply connecting to two analog inputs, the robot is at your commands with X, Y control. It also has a switch that is connected to a digital pin. This joystick module can be easily connected to Arduino by IO Shield. This module is for Arduino (V5) with cables supplied.

Specification

Supply Voltage: 3.3V to 5V Interface:

Analog x2, Digital x1 Size: 40*28 mm

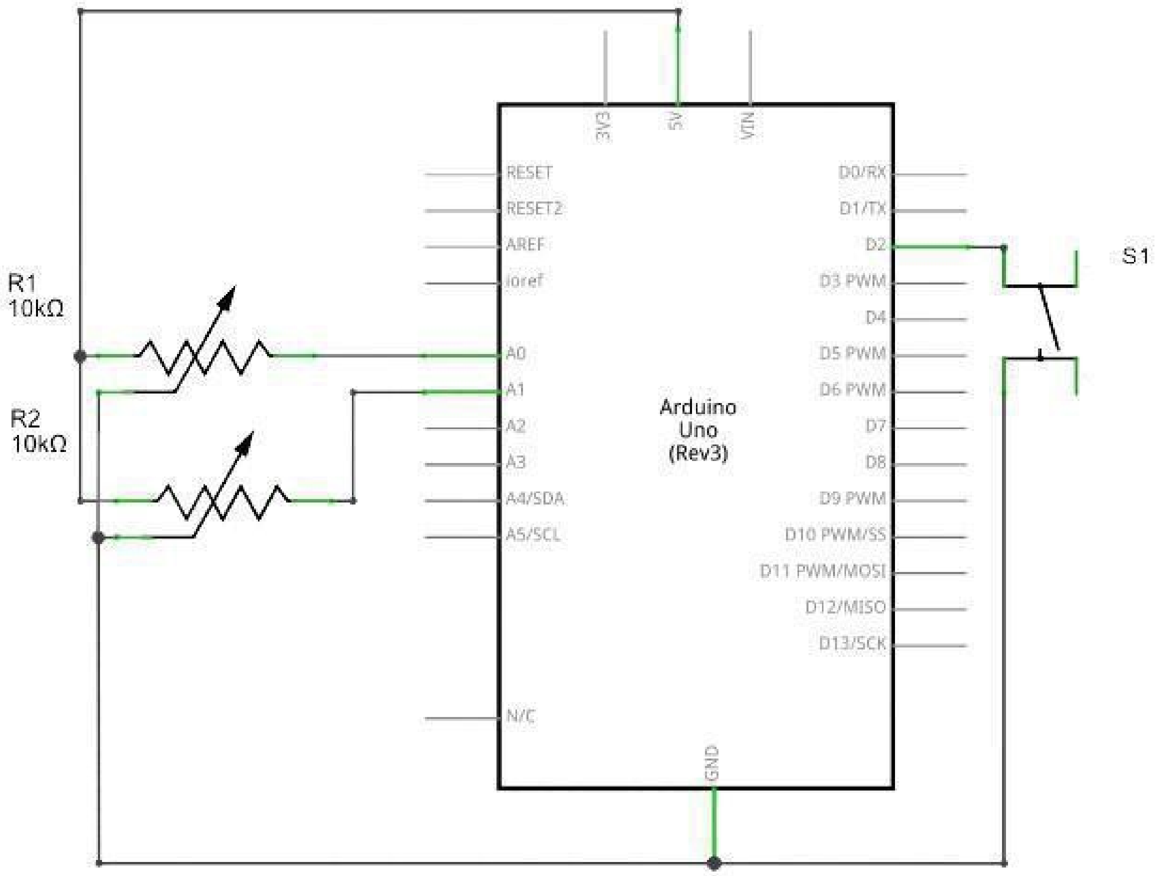
Weight: 12g

The module has 5 pins: Vcc, Ground, X, Y, Key. Note that the labels on yours may be slightly different, depending on where you got the module from. The thumb stick is analog and should provide more accurate readings than simple 'directional' joysticks tact use some forms of buttons, or mechanical switches. Additionally, you can press the joystick down (rather hard on mine) to activate a 'press to select' push-button.

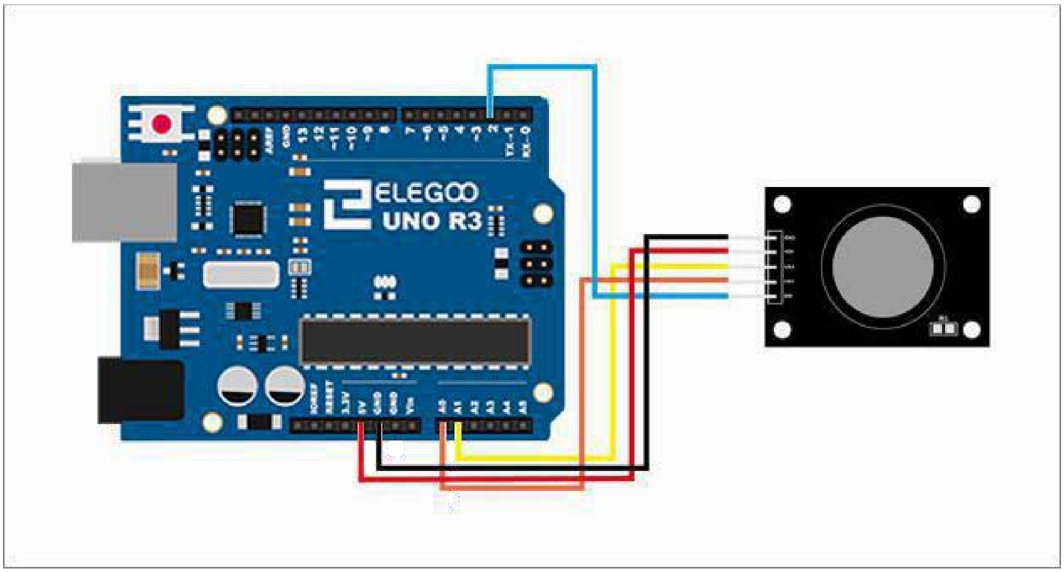
We have to use analog Arduino pins to read the data from the X/Y pins, and a digital pin to read the button. The Key pin is connected to ground, when the joystick is pressed down, and is floating otherwise. To get stable readings from the Key /Select pin, it needs to be connected to Vcc via a pull-up resistor. The built in resistors on the Arduino digital pins can be used. For a tutorial on how to activate the pull-up resistors for Arduino pins, configured as inputs

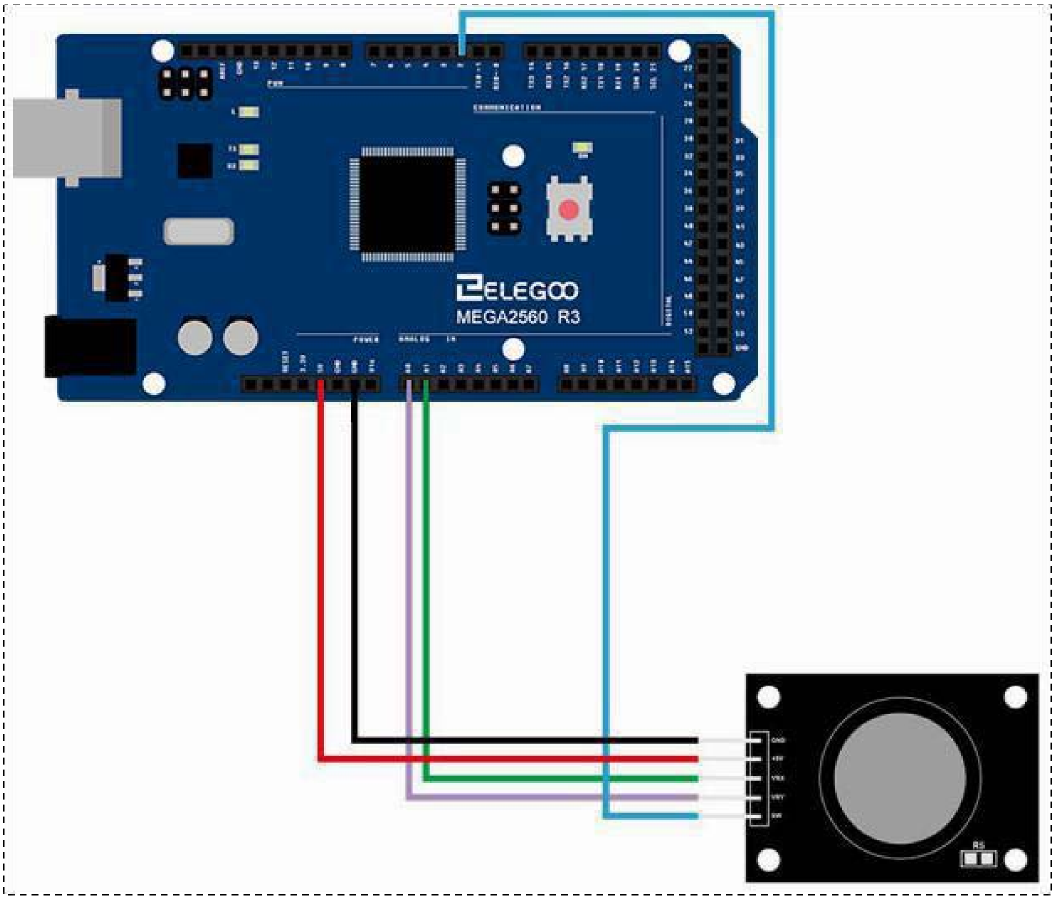
Connection

Schematic



Wiring diagram

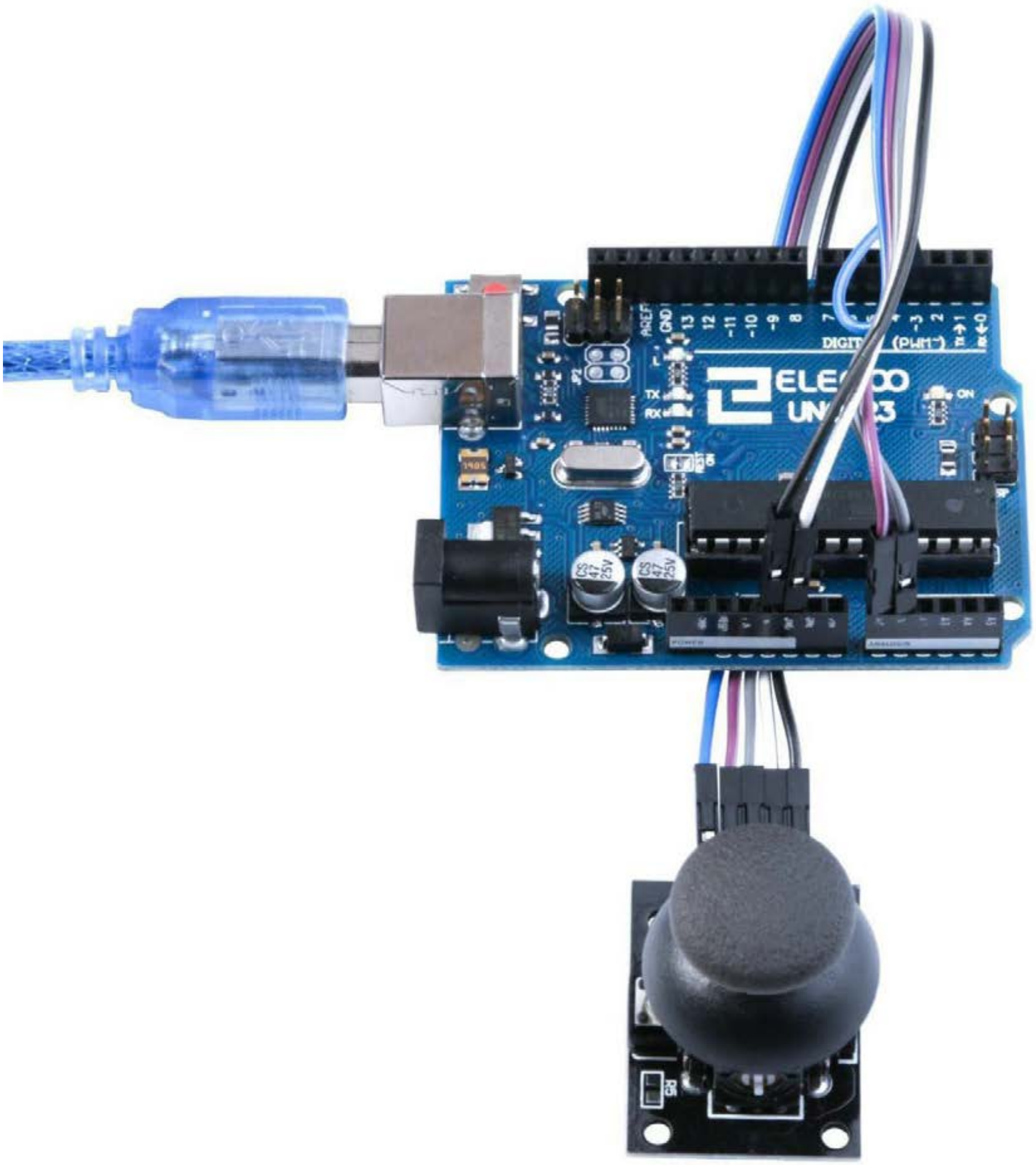


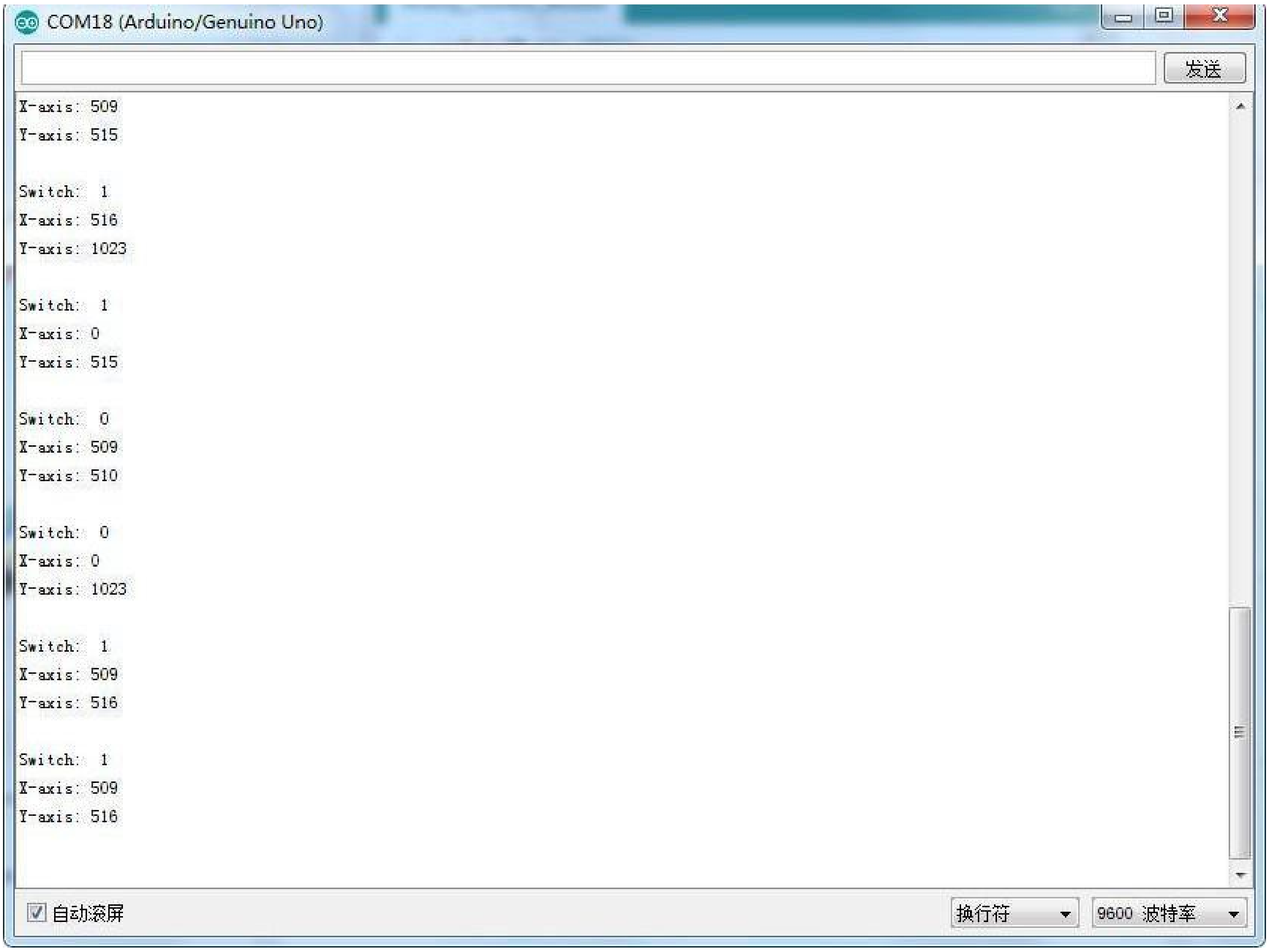


Result

Analog joysticks are basically potentiometers so they return analog values.

When the joystick is in the resting position or middle, it should return a value of about 512. The range of values go from 0 to 1024.




COM18 (Arduino/Genuino Uno)
发送
X-axis: 509
Y-axis: 515
Switch: 1
X-axis: 516
Y-axis: 1023
Switch: 1
X-axis: 0
Y-axis: 515
Switch: 0
X-axis: 509
Y-axis: 510
Switch: 0
X-axis: 0
Y-axis: 1023
Switch: 1
X-axis: 509
Y-axis: 516
Switch: 1
X-axis: 509
Y-axis: 516
自动滚屏
换行符
9600 波特率

Lesson 24 Line Tracking Module

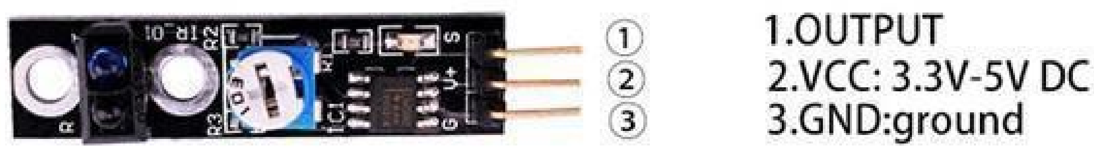
Overview

In this experiment, we will learn how to use the tracking module and avoidance module.

Infrared obstacle avoidance sensor is designed for the design of a wheeled robot obstacle avoidance sensor distance adjustable. This ambient light sensor Adaptable, high precision, having a pair of infrared transmitter and receiver, transmitter tubes emit a certain frequency of infrared, When detecting the direction of an obstacle (reflector), the infrared receiver tube receiver is reflected back, when the indicator is lit, Through the circuit, the signal output interface output digital signal that can be detected by means of potentiometer knob to adjust the distance, the effective distance From 2 ~ 40 cm, working voltage of 3.3V-5V, operating voltage range as broad, relatively large fluctuations in the power supply voltage of the situation Stable condition and still work for a variety of microcontrollers, Arduino controller, BS2 controller, attached to the robot that can sense changes in the ir surroundings

Tracking

IR light reflection switch, useful for obstacle avoidance or line following on models that move around the floor. An obstacle in front of the sender/receiver diodes will cause the 'out' pin to be pulled low (active low). A pot allows adjustment of the circuit's sensitivity. The detection distance can be up to approximately 1cm.



Component Required:

(1)x Elegoo Uno R3

(1)x USB cable

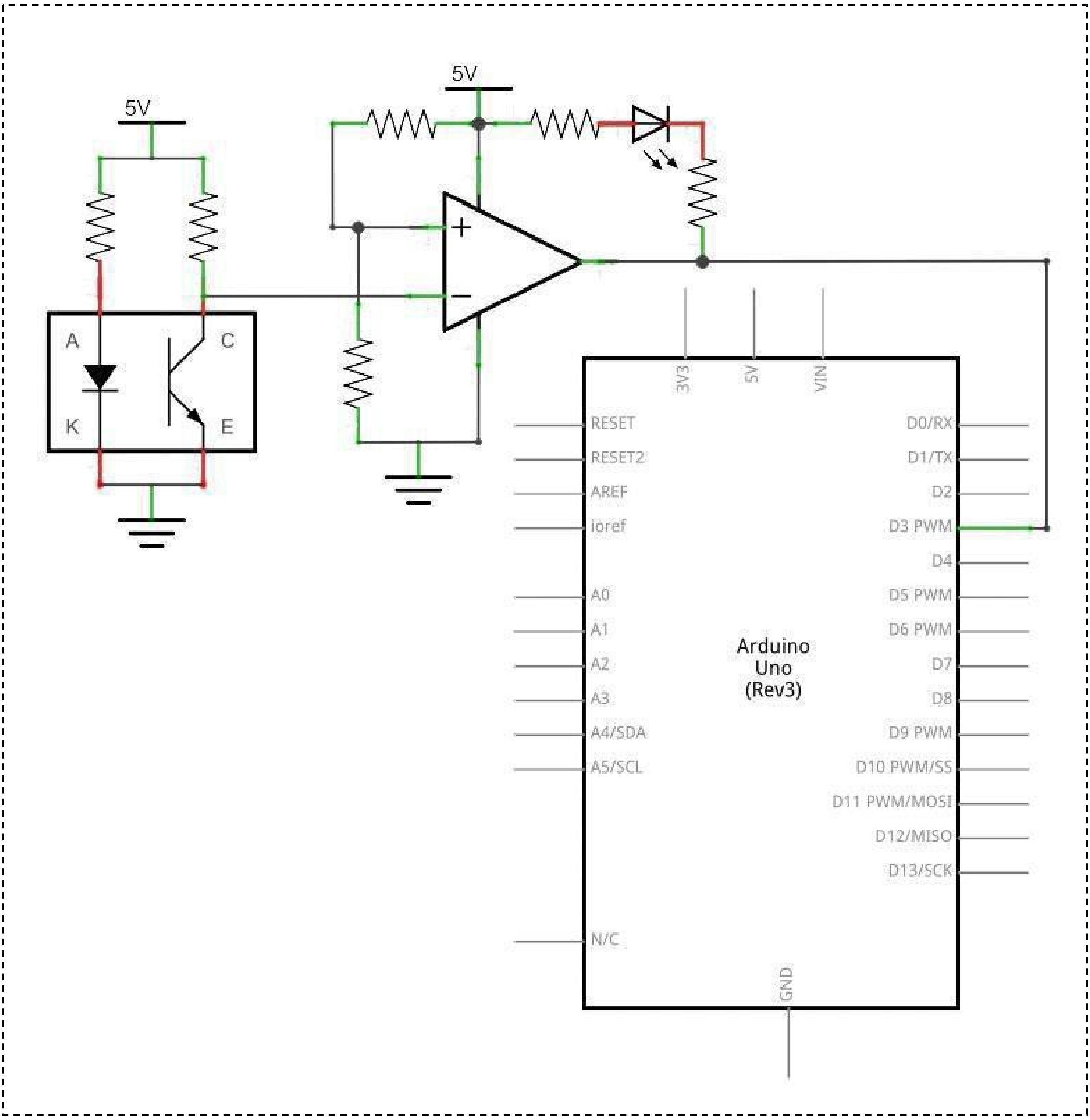
(1) x Tracking module

(x) x F-M wires

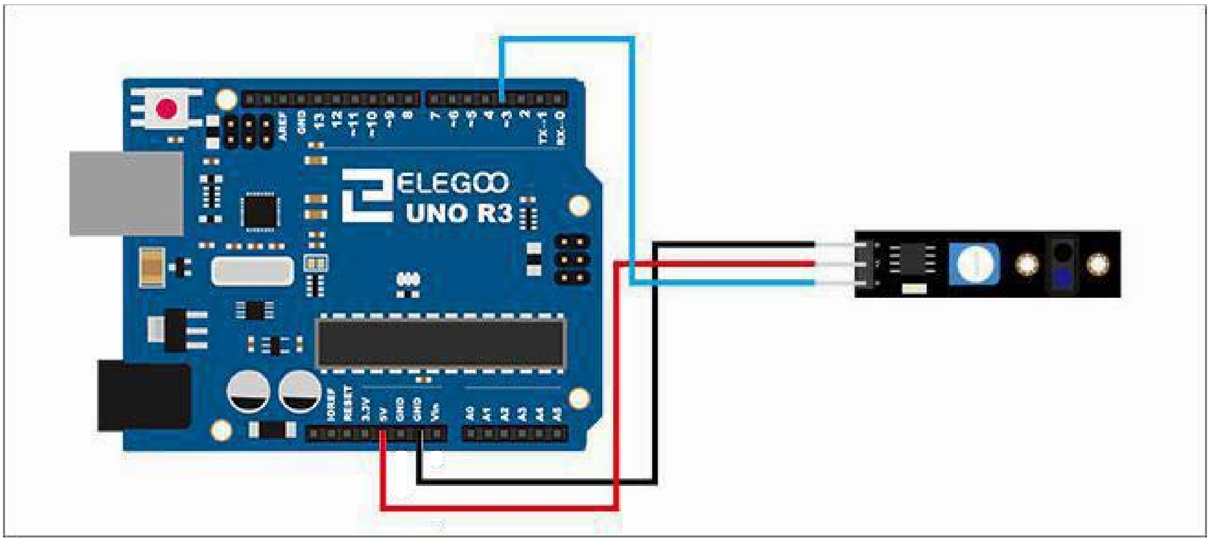
Component Introduction

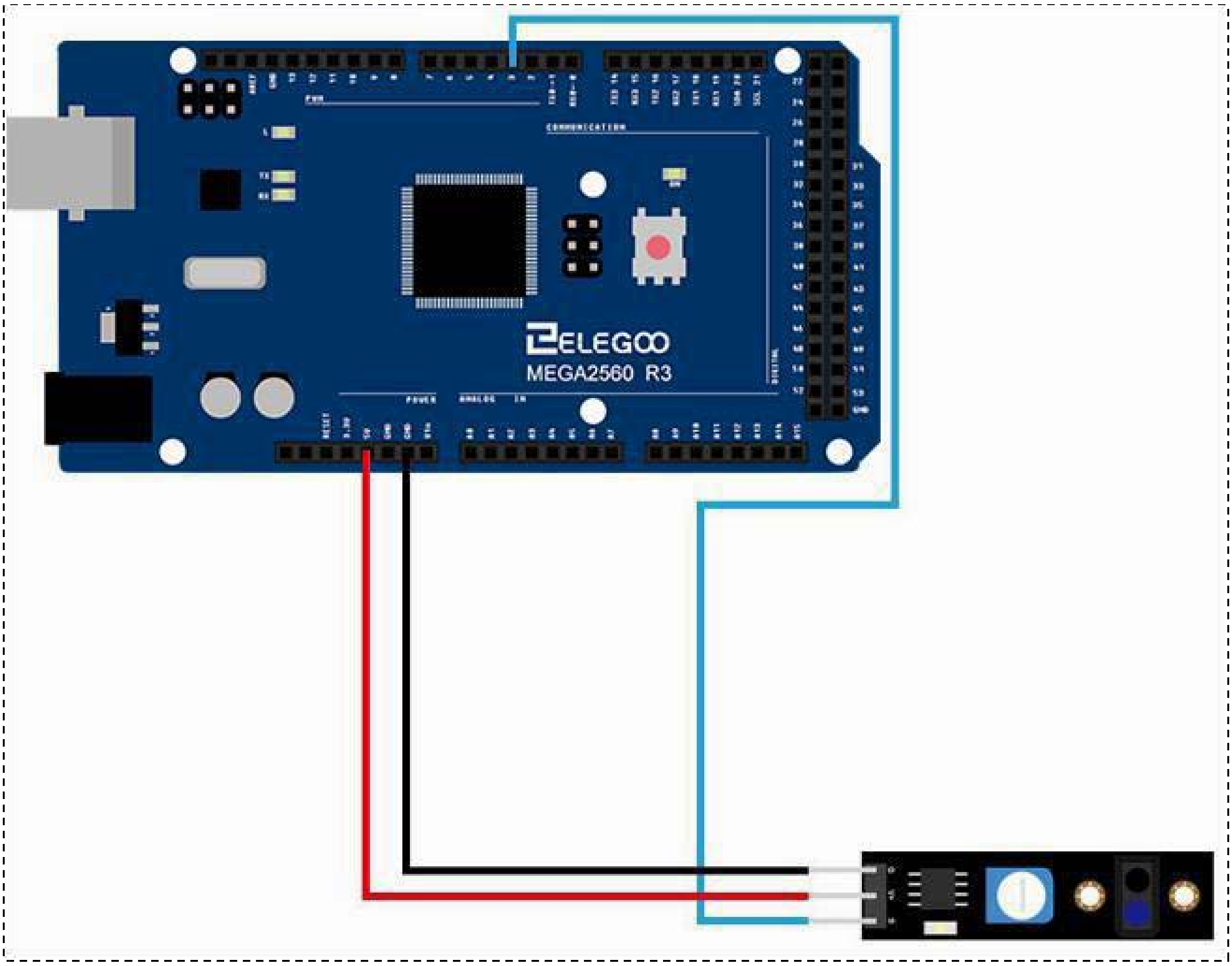
Connection

Schematic

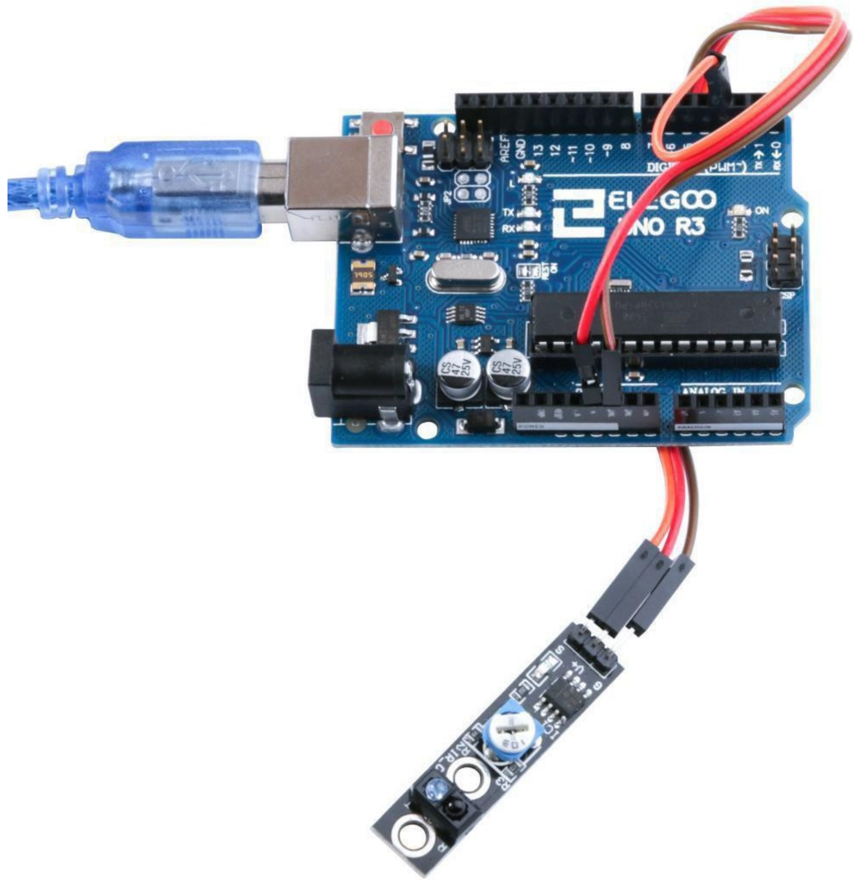


Wiring diagram





Result



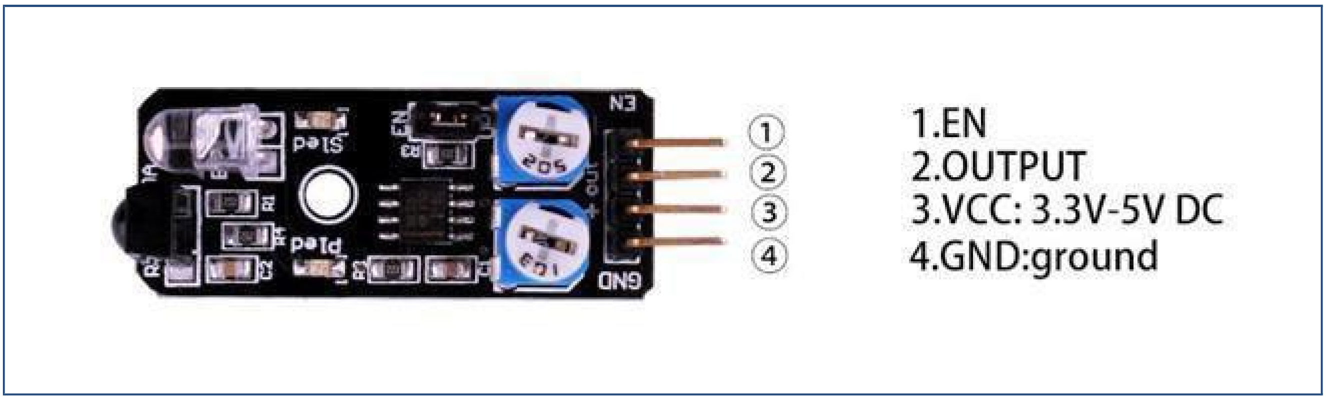
Lesson 25 Obstacle Avoidance Sensor Module

Overview

Non logic chip oscillation frequency regulation 38 KHz detection circuit. This infrared 38 KHZ obstacle avoidance module can completely block the traditional photoelectric obstacle detection distance and a single normally open or normally closed signal out put function.

Avoid

IR-reflection sensor, useful for obstacle avoidance applications. When an obstacle is in front of the IR sender/receiver the ‘Out’ pin is switched low (active low). The circuit sensitivity can be adjusted with a pot. The obstacle detection distance can be adjusted up to approximately 7 cm. An enable (EN) jumper can be fitted for continuous operation. Removal of the EN jumper allows an external logic signal (at the EN pin) to switch the detector on and off (low = active, high = off).



Component Required:

(1)x Elegoo Uno R3

(1)x USB cable

(1) x Obstacle Avoidance Sensor

(x) x F-M wires

Component Introduction

Obstacle avoidancemodule:

Specifications

Working voltage: DC 3.3V-5V
Working current: ⩾ 20 mA Operating
temperature:-10 ℃-+50℃

Detection distance: 2-40 cm

IO Interface: 4-wire interfaces (- / + / S / EN)

Output signal: TTL level (low level there is an obstacle, no obstacle high)

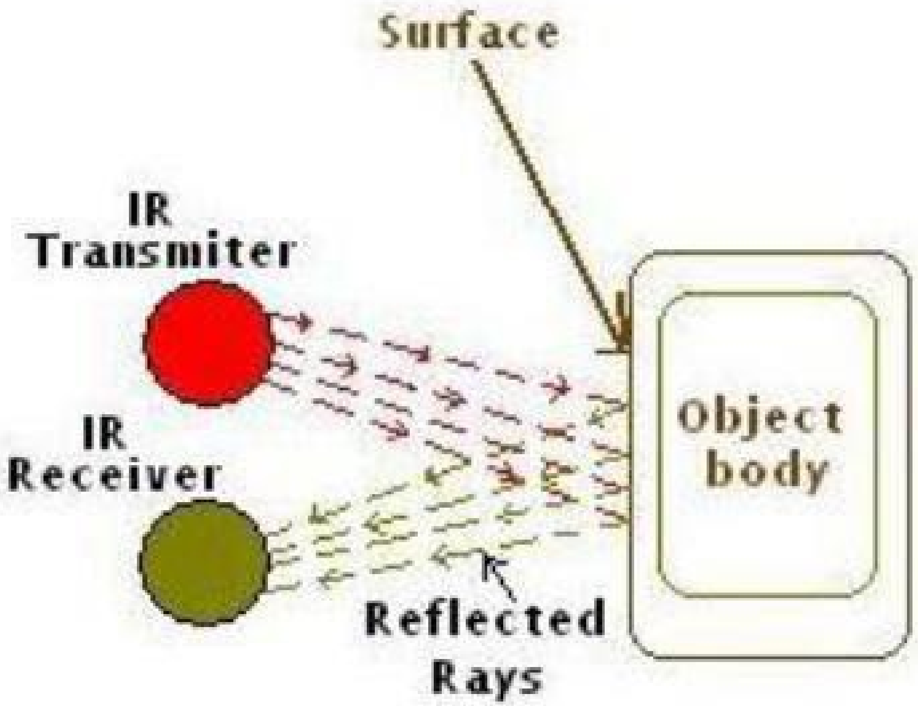
Adjustment: adjust multi- turn resistance

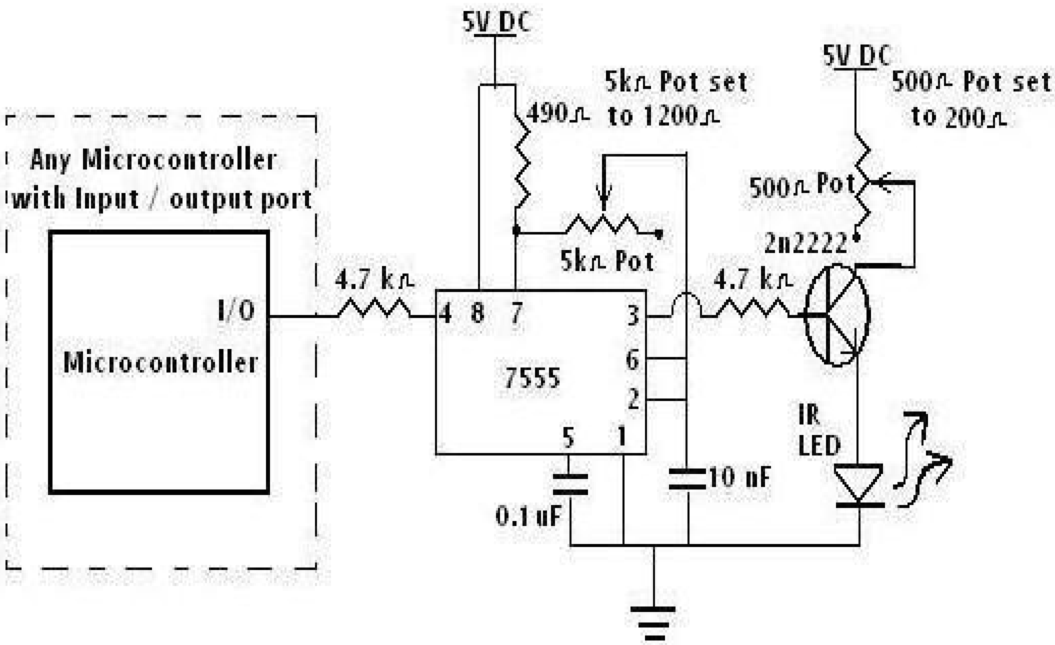
Effective angle: 35°

Size: 28 mm × 23 mm Weight

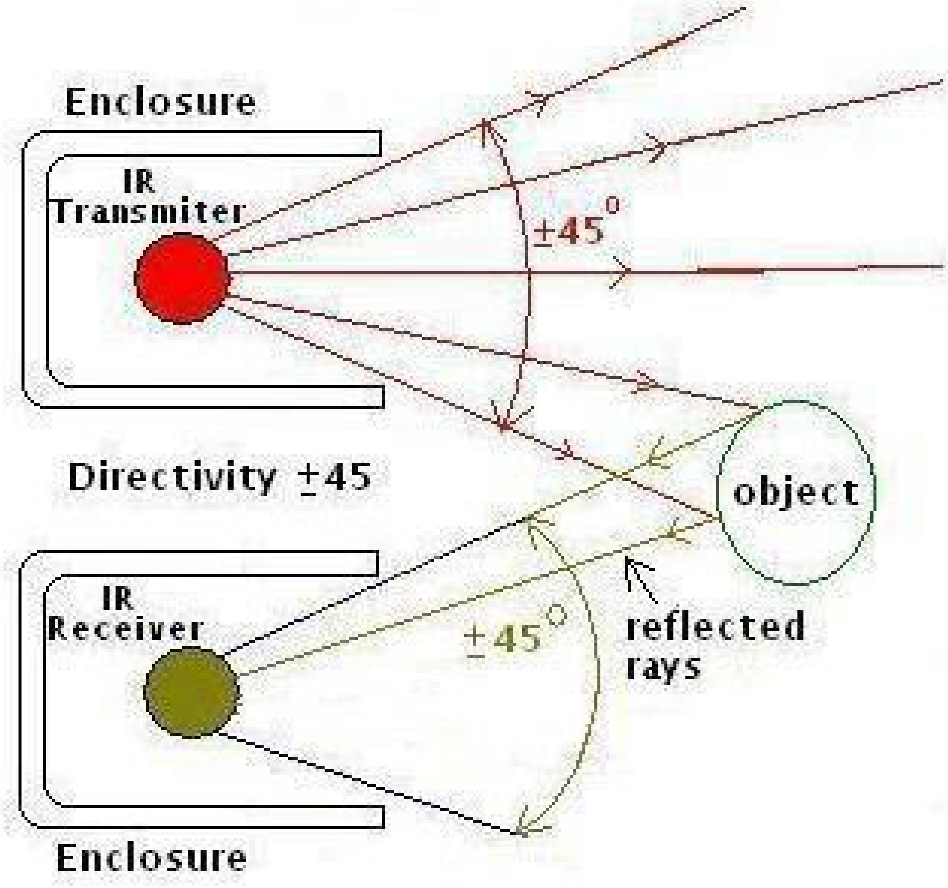
Size: 9 g

The basic concept of IR (infrared) obstacle detection is to transmit the IR signal (radiation) in a direction and a signal is received at the IR receiver when the IR radiation bounces back from a surface of the object.


Surface
IR
Transmiter
IR
Receiver
Object
body
Reflected
Rays


5V DC
5V DC
5kΩ Pot set
to 1200Ω
500Ω Pot set
to 200Ω
490Ω
500Ω Pot
2n2222
5kΩ Pot
Any Microcontroller
with Input / output port
4.7 kΩ
4.7 kΩ
I/O
Microcontroller
4 8 7
3
6
2
7555
5 1
IR
LED
0.1 uF
10 nF

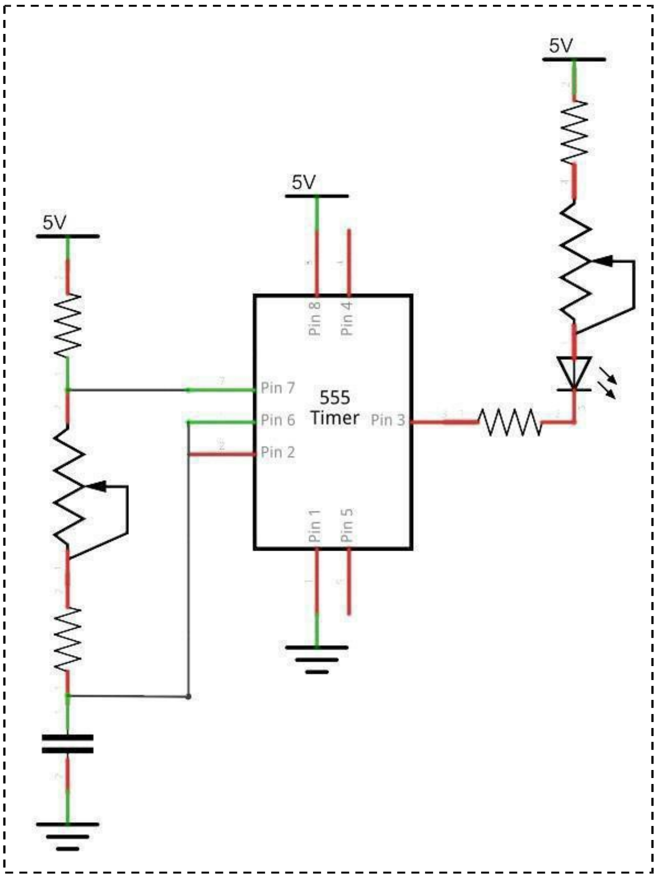
There are two potentiometers on the module one controlling operating frequency (centered at 38 kHz) the other controlling intensity. The detector was designed for 38 kHz and the onboard oscillator circuit is based on a 555 timer. Tweaking gives a little better range but I'd suggest leaving it alone because the useful range is narrow.

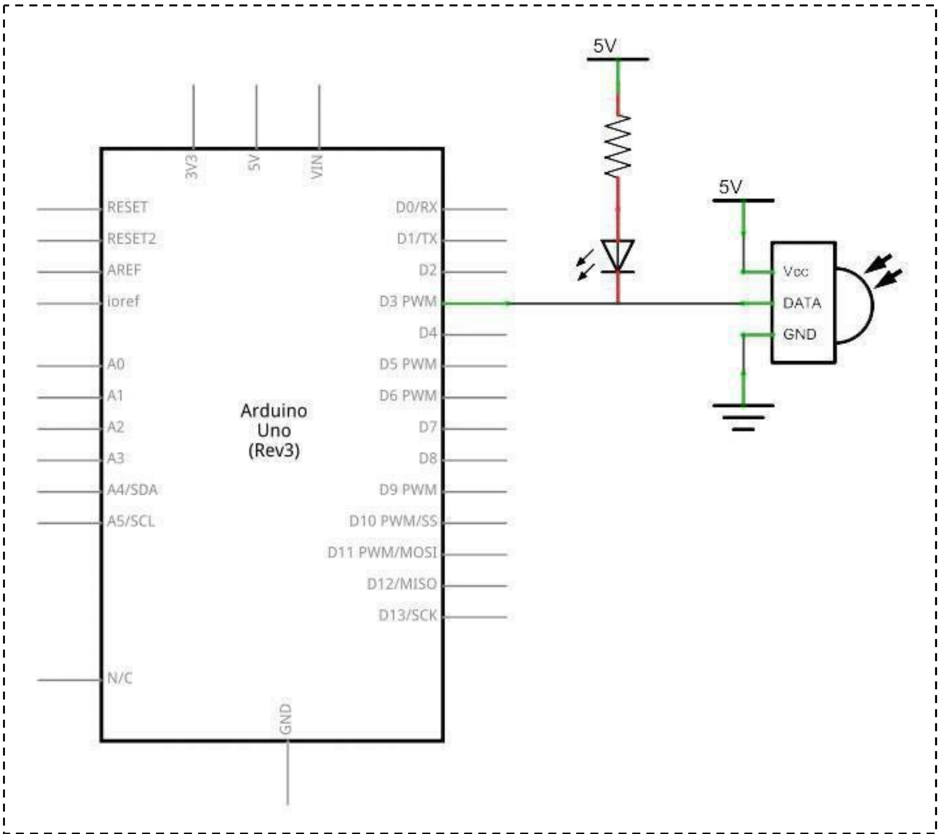


Infrared obstacle avoidance sensor is designed for the design of a wheeled robot obstacle avoidance sensor distance adjustable. This ambient light sensor Adaptable, high precision, having a pair of infrared transmitter and receiver, transmitter tubes emit a certain frequency of infrared, When detecting the direction of an obstacle (reflector), the infrared receiver tube receiver is reflected back, when the indicator is lit, Through the circuit, the signal output interface output digital signal that can be detected by means of potentiometer knob to adjust the distance, the effective distance From 2 ~ 40 cm, working voltage of 3.3V-5V, operating oltage range as broad, relatively large fluctuations in the power supply voltage of the situation Stable condition and still work for a variety of microcontrollers, Arduino controller, BS2 controller, attached to the robot that Can sense changes in their surroundings.

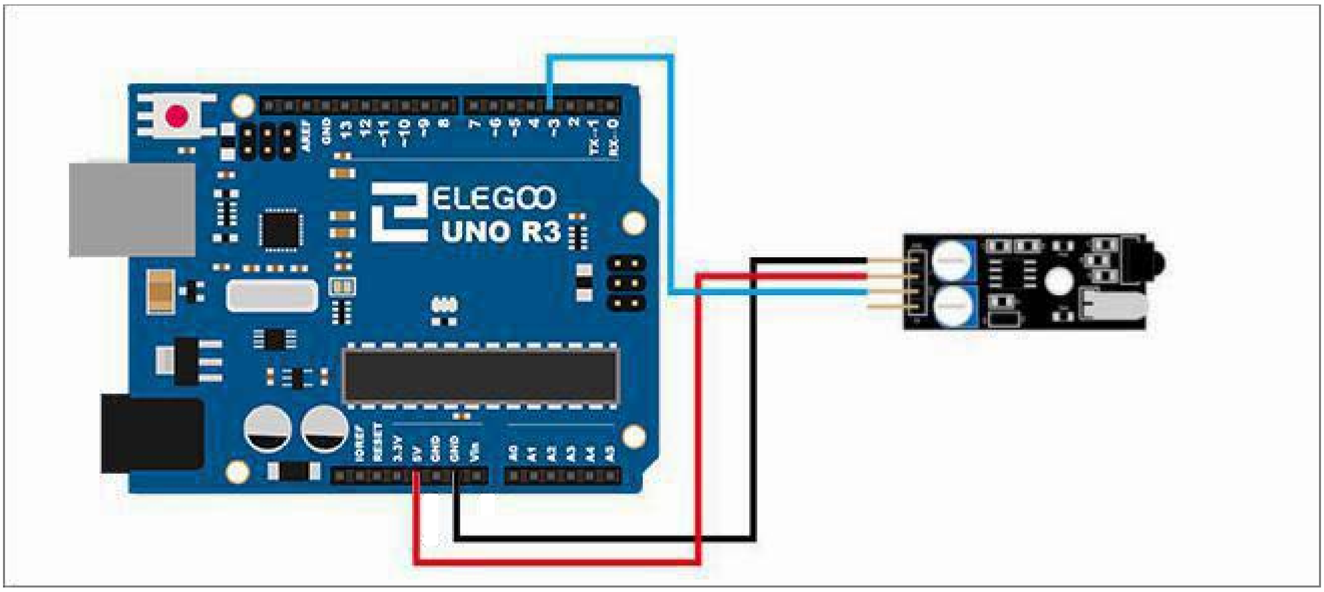
Connection

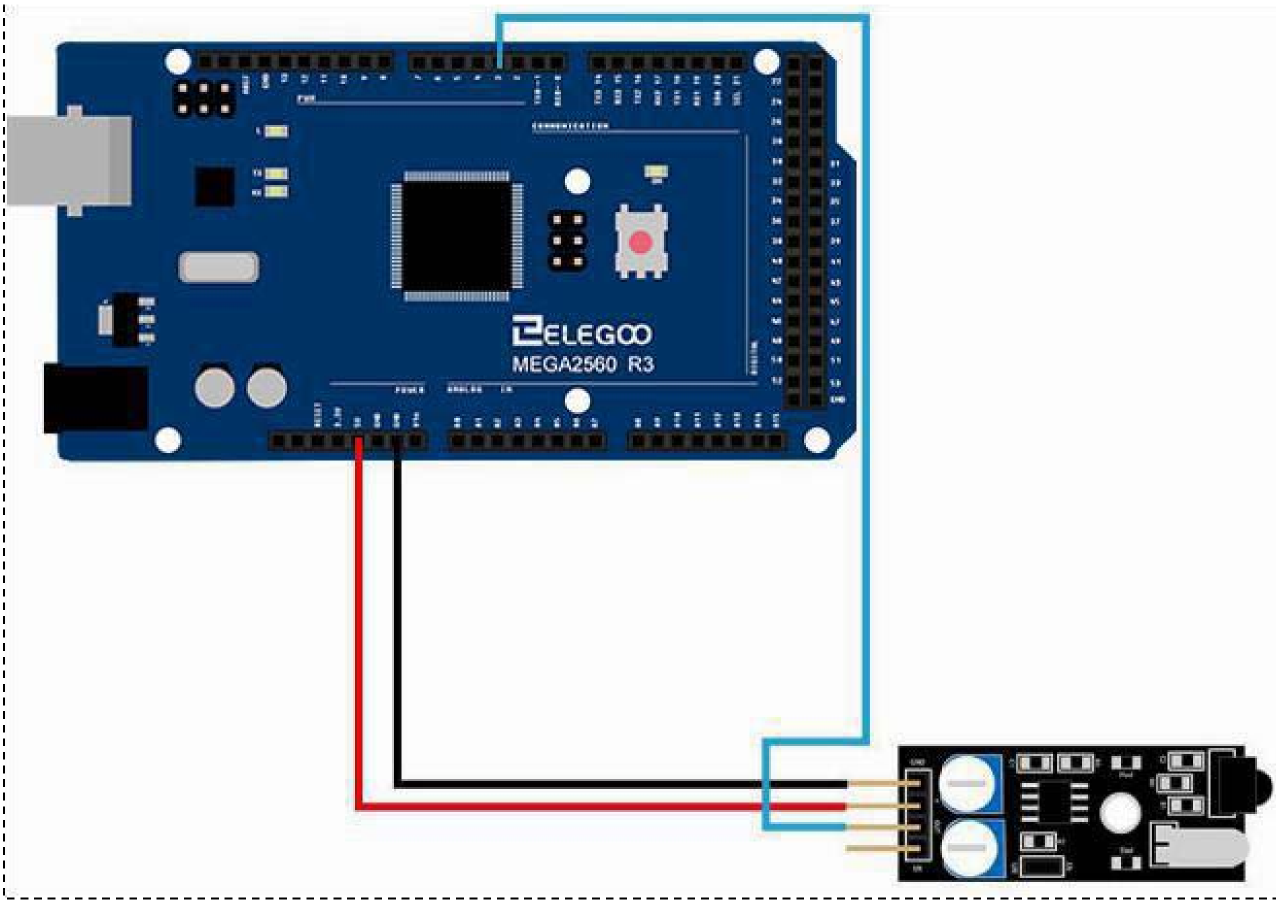
Schematic



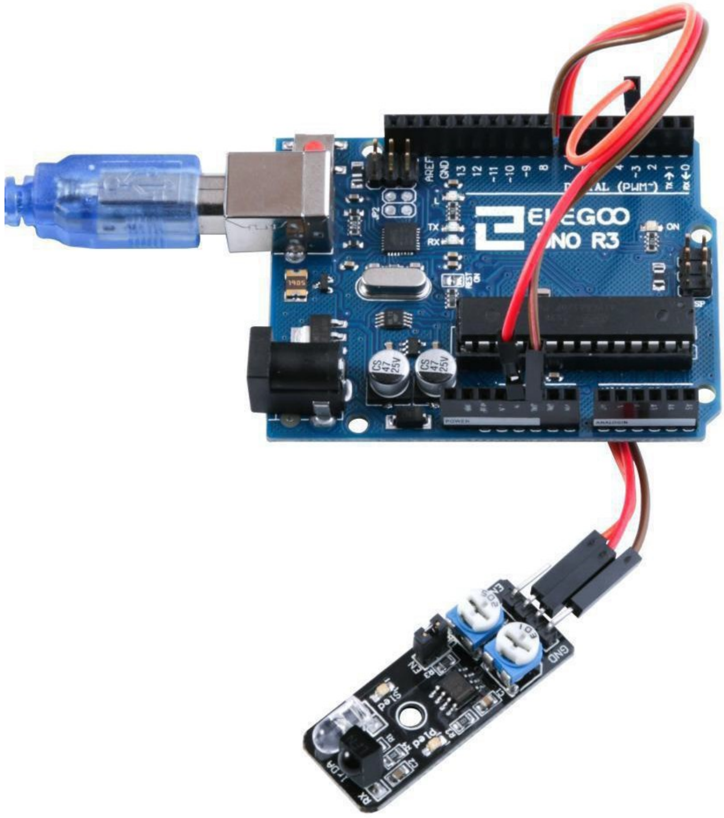


Wiring diagram





Result



Here we use the obstacle avoidance module and a digital interface, built-in 13 LED build a simple circuit, making avoidance warning lamp, the obstacle avoidance Sensor Access Digital 3 interface, when obstacle avoidance sensor senses a signal, LED light, and vice versa off.

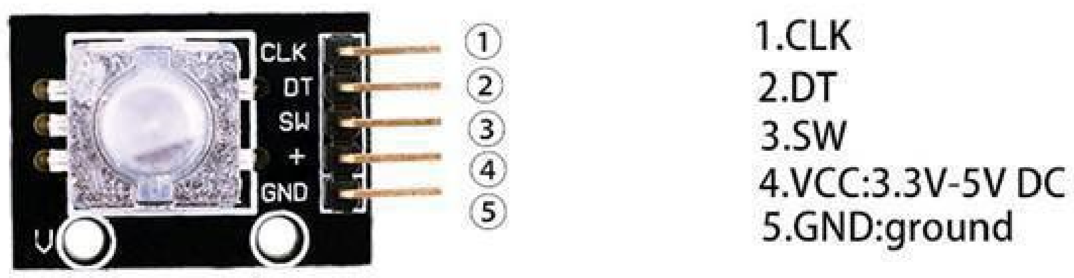
Lesson 26 Rotary Encode Module

Overview

In this experiment, we will learn how to use Rotary Encode Module

Rotary encoder

Rotary encoder useful for making an electronic pot etc. Connection identifications are Printed on the PCB.



Component Required:

(1) x Elegoo Uno R3

(1)x USB cable

(1) x Rotary Encode Module

(x) x F-M wires

Component Introduction

Rotary Encoders:

Mechanical Specifications:

- Operating Temp: -10°C to 70°C
- Storage Temp: -40°C to 85°C
- Rotational Torque: 50gf.cm max.
- No. and Pos. of detents: 12 detents (Step angle 30°±3°)
- Terminal Strength: A static load for 300gf.cm shall be applied to the tip of the terminals for 10 sec. in any direction
- Shaft push-pull strength: 5.1kgf
- Rotational life: 30,000 cycles

Note:

- RoHS Compliant

Electrical Specifications:

- Rating: 1mA/10VDC
- Insulation Resistance: 50VDC 10MΩ Min.
- Dielectric Strength: 50VAC for 1 min.
- Resolution: 12 pulses/360° for each phase

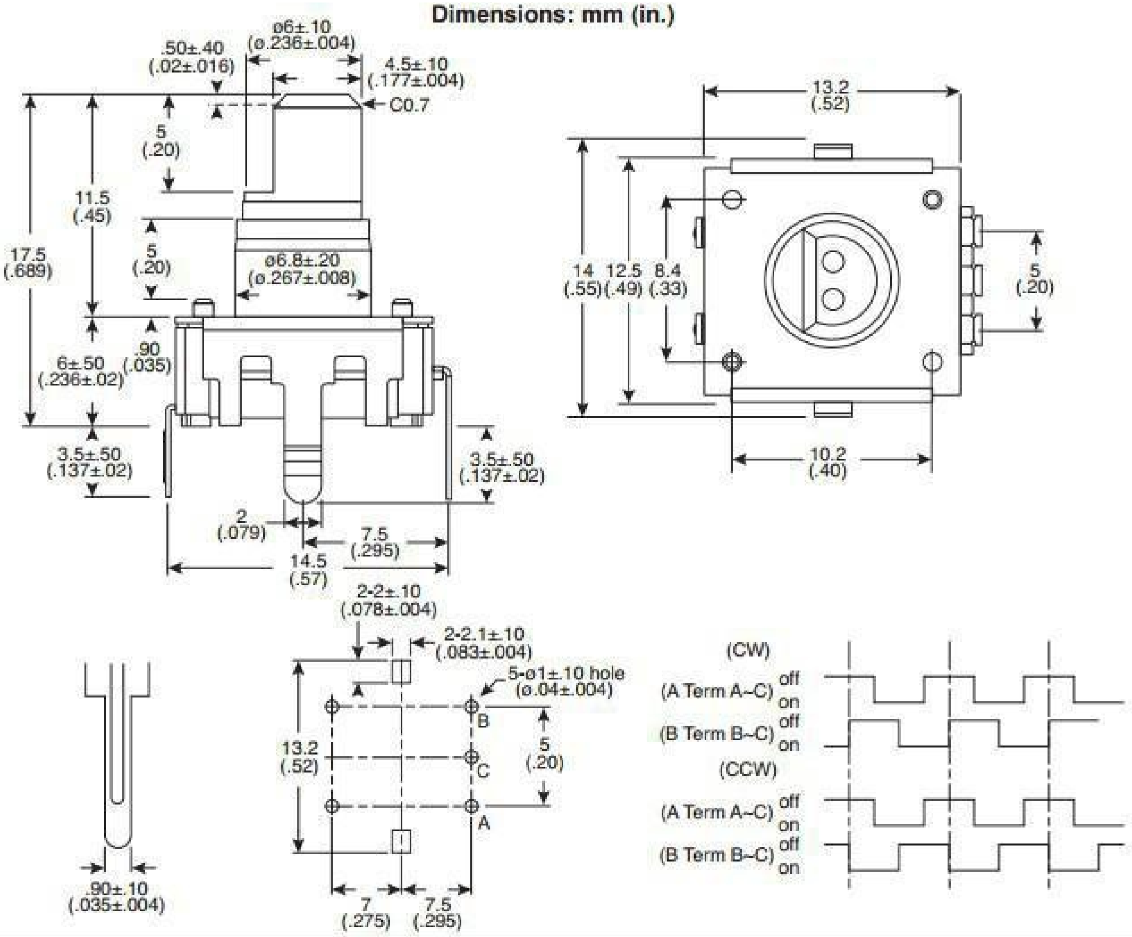
Soldering Specifications:

- Soldering: To be performed in 5 seconds within 260±5°C
- Manual Soldering: To be performed in 3 seconds within 350±5°C
- Preheating: The entire flow duration should not exceed 2 min., and soldering surface temperature (undersurface of PCB) shall be settled within 100°C

Push-on Switch Specifications:

- Type: Single Pole Single Throw (Push on)
- Rating: 10mA/5VDC
- Switch Travel (mm): 0.5±0.4
- Operating Force: 200~460gf
- Operating Life: 20,000 times

Dimensions: mm (in.)



Principle

Incremental encoder

Incremental encoders give two-phase square wave, the phase difference between them 90°, often referred to as A and B channels. One of the channels is given and speed-related Information, at the same time, by sequentially comparing two channel signals, the direction of rotation of the information obtained. There is also a special signal called Z or Zero channel, whichgives the absolute zero position encoder, the signal is a square wave with the center line of channel a square wave coincide.

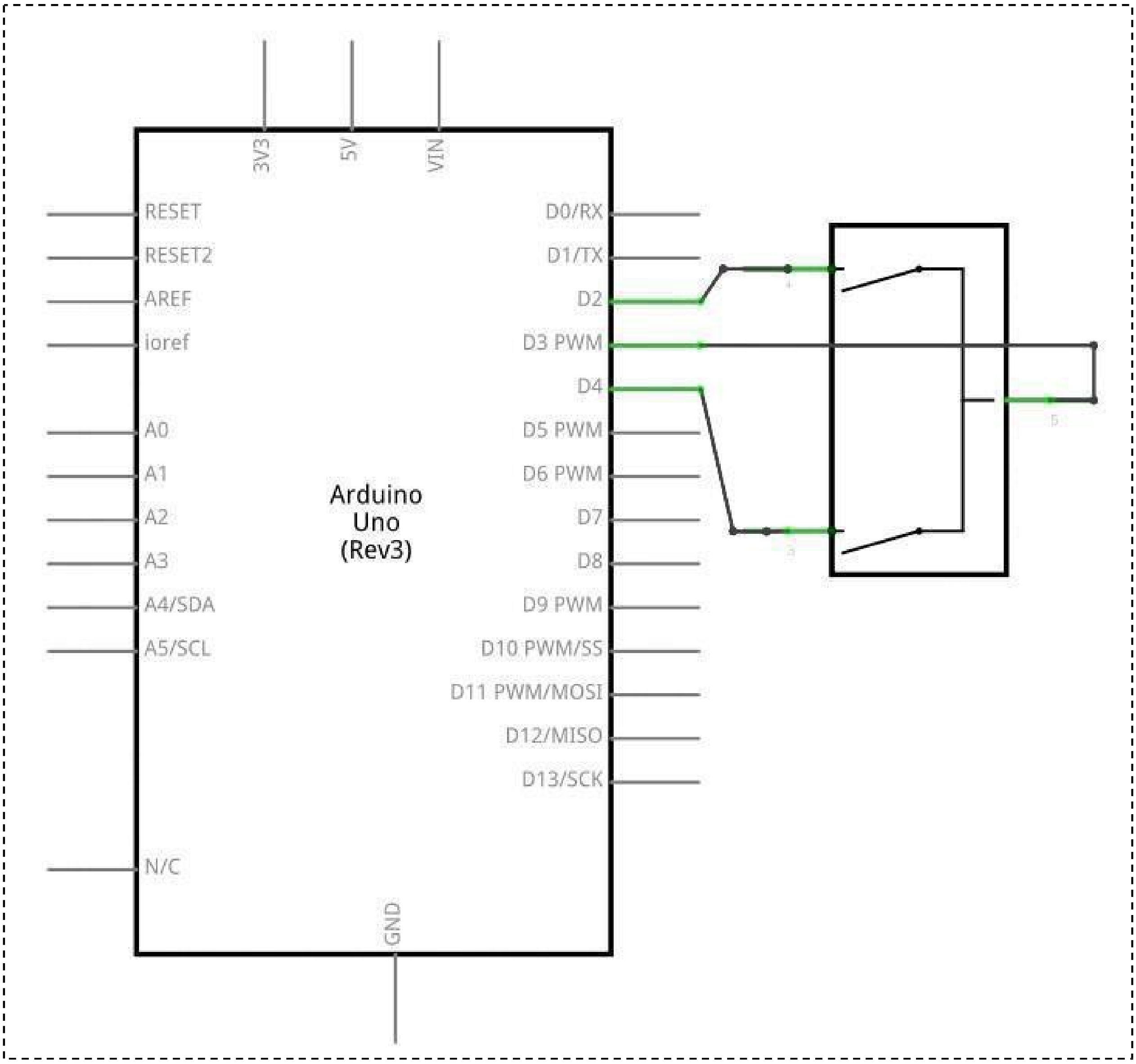
Clockwise counterclockwise

A	B
1	1
0	1
0	0
1	0
1	1
1	0
0	0
0	1

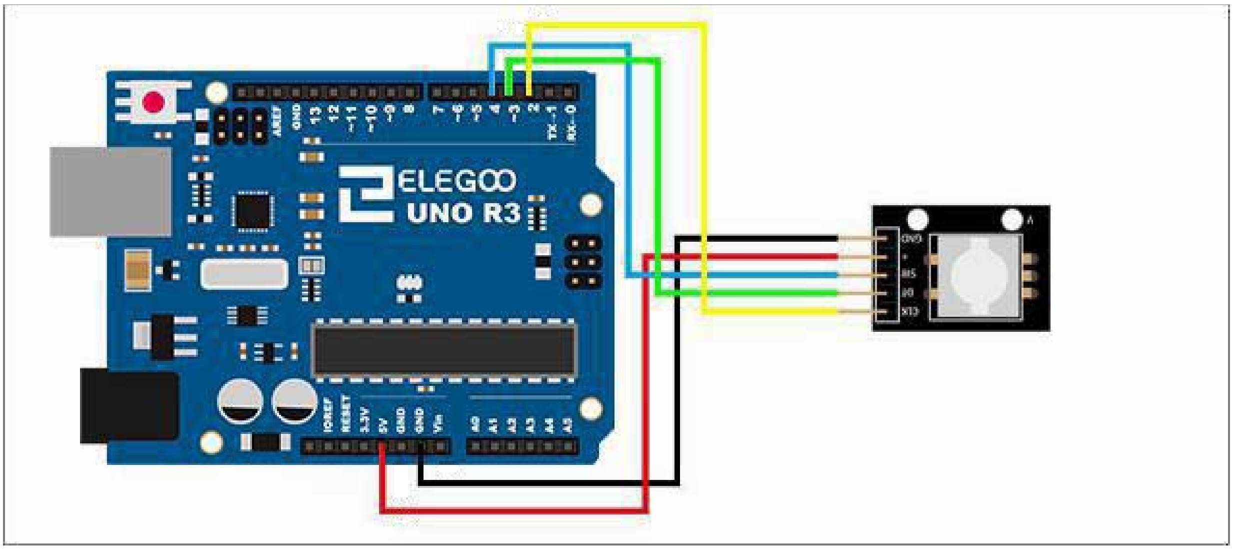
Incremental encoder accuracy depends on the mechanical and electrical two factors, these factors are: Raster indexing error, disc eccentricity, bearing eccentricity, e-reading Several means into the optical portion of the errors and inaccuracies. Determine the encoder resolution is measured in electrical degrees, the encoder accuracy depends Set the pulse encoder generates indexing. The following electrical degrees with a 360 ° rotation of the shaft to said machine, and rotation of the shaft must be a full week of Period. To know how much electrical equivalent of the mechanical angle of 360 degrees can be calculated with the following formula: Electrical 360 = Machine 360 ° / n ° pulses / revolution encoder indexing error is the electrical angle of the unit two successive pulse maximum offset to represent. Error exists in any encoder, which is caused by the aforementioned factors. Eltra encoder maximum error is ± 25 electrical degrees (declared in any condition), equivalent to the rated Offset values ± 7%, as the phase difference 90 ° (electrical) of the two channels of the maximum deviation ± 35 electrical degrees is equal to ±10% deviation left ratings Right.

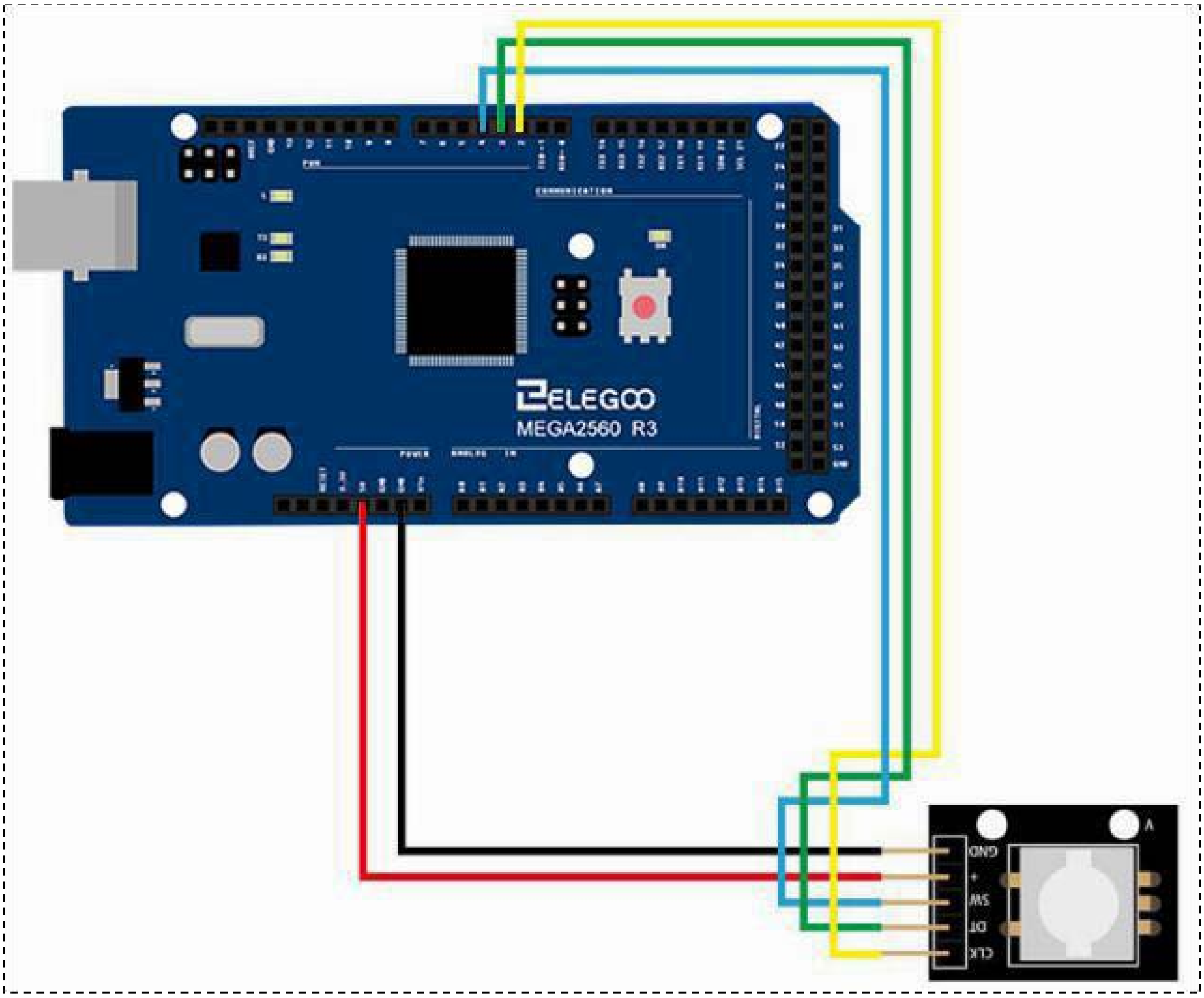
Connection

Schematic

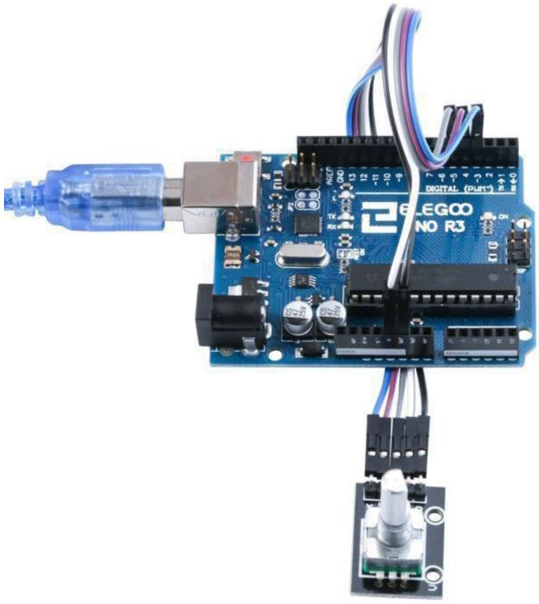


Wiring diagram

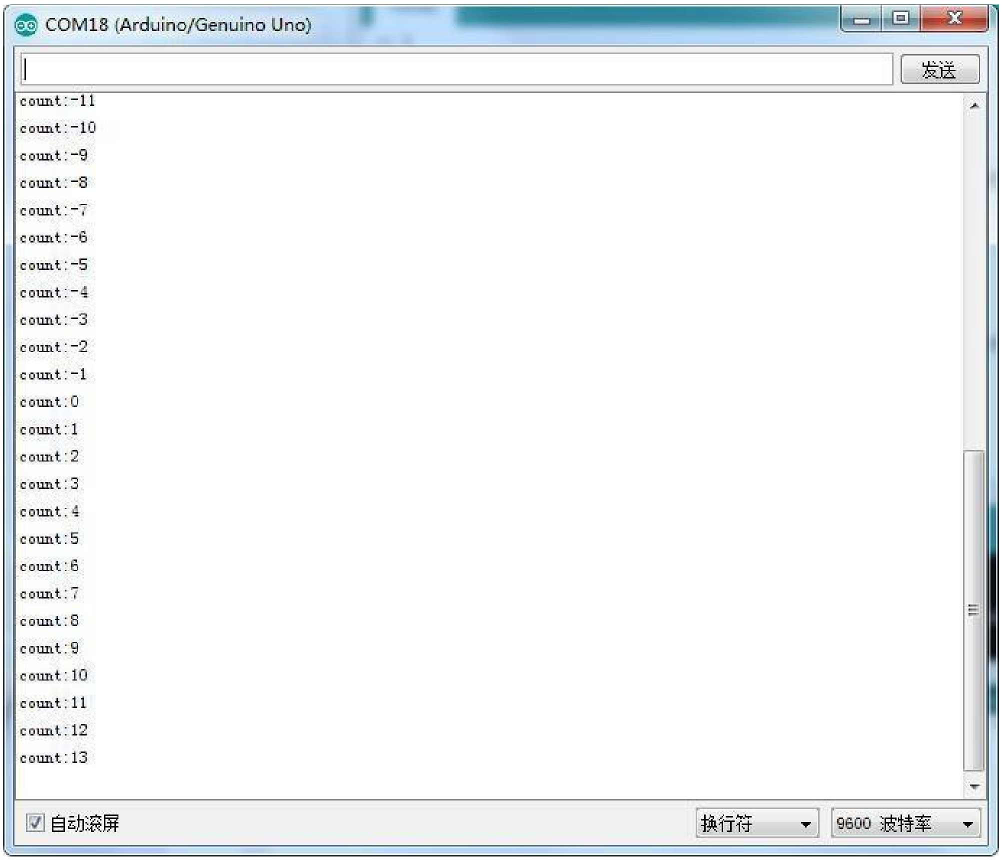




Result



Upload the program, rotate the encodes, and then open the monitor, and we can see the data as below:



Lesson 27 Relay Module

Overview

In this experiment, we will learn how to use the Relay Module.

Relay is a kind of component when the change of the input variables (incentive) to specified requirements, the output electric circuits of the charged amount occurs due to the step change of a kind of electrical appliances. This company produces the relay module can meet in 28 v to 240 v ac or dc power to control all kinds of other electric parts. MCU can be used to achieve the goal of timing control switch. Can be applied to guard against theft and alarm, toys, construction and other fields. Relay is an electrical control device. It has a control system (also called input circuit) and control system (also called the output circuit), the interaction between. Usually used in automatic control circuit, it is actuallywithasmallcurrenttocontrollargecurrentoperationofakindof"automatics.

Relay

A relay module suitable for direct connection to an Arduino board. The module requires 5 V power supply. The input control signal is identified with an 'S'. The relay has one change-over contact. It is capable of switching resistive loads up to 10 A at 250 V AC and up to 10 A at 30V maximum.
Don't forget to provide interference suppression for the switched load!

1.GND:ground
2.VCC:5V DC
3.OUTPUT

Component Required:

(1) x Elegoo Uno R3

(1)x USB cable

(1)x Relay Module

(x)x F-M wires

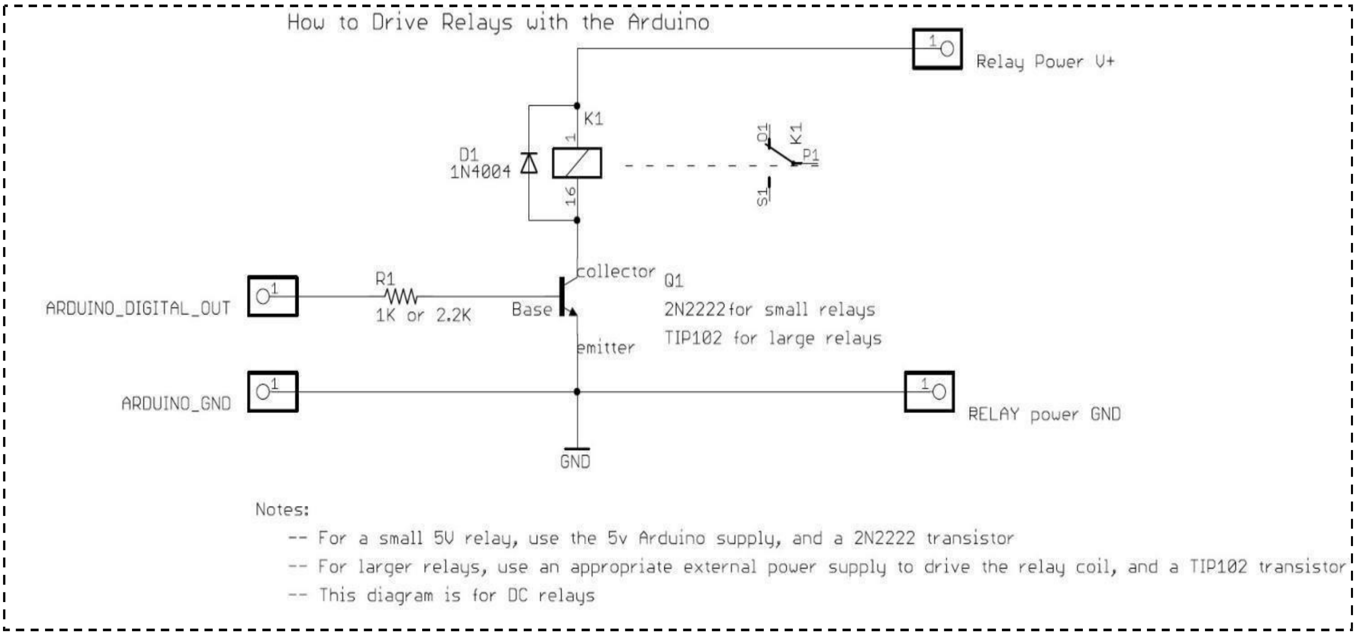
Component Introduction

Relay:

A relay is an electrically operated switch. Many relays use an electromagnet to mechanically operate a switch, but other operating principles are also used, such as solid-state relays. Relays are used where it is necessary to control a circuit by a low-power signal (with complete electrical isolation between control and controlled circuits), or where several circuits must be controlled by one signal. The first relays were used in long-distance telegraph circuits as amplifiers: they repeated the signal coming in from one circuit and re-transmitted it on another circuit. Relays were used extensively in telephone exchanges and early computers to perform logical operations.

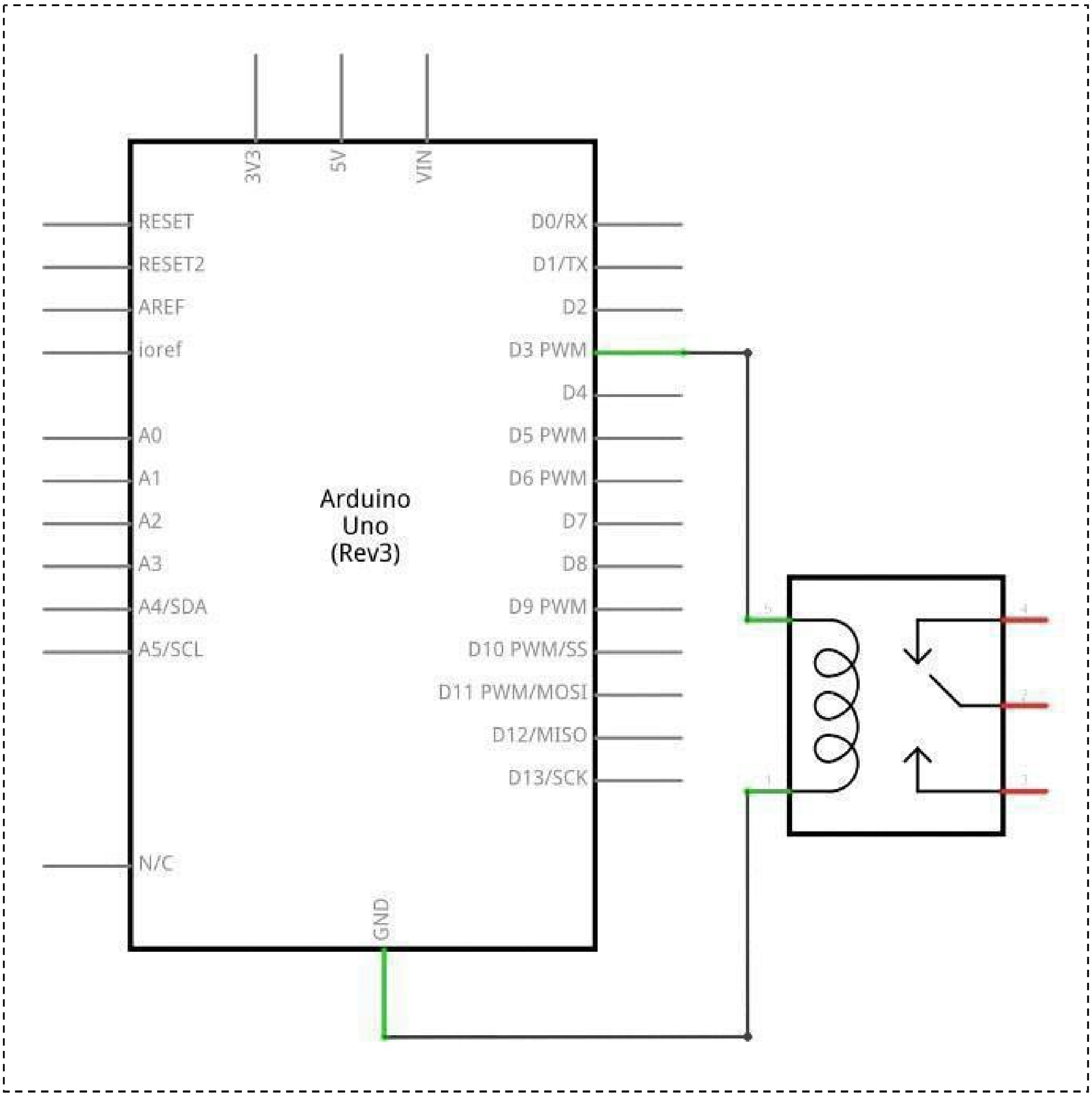
A type of relay that can handle the high power required to directly control an electric motor or other loads is called a contactor. Solid-state relays control power circuits with no moving parts, instead using a semiconductor device to perform switching. Relays with calibrated operating characteristics and sometimes multiple operating coils are used to protect electrical circuits from over load or faults; in modern electric power systems these functions are performed by digital instruments still called "protectiverelays".

Bellow is the schematic of how to drive relay with Arduino (download from the Arduino.cc)

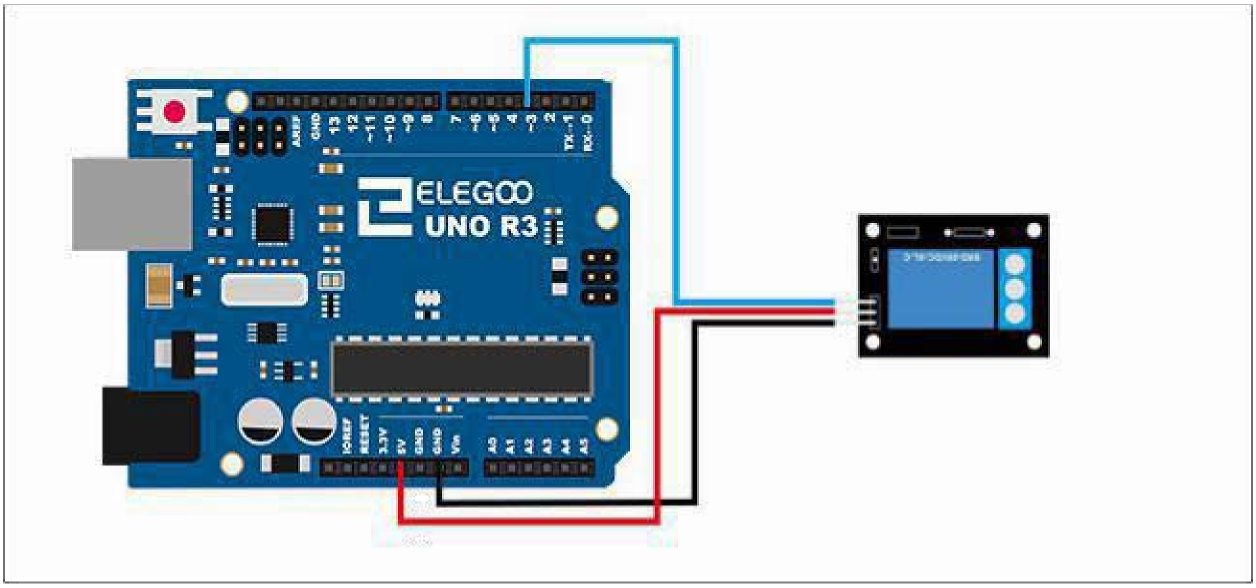


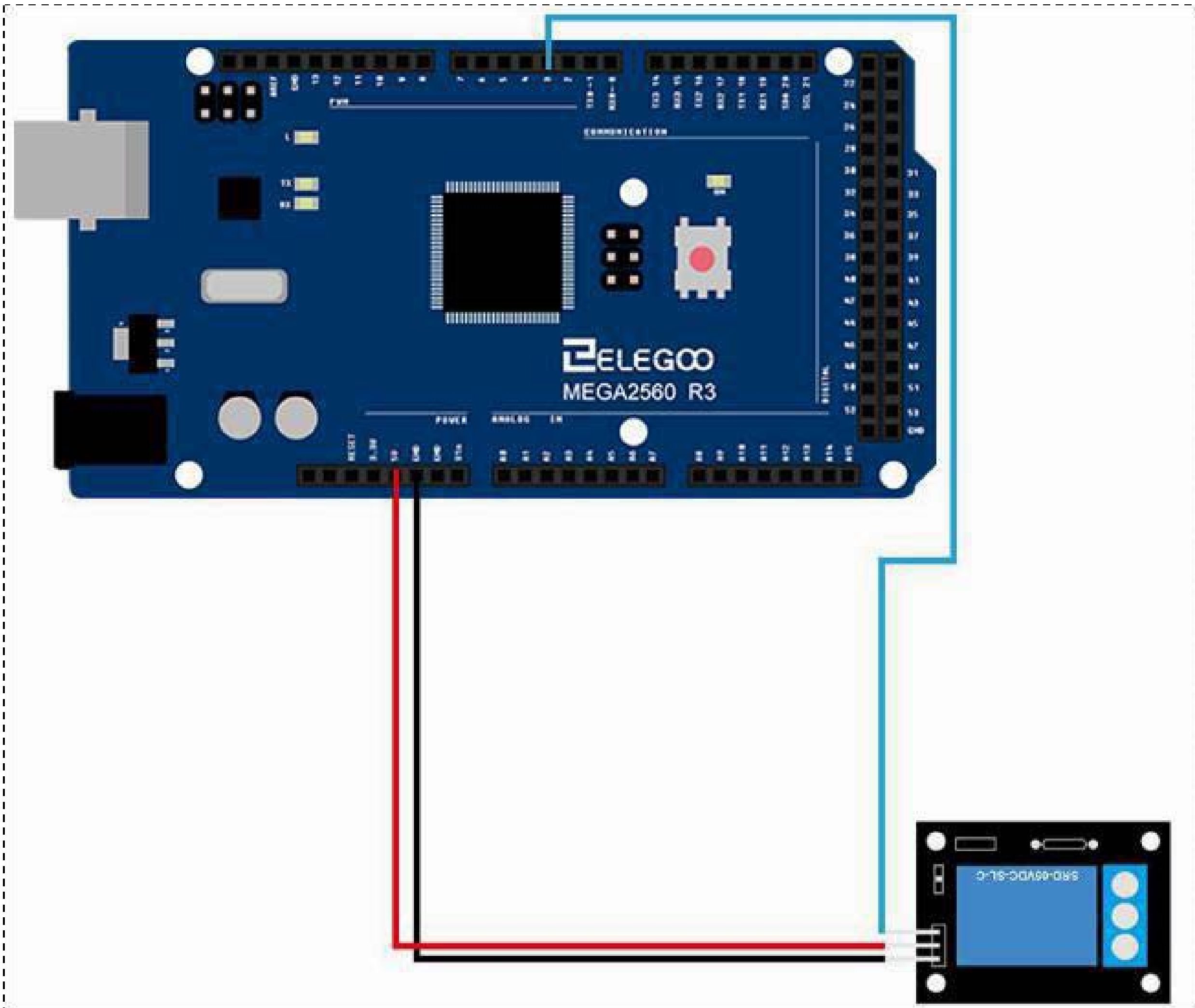
Connection

Schematic

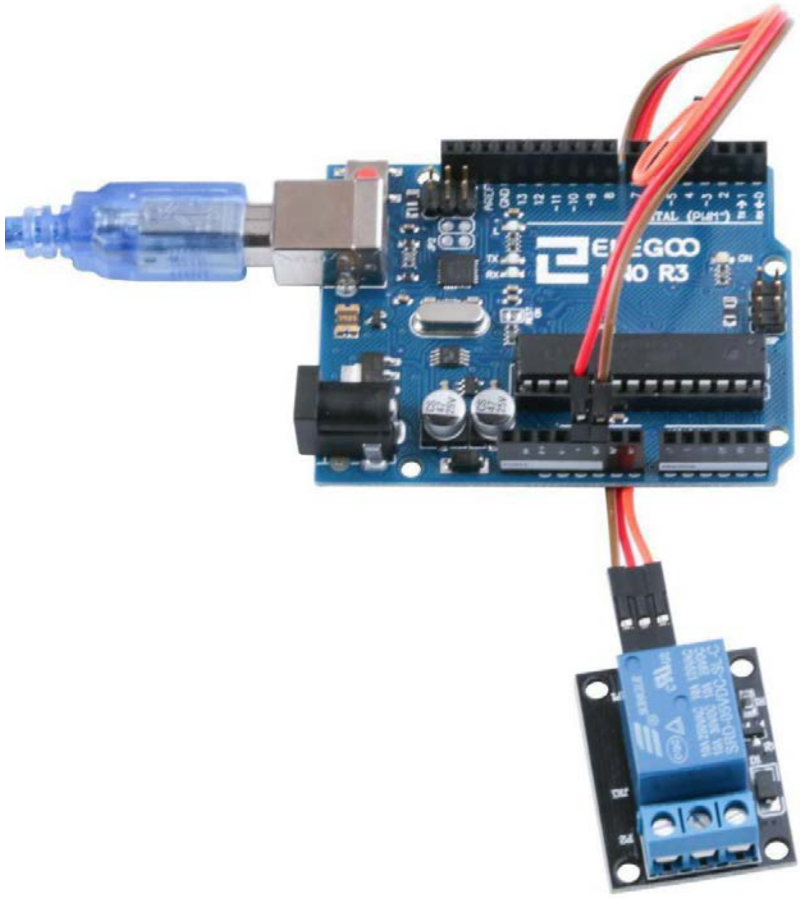


Wiring diagram


ELEGOO
UNO R3


ELEGOO
MEGA2560 R3

Result



Lesson 28 Heartbeat Sensor Module

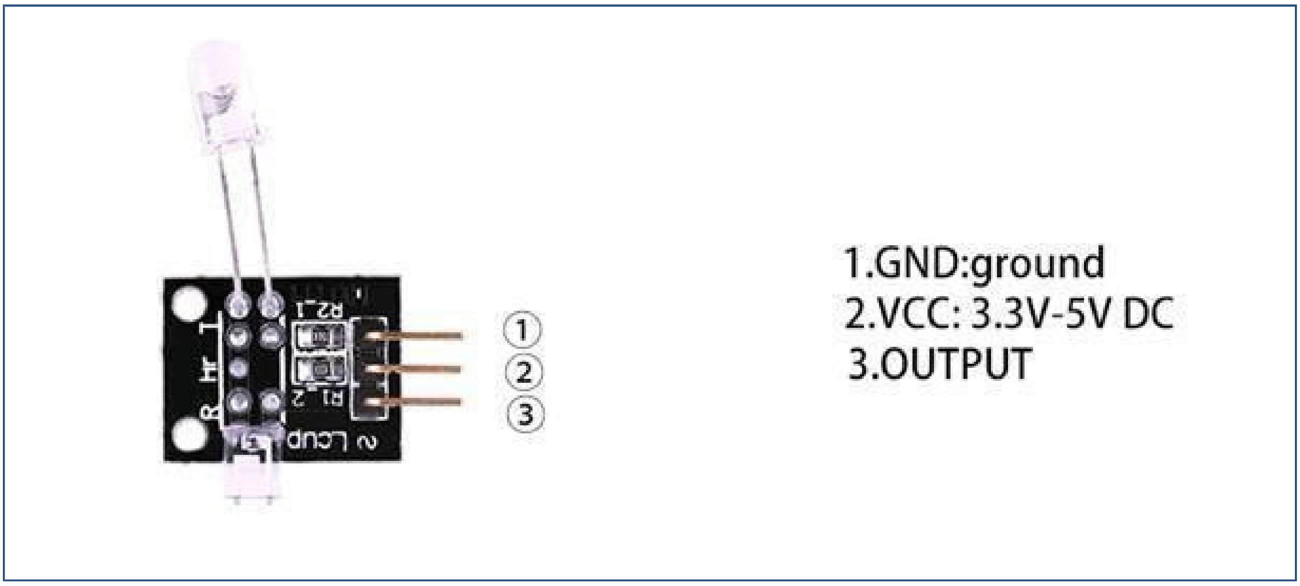
Overview

This project uses bright infrared (IR) LED and a phototransistor to detect the pulse of the finger, are LED flashes with each pulse. Pulse monitor works as follows: The LED is the light side of the finger, and photo transistor on the other side of the finger, phototransistor used to obtain the flux emitted, when the blood pressure pulse by the finger when the resistance of the photo transistor will be slightly changed. The project's schematic circuit as shown, we chose a very high resistance resistor R1, because most of the light through the finger is absorbed, it is desirable that the photo transistor is sensitive enough. Resistance can be selected by experiment to get the best results. The most important is to keep the shield stray light into the phototransistor. For home lighting that is particularly important because the lights at home mostly based 50 HZ or 60 HZ fluctuate, so far the heartbeat will add considerable noise.

When running the program the measured values are printed. To get a real heartbeat from this could be challenging.

Tilt Switch

Mercury tilt switch which makes or breaks depending on its attitude.



Component Required:

(1)xElegoo Uno R3

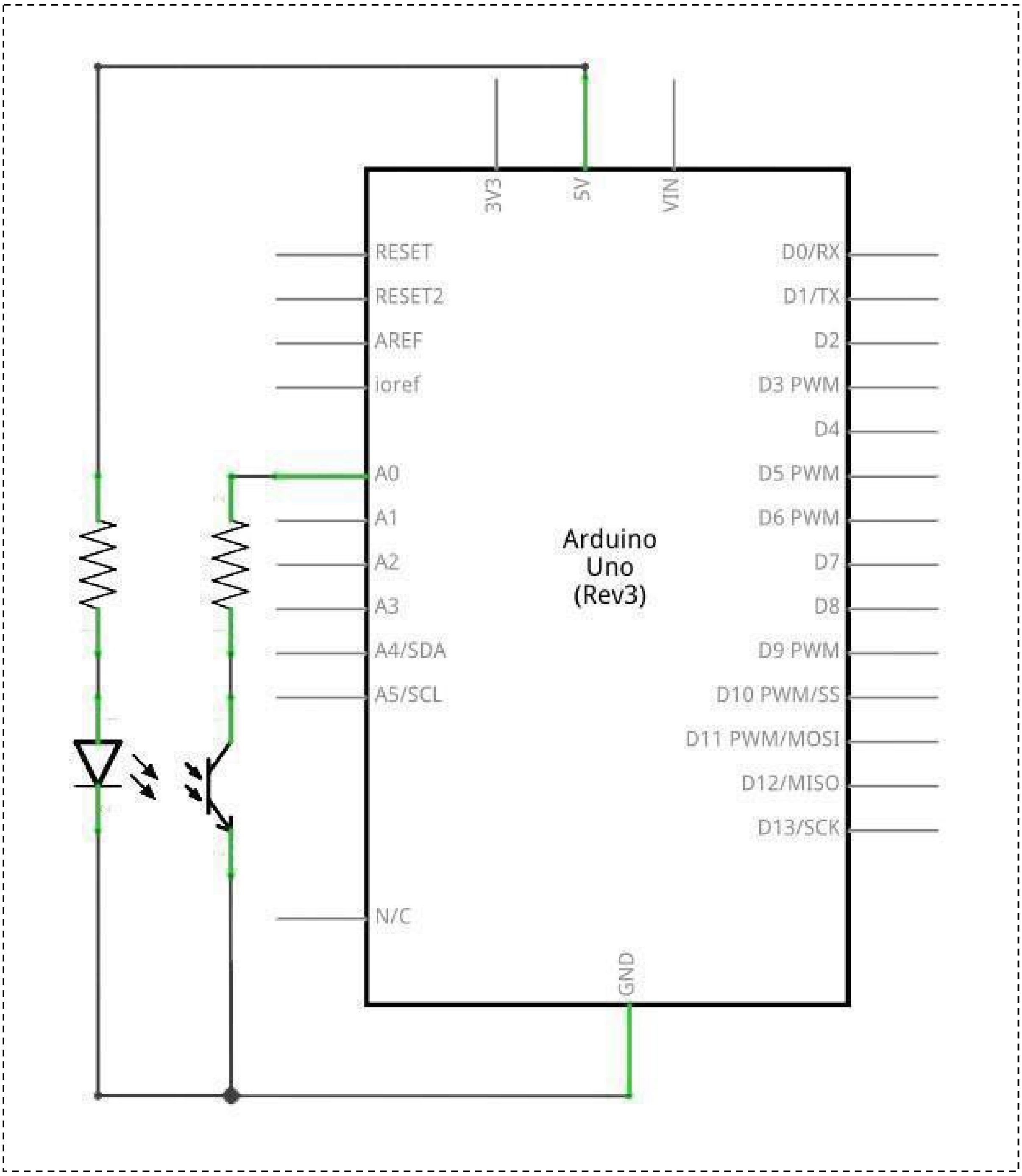
(1)x USB cable

(1)x Heartbeat Sensor

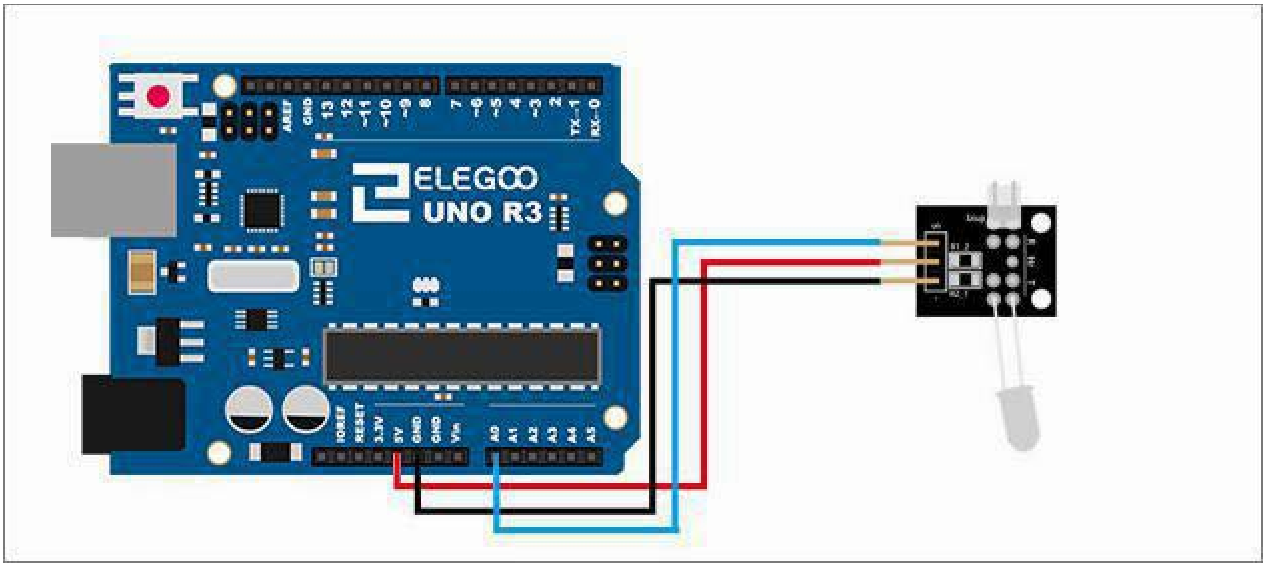
(x)x F-M wires

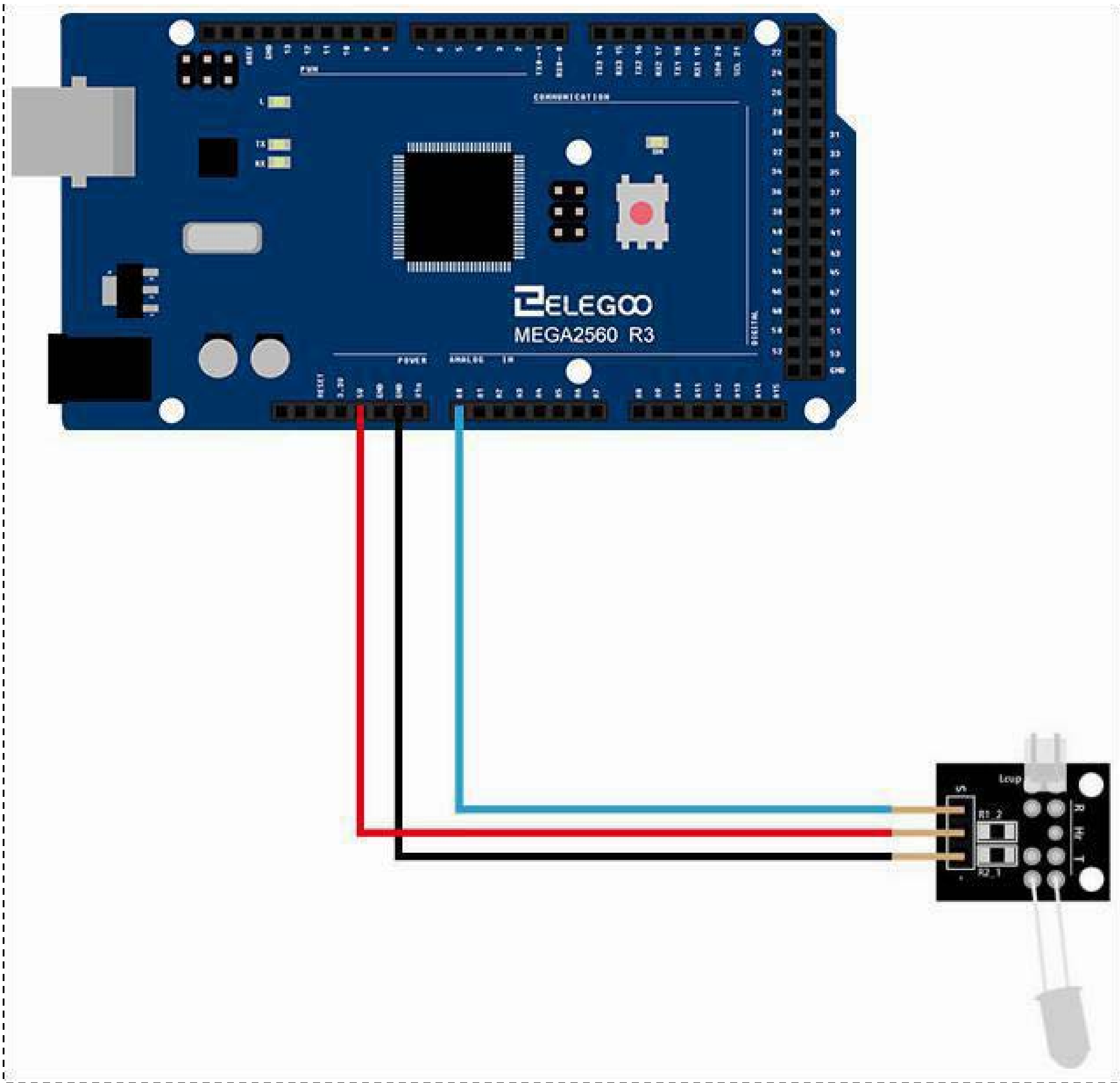
Connection

Schematic

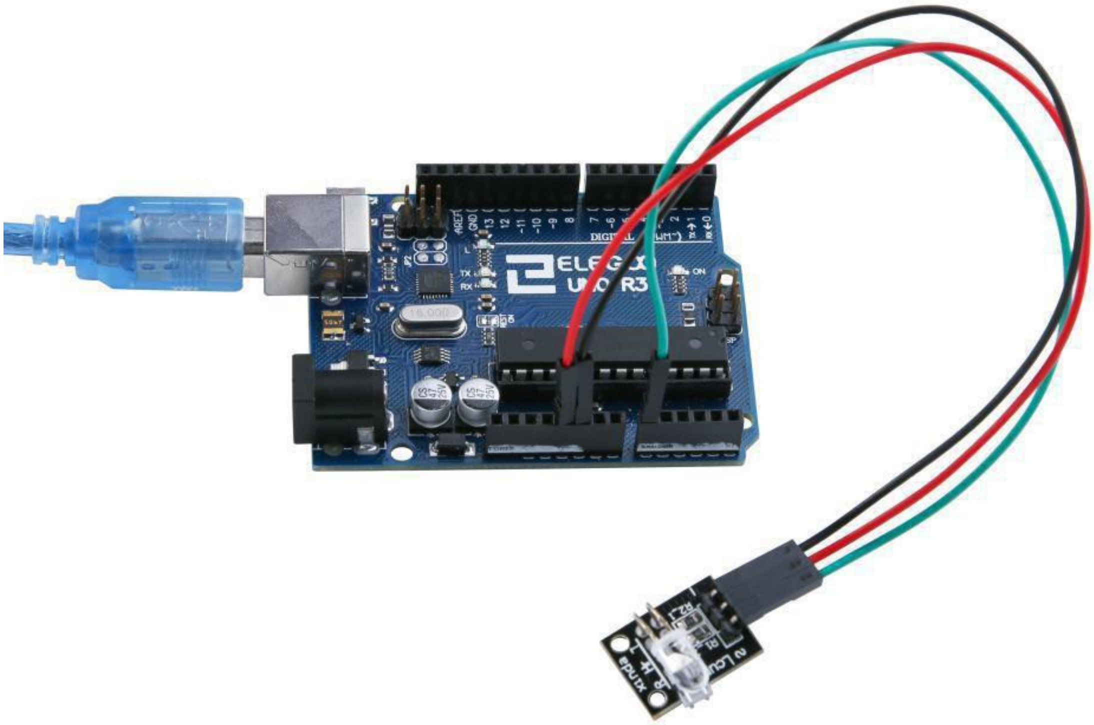


Wiring diagram

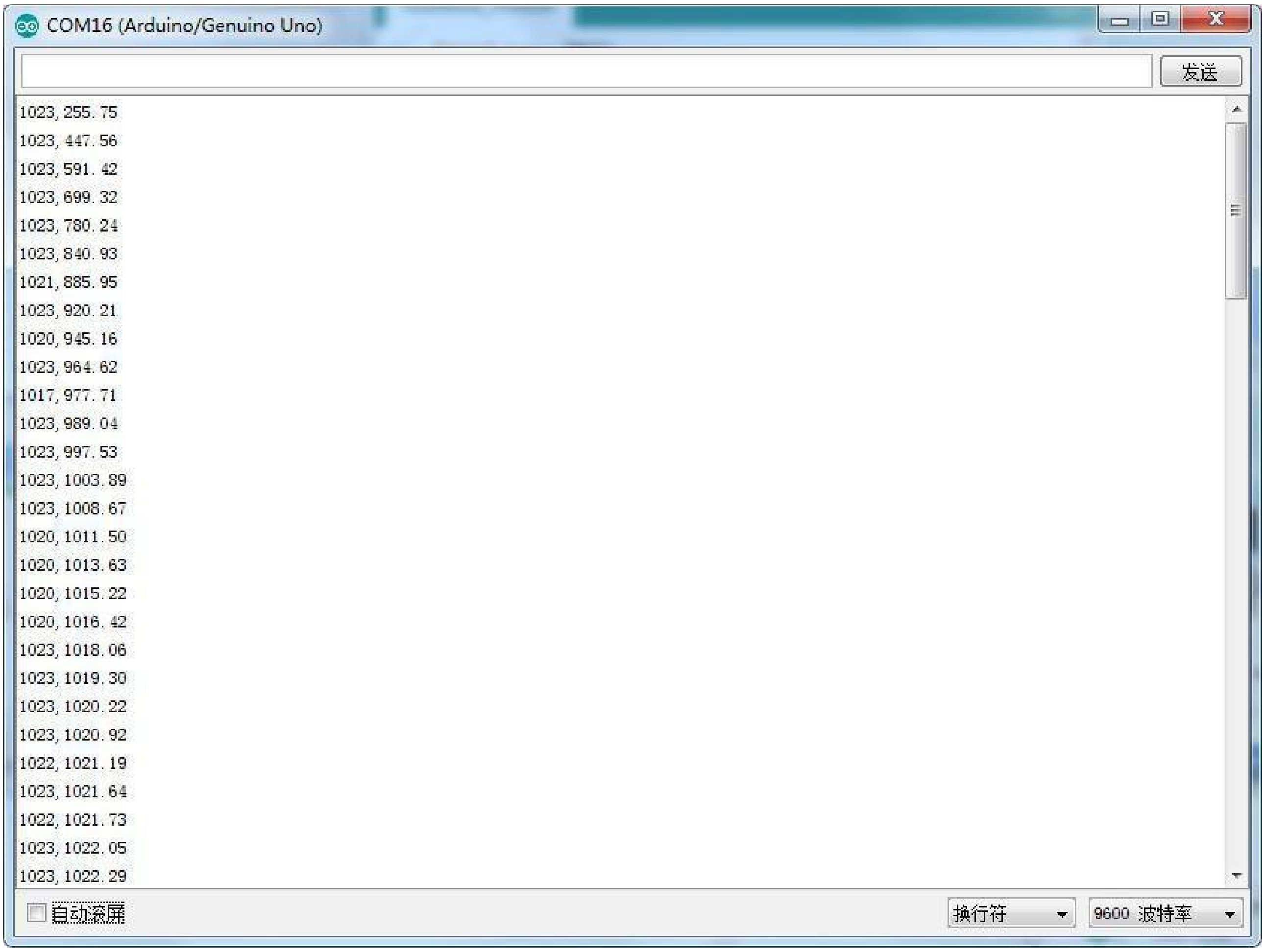




Result



Uploadthe program then open the monitor, we can see the data as below:


COM16 (Arduino/Genuino Uno)
发送
1023, 255. 75
1023, 447. 56
1023, 591. 42
1023, 699. 32
1023, 780. 24
1023, 840. 93
1021, 885. 95
1023, 920. 21
1020, 945. 16
1023, 964. 62
1017, 977. 71
1023, 989. 04
1023, 997. 53
1023, 1003. 89
1023, 1008. 67
1020, 1011. 50
1020, 1013. 63
1020, 1015. 22
1020, 1016. 42
1023, 1018. 06
1023, 1019. 30
1023, 1020. 22
1023, 1020. 92
1022, 1021. 19
1023, 1021. 64
1022, 1021. 73
1023, 1022. 05
1023, 1022. 29
自动滚屏
换行符
9600 波特率